

[54], but this becomes less important when a chemical reaction occurs, as in the case of the sulfur dioxide absorbers. Multistage countercurrent effects can be obtained by using several venturists [3]. The device is also used for removing dust particles from gases [12].

WETTED-WALL TOWERS

A thin film of liquid running down the inside of a vertical pipe, with gas flowing either cocurrently or countercurrently, constitutes a wetted-wall tower. Such devices have been used for theoretical studies of mass transfer, as described in Chap. 3, because the interfacial surface between the phases is readily kept under control and is measurable. Industrially, they have been used as absorbers for hydrochloric acid, where absorption is accompanied by a very large evolution of heat [63]. In this case the wetted-wall tower is surrounded with rapidly flowing cooling water. Multitube devices have also been used for distillation, where the liquid film is generated at the top by partial condensation of the rising vapor. Gas-pressure drop in these towers is probably lower than in any other gas-liquid contacting device, for a given set of operating conditions.

SPRAY TOWERS AND SPRAY CHAMBERS

The liquid can be sprayed into a gas stream by means of a nozzle which disperses the liquid into a fine spray of drops. The flow may be countercurrent, as in vertical towers with the liquid sprayed downward, or parallel, as in horizontal spray chambers (see Chap. 7). These devices have the advantage of low pressure drop for the gas but also have a number of disadvantages. There is a relatively high pumping cost for the liquid, owing to the pressure drop through the spray nozzle. The tendency for entrainment of liquid by the gas leaving is considerable, and mist eliminators will almost always be necessary. Unless the diameter/length ratio is very small, the gas will be fairly thoroughly mixed by the spray and full advantage of countercurrent flow cannot be taken. Ordinarily, however, the diameter/length ratio cannot be made very small since then the spray would quickly reach the walls of the tower and become ineffective as a spray.

PACKED TOWERS

Packed towers, used for continuous contact of liquid and gas in both countercurrent and cocurrent flow, are vertical columns which have been filled with packing or devices of large surface, as in Fig. 6.27. The liquid is distributed over, and trickles down through, the packed bed, exposing a large surface to contact the gas.

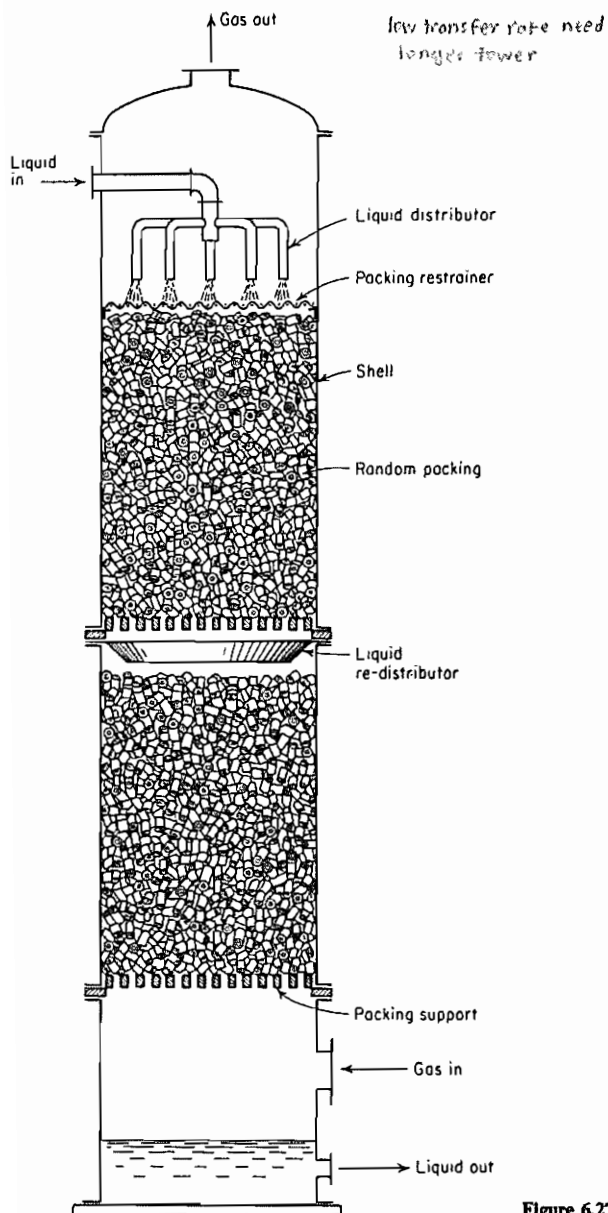


Figure 6.27 Packed tower.

Packing

The tower packing, or *fill*, should offer the following characteristics:

1. Provide for large interfacial surface between liquid and gas. The surface of the packing per unit volume of packed space a_p should be large but not in a microscopic sense. Lumps of coke, for example, have a large surface thanks to their porous structure, but most of this would be covered by the trickling film of liquid. The specific packing surface a_p in any event is almost always larger than the interfacial liquid-gas surface.
2. Possess desirable fluid-flow characteristics. This ordinarily means that the fractional void volume ϵ , or fraction of empty space, in the packed bed should be large. The packing must permit passage of large volumes of fluid through small tower cross sections without loading or flooding (see below) and with low pressure drop for the gas. Furthermore, gas-pressure drop should be largely the result of skin friction if possible, since this is more effective than form drag in promoting high values of the mass-transfer coefficients (see Wetted-wall towers).
3. Be chemically inert to fluids being processed.
4. Have structural strength to permit easy handling and installation.
5. Represent low cost.

Packings are of two major types, random and regular.

Random Packings

Random packings are simply dumped into the tower during installation and allowed to fall at random. In the past such readily available materials as broken stone, gravel, or lumps of coke were used, but although inexpensive, they are not desirable for reasons of small surface and poor fluid-flow characteristics. Random packings most frequently used at present are manufactured, and the common types are shown in Fig. 6.28. Raschig rings are hollow cylinders, as shown, of diameters ranging from 6 to 100 mm ($\frac{1}{4}$ to 4 in) or more. They may be made of chemical stoneware or porcelain, which are useful in contact with most liquids except alkalis and hydrofluoric acid; of carbon, which is useful except in strongly oxidizing atmospheres; of metals; or of plastics. Plastics must be especially carefully chosen, since they may deteriorate rapidly with certain organic solvents and with oxygen-bearing gases at only slightly elevated temperatures. Thin-walled metal and plastic packings offer the advantage of lightness in weight, but in setting floor-loading limits it should be anticipated that the tower may inadvertently fill with liquid. Lessing rings and others with internal partitions are less frequently used. The saddle-shaped packings, Berl and Intalox saddles, and variants of them, are available in sizes from 6 to 75 mm ($\frac{1}{4}$ to 3 in), made of chemical stoneware or plastic. Pall rings, also known as Flexirings, Cascade rings, and as a variant, Hy-Pak, are available in metal and plastic. Tellerettes are available in the shape shown and in several modifications, made

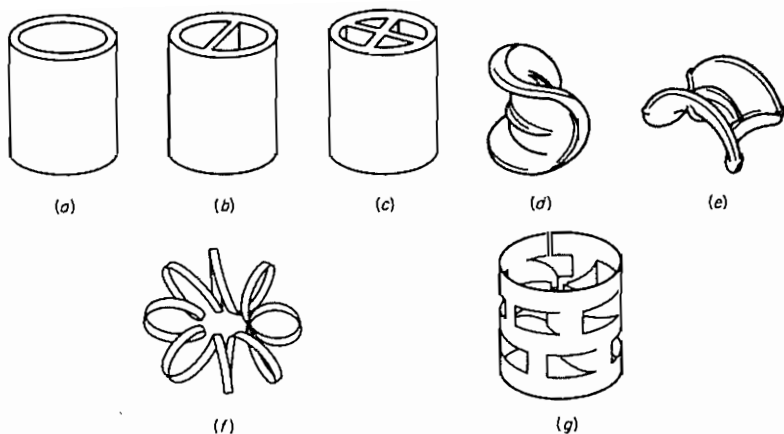


Figure 6.28 Some random tower packings: (a) Raschig rings, (b) Lessing ring, (c) partition ring, (d) Berl saddle (courtesy of Maurice A. Knight), (e) Intalox saddle (Chemical Processing Products Division, Norton Co.), (f) Tellerette (Ceilcote Company, Inc.), and (g) pall ring (Chemical Processing Products Division, Norton Co.).

of plastic. Generally the random packings offer larger specific surface (and larger gas-pressure drop) in the smaller sizes, but they cost less per unit volume in the larger sizes. As a rough guide, packing sizes of 25 mm or larger are ordinarily used for gas rates of $0.25 \text{ m}^3/\text{s}$ ($\approx 500 \text{ ft}^3/\text{min}$), 50 mm or larger for gas rates of $1 \text{ m}^3/\text{s}$ ($2000 \text{ ft}^3/\text{min}$). During installation the packings are poured into the tower to fall at random, and in order to prevent breakage of ceramic or carbon packings, the tower may first be filled with water to reduce the velocity of fall.

Regular Packings

These are of great variety. The counterflow trays already considered are a form of regular packing, as are the arrangements of Fig. 6.29. The regular packings offer the advantages of low pressure drop for the gas and greater possible fluid flow rates, usually at the expense of more costly installation than random packing. Stacked Raschig rings are economically practical only in very large sizes. There are several modifications of the expanded metal packings [105]. Wood grids, or *hurdles*, are inexpensive and frequently used where large void volumes are required, as with tar-bearing gases from coke ovens or liquids carrying suspended solid particles. Knitted or otherwise woven wire screen, rolled as a fabric into cylinders (Neo-Kloss) or other metal gauzelike arrangements (Koch-Sulzer, Hyperfil, and Goodloe packings) provide a large interfacial surface of contacted liquid and gas and a very low gas-pressure drop, especially useful for vacuum distillation [10, 77]. *Static mixers* were originally designed as *line mixers*, for mixing two fluids in cocurrent flow. There are several designs,

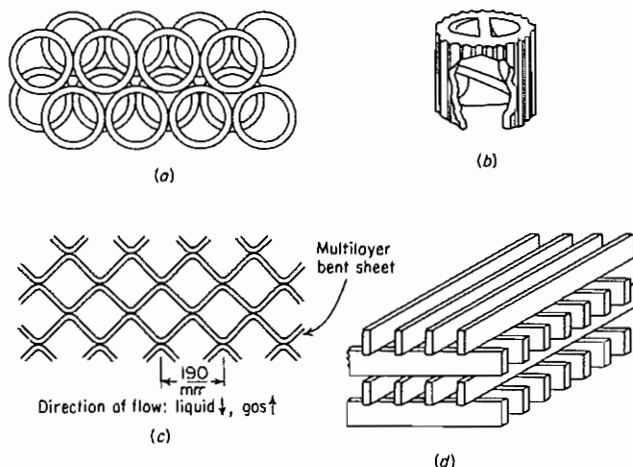


Figure 6.29 Regular, or stacked, packings: (a) Raschig rings, stacked staggered (top view), (b) double spiral ring (*Chemical Processing Products Division, Norton Co.*), (c) section through expanded-metal-lath packing, (d) wood grids.

but in general they consist of metal eggcratelike devices installed in a pipe to cause a multitude of splits of cocurrently flowing fluids into left- and right-hand streams, breaking each stream down into increasingly smaller streams [85]. These devices have been shown to be useful for countercurrent gas-liquid contact [121], with good mass-transfer characteristics at low gas-pressure drop.

Tower Shells

These may be of wood, metal, chemical stoneware, acidproof brick, glass, plastic, plastic- or glass-lined metal, or other material depending upon the corrosion conditions. For ease of construction and strength they are usually circular in cross section.

Packing Supports

An open space at the bottom of the tower is necessary for ensuring good distribution of the gas into the packing. Consequently the packing must be supported above the open space. The support must, of course, be sufficiently strong to carry the weight of a reasonable height of packing, and it must have ample free area to allow for flow of liquid and gas with a minimum of restriction. A bar grid, of the sort shown in Fig. 6.27, can be used [112], but specially designed supports which provide separate passageways for gas and liquid are preferred. Figure 6.30 shows one variety, whose free area for flow is of the order of 85 percent and which can be made in various modifications and of

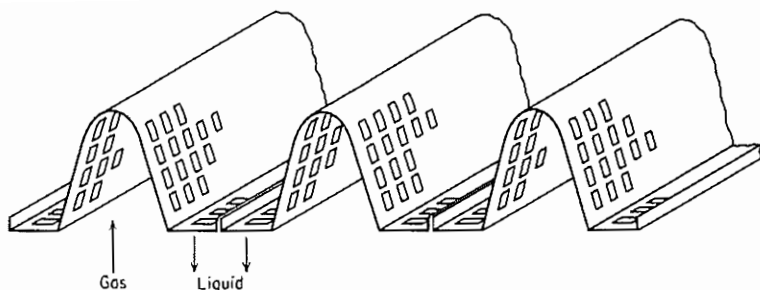


Figure 6.30 Multibeam support plate. (*Chemical Process Products Division, Norton Co.*)

many different materials including metals, expanded metals, ceramics, and plastics.

Liquid Distribution

The importance of adequate initial distribution of the liquid at the top of the packing is indicated in Fig. 6.31. Dry packing is of course completely ineffective for mass transfer, and various devices are used for liquid distribution. Spray nozzles generally result in too much entrainment of liquid in the gas to be useful. The arrangement shown in Fig. 6.27 or a ring of perforated pipe can be used in small towers. For large diameters, a distributor of the type shown in Fig. 6.32 can be used, and many other arrangements are available. It is generally considered necessary to provide at least five points of introduction of liquid for each 0.1 m^2 (1 ft^2) of tower cross section for large towers ($d \geq 1.2 \text{ m} = 4 \text{ ft}$) and a greater number for smaller diameters.

Random Packing Size and Liquid Redistribution

In the case of random packings, the packing density, i.e., the number of packing pieces per unit volume, is ordinarily less in the immediate vicinity of the tower

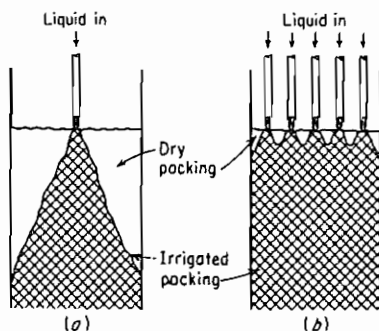


Figure 6.31 Liquid distribution and packing irrigation: (a) inadequate, (b) adequate.

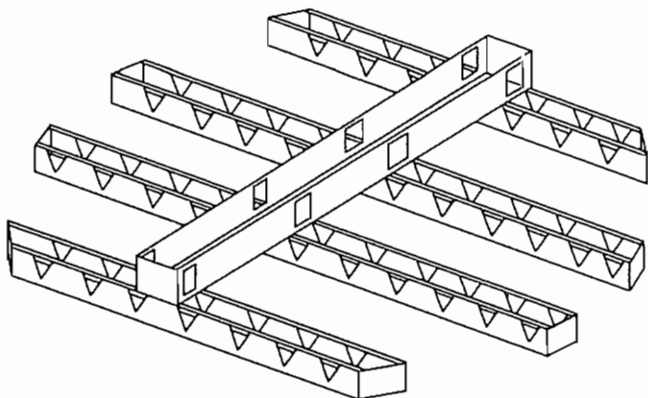


Figure 6.32 Weir-trough liquid distributor. (*Chemicals Process Products Division, Norton Co.*)

walls, and this leads to a tendency of the liquid to segregate toward the walls and the gas to flow in the center of the tower (channeling). This tendency is much less pronounced when the diameter of the individual packing pieces is smaller than at least one-eighth the tower diameter, but it is recommended that, if possible, the ratio $d_p/T = 1 : 15$. Even so it is customary to provide for redistribution of the liquid at intervals varying from 3 to 10 times the tower diameter, but at least every 6 or 7 m. Knitted mesh packings placed under a packing support (Fig. 6.30) make good redistributors [41]. With proper attention to liquid distribution, packed towers are successfully built to diameters of 6 or 7 m or more.

Packing Restrainers

These are necessary when gas velocities are high, and they are generally desirable to guard against lifting of packing during a sudden gas surge. Heavy screens or bars may be used. For heavy ceramic packing, heavy bar plates resting freely on the top of the packing may be used. For plastics and other lightweight packings, the restrainer is attached to the tower shell.

Entrainment Eliminators

Especially at high gas velocities, the gas leaving the top of the packing may carry off droplets of liquid as a mist. This can be removed by mist eliminators, through which the gas must pass, installed above the liquid inlet. A layer of mesh (of wire, Teflon, polyethylene, or other material) especially knitted with 98 to 99 percent voids, roughly 100 mm thick, will collect virtually all mist particles [18, 124]. Other types of eliminators include cyclones and venetian-blind arrangements [53]. A meter of dry random packing is very effective.

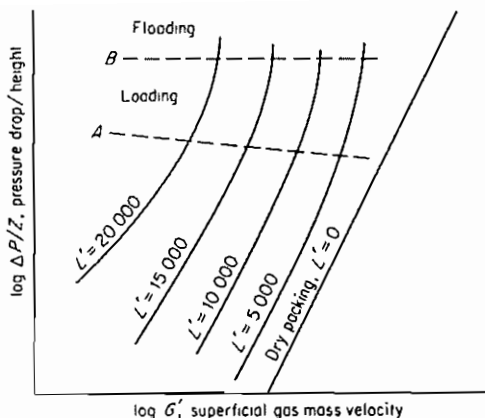


Figure 6.33 Typical gas pressure drop for counterflow of liquid and gas in random packings.

Countercurrent Flow of Liquid and Gas through Packing

For most random packings, the pressure drop suffered by the gas is influenced by the gas and liquid flow rates in a manner similar to that shown in Fig. 6.33. The slope of the line for dry packing is usually in the range 1.8 to 2.0, indicating turbulent flow for most practical gas velocities.

At a fixed gas velocity, the gas-pressure drop increases with increased liquid rate, principally because of the reduced free cross section available for flow of gas resulting from the presence of the liquid. In the region of Fig. 6.33 below *A*, the liquid holdup, i.e., the quantity of liquid contained in the packed bed, is reasonably constant with changing gas velocity, although it increases with liquid rate. In the region between *A* and *B*, the liquid holdup increases rapidly with gas rate, the free area for gas flow becomes smaller, and the pressure drop rises more rapidly. This is known as *loading*. As the gas rate is increased to *B* at fixed liquid rate, one of a number of changes occurs: (1) a layer of liquid, through which the gas bubbles, may appear at the top of the packing; (2) liquid may fill the tower, starting at the bottom or at any intermediate restriction such as a packing support, so that there is a change from gas-continuous liquid-dispersed to liquid-continuous gas-dispersed (inversion); or (3) slugs of foam may rise rapidly upward through the packing. At the same time, entrainment of liquid by the effluent gas increases rapidly, and the tower is *flooded*. The gas pressure drop then increases very rapidly. The change in conditions in the region *A* to *B* of Fig. 6.33 is gradual, and initial loading and flooding are frequently determined by the change in slope of the pressure-drop curves rather than through any visible effect. It is not practical to operate a tower in a flooded condition; most towers operate just below, or in the lower part of, the loading region.

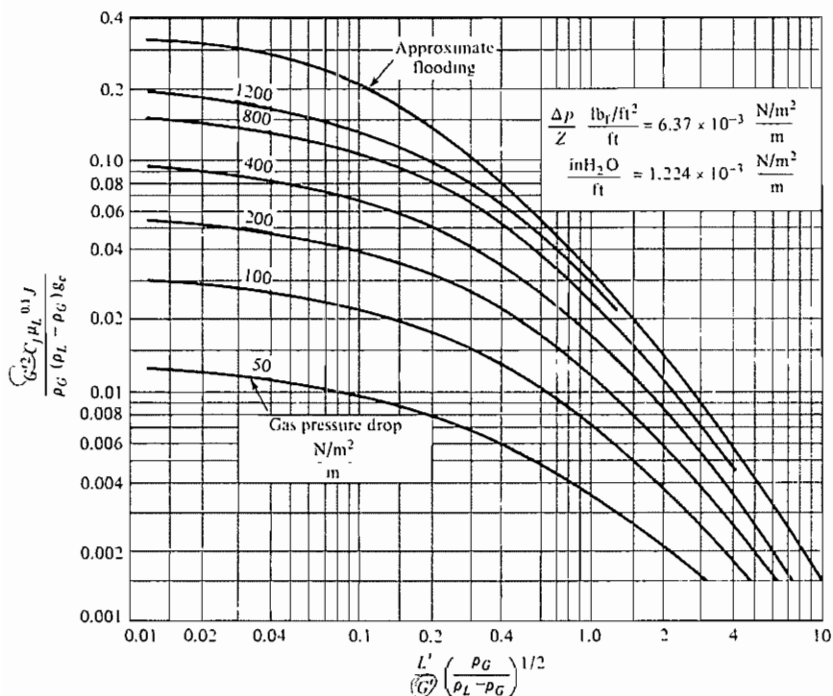


Figure 6.34 Flooding and pressure drop in random-packed towers. For SI units $g_c = 1$, C_f from Table 6.3, and use $J = 1$. For $G' = \text{lb}/\text{ft}^2 \cdot \text{h}$, $\rho = \text{lb}/\text{ft}^3$, $\mu_L = \text{cP}$, $g_c = 4.18 \times 10^8$, C_f from Table 6.3, and use $J = 1.502$. [Coordinates of Eckert [38], Chemical Process Products Division, Norton Co.]

Flooding and Loading

Flooding conditions in random packings depend upon the method of packing (wet or dry) and settling of the packing [76]. The upper curve of Fig. 6.34 correlates the flooding data for most random packings reasonably well. More specific data are in the handbooks [47] or are available from the manufacturers. The limit of loading cannot readily be correlated.

Typically, absorbers and strippers are designed for gas-pressure drops of 200 to 400 N/m^2 per meter of packed depth (0.25 to 0.5 $\text{inH}_2\text{O}/\text{ft}$), atmospheric-pressure fractionators from 400 to 600 (N/m^2)/m, and vacuum stills for 8 to 40 (N/m^2)/m (0.01 to 0.05 $\text{inH}_2\text{O}/\text{ft}$) [38]. Values of C_f which characterize the packings are given in Table 6.3 (these and other values in Table 6.3 change with changes in manufacturing procedures, so that the manufacturers should be consulted before completing final designs). Flooding velocities for regular or stacked packings will generally be considerably greater than for random packing.

Table 6.3 Characteristics of random packings†

Packing	Nominal size, mm (in)										
	6 (1/4)	9.5 (3/8)	13 (1/2)	16 (5/8)	19 (3/4)	25 (1)	32 (1 1/4)	38 (1 1/2)	50 (2)	76 (3)	89 (3 1/2)
Raschig ring											
Ceramic:											
Wall thickness, mm	0.8	1.6	2.4	2.4	2.4	3	4.8	4.8	6	9.5	
C_f	1600	1000	580	380	255	155	125	95	65	37	
C_D			909	749	457	301		181.8	135.6		
ϵ	0.73	0.68	0.63	0.68	0.73	0.73	0.74	0.71	0.74	0.78	
a_p , m ² /m ³ (ft ² /ft ³)	787 (240)	508 (155)	364 (111)	328 (100)	262 (80)	190 (58)	148 (45)	125 (38)	92 (28)	62 (19)	
Metal:											
0.8-mm wall:											
C_f	700	390	300	170	155	115					
ϵ	0.69		0.84		0.88	0.92					
a_p , m ² /m ³ (ft ² /ft ³)	774 (236)		420 (128)		274 (83.5)	206 (62.7)					
1.6-mm wall:											
C_f			410	290	220	137	110	83	57	32	
C_D			688	431	485	304		172.9	133.5		
ϵ			0.73		0.78	0.85	0.87	0.90	0.92	0.95	
a_p , m ² /m ³ (ft ² /ft ³)			387 (118)		236 (71.8)	186 (56.7)	162 (49.3)	135 (41.2)	103 (31.4)	68 (20.6)	

Pall rings

[illegible]

Table 6.3 Continued

Packing	Nominal size, mm (in)										
	6 ($\frac{1}{4}$)	9.5 ($\frac{3}{8}$)	13 ($\frac{1}{2}$)	16 ($\frac{3}{4}$)	19 ($\frac{7}{8}$)	25 (1)	32 (1 $\frac{1}{4}$)	38 (1 $\frac{1}{2}$)	50 (2)	76 (3)	89 (3 $\frac{1}{2}$)
Berl saddles											
Ceramic:											
C_f	900		240		170	110		65	45		
C_D			508		295	184					
ϵ	0.60		0.63		0.66	0.69		0.75	0.72		
q_p , m ² /m ³ (ft ² /ft ³)	899 (274)		466 (142)		269 (82)	249 (76)		144 (44)	105 (32)		
Intalox saddles											
Ceramic:											
C_f	725	330	200		145	98		52	40	22	
C_D			399		256	241.5		96.2	71.3	40.6	
ϵ	0.75		0.78		0.77	0.775		0.81	0.79		
q_p , m ² /m ³ (ft ² /ft ³)	984 (300)		623 (190)		335 (102)	256 (78)		195 (59.5)	118 (36)		
Plastic:											
C_f						33			21	16	
C_D						96.7			56.5	30.1	
ϵ						0.91			0.93	0.94	
q_p , m ² /m ³ (ft ² /ft ³)						207 (63)			108 (33)	89 (27)	

Super Intalox

Ceramic:†									
C_f									
C_D									
ϵ									
$a_p, \text{m}^2/\text{m}^3 (\text{ft}^2/\text{ft}^3)$									
Plastic:†									
C_f									
C_D									
ϵ									
$a_p, \text{m}^2/\text{m}^3 (\text{ft}^2/\text{ft}^3)$									

Tellerettes

Plastic:									
C_f									
ϵ									
$a_p, \text{m}^2/\text{m}^3 (\text{ft}^2/\text{ft}^3)$									

† Data are for wet-dumped packing from Chemical Process Products Division, Norton Co.; Koch Engineering Co.; Celcote Co.; Chemical Engineering Progress, and Chemical Engineering.

‡ Sizes, 1, 2, and 3.

Pressure Drop for Single-Phase Flow

The pressure drop suffered by a single fluid in flowing through a bed of packed solids such as spheres, cylinders, gravel, sand, etc., when it alone fills the voids in the bed, is reasonably well correlated by the Ergun equation [42]

$$\frac{\Delta p}{Z} \frac{g_c \varepsilon^3 d_p \rho_g}{(1 - \varepsilon) G'^2} = \frac{150(1 - \varepsilon)}{\text{Re}} + 1.75 \quad (6.66)$$

This is applicable equally well to flow of gases and liquids. The term on the left is a friction factor. Those on the right represent contributions to the friction factor, the first for purely laminar flow, the second for completely turbulent flow. There is a gradual transition from one type of flow to the other as a result of the diverse character of the void spaces, the two terms of the equation changing their relative importance as the flow rate changes. $\text{Re} = d_p G' / \mu$, and d_p is the effective diameter of the particles, the diameter of a sphere of the same surface/volume ratio as the packing in place. If the specific surface is a_p , the surface per unit volume of the particles is $a_p / (1 - \varepsilon)$, and from the properties of a sphere

$$d_p = \frac{6(1 - \varepsilon)}{a_p} \quad (6.67)$$

This will not normally be the same as the nominal size of the particles.

For flow of gases at G' greater than about $0.7 \text{ kg/m}^2 \cdot \text{s}$ ($500 \text{ lb/ft}^2 \cdot \text{h}$), the first term on the right of Eq. (6.66) is negligible. For a specific type and size of manufactured tower packing, Eq. (6.66) can then be simplified to the empirical expression

$$\frac{\Delta p}{Z} = C_D \frac{G'^2}{\rho_g} \quad (6.68)$$

Values of C_D for SI units are listed in Table 6.3.†

Pressure Drop for Two-Phase Flow

For simultaneous countercurrent flow of liquid and gas, the pressure-drop data of various investigators show wide discrepancies due to differences in packing density and manufacture, such as changes in wall thickness. Estimates therefore cannot be expected to be very accurate. For most purposes the generalized correlation of Fig. 6.34 will serve. The values of C_f are found in Table 6.3. More elaborate data for particular packings are available from the manufacturers.

Illustration 6.5 Sulfur dioxide is to be removed from gas with the characteristics of air by scrubbing with an aqueous ammonium salt solution in a tower packed with 25-mm ceramic Intalox saddles. The gas, entering at the rate $0.80 \text{ m}^3/\text{s}$ ($1690 \text{ ft}^3/\text{min}$) at 30°C , 1 bar, contains 7.0% SO_2 , which is to be nearly completely removed. The washing solution will enter at the rate

† For G' in $\text{lb/ft}^2 \cdot \text{h}$, ρ_g in lb/ft^3 , and $\Delta p/Z$ in $\text{inH}_2\text{O/ft}$, multiply the values of C_D from Table 6.3 by 1.405×10^{-10} .

3.8 kg/s (502 lb/min) and has a density 1235 kg/m³ (77 lb/ft³), viscosity 2.5×10^{-3} kg/m · s (2.5 cP). (a) Choose a suitable tower diameter. (b) If the irrigated packed height is 8.0 m (26.25 ft), and if 1 m of 25-mm ceramic Intalox saddles is used above the liquid inlet as an entrainment separator, estimate the power requirement to overcome the gas-pressure drop. The overall efficiency of fan and motor is 60%.

SOLUTION (a) Since the larger flow quantities are at the bottom for an absorber, the diameter will be chosen to accommodate the bottom conditions.

$$\text{Av mol wt gas in} = 0.07(64) + 0.93(29) = 31.45 \text{ kg/kmol}$$

$$\text{Gas in} = 0.80 \frac{273}{303} \frac{0.987}{1.0} \frac{1}{22.41} = 0.0317 \text{ kmol/s}$$

$$\text{or} \quad 0.0317(31.45) = 0.998 \text{ kg/s}$$

$$\rho_G = \frac{0.998 \text{ kg/s}}{0.80 \text{ m}^3/\text{s}} = 1.248 \text{ kg/m}^3$$

Assuming essentially complete absorption, SO₂ removed = 0.0317(0.07)(64) = 0.1420 kg/s.

$$\text{Liquid leaving} = 3.8 + 0.1420 = 3.94 \text{ kg/s}$$

$$\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = \frac{3.94}{0.998} \left(\frac{1.248}{1235} \right)^{0.5} = 0.125$$

In Fig. 6.34 use a gas-pressure drop of 400 (N/m²)/m. The ordinate is therefore 0.061. For 25-mm ceramic Intalox saddles, $C_f = 98$ (Table 6.3). Therefore,

$$G' = \left[\frac{0.061 \rho_G (\rho_L - \rho_G) g_c}{C_f \mu_L^{0.1 J}} \right]^{0.5} = \left[\frac{0.061(1.248)(1235 - 1.248)}{98(2.5 \times 10^{-3})^{0.1}(1.0)} \right]^{0.5}$$

$$= 1.321 \text{ kg/m}^2 \cdot \text{s}$$

The tower cross-sectional area is therefore (0.998 kg/s)/(1.321 kg/m² · s) = 0.756 m².

$$T = [4(0.756)/\pi]^{0.5} = 0.98 \text{ m, say 1.0 m diameter} \quad \text{Ans.}$$

The corresponding cross-sectional area is 0.785 m².

(b) The pressure drop for the 8.0 m of irrigated packing is approximately 400(8) = 3200 N/m².

For the dry packing, gas flow rate = $G' = (0.998 - 0.1420)/0.785 = 1.09 \text{ kg/m}^2 \cdot \text{s}$ at a pressure of 100 000 - 3200 = 96 800 N/m². The gas density is

$$\rho_G = \frac{29}{22.41} \frac{273}{303} \frac{96\,800}{101\,330} = 1.114 \text{ kg/m}^3$$

Table 6.3: C_D for this packing = 241.5 for SI units. Eq. (6.68):

$$\frac{\Delta p}{Z} = \frac{241.5(1.09)^2}{1.114} = 258 \text{ N/m}^2 \text{ for 1 m of packing}$$

The total pressure drop for packing is therefore 3200 + 258 = 3458 N/m². To estimate fan power, to this must be added the pressure drop for the packing supports and liquid distributor (negligible for a well-designed apparatus) and inlet expansion and outlet contraction losses for the gas. For a gas velocity in the inlet and outlet pipe of 7.5 m/s, the expansion and outlet-contraction losses would amount at most to 1.5 velocity heads, or $1.5V^2/2g_c = 1.5(7.5)^2/2(1) = 42 \text{ N} \cdot \text{m/kg}$ or $(42 \text{ N} \cdot \text{m/kg})(1.24 \text{ kg/m}^3) = 52.1 \text{ N/m}^2$. The fan power output for the tower is therefore estimated as

$$\frac{[(3458 + 52) \text{ N/m}^2][(0.998 - 0.147) \text{ kg/s}]}{1.114 \text{ kg/m}^3} = 2681 \text{ N} \cdot \text{m/s} = 2.681 \text{ kW}$$

The power for the fan motor is then 2.681/0.6 = 4.469 kW = 6 hp.

MASS-TRANSFER COEFFICIENTS FOR PACKED TOWERS

When a packed tower is operated in the usual manner as a countercurrent absorber or stripper for transfer of solute between the gas and liquid, the rate of solute transfer can be computed from measured values of the rate of gas and liquid flow and the bulk concentrations of solute in the entering and leaving streams. As explained in Chap. 5, because of the impossibility of measuring solute concentrations at the gas-liquid interface, the resulting rates of mass transfer can be expressed only as overall coefficients, rather than as coefficients for the individual fluids. Further, since the interfacial area between gas and liquid is not directly measured in such experiments, the *flux* of mass transfer cannot be determined, but instead only the rate as the product of the flux and the total interfacial area. By dividing these rates by the volume of the packing, the results appear as *volumetric overall coefficients*, $K_x a$, $K_y a$, $K_G a$, $F_{OG} a$, $F_{OL} a$, etc., where a is the interfacial surface per unit packed volume.

The individual fluid mass-transfer coefficients (k_x , k_y , F_L , F_G) and the interfacial area a which make up these overall volumetric coefficients are differently dependent upon fluid properties, flow rates, and type of packing. The overall volumetric coefficients are therefore useful only in the design of towers filled with the same packing and handling the same chemical system at the same flow rates and concentrations as existed during the measurements. For general design purposes, the individual coefficients and the interfacial area are necessary.

To obtain individual coefficients, the general approach has been to choose experimental conditions such that the resistance to mass transfer in the gas phase is negligible in comparison with that in the liquid. This is the case for the absorption or desorption of very insoluble gases, e.g., oxygen or hydrogen in water [see Eq. (5.14)] [106]. Measurements in such systems then lead to values of $k_x a$, $k_L a$, and $F_L a$, which can be correlated in terms of system variables. There are evidently no systems involving absorption or desorption where the solute is so soluble in the liquid that the liquid-phase resistance is entirely negligible. But by subtracting the known liquid resistance from the overall resistances (see Chap. 5), it is possible to arrive at the gas-phase coefficients $k_y a$, $k_G a$, and $F_G a$ and to correlate them in terms of system variables.

Another approach to obtaining pure gas-phase coefficients is to make measurements when a pure liquid evaporates into a gas. Here there is no liquid resistance since there is no concentration gradient within the liquid. The resulting volumetric coefficients $k_y a$ and $F_G a$, however, do not agree with those obtained in the manner first described above. The reason is the different effective interfacial areas, as explained below.

Liquid Holdup

Holdup refers to the liquid retained in the tower as films wetting the packing and as pools caught in the crevices between packing particles. It is found that

the total holdup ϕ_{L_t} is made up of two parts,

$$\phi_{L_t} = \phi_{L_0} + \phi_{L_s} \quad (6.69)$$

where ϕ_{L_s} is the static and ϕ_{L_0} the operating, or moving, holdup, each expressed as volume liquid/packed volume. The moving holdup consists of liquid which continuously moves through the packing, continuously replaced regularly and rapidly by new liquid flowing from above. On stopping the gas and liquid flow, the moving holdup drains from the packing. The static holdup is liquid retained as pools in protected interstices in the packing, largely stagnant, and only slowly replaced by fresh liquid. On stopping the flows, the static holdup does not drain.

When absorption or desorption of a solute occurs, involving transfer of a solute between the bulk liquid and gas, the liquid of the static holdup rapidly comes to equilibrium with the adjacent gas, and thereafter its interfacial surface contributes nothing to mass transfer except as it is slowly replaced. For absorption and desorption, therefore, the smaller area offered by the moving holdup is effective. When evaporation or condensation occurs with the liquid phase a single, pure component, however, the area offered by the total holdup is effective since the liquid then offers no resistance to mass transfer.

Mass Transfer

For most of the standard packings, data for $K_G a$ or the equivalent H_{OG} are available in handbooks [47] or from manufacturers' bulletins for specific systems. For some of the packings, data for $k_G a$ (or H_{LG}) and $k_L a$ (or H_{LL}) are also available, and attempts have been made to correlate these generally in terms of operating conditions [30, 46, 47, 94].

For *Raschig rings* and *Berl saddles*, Shulman and coworkers [107] have established the nature of the area-free mass-transfer coefficients k_G by passing gases through beds filled with packings made with naphthalene, which sublimates into the gas. By comparing these with $k_G a$'s from aqueous absorption [48] and other systems the interfacial areas for absorption and vaporization were obtained. The data on $k_L a$'s [106] then provided the correlation for k_L , the liquid-phase coefficient. Their work is summarized as follows.

For Raschig rings and Berl saddles, the gas-phase coefficient is given by†

$$\frac{F_G \text{Sc}_G^{2/3}}{G} = \frac{k_G p_{B,M} \text{Sc}_G^{2/3}}{G} = 1.195 \left[\frac{d_s G'}{\mu_G (1 - \epsilon_{L_0})} \right]^{-0.36} \quad (6.70)$$

where ϵ_{L_0} , the operating void space in the packing, is given by

$$\epsilon_{L_0} = \epsilon - \phi_{L_t} \quad (6.71)$$

and d_s is the diameter of a sphere of the same surface as a single packing particle

† The j form of Eq. (6.70), with the $\frac{2}{3}$ exponent on Sc_G , correlates the data well, although it is inconsistent with the surface-renewal-theory approach, which is generally accepted for fluid interfaces, and with the correlation of Eq. (6.72).

(not the same as d_p). The fluid properties should be evaluated at average conditions between interface and bulk gas. The liquid coefficient is given by

$$\frac{k_L d_s}{D_L} = 25.1 \left(\frac{d_s L'}{\mu_L} \right)^{0.45} Sc_L^{0.5} \quad (6.72)$$

Since the liquid data were obtained at very low solute concentrations, k_L can be converted to F_L through $F_L = k_L c$, where c is the molar density of the solvent liquid. There may be an additional effect not included in Eq. (6.72). If concentration changes result in an increase in surface tension as the liquid flows down the column, the liquid film on the packing becomes *stabilized* and the mass-transfer rate may become larger. If the surface tension decreases, the film may break up into rivulets and the mass-transfer rate decreases [127]. Definite conclusions cannot yet be drawn [16].

The interfacial areas for absorption and desorption with water or very dilute aqueous solutions a_{AW} are given for conditions below loading by Shulman in an extensive series of graphs, well represented by the empirical expressions of Table 6.4. For absorption or desorption with nonaqueous liquids, the area is a_A , given by

$$a_A = a_{AW} \frac{\varphi_{Lo}}{\varphi_{LoW}} \quad (6.73)$$

and for contact of a gas with a pure liquid, as in vaporization, the areas are

$$a_V = 0.85 a_A \frac{\varphi_{Li}}{\varphi_{Lo}} \quad (6.74)$$

$$a_{VW} = 0.85 a_{AW} \frac{\varphi_{LiW}}{\varphi_{LoW}} \quad (6.75)$$

The subscript W indicates water as the liquid. The holdup data are summarized in Table 6.5. Although these data are based on work with absorption, desorption, and vaporization, the evidence is that the same coefficients apply to distillation in packed towers [89]. In distillation, a_A 's should be used.

Illustration 6.6 A tower packed with 38-mm (1.5-in) Berl saddles is to be used for absorbing benzene vapor from a dilute mixture with inert gas. The circumstances are:

Gas Av mol wt = 11, viscosity = 10^{-5} kg/m · s (0.01 cP); 27°C , $p_i = 107$ kN/m² (803 mmHg), $D_G = 1.30 \times 10^{-5}$ m²/s (0.504 ft²/h), flow rate $G' = 0.716$ kg/m² · s (528 lb/ft² · h).

Liquid Av mol wt = 260, viscosity = 2×10^{-3} kg/m · s (2.0 cP), density = 840 kg/m³ (52.4 lb/ft³), surface tension = 3.0×10^{-2} N/m (30 dyn/cm), $D_L = 4.71 \times 10^{-10}$ m²/s (1.85×10^{-5} ft²/h), flow rate $L' = 2.71$ kg/m² · s (2000 lb/ft² · h).

Compute the volumetric mass-transfer coefficients.

Table 6.4 Interfacial area for absorption and desorption, aqueous liquids†

For conditions below loading, $a_{AW} = m(808 G' / \rho_G^{0.5} L'^p)$; use only for SI units: L' and G' $\text{kg/m}^2 \cdot \text{s}$ [or $(\text{lb/h ft}^2)(0.001356)$] and ρ_G kg/m^3 [or $(\text{lb/ft}^3)(16.019)$]; $a_{AW} = \text{m}^2/\text{m}^3$; divide by 3.281 for ft^2/ft^3 ; the original data cover L' up to $6.1 \text{ kg/m}^2 \cdot \text{s}$ ($4500 \text{ lb/ft}^2 \cdot \text{h}$); extrapolation to $L' = 10.2 \text{ kg/m}^2 \cdot \text{s}$ ($7500 \text{ lb/ft}^2 \cdot \text{h}$) has been suggested [107]

Packing	Nominal size		Range of L'		m	n	p
	mm	in	$\text{kg/m}^2 \cdot \text{s}$	$\text{lb/ft}^2 \cdot \text{h}$			
Raschig rings	13	0.5	0.68–2.0	500–1500	28.01	$0.2323 L' - 0.30$	-1.04
			2.0–6.1	1500–4500	14.69	$0.01114 L' + 0.148$	-0.111
	25	1	0.68–2.0	500–1500	34.42	0	0.552
			2.0–6.1	1500–4500	68.2	$0.0389 L' - 0.0793$	-0.47
	38	1.5	0.68–2.0	500–1500	36.5	$0.0498 L' - 0.1013$	0.274
			2.0–6.1	1500–4500	40.11	$0.01091 L' - 0.022$	0.140
Berl saddles	50	2	0.68–2.0	500–1500	31.52	0	0.481
			2.0–6.1	1500–4500	34.03	0	0.362
	13‡	0.5‡	0.68–2.0	500–1500	16.28	0.0529	0.761
			2.0–6.1	1500–4500	25.61	0.0529	0.170
	25§	1§	0.68–2.0	500–1500	52.14	$0.0506 L' - 0.1029$	0
			2.0–6.1	1500–4500	73.0	$0.0310 L' - 0.0630$	-0.359
	38§	1.5§	0.68–2.0	500–1500	40.6	-0.0508	0.455
			2.0–6.1	1500–4500	62.4	$0.0240 L' - 0.0996$	-0.1355

† Data of Shulman, et al. [107].

‡ For $G' < 1.08 \text{ kg/m}^2 \cdot \text{s}$ ($800 \text{ lb/ft}^2 \cdot \text{h}$) only. For higher values, see Shulman, *AIChE J.*, 1, 253 (1955), figs. 16 and 17.

§ It has been shown that constants for $L' = 0.68$ to 2.0 may apply over the range $L' = 0.68$ to 6.1 [99]. This may reflect some change in the shape of the packing over the years.

SOLUTION For the gas,

$$\rho_G = \frac{11}{22.41} \frac{1.07 \times 10^5}{1.0133 \times 10^5} \frac{273}{273 + 27} = 0.472 \text{ kg/m}^3$$

$$\text{Sc}_G = \frac{\mu_G}{\rho_G D_G} = \frac{1 \times 10^{-5}}{0.472(1.30 \times 10^{-5})} = 1.630$$

$$G' = 0.716 \text{ kg/m}^2 \cdot \text{s} \quad G = \frac{0.716}{11} = 0.0651 \text{ kmol/m}^2 \cdot \text{s}$$

For the liquid,

$$\text{Sc}_L = \frac{\mu_L}{\rho_L D_L} = \frac{2 \times 10^{-3}}{840(4.77 \times 10^{-10})} = 4990$$

Table 6.5 Liquid holdup in packed towers

$\varphi_{Ls} = \varphi_{Lo} + \varphi_{Ls}$ $\varphi_{LsW} = \varphi_{LoW} + \varphi_{LsW}$ $\varphi_{Lo} = \varphi_{LoW}H$ Use only SI units: L' in $\text{kg}/\text{m}^2 \cdot \text{s}$, $s = (\text{lb}/\text{ft}^2 \cdot \text{h})(0.001356)$;
 ρ_L in $\text{kg}/\text{m}^3 = (\text{lb}/\text{ft}^3)(16.019)$; μ_L in $\text{kg}/\text{m} \cdot \text{s} = \text{cP}(0.001)$; σ in $\text{N}/\text{m} = (\text{dyn}/\text{cm})(0.001)$; φ is dimensionless

Packing	Nominal size		d_p , m	φ_{Ls}	Water, ordinary temperatures		μ_L , kg/m · s	H
	mm	in						
Ceramic Raschig rings	13	0.5	0.01774	$\frac{0.0486\mu_L^{0.02}\varphi^{0.99}}{d_p^{1.21}\rho_L^{0.37}}$	$\beta = 1.508d_p^{0.376}$	<0.012	$\frac{975.7L'^{0.57}\mu_L^{0.13}}{\rho_L^{0.84}(2.024L'^{0.430} - 1)}\left(\frac{\sigma}{0.073}\right)^{0.1737 - 0.262 \log L'}$	
	25	1	0.0356		$\varphi_{LsW} = \frac{2.47 \times 10^{-4}}{d_p^{1.21}}$			
	38	1.5	0.0530					
	50	2	0.0725		$\varphi_{LsW} = \frac{(2.09 \times 10^{-6})(737.5L')^\beta}{d_p^2}$			
Carbon Raschig rings	25	1	0.01301	$\frac{0.0237\mu_L^{0.02}\varphi^{0.23}}{d_p^{1.21}\rho_L^{0.37}}$	$\beta = 1.104d_p^{0.376}$	<0.012	$\frac{407.9L'^{0.57}\mu_L^{0.13}}{\rho_L^{0.84}(1.393L'^{0.315} - 1)}\left(\frac{\sigma}{0.073}\right)^{0.1737 - 0.262 \log L'}$	
	38	1.5	0.0543		$\varphi_{LsW} = \frac{5.94 \times 10^{-4}}{d_p^{1.21}}$			
	50	2	0.0716		$\varphi_{LsW} = \frac{(7.34 \times 10^{-6})(737.5L')^\beta}{d_p^2}$			
Ceramic Berl saddles	13	0.5	0.31622	$\frac{4.23 \times 10^{-3}\mu_L^{0.04}\varphi^{0.55}}{d_p^{1.56}\rho_L^{0.37}}$	$\beta = 1.508d_p^{0.376}$	<0.020	$\frac{1404L'^{0.57}\mu_L^{0.13}}{\rho_L^{0.84}(3.24L'^{0.413} - 1)}\left(\frac{\sigma}{0.073}\right)^{0.2817 - 0.262 \log L'}$	
	25	1	0.0320		$\varphi_{LsW} = \frac{5.014 \times 10^{-5}}{d_p^{1.56}}$			
	38	1.5	0.0472					
					$\varphi_{LsW} = \frac{(2.32 \times 10^{-6})(737.5L')^\beta}{d_p^2}$			

† Data of Shulman, et al. [107].

Holdup Table 6.5:

$$\begin{aligned}
 L' &= 2.71 \text{ kg/m}^2 \cdot \text{s} & d_s &= 0.0472 \text{ m} & \beta &= 1.508 d_s^{0.376} = 0.478 \\
 \Phi_{LsW} &= \frac{5.014 \times 10^{-5}}{d_s^{1.56}} = 5.86 \times 10^{-3} \text{ m}^3/\text{m}^3 \\
 \Phi_{LiW} &= \frac{(2.32 \times 10^{-6})(737.5 L')^\beta}{d_s^2} = 0.0394 \text{ m}^3/\text{m}^3 \\
 \Phi_{LoW} &= \Phi_{LiW} - \Phi_{LsW} = 0.0335 \text{ m}^3/\text{m}^3 \\
 H &= \frac{1404 L'^{0.57} \mu_L^{0.13}}{\rho_L^{0.84} (3.24 L'^{0.413} - 1)} \left(\frac{\sigma}{0.073} \right)^{0.2817 - 0.262 \log L'} \\
 &= 0.855 \\
 \Phi_{Lo} &= \Phi_{LoW} H = 0.0335(0.855) = 0.0286 \\
 \Phi_{Ls} &= \frac{4.23 \times 10^{-3} \mu_L^{0.04} \sigma^{0.55}}{d_s^{1.56} \rho_L^{0.37}} = 4.65 \times 10^{-3} \text{ m}^3/\text{m}^3 \\
 \Phi_{Li} &= \Phi_{Lo} + \Phi_{Ls} = 0.0286 + 4.65 \times 10^{-3} = 0.0333 \text{ m}^3/\text{m}^3
 \end{aligned}$$

Interfacial area Table 6.4:

$$\begin{aligned}
 m &= 62.4 & n &= 0.0240 L' - 0.0996 = -0.0346 \\
 p &= -0.1355 & a_{AW} &= m \left(\frac{808 G'}{\rho_G^{0.5}} \right)^n L'^p = 43.8 \text{ m}^2/\text{m}^3
 \end{aligned}$$

Eq. (6.73):

$$a_A = a_{AW} \frac{\Phi_{Lo}}{\Phi_{LoW}} = \frac{43.8(0.0286)}{0.0335} = 37.4 \text{ m}^2/\text{m}^3$$

Table 6.3:

$$\epsilon = 0.75$$

Eq. (6.71):

$$\epsilon_{Lo} = \epsilon - \Phi_{Li} = 0.75 - 0.0333 = 0.717$$

Eq. (6.70):

$$\frac{F_G (1.630)^{2/3}}{0.0651} = 1.195 \left[\frac{0.0472(0.716)}{(1 \times 10^{-3})(1 - 0.717)} \right]^{-0.36}$$

$$F_G = 1.914 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s} = k_G p_{B,M}$$

Eq. (6.72):

$$\frac{k_L (0.0472)}{4.77 \times 10^{-10}} = 25.1 \left[\frac{0.0472(2.71)}{0.002} \right]^{0.45} (4990)^{0.5}$$

$$k_L = 1.156 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

Since the data for k_L were taken at very low concentrations, this can be converted into $F_L = k_L c$, where $c = 840/260 = 3.23 \text{ kmol/m}^3$ molar density. Therefore $F_L = (1.156 \times 10^{-4})(3.23) = 3.73 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} = k_L x_{B,M} c$. The volumetric coefficients are

$$F_G a_A = (1.914 \times 10^{-3})(37.4) = 0.0716 \text{ kmol/m}^3 \cdot \text{s} = k_s a_A p_{B,M}$$

$$F_L a_A = (3.73 \times 10^{-4})(37.4) = 0.0140 \text{ kmol/m}^3 \cdot \text{s} = k_L a_A x_{B,M} c$$

Illustration 6.7 A tower packed with 50-mm (2-in) ceramic Raschig rings is to be used for dehumidification of air by countercurrent contact with water. At the top of the tower, the

conditions are to be:

Water: flow rate = $5.5 \text{ kg/m}^2 \cdot \text{s}$ ($4060 \text{ lb/ft}^2 \cdot \text{h}$), temp = 15°C .

Air: flow rate = $1.10 \text{ kg/m}^2 \cdot \text{s}$ ($811 \text{ lb/ft}^2 \cdot \text{h}$), temp = 20°C , 1 std atm, essentially dry.

Compute the transfer coefficients for the top of the tower.

SOLUTION For the gas, $G' = 1.10 \text{ kg/m}^2 \cdot \text{s}$, $G = 1.10/29 = 0.0379 \text{ kmol/m}^2 \cdot \text{s}$, $\mu_G = 1.8 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, $Sc_G = 0.6$ for air-water vapor, and

$$\rho_G = \frac{29}{22.41} \frac{273}{293} = 1.206 \text{ kg/m}^3$$

For the liquid, $L' = 5.5 \text{ kg/m}^2 \cdot \text{s}$. Table 6.5:

$$d_s = 0.0725 \text{ m} \quad \beta = 1.508 d_s^{0.376} = 0.562$$

$$\phi_{L,W} = \frac{(2.09 \times 10^{-6})(737.5 L')^\beta}{d_s^2} = 0.0424 \text{ m}^3/\text{m}^3$$

$$\phi_{L,W} = \frac{2.47 \times 10^{-8}}{d_s^{1.21}} = 5.91 \times 10^{-3} \text{ m}^3/\text{m}^3$$

$$\phi_{L,oW} = \phi_{L,W} - \phi_{L,W} = 0.0424 - 5.91 \times 10^{-3} = 0.0365 \text{ m}^3/\text{m}^3$$

Table 6.4:

$$m = 34.03, n = 0, p = 0.362.$$

$$a_{LW} = m \left(\frac{808 G'}{\rho_G^{0.5}} \right)^n L'^p = 63.1 \text{ m}^2/\text{m}^3$$

Eq. (6.75):

$$a_{VW} = 0.85 a_{LW} \frac{\phi_{L,W}}{\phi_{L,oW}} = 0.85(63.1) \frac{0.0424}{0.0365} = 62.3 \text{ m}^2/\text{m}^3$$

Table 6.3:

$$\varepsilon = 0.74, \varepsilon_{L,o} = \varepsilon - \phi_{L,W} = 0.74 - 0.0424 = 0.698.$$

Eq. (6.70):

$$\frac{F_G(0.6)^{2/3}}{0.0379} = 1.195 \left[\frac{0.0725(1.10)}{(1.8 \times 10^{-5})(1 - 0.698)} \right]^{-0.36} = 0.0378$$

$$F_G = 2.01 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$$

$$F_G a_{VW} = (2.01 \times 10^{-3})(62.3) = 0.125 \text{ kmol/m}^3 \cdot \text{s} = k_G a_{VW} p_{B,M}$$

Since the liquid is pure water, it has no mass-transfer coefficient. However, for such processes (see Chap. 7), we need convection heat-transfer coefficients for both gas and liquid. These can be estimated, in the absence of directly applicable data, through the heat-mass-transfer analogy. Thus, from Eq. (6.70), and assuming $j_D = j_H$,

$$j_H = \frac{h_G}{C_p G'} \text{Pr}_G^{2/3} = 0.0378$$

where h_G is the gas-phase heat-transfer coefficient. For air, $C_p = 1005 \text{ N} \cdot \text{m/kg} \cdot \text{K}$, $\text{Pr}_G = 0.74$.

$$h_G = \frac{0.0378 C_p G'}{\text{Pr}_G^{2/3}} = \frac{0.0378(1005)(1.10)}{(0.74)^{2/3}} = 51.1 \text{ W/m}^2 \cdot \text{K}$$

Similarly, the heat-transfer analog of Eq. (6.72) is

$$\text{Nu} = \frac{h_L d_s}{k_{Lb}} = 25.1 \left(\frac{d_s L'}{\mu_L} \right)^{0.45} \text{Pr}_L^{0.5}$$

where h_L is the liquid-phase heat-transfer coefficient. $k_{th} = 0.587 \text{ W/m} \cdot \text{K}$ and $\text{Pr}_L = C_p \mu_L / k_{th} = 4187(1.14 \times 10^{-3}) / 0.587 = 8.1$. Therefore

$$h_L = 25.1 \frac{k_{th}}{d_p} \left(\frac{d_p L'}{\mu_L} \right)^{0.45} \text{Pr}_L^{0.5} = \frac{25.1(0.587)}{0.0725} \left[\frac{0.0725(5.5)}{1.14 \times 10^{-4}} \right]^{0.45} (8.1)^{0.5} \\ = 8071 \text{ W/m}^2 \cdot \text{K}$$

The corresponding volumetric coefficients are

$$h_G a_{yw} = 51.1(62.3) = 3183 \text{ W/m}^3 \cdot \text{K}$$

$$h_L a_{yw} = 8071(62.3) = 503\,000 \text{ W/m}^3 \cdot \text{K}$$

COCURRENT FLOW OF GAS AND LIQUID

Cocurrent flow of gas and liquid (usually downward) through packed beds is used for catalytic chemical reaction between components of the fluids, the catalyst usually being some active substance supported upon granular ceramic material. Such arrangements are known as *trickle-bed reactors*. For the strictly diffusional separation processes, where the packing is inactive and merely provides a means of producing interfacial surface, it should be recalled that cocurrent flow can produce no better than one theoretical stage of separation. This is sufficient for substantially complete separation where chemical reaction accompanies the diffusional separation, e.g., in the absorption of hydrogen sulfide into a strongly basic solution such as aqueous sodium hydroxide. For these situations, there is the advantage that flooding is impossible: there is no upper limit for the permissible flow rates. In the absence of reaction, however, cocurrent flow rarely is useful.

Practically all the available data concern conditions suitable for catalysis, with small packing particles (small spheres, cylinders, and the like) which are the catalyst supports. Very complete reviews of pressure drop and mass-transfer for such situations are available [100, 103].

END EFFECTS AND AXIAL MIXING

In addition to that occurring within the packed volume, there will be mass transfer where the liquid is introduced by sprays or other distributors and where it drips off the lower packing support. These constitute the *end effects*. For proper evaluation of the packing itself, correction of experimental data must be made for these effects, usually by operating at several packing heights followed by extrapolation of the results to zero height. New equipment designed with uncorrected data will be conservatively designed if shorter than the experimental equipment but underdesigned if taller. Alternatively, equipment designed with corrected data will include a small factor of safety represented by the end effects. The data represented by Eqs. (6.69) to (6.75) and Tables 6.4 and 6.5 are believed to be reasonably free of end effects.

As will be clear in later chapters, the equations generally used for design, which are the same as those used for computing mass-transfer coefficients from laboratory measurements, are based on the assumption that the gas and liquid flow in piston, or plug, flow, with a flat velocity profile across the tower and with each portion of the gas or liquid having the same residence time in the tower. Such is not actually the case. Nonuniformity of packing and maldistribution of liquid may lead to channeling, or regions where the liquid flow is abnormally great; the liquid in the static holdup moves forward much more slowly than that in the moving holdup; drops of liquid falling from a packing piece may be blown upward by the gas. Similarly the resistance to gas flow at the tower walls and in relatively dry regions of the packing is different from that elsewhere; the downward movement of liquid induces downward movement of the gas. As a result, the purely piston-type flow does not occur, and there is a relative movement within each fluid parallel to the axis of the tower, variously described as axial mixing, axial dispersion, and back mixing. *Axial dispersion* is the spreading of the residence time in unidirectional flow owing to the departure from purely piston-type, or plug, flow; the fluid particles move forward but at different speeds. *Back mixing* is the backward flow in a direction opposite to that of the net flow, caused by frictional drag of one fluid upon the other, spray of liquid into the gas (entrainment), and the like. The transport of solute by these mechanisms is usually described in terms of an eddy diffusivity. For gas-liquid packed towers [19, 28, 123] many of the data scatter badly. For 25-mm Berl saddles and Raschig rings, and 50-mm Raschig rings, axial mixing in the liquid is more important than that in the gas [36]. Such transport of solute reduces the concentration differences for interphase mass transfer. The true mass-transfer coefficients are undoubtedly larger than those we customarily use, since the latter have not been corrected for axial mixing. The effect is strong for spray towers, which are relatively poor devices for separation operations as a result. Fortunately the effects appear not to be large for packed towers under ordinary circumstances.

In tray towers, entrainment of liquid in the gas is a form of back mixing, and there are back mixing and axial mixing on the trays, which we have already considered. In sparged vessels, the liquid is essentially completely back-mixed to a uniform solute concentration. Both phases are largely completely mixed in mechanically agitated vessels.

TRAY TOWERS VS. PACKED TOWERS

The following may be useful in considering a choice between the two major types of towers.

1. *Gas-pressure drop.* Packed towers will ordinarily require a smaller pressure drop. This is especially important for vacuum distillation.
2. *Liquid holdup.* Packed towers will provide a substantially smaller liquid

holdup. This is important where liquid deterioration occurs with high temperatures and short holding times are essential. It is also important in obtaining sharp separations in batch distillation.

3. *Liquid/gas ratio.* Very low values of this ratio are best handled in tray towers. High values are best handled in packed towers.
4. *Liquid cooling.* Cooling coils are more readily built into tray towers; and liquid can more readily be removed from trays, to be passed through coolers and returned, than from packed towers.
5. *Side streams.* These are more readily removed from tray towers.
6. *Foaming systems.* Packed towers operate with less bubbling of gas through the liquid and are the more suitable.
7. *Corrosion.* Packed towers for difficult corrosion problems are likely to be less costly.
8. *Solids present.* Neither type of tower is very satisfactory. Agitated vessels and venturi scrubbers are best but provide only a single stage. If multistage countercurrent action is required, it is best to remove the solids first. Dust in the gas can be removed by a venturi scrubber at the bottom of a tower. Liquids can be filtered or otherwise clarified before entering a tower.
9. *Cleaning.* Frequent cleaning is easier with tray towers.
10. *Large temperature fluctuations.* Fragile packings (ceramic, graphite) tend to be crushed. Trays or metal packings are satisfactory.
11. *Floor loading.* Plastic packed towers are lighter in weight than tray towers, which in turn are lighter than ceramic or metal packed towers. In any event, floor loadings should be designed for accidental complete filling of the tower with liquid.
12. *Cost.* If there is no overriding consideration, cost is the major factor to be taken into account.

NOTATION FOR CHAPTER 6

Any consistent set of units may be used, except as noted.

a	average specific interfacial surface for mass transfer, area/volume, L^2/L^3
a_d	specific interfacial surface for absorption, desorption, and distillation, area/volume, L^2/L^3
a_p	specific surface of packing, area/volume, L^2/L^3
a_v	specific interfacial surface for contact of a gas with a pure liquid, area/volume, L^2/L^3
A	projected area, L^2
A_a	active area, area of perforated sheet, L^2
A_d	downspout cross-sectional area, L^2
A_{da}	smaller of the two areas, A_d or free area between downspout apron and tray, L^2
A_n	net tower cross-sectional area for gas flow = $A_t - A_d$ for cross-flow trays, L^2
A_o	area of perforations, L^2
A_t	tower cross-sectional area, L^2
b	baffle width, L
B	blade length, L
c	molar density of the liquid, mole/L^3
	solute concentration, mole/L^3

C	distance from impeller to tank bottom, L
C_D	empirical constant, Eq. (6.68) and Table 6.3
C_f	characterization factor of packing, two-phase flow, empirical constant, Table 6.3
C_F	flooding constant for trays, Eqs. (6.29) and (6.30), empirical
C_o	orifice coefficient, dimensionless
C_p	specific heat at constant pressure, FL/MT
d_d	disk diameter, L
d_i	impeller diameter, L
d_o	orifice or perforation diameter, L
d_b	average bubble diameter, L
	nominal diameter of tower-packing particle, L
d_s	diameter of sphere of same surface as a single packing particle, L
D	diffusivity, L ² /Θ
D_E	eddy diffusivity of back mixing, L ² /Θ
E	fractional entrainment, entrained liquid/(entrained liquid + net liquid flow), mole/mole or M/M
E_{MG}	Murphree gas-phase stage efficiency, fractional
E_{MGE}	Murphree gas-phase stage efficiency corrected for entrainment, fractional
E_O	overall tray efficiency of a tower, fractional
E_{OG}	point gas-phase tray efficiency, fractional
f	function
	Fanning friction factor, dimensionless
F	mass-transfer coefficient, mole/L ² Θ
F_D	drag force, F
Fr	impeller Froude number = $d_i N^2/g$, dimensionless
g	acceleration of gravity, L/Θ ²
g_c	conversion factor, ML/FΘ ²
G	superficial molar gas mass velocity, mole/L ² Θ
G'	superficial gas mass velocity, M/L ² Θ
h	heat-transfer coefficient, FL/L ² Θ
h_D	dry-plate gas-pressure drop as head of clear liquid, L
h_G	gas-pressure drop as head of clear liquid, L
h_L	gas-pressure drop due to liquid holdup on tray, as head of clear liquid, L
h_R	residual gas-pressure drop as head of clear liquid, L
h_W	weir height, L
h_1	weir crest, L
h_2	head loss owing to liquid flow under downspout apron, L
h_3	backup of liquid in downspout, L
H	correction factor (holdup, packed towers), Table 6.5
H_f	mechanical energy lost to friction, FL/M
H_{IG}	height of gas-phase transfer unit, L
H_{IL}	height of liquid-phase transfer unit, L
H_{IOG}	overall height of a transfer unit, L
j_D	mass-transfer group = $F_G Sc_G^{2/3}/G$, dimensionless
j_H	heat-transfer group = $h Pr_G^{2/3}/C_p G'$, dimensionless
J	conversion factor, Fig. 6.31
k_G	gas-phase mass-transfer coefficient, mole/ΘL ² (F/L ²)
k_L	liquid-phase mass-transfer coefficient, mole/ΘL ² (mole/L ³)
k_{th}	thermal conductivity, FL ² /ΘL ² T
k_x	liquid-phase mass-transfer coefficient, mole/ΘL ² (mole fraction)
k_y	gas-phase mass-transfer coefficient, mole/ΘL ² (mole fraction)
K_G	overall gas-phase mass-transfer coefficient, mole/ΘL ² (F/L ²)
K_y	overall gas-phase mass-transfer coefficient, mole/ΘL ² (mole fraction)
l	plate thickness, L

ln	natural logarithm
log	common logarithm
L	superficial liquid molar mass velocity, $\text{mole}/L^2\Theta$
L'	superficial liquid mass velocity, $M/L^2\Theta$
$\bar{L}$	characteristic length, L
m	average slope of a chord of equilibrium curve, Eqs. (6.52) to (6.59), mole fraction in gas/mole fraction in liquid
m'	empirical constant, Table 6.4
n	slope of a chord of equilibrium curve, Eq. (6.53), mole fraction gas/mole fraction in liquid
	number of baffles, dimensionless
	tray number, dimensionless
	empirical constant, Table 6.4
N	impeller speed, Θ^{-1}
N_{IG}	number of gas-phase transfer units, dimensionless
N_{LG}	number of liquid-phase transfer units, dimensionless
N_{IOG}	number of overall gas-phase transfer units, dimensionless
Nu	Nusselt number = hd_i/k_{th} , dimensionless
p	pressure, F/L^2
	empirical constant, Table 6.4
p'	pitch of perforations, L
Δp	pressure difference, F/L^2
Δp_R	residual gas-pressure drop, F/L^2
P	power delivered by an impeller, no gas flow, FL/Θ
P_G	power delivered by an impeller with gas flow, FL/Θ
Pe	Péclet number for liquid mixing, Eq. (6.59), dimensionless
Po	power number = $P_{Gc}/\rho LN^3 d_i^5$, dimensionless
Pr	Prandtl number = $C_p \mu/k_{th}$, dimensionless
q	volumetric liquid flow rate, L^3/Θ
Q	volumetric flow rate, L^3/Θ
Q_{Go}	volumetric gas flow rate per orifice, L^3/Θ
Ra	Rayleigh number = $d_p^3 \Delta p g / D_L \mu_L$, dimensionless
Re	impeller Reynolds number = $d^2 N \rho_L / \mu_L$, dimensionless
	Reynolds number for flow through packing = $d_p G' / \mu$, Eq. (6.66), dimensionless
Re_G	gas Reynolds number = $d_p V_s \rho_L / \mu_L$, dimensionless
Re_o	orifice Reynolds number = $d_o V_o \rho_G / \mu_G = 4w_o / \pi d_o \mu_G$, dimensionless
Sc	Schmidt number = $\mu / \rho D$, dimensionless
Sh	Sherwood number = $F d_p / c D$, dimensionless
T	tank diameter, tower diameter, L
t	tray spacing, L
u	average velocity, L/Θ
u'	root-mean-square fluctuating velocity, L/Θ
v_L	liquid volume, L^3
V	superficial velocity based on tower cross section; for tray towers, based on A_n , L/Θ
V_a	gas velocity based on A_a , L/Θ
V_f	flooding velocity based on A_n , L/Θ
V_o	velocity through an orifice, L/Θ
V_{ow}	minimum gas velocity through perforations below which excessive weeping occurs, L/Θ
V_S	slip velocity, L/Θ
V_t	terminal settling velocity of a single bubble, L/Θ
w	blade width, L
w_o	mass rate of flow per orifice, M/Θ
W	weir length, L
	work done by gas/unit mass, FL/M
We	Weber number = $\rho u^2 d_p / \sigma_G$, dimensionless

x	concentration in liquid, mole fraction
y	concentration in gas, mole fraction
z	average flow width for liquid on a tray, L
Z	liquid depth in an agitated vessel; depth of packing; length of travel on a tray, L
α	empirical constant, Table 6.2
β	empirical constant: for flooding velocity, Table 6.2; for holdup in packing, Table 6.5
ϵ	fractional void volume in a dry packed bed, volume voids/volume bed, dimensionless
η	defined by Eq. (6.58), dimensionless
θ_L	time of residence of a liquid on a tray, Θ
μ	viscosity, $M/L\Theta$
ρ	density, M/L^3
$\Delta\rho$	difference in density, M/L^3
σ	surface tension F/L
φ_G	gas holdup, volume fraction, dimensionless
φ_L	liquid holdup, fraction of packed volume, dimensionless

Subscripts

av	average
AW	air-water
F	at flooding
G	gas
L	liquid
n	tray number
o	orifice; operating or moving (holdup and packing void space)
s	surface; static (holdup in packing)

Superscript

*	in equilibrium with bulk liquid
---	---------------------------------

REFERENCES

1. Akita, K., and F. Yoshida: *Ind. Eng. Chem. Process Des. Dev.*, **12**, 76 (1973).
2. Anderson, R. H., G. Garrett, and M. Van Winkle: *Ind. Eng. Chem. Process Des. Dev.*, **15**, 96 (1976).
3. Anon.: *Chem. Eng.*, **72**, 86 (Aug. 16, 1965).
4. Bates, R. L., P. L. Fondy, and R. R. Corpstein: *Ind. Eng. Chem. Process Des. Dev.*, **2**, 310 (1963).
5. Bates, R. L., P. L. Fondy, and J. G. Fenic: in V. W. Uhl and J. B. Gray (eds.), "Mixing," vol. 1, chap. 3, Academic, New York, 1966.
6. Bell, R. L.: *AIChE J.*, **18**, 491, 498 (1972).
7. Bell, R. L., and R. B. Solari: *AIChE J.*, **20**, 688 (1974).
8. Biddulph, M. W.: *AIChE J.*, **21**, 41 (1975).
9. Biddulph, M. W., and D. J. Stephens: *AIChE J.*, **20**, 60 (1974).
10. Billet, R.: *Br. Chem. Eng.*, **14**, 489 (1969).
11. Billet, R.: *Chem. Eng.*, **79**, 68 (Feb. 21, 1972); *Chem. Eng. Prog.*, **63**(9), 55 (1967).
12. Boll, R. H.: *Ind. Eng. Chem. Fundam.*, **12**, 40 (1973).
13. Bolles, W. L.: in B. D. Smith, "Design of Equilibrium Stage Processes," chap. 14, McGraw-Hill Book Company, New York, 1963.
14. Bolles, W. L.: *Chem. Eng. Prog.*, **72**(9), 43 (1976).
15. Boucher, D. F., and G. E. Alves: in R. H. Perry, and C. Chilton (eds.), "The Chemical Engineers' Handbook," 5th ed., p. 5-22, McGraw-Hill Book Company, New York, 1973.

16. Boyes, A. P., and A. B. Ponter: *Ind. Eng. Chem. Process Des. Dev.*, **10**, 140 (1971).
17. Bridge, A. G., L. Lapidus, and J. C. Elgin: *AIChE J.*, **10**, 819 (1964).
18. Brinck, J. A., W. F. Burggrabe, and C. E. Greenwell: *Chem. Eng. Prog.*, **64**(11), 82 (1968).
19. Brittan, M. I., and E. T. Woodburn: *AIChE J.*, **12**, 541 (1966).
20. Bruijn, W., K. van't Riet, and J. M. Smith: *Trans. Inst. Chem. Eng. Lond.*, **52**, 88 (1974).
21. "Bubble-Tray Design Manual," American Institute of Chemical Engineers, New York, 1958.
22. Butcher, K. L., and M. S. Medani: *Trans. Inst. Chem. Eng. Lond.*, **49**, 225 (1971).
23. Calderbank, P. H.: *Trans. Inst. Chem. Eng. Lond.*, **36**, 443 (1958); **37**, 173 (1959).
24. Calderbank, P. H.: in V. W. Uhl and J. B. Gray (eds.), "Mixing," vol. 2, chapt. 6, Academic, New York, 1967.
25. Chase, J. D.: *Chem. Eng.* **74**(10), 105 (July 31, 1967); (18), 139 (Aug. 28, 1967).
26. Cicalese, J. J., J. A. Davies, P. J. Harrington, G. S. Houghland, A. J. L. Hutchinson, and T. G. Walsh: *Proc. Am. Petrol. Inst.*, **26**(III), 180 (1946).
27. Clark, M. W., and T. Vermeulen: *AIChE J.*, **10**, 420 (1964).
28. Co, P., and R. Bibaud: *Can. J. Chem. Eng.*, **49**, 727 (1971).
29. Colburn, A. P.: *Ind. Eng. Chem.*, **28**, 526 (1936).
30. Cornell, D., W. G. Knapp, and J. R. Fair: *Chem. Eng. Prog.*, **56**(7), 68 (1960).
31. Cutter, L. A.: *AIChE J.*, **12**, 35 (1966).
32. Davidson, J. F., and B. G. Schuler: *Trans. Inst. Chem. Eng. Lond.*, **38**, 144 (1960).
33. Davy, C. A. E., and G. G. Haselden: *AIChE J.*, **21**, 1218 (1975).
34. Delnicki, W. V.: private communication, May 23, 1975.
35. Diener, D. A.: *Ind. Eng. Chem. Process Des. Dev.*, **6**, 499 (1967).
36. Dunn, W. E., T. Vermeulen, C. R. Wilke, and T. T. Word: *Ind. Eng. Chem. Fundam.*, **16**, 116 (1977).
37. Eckert, J. S.: *Chem. Eng. Prog.*, **66**(3), 39 (1970).
38. Eckert, J. S.: *Chem. Eng.*, **82**, 70 (Apr. 14, 1975).
39. Edmister, W. C.: *Petr. Eng.*, **1948**(12), 193.
40. Eissa, S. H., and K. Schügerl: *Chem. Eng. Sci.*, **30**, 1251 (1975).
41. Ellis, S. R. M., and D. P. Bayley: *Chem. Eng.*, **1974**(291), 714.
42. Ergun, S.: *Chem. Eng. Prog.*, **48**, 89 (1952).
43. Eversole, W. G., G. H. Wagner, and E. Stackhouse: *Ind. Eng. Chem.*, **33**, 1459 (1941).
44. Fair, J. R.: *Petro/Chem. Eng.*, **33**, 210 (September 1961).
45. Fair, J. R.: *Chem. Eng.*, **72**, 116 (July 5, 1965).
46. Fair, J. R.: *Chem. Eng.*, **76**, 90 (July 14, 1970).
47. Fair, J. R.: in R. H. Perry and C. H. Chilton (eds.), "The Chemical Engineers' Handbook," 5th ed., sec. 18, McGraw-Hill Book Company, New York, 1973.
48. Fellinger, L.: reported by T. K. Sherwood and R. L. Pigford, "Absorption and Extraction," 2d ed., McGraw-Hill Book Company, New York, 1952.
49. Foss, A. S., and J. A. Gerster: *Chem. Eng. Prog.*, **59**, 35 (1963).
50. Gerster, J. A.: *Chem. Eng. Prog.*, **59**, 28 (1956).
51. Gleason, R. J., and J. D. McKenna: *Chem. Eng. Prog. Symp. Ser.*, **68**(126), 119 (1972).
52. Güntel, A., and M. E. Weber: *AIChE J.*, **21**, 931 (1975).
53. Hanf, E. B.: *Chem. Eng. Prog.*, **67**(11), 54 (1971).
54. Harris, L. S., and G. S. Haun: *Chem. Eng. Prog.*, **60**(5), 100 (1964).
55. Hasson, I. T. M., and C. W. Robinson: *AIChE J.*, **23**, 48 (1977).
56. Haug, H. F.: *Chem. Eng. Sci.*, **31**, 295 (1976).
57. Hicks, R. W., and L. E. Gates: *Chem. Eng.*, **83**, 141 (July 19, 1976).
58. Holland, C. D., et al.: *Chem. Eng. Sci.*, **25**, 431 (1970); **26**, 1723 (1971).
59. Hollands, K. G. T., and K. C. Goel: *Ind. Eng. Chem. Fundam.*, **14**, 16 (1975).
60. Hoppe, K., G. T., and H. Kier: *Br. Chem. Eng.*, **12**, 715, 1381 (1967).
61. Hughmark, G. A.: *Ind. Eng. Chem. Process Des. Dev.*, **6**, 218 (1967).
62. Hughmark, G. A., and H. E. O'Connell: *Chem. Eng. Prog.*, **53**, 127 (1957).
63. Hulswitt, C., and J. A. Mrasz: *Chem. Eng.*, **79**, 80 (May 15, 1972).
64. Hunt, C. d'A., D. N. Hanson, and C. R. Wilke: *AIChE J.*, **1**, 441 (1955).

65. Hutchinson, M. H., and R. F. Baddour: *Chem. Eng. Prog.*, **52**, 503 (1956).
66. Interest, E.: *Chem. Eng.*, **78**, 167 (Nov. 15, 1971).
67. Jamison, R. H.: *Chem. Eng. Prog.*, **65** (3), 46 (1969).
68. Kerr, C. P.: *Ind. Eng. Chem. Process Des. Dev.*, **13**, 222 (1974).
69. Koetsier, W. T., D. Thoenes, and J. F. Frankena: *Chem. Eng. J.*, **5**, 61, 71 (1973).
70. Laity, D. S., and R. E. Treybal: *AIChE J.*, **3**, 176 (1957).
71. Leibson, I., E. G. Holcomb, A. G. Cocosco, and J. J. Jacmie: *AIChE J.*, **2**, 296 (1956).
72. Lemieux, E. J., and L. J. Scotti: *Chem. Eng. Prog.*, **65**(3), 52 (1969).
73. Leva, M.: *Chem. Eng. Prog.*, **67**(3), 65 (1971).
74. Levins, D. M., and J. R. Glastonbury: *Trans. Inst. Chem. Eng. Lond.*, **50**, 32 (1972).
75. Lewis, W. K., Jr.: *Ind. Eng. Chem.*, **28**, 399 (1936).
76. Lobo, W. E., L. Friend, F. Hashmall, and F. Zenz: *Trans. AIChE*, **41**, 693 (1945).
77. Lubowicz, R. E., and P. Reich: *Chem. Eng. Prog.*, **67**(3), 59 (1971).
78. Mack, D. E., and A. E. Kroll: *Chem. Eng. Prog.*, **44**, 189 (1948).
79. Madigan, C. M.: M. Ch. Eng. dissertation, New York University, 1964.
80. Maneri, C. C., and H. D. Mendelson: *AIChE J.*, **14**, 295 (1968).
81. Martin, G. Q.: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 397 (1972).
82. McAllister, R. A., P. H. McGinnis, and C. A. Planck: *Chem. Eng. Sci.*, **9**, 25 (1958).
83. Mendelson, H. D.: *AIChE J.*, **13**, 250 (1967).
84. Miller, D. N.: *AIChE J.*, **20**, 445 (1974).
85. Morris, W. D., and J. Benyan: *Ind. Eng. Chem. Process Des. Dev.*, **15**, 338 (1976).
86. Nagata, S.: "Mixing, Principles and Applications," Halsted, New York, 1975.
87. Nemunaitis, R. R.: *Hydrocarbon Process.*, **55**(11), 235 (1971).
88. Nienow, A. W., and D. Miles: *Ind. Eng. Chem. Process Des. Dev.*, **10**, 41 (1971).
89. Norman, W. S., T. Cakaloz, A. Z. Fiesco, and D. H. Sutcliffe: *Trans. Inst. Chem. Eng. Lond.*, **41**, 61 (1963).
90. O'Connell, H. E.: *Trans. AIChE*, **42**, 741 (1946).
91. Oldshue, J. Y.: *Biotech. Bioeng.*, **8**, 3 (1966).
92. Oldshue, J. Y., and F. L. Connelly: *Chem. Eng. Prog.*, **73**(3), 85 (1977).
93. Oldshue, J. Y., and J. H. Rushton: *Chem. Eng. Prog.*, **49**, 161, 267 (1953).
94. Onda, K., H. Takeuchi, and Y. Okumoto: *J. Chem. Eng. Jap.*, **1**, 56 (1968).
95. Petrick, M.: *U.S. At. Energy Comm. ANL-658*, 1962.
96. Pinczewski, W. V., N. D. Benke, and C. J. D. Fell: *AIChE J.*, **21**, 1019, 1210 (1975).
97. Plank, C. A., and E. R. Gerhard: *Ind. Eng. Chem. Process Des. Dev.*, **2**, 34 (1963).
98. Pollard, B.: *Chem. Ind. Lond.*, **1958**, 1414.
99. Raal, J. D., and M. K. Khurana: *Can. J. Chem. Eng.*, **51**, 162 (1973).
100. Reiss, L. P.: *Ind. Eng. Chem. Process Des. Dev.*, **6**, 486 (1967).
101. Resnick, W., and B. Gal-Or: *Adv. Chem. Eng.*, **7**, 295 (1968).
102. Rushton, J. H., E. W. Costich, and H. J. Everett: *Chem. Eng. Prog.*, **46**, 395, 467 (1950).
103. Satterfield, C. N.: *AIChE J.*, **21**, 209 (1975).
104. Schwartzberg, H. G., and R. E. Treybal: *Ind. Eng. Chem. Fundam.*, **7**, 1, 6 (1968).
105. Scofield, R. C.: *Chem. Eng. Prog.*, **46**, 405 (1950).
106. Sherwood, T. K., and F. A. L. Holloway: *Trans. AIChE*, **36**, 39 (1940).
107. Shulman, H. L., et al.: *AIChE J.*, **1**, 247, 253, 259 (1955); **3**, 157 (1957); **5**, 280 (1959); **6**, 175, 469 (1960); **9**, 479 (1963); **13**, 1137 (1967); **17**, 631 (1971).
108. Smith, V. C., and W. V. Delnicki: *Chem. Eng. Prog.*, **71**(8), 68 (1975).
109. Staff, Shell Development Co.: *Chem. Eng. Prog.*, **50**, 57 (1954).
110. Stahl, Apparate- und Gerätebau GmbH, Viernheim, Germany, *Bull.*, 1975.
111. Standart, G. L.: *Chem. Eng. Sci.*, **26**, 985 (1971).
112. Strek, F., J. Werner, and A. Paniuticz: *Int. Chem. Eng.*, **9**, 464 (1969).
113. Sullivan, G. A., and R. E. Treybal: *Chem. Eng. J.*, **1**, 302 (1970).
114. Thorngren, J. T.: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 428 (1972).
115. Tupuwala, H. H., and G. Hamer: *Trans. Inst. Chem. Eng. Lond.*, **52**, 113 (1974).
116. Uchida, S., and C. Y. Wen: *Ind. Eng. Chem. Process Des. Dev.*, **12**, 437 (1973).

117. Urza, I. J., and M. J. Jackson: *Ind. Eng. Chem. Process Des. Dev.*, **14**, 106 (1975).
118. Van Krevelen, W., and P. J. Hoftijzer: *Chem. Eng. Prog.*, **46**(1), 29 (1950).
119. Van Winkle, M.: "Distillation," McGraw-Hill Book Company, New York, 1967.
120. Weiler, D. W., B. L. England, and W. V. Delnicki: *Chem. Eng. Prog.*, **69**(10), 67 (1973).
121. Weinstein, B., and A. H. Williams: *79th Natl. Meet. AIChE, Houston, 1975*.
122. Westerterp, K. R., L. L. van Dierendonck, and J. A. de Kraa: *Chem. Eng. Sci.*, **18**, 157, 495 (1963).
123. Woodburn, E. T.: *AIChE J.*, **20**, 1003 (1974).
124. York, O. H., and E. W. Popple: *Chem. Eng. Prog.*, **59**(6), 45 (1963).
125. Young, G. C., and J. H. Weber: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 440 (1972).
126. Zenz, F. A.: *Chem. Eng.*, **79**, 120 (Nov. 13, 1972).
127. Zuiderweg, F. J., and A. Harmens: *Chem. Eng. Sci.*, **9**, 89 (1958).

PROBLEMS

6.1 A gas bubble released below the surface of a deep pool of liquid will expand as it rises to the surface because of the reduction of pressure to which it is subject.

(a) Derive an expression for the time of rise of a bubble of initial diameter d_{00} released at a depth Z below the surface of a liquid of density ρ_L . Atmospheric pressure is p_A . Assume that the bubble assumes the terminal velocity corresponding to its diameter instantly.

(b) If an air bubble is initially 0.50 mm in diameter and released at a depth of 10 m in water at 25°C, compute the time of rise to the surface. Atmospheric pressure is standard.

Ans.: 54 s.

(c) Repeat for a 2.0-mm-diameter bubble.

6.2 Calculate the power imparted to the vessel contents by the gas in the case of the sparged vessel of Illustration 6.1.

Ans.: 1.485 kW.

6.3 Calculate the power imparted to the vessel by the gas and the total power for the agitated vessel of Illustration 6.2.

6.4 Petroleum oils which contain small amounts of suspended water droplets, resulting from washing the oils with water after chemical treatment, are cloudy in appearance. Sparging with air evaporates the moisture and the oil is "brightened," i.e., made clear. Presumably the air bubbles pick up water after diffusion of dissolved water through the oil.

A batch of a petroleum-oil product is to be brightened at 80°C with air. The oil viscosity at that temperature is 10 cP, its specific gravity 0.822, and its surface tension 20 dyn/cm; av mol wt = 320. The oil will be contained in a vessel 1.25 m in diameter, liquid depth 2.5 m. Air will be introduced at the rate of 0.06 kg/s at the bottom through a sparger ring 400 mm in diameter, containing 50 holes each 6.5 mm in diameter.

Estimate the gas-bubble size, the gas holdup, the specific interfacial area, and the mass-transfer coefficient for water in the liquid at the gas-liquid surface.

Ans.: $a = 216.6 \text{ m}^2/\text{m}^3$, $k_L = 5.71 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$.

6.5 A baffled fermenter tank, 5.0 ft (1.525 m) in diameter, which will contain 1600 gal (6.056 m³) of a beet-sugar solution, will be used to produce citric acid by action of the microorganism *Aspergillus niger* at 25°C. The vessel will be agitated with two 28-in-diameter (0.71-m) flat-blade disk turbines, one located 2 ft (0.61 m) above the bottom of the tank, the other 6 ft (1.83 m) from the bottom, on the same shaft, turning at 92 r/min. Sterilized air will be introduced below the lower impeller at 2.1 ft/min (0.01067 m/s) superficial velocity based on the tank cross-sectional area. The solution sp gr = 1.038, the viscosity 1.4 cP ($1.4 \times 10^{-3} \text{ kg/m} \cdot \text{s}$).

(a) Estimate the agitator power required.

Ans.: 5.4 kW.

(b) If the air is introduced through five $\frac{1}{2}$ -in. schedule 40, open pipes (ID = 0.622 in = 15.8 mm), estimate the power imparted to the tank contents by the air.

6.6 A baffled fermenter [10 ft (3.05 m) diameter, liquid depth 10 ft (3.05 m)] contains nutrient liquid of sp gr 1.1, viscosity 5 cP, surface tension 50 dyn/cm. The liquid contains dissolved electrolytes. It will be aerated at the rate 360 ft³/h (0.0305 m³/s) superficial air velocity based on the tank cross section and agitated with a flat-blade disk turbine 4.5 ft (1.37 m) in diameter turning at 60 r/min and located 3 ft (0.91 m) from the bottom of the tank. The temperature will be 80°F (26.7°C). The diffusivity of oxygen in water at 25°C = 2.5×10^{-5} cm²/s. Estimate the diffusivity of oxygen in the solution by assuming that the quantity $D_L \mu_L / t = \text{const}$, where t = absolute temperature.

Compute the agitator power required, the interfacial area, and the mass-transfer coefficient for oxygen transfer.

Ans.: $P = 16.2$ hp = 12.1 kW, $k_L = 0.31$ lb mol/ft² · h · (lb mol/ft³) = 2.63×10^{-5} kmol/m² · s · (kmol/m³).

6.7 A baffled fermentation tank for a pilot plant is to be 1.5 m in diameter and will contain liquid to a depth 2.0 m. The flat-blade disk turbine impeller, 0.5 m diameter, will be located $\frac{1}{3}$ m from the bottom of the vessel. Air at a superficial velocity of 0.06 m/s will be introduced beneath the impeller. The temperature will be 27°C, and the liquid properties may be taken as those of water. Small-scale tests indicate that a suitable agitator power will be 0.5 kW/m³ of liquid. At what speed should the impeller be turned?

6.8 A sieve-tray tower is to be designed for stripping an aniline-water solution with steam. The circumstances at the top of the tower, which are to be used to establish the design, are

$$\text{Temperature} = 98.5^\circ\text{C} \quad \text{pressure} = 745 \text{ mmHg abs}$$

Liquid:

$$\text{Composition} = 7.00 \text{ mass \% aniline}$$

$$\text{Rate} = 6.3 \text{ kg/s} = 50\,000 \text{ lb/h} \quad \text{density} = 961 \text{ kg/m}^3 = 60 \text{ lb/ft}^3$$

$$\text{Viscosity} = 3 \times 10^{-4} \text{ kg/m} \cdot \text{s} = 0.3 \text{ cP}$$

$$\text{Surface tension} = 0.058 \text{ N/m} = 58 \text{ dyn/cm}$$

$$\text{Aniline diffusivity} = 52 \times 10^{-10} \text{ m}^2/\text{s} = 52 \times 10^{-6} \text{ cm}^2/\text{s} \text{ (est)}$$

Vapor:

$$\text{Composition} = 3.6 \text{ mole \% aniline} \quad \text{rate} = 3.15 \text{ kg/s} = 25\,000 \text{ lb/h}$$

$$\text{Aniline diffusivity} = 1.261 \times 10^{-5} \text{ m}^2/\text{s} = 0.1261 \text{ cm}^2/\text{s} \text{ (est)}$$

The equilibrium data [Griswold, et al.: *Ind. Eng. Chem.*, 32, 878 (1940)] indicate that $m = 0.0636$ at this concentration.

(a) Design a suitable cross-flow tray for such a tower. Report details respecting perforation size and arrangement, tower diameter, tray spacing, weir length and height, downspout seal, pressure drop for the gas, height of liquid in the downspout, and entrainment in the gas. Check for excessive weeping.

(b) Estimate the tray efficiency for the design reported in part (a).

6.9 A gas containing methane, propane, and butane is to be scrubbed countercurrently in a sieve-tray tower with a hydrocarbon oil to absorb principally the butane. It is agreed to design a tray for the circumstances existing at the bottom of the tower, where the conditions are

$$\text{Pressure} = 350 \text{ kN/m}^2 \text{ (51 lb}_f\text{/in}^2\text{ abs)} \quad \text{temperature} = 38^\circ\text{C}$$

Gas: 0.25 kmol/s (1984 lb mol/h), containing 85% methane, 10% propane, and 5% butane by volume.

Liquid: 0.15 kmol/s (1190 lb mol/h), av mol wt = 150, density = 849 kg/m³ (53 lb/ft³), surface tension = 0.025 N/m (25 dyn/cm), viscosity = 0.00160 kg/m · s (1.60 cP).

(a) Design a suitable sieve tray, and check for weeping and flooding by downspout backup.

(b) Estimate the tray efficiency for butane absorption, corrected for entrainment. The average molecular diffusivities may be taken as 3.49×10^{-6} m²/s for the gas and 1.138×10^{-9} m²/s for the liquid.

6.10 A packed tower is to be designed for the countercurrent contact of a benzene-nitrogen gas mixture with kerosene to wash out the benzene from the gas. The circumstances are:

Gas in: $1.50 \text{ m}^3/\text{s}$ ($53 \text{ ft}^3/\text{s}$), containing 5 mol % benzene, at 25°C , $1.1 \times 10^5 \text{ N/m}^2$ ($16 \text{ lb}_f/\text{in}^2$).

Gas out: substantially pure nitrogen.

Liquid in: 4.0 kg/s (8.82 lb/s), density = 800 kg/m^3 (50 lb/ft^3), viscosity = $0.0023 \text{ kg/m} \cdot \text{s}$.

The packing will be 50-mm (2-in) metal Pall rings, and the tower diameter will be set to produce 400 N/m^2 per meter of gas-pressure drop ($0.5 \text{ in H}_2\text{O/ft}$) for irrigated packing.

(a) Calculate the tower diameter to be used.

(b) Assume that, for the diameter chosen, the irrigated packed depth will be 6 m (19.7 ft) and that 1 m of unirrigated packing will be placed over the liquid inlet to act as entrainment separator. The blower-motor combination to be used at the gas inlet will have an overall efficiency of 60%. Calculate the power required to blow the gas through the packing.

6.11 A small water-cooling tower, 1 m diameter, packed with 76-mm (3-in) ceramic Raschig rings, is fed with water at the rate $28 \text{ kg/m}^2 \cdot \text{s}$ ($20\,600 \text{ lb/ft}^2 \cdot \text{h}$), in at 40°C and out at 25°C . The water is contacted with air (30°C , 1 std atm, essentially dry) drawn upward countercurrently to the water flow. Neglecting evaporation of the water, estimate the rate of airflow which would flood the tower.

CHAPTER SEVEN

HUMIDIFICATION OPERATIONS 16/5/09

MASS & ENERGY TRANSFER

The operations considered in this chapter are concerned with the interphase transfer of mass and of energy which result when a gas is brought into contact with a pure liquid in which it is essentially insoluble. While the term *humidification operations* is used to characterize these in a general fashion, the purpose of such operations may include not only humidification of the gas but dehumidification and cooling of the gas, measurement of its vapor content, and cooling of the liquid as well. The matter transferred between phases in such cases is the substance constituting the liquid phase, which either vaporizes or condenses. As in all mass-transfer problems, it is necessary for a complete understanding of the operation to be familiar with the equilibrium characteristics of the systems. But since the mass transfer in these cases will invariably be accompanied by a simultaneous transfer of heat energy as well, some consideration must also be given to the enthalpy characteristics of the systems.

VAPOR-LIQUID EQUILIBRIUM AND ENTHALPY FOR A PURE SUBSTANCE

As indicated above, the substance undergoing interphase transfer in these operations is the material constituting the liquid phase, which diffuses in the form of a vapor. The equilibrium vapor-pressure characteristics of the liquid are therefore of importance.

is not always strictly adhered to, and the term "gas" is frequently used to designate merely a condition relatively far removed from the vapor-pressure curve. The temperature corresponding to each pressure on the curve is termed the *boiling point* of the liquid at the pressure in question, and that corresponding to 1 std atm in particular is known as the *normal boiling point*, as at t_{nbp} in Fig. 7.1.

Whenever a process involves bringing a sample of fluid across the vapor-pressure curve, such as the isobaric process ADE or the isothermal process ABF , there will be a change of phase. This will be accompanied by the evolution (for condensation) or absorption (for vaporization) of the *latent heat of vaporization* at constant temperature, for example, at points B or D in the processes mentioned above. Heat added or given up with changing temperatures is called *sensible heat*.

Interpolation between data For such common liquids as water, many refrigerants, and others, the vapor-pressure-temperature curve has been established at many points. For most liquids, however, only relatively few data are available, so that it is necessary frequently to interpolate between, or extrapolate beyond, the measurements. The curve on arithmetic coordinates (Fig. 7.1) is very inconvenient for this because of the curvature, and some method of linearizing the curve is needed. Most of the common methods stem from the Clausius-Clapeyron equation, which relates the slope of the vapor-pressure curve to the latent heat of vaporization

$$\frac{dp}{dT} = \frac{\lambda'}{T(v_G - v_L)} = \frac{\lambda}{T v_m} = \frac{\lambda p}{T R T} = \frac{p \lambda}{R T^2} \quad (7.1)$$

where v_G and v_L are molal specific volumes of the saturated vapor and liquid, respectively, and λ' is the molal latent heat in units consistent with the rest of the equation. As a simplification, we can neglect v_L in comparison with v_G and express the latter by the ideal-gas law, to obtain

$$\frac{d \ln p}{dp} = \frac{dp}{p} = \frac{\lambda' dT}{R T^2} \quad (7.2)$$

and if λ' can be considered reasonably constant over a short range of temperature,

$$\ln p = -\frac{\lambda'}{RT} + \text{const} \quad (7.3)$$

Equation (7.3) suggests that a plot of $\log p$ against $1/T$ will be straight for short temperature ranges. It also suggests a method of interpolating between points listed in a table of data.

Illustration 7.1 A table lists the vapor pressure of benzene to be 100 mmHg at 26.1°C and 400 mmHg at 60.6°C. At what temperature is the vapor pressure 200 mmHg?

SOLUTION At 26.1°C, $1/T = 1/299.1 \text{ K}^{-1}$; at 60.6°C, $1/T = 1/333.6 \text{ K}^{-1}$.

$$\frac{1/299.1 - 1/T}{1/299.1 - 1/333.6} = \frac{\log 100 - \log 200}{\log 100 - \log 400}$$

$$T = 315.4 \text{ K} = 42.4^\circ\text{C}$$

The correct value is 42.2°C. Linear interpolation would have given 37.6°C.

Reference-Substance Plots [21]

Equation (7.2) can be rewritten for a second substance, a *reference substance*, at the same temperature,

$$d \ln p_r = \frac{\lambda'_r dT}{RT^2} \quad (7.4)$$

where the subscript *r* denotes the reference substance. Dividing Eq. (7.2) by (7.4) provides

$$\frac{d \ln p}{d \ln p_r} = \frac{\lambda'}{\lambda'_r} = \frac{M\lambda}{M_r\lambda_r} \quad (7.5)$$

which, upon integration, becomes

$$\log p = \frac{M\lambda}{M_r\lambda_r} \log p_r + \text{const} \quad (7.6)$$

Equation (7.6) suggests that a linear graph will result if $\log p$ as ordinate is plotted against $\log p_r$ for the reference substance as abscissa, where for each plotted point the vapor pressures are taken at the same temperature. Such a plot is straight over larger temperature ranges (but not near the critical temperature) than that based on Eq. (7.3), and, moreover, the slope of the curve gives the ratio of the latent heats at the same temperature. The reference substance chosen is one whose vapor-pressure data are well known.

Illustration 7.2 (a) Plot the vapor pressure of benzene over the range 15 to 180°C using water as the reference substance according to Eq. (7.6). (b) Determine the vapor pressure of benzene at 100°C. (c) Determine the latent heat of vaporization of benzene at 25°C.

SOLUTION (a) Logarithmic graph paper is marked with scales for the vapor pressure of benzene and for water, as in Fig. 7.2. The vapor pressure of benzene at 15.4°C is 60 mmHg, and that for water at this temperature is 13.1 mmHg. These pressures provide the coordinates of the lowest point on the plot. In similar fashion, additional data for benzene are plotted with the help of a steam table, thus providing the curve shown. The line is very nearly straight over the temperature range used.

(b) At 100°C, the vapor pressure of water is 760 mmHg. Entering the plot at this value for the abscissa, the vapor pressure of benzene is read as 1400 mmHg. Alternatively the abscissa can be marked with the temperatures corresponding to the vapor pressures of water, as shown, thus eliminating the necessity of referring to the steam table.

(c) The slope of the curve at 25°C is 0.775. (Note: This is most conveniently determined with a millimeter rule. If the coordinates are used, the slope will be $\Delta \log p / \Delta \log p_r$.) At 25°C,

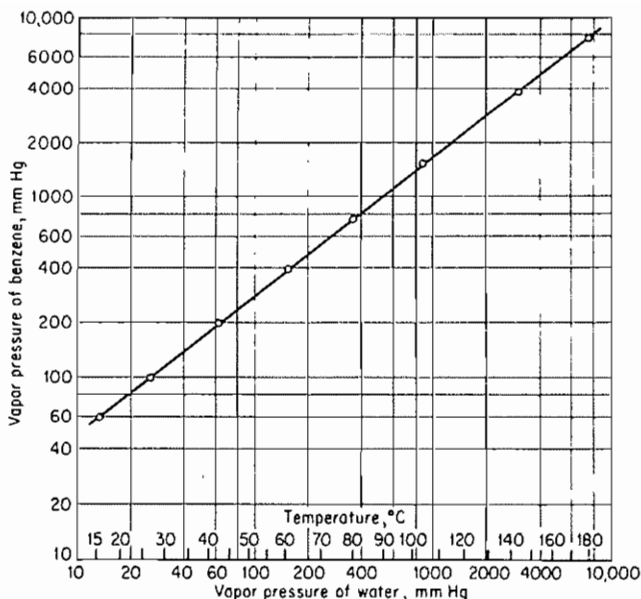


Figure 7.2 Reference-substance plot for the vapor pressure of benzene.

the latent heat of vaporization of water is $2443 \text{ kN} \cdot \text{m/kg}$. From Eq. (7.6)

$$\frac{\lambda M}{\lambda_r M_r} = \frac{78.05}{2443000(18.02)} = 0.775$$

$$\lambda = 437 \text{ kN} \cdot \text{m/kg for benzene at } 25^\circ\text{C}$$

(The accepted value is $434 \text{ kN} \cdot \text{m/kg}$.)

$\rightarrow f(T, P)$
Enthalpy weak funⁿ of p
at low P can neglect

The internal energy U of a substance is the total energy residing in the substance owing to the motion and relative position of the constituent atoms and molecules. Absolute values of internal energy are not known, but numerical values relative to some arbitrarily defined standard state for the substance can be computed. The sum of the internal energy and the product of pressure and volume of the substance, when both quantities are expressed in the same units, is defined as the *enthalpy* of the substance.

$\leftarrow \text{functⁿ of } T \text{ and } P$

$$H = U + pv$$

In a batch process at constant pressure, where work is done only in expansion against the pressure, the heat absorbed by the system is the gain in enthalpy,

$$Q = \Delta H = \Delta(U + pv) \quad (7.7)$$

In a steady-state continuous-flow process, the net transfer of energy to the system as heat and work will be the sum of its gains in enthalpy and potential and kinetic energies. It frequently happens that the changes in potential and kinetic energies are insignificant in comparison with the enthalpy change and that there is no mechanical work done. In such cases, Eq. (7.7) can be used to compute the heat added to the system, and such a calculation is termed a *heat balance*. In *adiabatic* operations, where no exchange of heat between the system and its surroundings occurs, the heat balance becomes simply an equality of enthalpies in the initial and final condition.

Absolute values of the enthalpy of a substance, like the internal energy, are not known. However, by arbitrarily setting the enthalpy of a substance at zero when it is in a convenient reference state, relative values of enthalpy at other conditions can be calculated. To define the reference state, the temperature, pressure, and state of aggregation must be established. For the substance water, the ordinary steam tables list the relative enthalpy at various conditions referred to the enthalpy of the substance at 0°C , the equilibrium vapor pressure at this temperature, and in the liquid state. For other substances, other reference conditions may be more convenient.

Figure 7.3 is a graphical representation of the relative enthalpy of a typical substance where the liquid, vapor, and gaseous states are shown. The data are

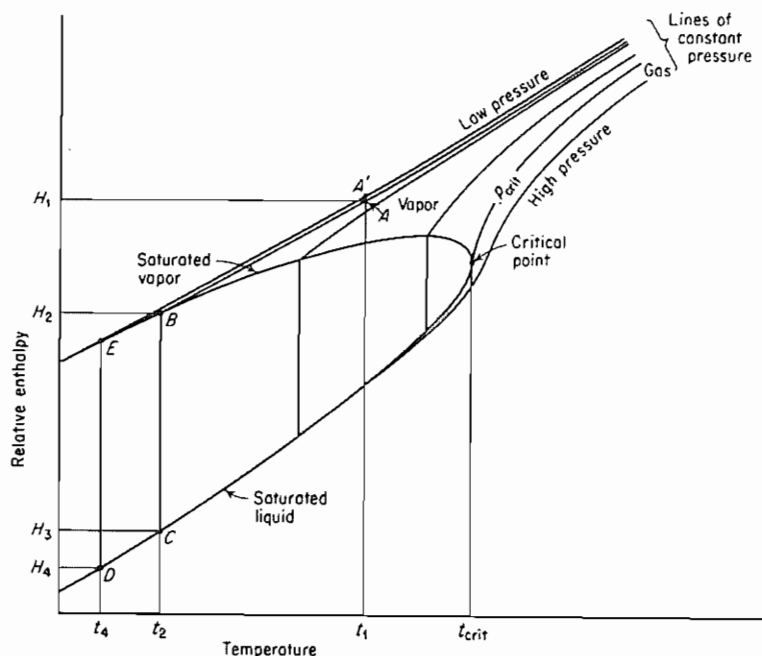


Figure 7.3 Typical enthalpy-temperature diagram for a pure substance.

most conveniently shown on lines of constant pressure. The curves marked "saturated liquid" and "saturated vapor," however, cut across the constant-pressure lines and show the enthalpies for these conditions at temperatures and pressures corresponding to the equilibrium vapor-pressure relationship for the substance. The vertical distance between the saturated-vapor and -liquid curves, such as the distance BC , represents the latent heat of vaporization at the corresponding temperature. The latent heat thus decreases with increased temperature, becoming zero at the critical point. In the vapor state at low pressures, the enthalpy is essentially a function of temperature; at all pressures where the ideal-gas law can be used to describe the pvt relation, the lines of constant pressure are superimposed and the enthalpy is independent of pressure. Except near the critical temperature, the enthalpy of the liquid is also substantially independent of pressure until exceedingly high pressures are reached.

The change in enthalpy between two conditions, such as those at A and D , may be taken simply as the difference in ordinates corresponding to the points. Thus, to calculate the enthalpy of the substance in the superheated condition at point A relative to the saturated liquid at D , or $H_1 - H_4$, we can add the enthalpy change $H_1 - H_2$, the sensible heat of the vapor from the saturation temperature t_2 at the same pressure to the superheated condition at A ; $H_2 - H_3$, the latent heat of vaporization at t_2 ; and $H_3 - H_4$, the sensible heat of the liquid from the final condition at D to the boiling point at the prevailing pressure t_2 . For a liquid or vapor, the slope of the constant-pressure lines at any temperature is termed the *heat capacity*. The lines are not strictly straight, so that the heat capacity changes with temperature. By use of an average heat capacity or average slope, however, sensible heats are readily calculated. Thus, referring again to Fig. 7.3,

$$\text{For no change in state:} \quad \frac{dH}{dt} = C_p$$

$$H_1 - H_2 = C(t_1 - t_2) \quad H_1 - H_1 = \int_{t_2}^{t_1} C_p dt$$

where C is the average heat capacity of the vapor at constant pressure over the indicated temperature range.

Illustration 7.3 Compute the heat evolved when 10 kg of benzene as a superheated vapor at 94 mmHg, 100°C, is cooled and condensed to a liquid at 10°C. The average heat capacity for the vapor may be taken as 1.256 and for the liquid 1.507 kJ/kg · K.

SOLUTION Refer to Fig. 7.2. When the pressure is 94 mmHg, the saturation temperature for benzene is 25°C. The latent heat of vaporization at this temperature is 434 kJ/kg (Illustration 7.2). The initial condition corresponds to a point such as A on Fig. 6.3, the final condition to point D , the path of the process to $ABCD$. Using the notation of Fig. 7.3,

$$H_1 - H_2 = C(t_1 - t_2) = 1.256(100 - 25) = 94.2 \text{ kJ/kg}$$

$$H_2 - H_3 = 434 \text{ kJ/kg}$$

$$H_3 - H_4 = C(t_2 - t_4) = 1.507(25 - 10) = 22.6 \text{ kJ/kg}$$

$$H_1 - H_4 = 94.2 + 434 + 22.6 = 550.8 \text{ kJ/kg}$$

$$\text{Heat evolved for 10 kg benzene} = 10(550.8) = 5508 \text{ kJ/kg}$$

VAPOR-GAS MIXTURES

In what follows, the term *vapor* will be applied to that substance, designated as substance A, in the vaporous state which is relatively near its condensation temperature at the prevailing pressure. The term *gas* will be applied to substance B, which is a relatively highly superheated gas.

Absolute Humidity

While the common concentration units (partial pressure, mole fraction, etc.) which are based on total quantity are useful, when operations involve changes in vapor content of a vapor-gas mixture without changes in the gas content, it is more convenient to use a unit based on the unchanging amount of gas. The ratio mass of vapor/mass of gas is the *absolute humidity* Y' . If the quantities are expressed in moles, the ratio is the *molal absolute humidity* Y . Under conditions where the ideal-gas law applies,

$$Y = \frac{y_A}{y_B} = \frac{\bar{p}_A}{\bar{p}_B} = \frac{\bar{p}_A}{p_t - \bar{p}_A} \frac{\text{moles A}}{\text{moles B}} \quad \leftarrow \text{molal absolute humidity}$$

$$Y' = Y \frac{M_A}{M_B} = \frac{\bar{p}_A}{p_t - \bar{p}_A} \frac{M_A}{M_B} \frac{\text{mass A}}{\text{mass B}} \quad \leftarrow \text{absolute humidity} \quad (7.8)$$

In many respects the molal ratio is the more convenient, thanks to the ease with which moles and volumes can be interrelated through the gas law, but the mass ratio has nevertheless become firmly established in the humidification literature. The mass absolute humidity was first introduced by Grosvenor [7] and is sometimes called the *Grosvenor humidity*.

Illustration 7.4 In a mixture of benzene vapor (A) and nitrogen gas (B) at a total pressure of 800 mmHg and a temperature of 60°C, the partial pressure of benzene is 100 mmHg. Express the benzene concentration in other terms.

SOLUTION $\bar{p}_A = 100$, $\bar{p}_B = 800 - 100 = 700$ mmHg.

(a) Mole fraction. Since the pressure fraction and mole fraction are identical for gas mixtures, $y_A = \bar{p}_A/p_t = 100/800 = 0.125$ mole fraction benzene. The mole fraction nitrogen = $y_B = 1 - 0.125 = 700/800 = 0.875$.

(b) Volume fraction of benzene equals the mole fraction, 0.125.

(c) Absolute humidity.

$$Y = \frac{y_A}{y_B} = \frac{\bar{p}_A}{\bar{p}_B} = \frac{0.125}{0.875} = \frac{100}{700} = 0.143 \text{ mol benzene/mol nitrogen}$$

$$Y' = Y \frac{M_A}{M_B} = 0.143 \frac{78.05}{28.08} = 0.398 \text{ kg benzene/kg nitrogen}$$

Saturated Vapor-Gas Mixtures

If an insoluble dry gas B is brought into contact with sufficient liquid A, the liquid will evaporate into the gas until ultimately, at equilibrium, the partial

pressure of A in the vapor-gas mixture reaches its saturation value, the vapor pressure p_A at the prevailing temperature. So long as the gas can be considered insoluble in the liquid, the partial pressure of vapor in the saturated mixture is independent of the nature of the gas and total pressure (except at very high pressures) and is dependent only upon the temperature and identity of the liquid. However, the saturated molal absolute humidity $Y_s = p_A/(p_t - p_A)$ will depend upon the total pressure, and the saturated absolute humidity $Y'_s = Y_s M_A/M_B$ upon the identity of the gas as well. Both saturated humidities become infinite at the boiling point of the liquid at the prevailing total pressure.

Illustration 7.5 A gas (B)–benzene (A) mixture is saturated at 1 std atm, 50°C. Calculate the absolute humidity if B is (a) nitrogen and (b) carbon dioxide.

SOLUTION Since the mixture is saturated, the partial pressure of benzene, $\bar{p}_A$, equals the equilibrium vapor pressure p_A of benzene at 50°C. From Fig. 7.2, $p_A = 275$ mmHg, or 0.362 std atm.

(a)

$$Y_s = \frac{p_A}{p_t - p_A} = \frac{0.362}{1 - 0.362} = 0.567 \text{ kmol C}_6\text{H}_6/\text{kmol N}_2$$

$$Y'_s = \frac{Y_s M_A}{M_B} = \frac{0.567(78.05)}{28.02} = 1.579 \text{ kg C}_6\text{H}_6/\text{kg N}_2$$

(b)

$$Y_s = \frac{p_A}{p_t - p_A} = \frac{0.362}{1 - 0.362} = 0.567 \text{ kmol C}_6\text{H}_6/\text{kmol CO}_2$$

$$Y'_s = \frac{Y_s M_A}{M_B} = \frac{0.567(78.05)}{44.01} = 1.006 \text{ kg C}_6\text{H}_6/\text{kg CO}_2$$

Unsaturated Vapor-Gas Mixtures

If the partial pressure of the vapor in a vapor-gas mixture is for any reason less than the equilibrium vapor pressure of the liquid at the same temperature, the mixture is unsaturated.

Dry-bulb temperature This is the temperature of a vapor-gas mixture as ordinarily determined by immersion of a thermometer in the mixture.

Relative saturation *Relative saturation*, also called *relative humidity*, expressed as a percentage is defined as $100\bar{p}_A/p_A$, where p_A is the vapor pressure at the dry-bulb temperature of the mixture. For any vapor, the graphical representation of conditions of constant relative saturation can easily be constructed on a vapor-pressure–temperature chart, as in Fig. 7.4a, by dividing the ordinates of the vapor-pressure curve into appropriate intervals. Thus the curve for 50 percent relative saturation shows a vapor partial pressure equal to one-half the equilibrium vapor pressure at any temperature. A reference-substance plot, such as Fig. 7.2, could also be used for this.

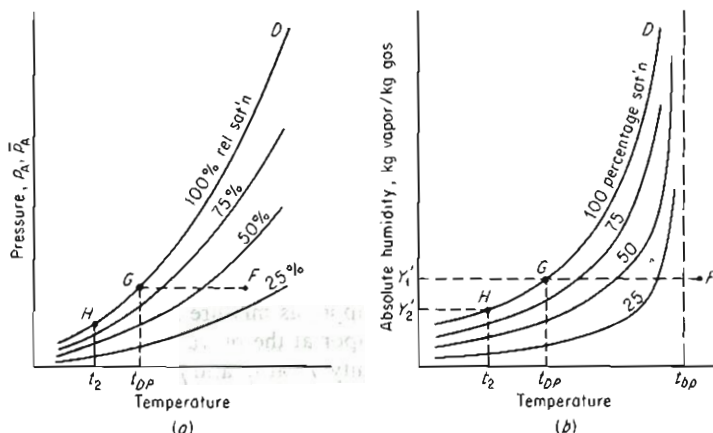


Figure 7.4 Forms of psychrometric charts.

Percentage saturation. Percentage saturation, or percentage absolute humidity, is defined as $100Y/Y_s$ and $100Y'/Y'_s$, where the saturated values are computed at the dry-bulb temperature of the mixture. Graphical representation of the quantity for any vapor can be made on a chart of Y vs. t (in which case the chart must be limited to a single total pressure) or one of Y' vs. t (for a single total pressure and a specific gas), as in Fig. 7.4b. On this chart, the saturation humidities are plotted from vapor-pressure data with the help of Eq. (7.8), to give curve GD . The curve for humidities at 50 percent saturation is plotted at half the ordinate of curve GD , etc. All the curves of constant percentage saturation reach infinity at the boiling point of the liquid at the prevailing pressure.†

Dew point This is the temperature at which a vapor-gas mixture becomes saturated when cooled at constant total pressure out of contact with a liquid. For example, if an unsaturated mixture such as that at F (Fig. 7.4) is cooled at constant pressure out of contact with liquid, the path of the cooling process follows the line FG , the mixture becoming more nearly saturated as the temperature is lowered, and fully saturated at t_{DP} , the dew-point temperature. All mixtures of absolute humidity Y_1' on this figure have the same dew point. If the temperature is reduced only an infinitesimal amount below t_{DP} , vapor will condense as a liquid dew. This is used as a method of humidity determination: a

† For this reason curves of constant relative saturation are sometimes drawn on absolute-humidity-temperature charts. Since relative saturation and percentage saturation are not numerically equal for an unsaturated mixture, the position of such curves must be computed by the methods of Illustration 7.6.

shiny metal surface is cooled in the presence of the gas mixture, and the appearance of a fog which clouds the mirrorlike surface indicates that the dew point has been reached.

If the mixture is cooled to a lower temperature, the vapor-gas mixture will continue to precipitate liquid, itself always remaining saturated, until at the final temperature t_2 (Fig. 7.4) the residual vapor-gas mixture will be at point H . The mass of vapor condensed per unit mass of dry gas will be $Y'_1 - Y'_2$. Except under specially controlled circumstances supersaturation will not occur, and no vapor-gas mixture whose coordinates lie to the left of curve GD will result.

Humid volume The humid volume v_H of a vapor-gas mixture is the volume of unit mass of dry gas and its accompanying vapor at the prevailing temperature and pressure. For a mixture of absolute humidity Y' at t_G and p_t , total pressure, the ideal-gas law gives the humid volume as

$$v_H = \left(\frac{1}{M_B} + \frac{Y'}{M_A} \right) 22.41 \frac{t_G + 273}{273} \frac{1.013 \times 10^5}{p_t} = 8315 \left(\frac{1}{M_B} + \frac{Y'}{M_A} \right) \frac{t_G + 273}{p_t} \quad (7.9)$$

where v_H is in m^3/kg , t_G in degrees Celsius, and $p_t = \text{N/m}^2$.† The humid volume of a saturated mixture is computed with $Y' = Y'_s$ and that for a dry gas with $Y' = 0$. These values can then be plotted against temperature on a psychrometric chart. For partially saturated mixtures, v_H can be interpolated between values for 0 and 100 percentage saturation at the same temperature according to percentage saturation. When the mass of *dry* gas in a mixture is multiplied by the humid volume, the volume of *mixture* results.

Humid heat The humid heat C_S is the heat required to raise the temperature of unit mass of gas and its accompanying vapor one degree at constant pressure. For a mixture of absolute humidity Y' ,

$$C_S = C_B + Y' C_A \quad (7.10)$$

Provided neither vaporation nor condensation occurs, the heat in Btu required to raise the temperature of a mass of W_B dry gas and its accompanying vapor an amount Δt will be

$$Q = W_B C_S \Delta t \quad (7.11)$$

Enthalpy The (relative) enthalpy of a vapor-gas mixture is the sum of the (relative) enthalpies of the gas and of the vapor content. Imagine unit mass of a gas containing a mass Y' of vapor at dry-bulb temperature t_G . If the mixture is unsaturated, the vapor is in a superheated state and we can calculate the

† For v_H in ft^3/lb , t_G in degrees Fahrenheit, and p_t in atmospheres Eq. (7.9) becomes

$$v_H = 0.730 \left(\frac{1}{M_B} + \frac{Y'}{M_A} \right) \frac{t_G + 460}{p_t}$$

enthalpy relative to the reference states gas and saturated liquid at t_0 . The enthalpy of the gas alone is $C_B(t_G - t_0)$. The vapor at t_G is at a condition corresponding to point A on Fig. 7.3, and its reference state corresponds to point D . If t_{DP} is the dew point of the mixture (t_2 in Fig. 7.3) and λ_{DP} the latent heat of vaporization of the vapor at that temperature, the enthalpy per unit mass of vapor will be $C_A(t_G - t_{DP}) + \lambda_{DP} + C_{A,L}(t_{DP} - t_0)$. Then the total enthalpy for the mixture, per unit mass of dry gas, is

$$H' = C_B(t_G - t_0) + Y'[C_A(t_G - t_{DP}) + \lambda_{DP} + C_{A,L}(t_{DP} - t_0)] \quad (7.12)$$

Refer again to Fig. 7.3. For the low pressures ordinarily encountered in humidification work, the point A which actually lies on a line of constant pressure corresponding to the partial pressure of the vapor in the mixture can, for all practical purposes, be considered as lying on the line whose pressure is the saturation pressure of the vapor at the reference temperature, or at A' . The vapor enthalpy can then be computed by following the path $A'ED$ and becomes, per unit mass of vapor, $C_A(t_G - t_0) + \lambda_0$, where λ_0 is the latent heat of vaporization at the reference temperature. The enthalpy of the mixture, per unit mass of dry gas, is then

$$H' = C_B(t_G - t_0) + Y'[C_A(t_G - t_0) + \lambda_0] = C_S(t - t_0) + Y'\lambda_0 \quad (7.13)$$

Occasionally different reference temperatures are chosen for the dry gas and for the vapor. Note that the enthalpy H' for a mixture can be increased by increasing the temperature at constant humidity, by increasing the humidity at constant temperature, or by increasing both. Alternatively, under certain conditions H' may remain constant as t and Y' vary in opposite directions.

By substitution of Y'_s and the appropriate humid heat in Eq. (7.13), the enthalpy of saturated mixtures H'_s can be computed and plotted against temperature on the psychrometric chart. Similarly H for the dry gas can be plotted. Enthalpies for unsaturated mixtures can then be interpolated between the saturated and dry values at the same temperature according to the percentage humidity.

The System Air-Water

While psychrometric charts for any vapor-gas mixture can be prepared when circumstances warrant, the system air-water occurs so frequently that unusually complete charts for this mixture are available. Figure 7.5a and 7.5b shows two versions of such a chart, for SI and English engineering units, respectively, prepared for a total pressure of 1 std atm. For convenient reference, the various equations representing the curves are listed in Table 7.1. It should be noted that all the quantities (absolute humidity, enthalpies, humid volumes) are plotted against temperature. For the enthalpies, gaseous air and saturated liquid water at 0°C (32°F) were the reference conditions, so that the chart can be used in conjunction with the steam tables. The data for enthalpy of saturated air were then plotted with two enthalpy scales to provide for the large range of values

At 50°C and 50% relative humidity if need 100% cool air along adiabatic curve

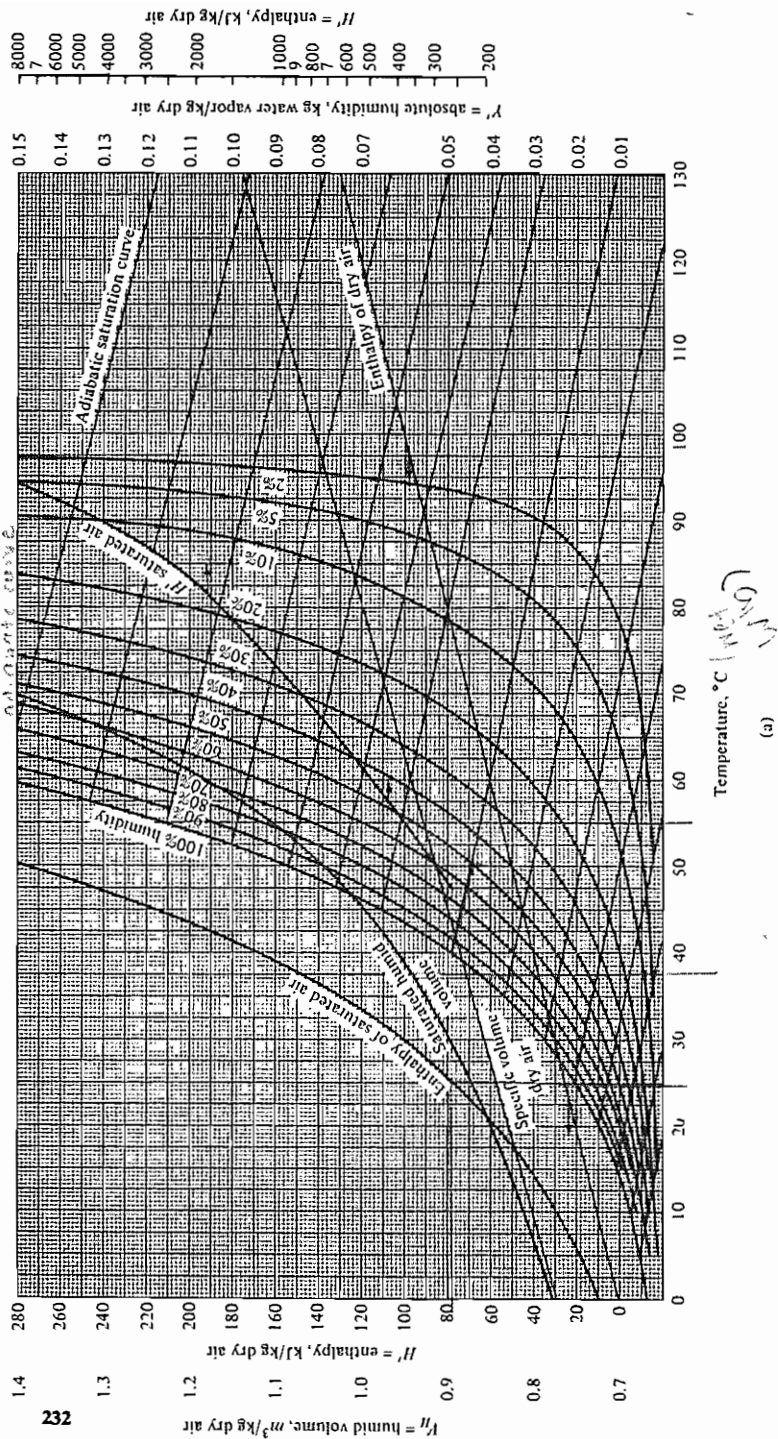


Figure 7.5 (a) Psychrometric chart for air-water vapor, 1 std atm abs, in SI units.

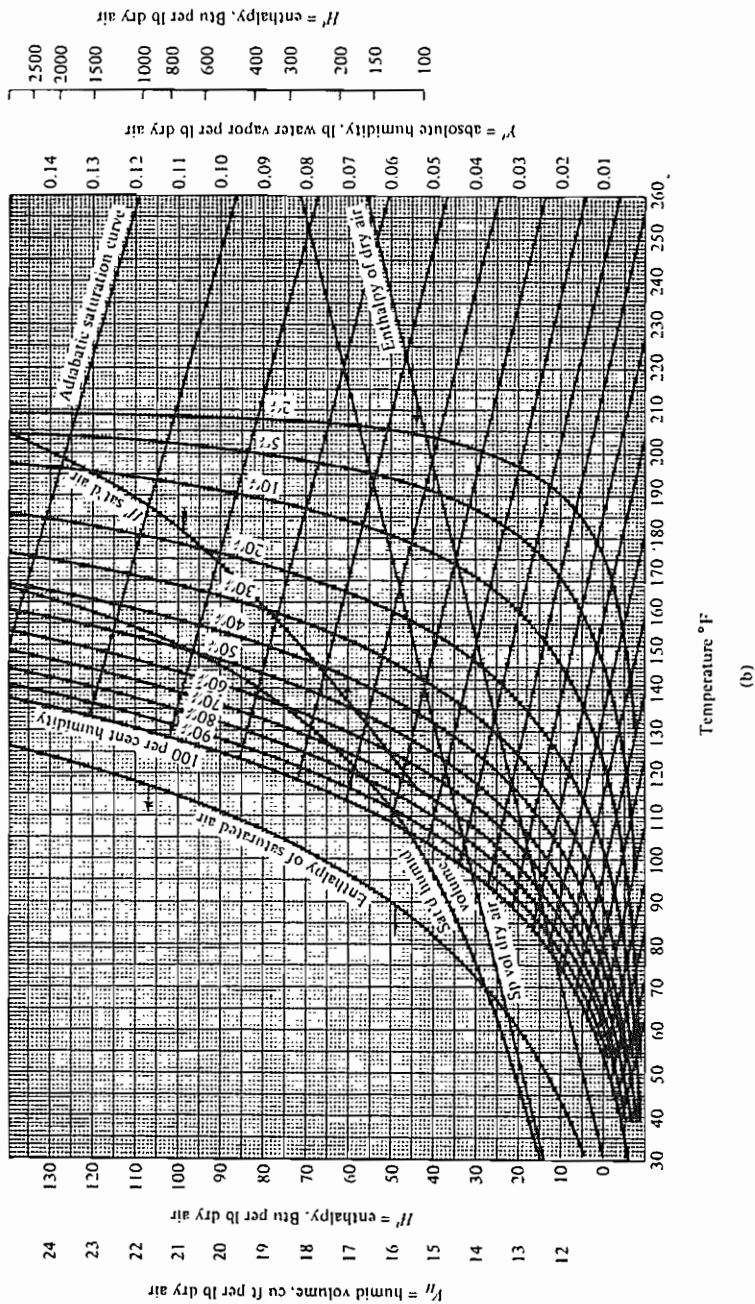


Figure 7.5 (b) Psychrometric chart for air-water vapor, 1 std atm abs, in English engineering units.

**Table 7.1 Psychrometric relations for the system air (B)–water (A)
at 1 std atm pressure**

	SI units (kg, m, N, °C)	English engineering units (Btu, ft ³ , lb, °F, lb _f /in ²)
M_A	18.02 kg/kmol, H ₂ O	18.02 lb/lb mol, H ₂ O
M_B	28.97 kg/kmol, air	28.97 lb/lb mol, air
Y'	$\frac{0.622\bar{p}_{H_2O}}{1.0133 \times 10^5 - \bar{p}_{H_2O}}$ kg H ₂ O/kg air	$\frac{0.622\bar{p}_{H_2O}}{14.696 - \bar{p}_{H_2O}}$ lb H ₂ O/lb air
Y'_s	$\frac{0.622p_{H_2O}}{1.0133 \times 10^5 - p_{H_2O}}$ kg H ₂ O/kg air	$\frac{0.622p_{H_2O}}{14.696 - p_{H_2O}}$ lb H ₂ O/lb air
v_H	$(0.00283 + 0.00456 Y')(t_G + 273)$ m ³ mixture/kg air	$(0.0252 + 0.0405 Y')(t_G + 460)$ ft ³ mixture/lb air
C_s	$1005 + 1884 Y'$ J for mixture/(kg air) · °C	$0.24 + 0.45 Y'$ Btu for mixture/(lb air) · °F
t_0	0°C	32°F
λ_0	2 502 300 J/kg	1075.8 Btu/lb
H	$(1005 + 1884 Y')t_G + 2 502 300 Y'$ J for mixture/kg air, referred to gaseous air and saturated liquid H ₂ O, 0°C	$(0.24 + 0.45 Y')(t_G - 32) + 1075.8 Y'$ Btu for mixture/lb air, referred to gaseous air and liquid H ₂ O, 32°F
H_s	$t, °C$	$t, °F$
	$H'_s, J/kg$	$H'_s, Btu/lb$
	0	32
	9 479	4.074
	10	40
	29 360	7.545
	20	60
	57 570	18.780
	30	80
	100 030	36.020
	40	100
	166 790	64.090
	50	120
	275 580	112.00
	60	140
	461 500	198.40
h_G/k_Y	950 J/kg · K	0.227 Btu/lb · °F

necessary. The series of curves marked “adiabatic-saturation curves” on the chart were plotted according to Eq. (7.21), to be considered later. For most purposes these can be considered as curves of constant enthalpy for the vapor-gas mixture per unit mass of gas.

Illustration 7.6 An air (B)–water-vapor (A) sample has a dry-bulb temperature 55°C and an absolute humidity 0.030 kg water/kg dry air at 1 std atm pressure. Tabulate its characteristics.

SOLUTION The point of coordinates $t_G = 55^\circ\text{C}$, $Y' = 0.030$ is located on the psychrometric chart (Fig. 7.5a), a schematic version of which is shown in Fig. 7.6. This is point *D* in Fig. 7.6.

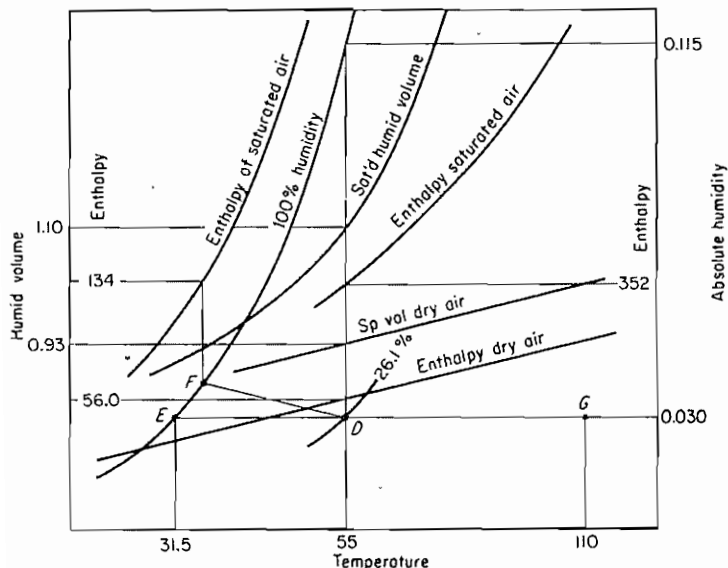


Figure 7.6 Solutions to Illustrations 7.6 and 7.7.

(a) By vertical interpolation between the adjacent curves of constant percent humidity, the sample has a percentage humidity = 26.1%. Alternatively, the saturation humidity at 55°C is $Y_s = 0.115$, and the percentage humidity at D is therefore $(0.030/0.115)100 = 26.1\%$.

(b) The molal absolute humidity = $Y = Y'(M_B/M_A) = 0.030(28.97/18.02) = 0.0482$ kmol water/kmol dry air.

(c) The partial pressure of water vapor in the sample, by Eq. (7.8), is

$$\bar{p}_A = \frac{Y p_i}{1 + Y} = \frac{0.0482(1.0133 \times 10^5)}{1.0482} = 4660 \text{ N/m}^2$$

(d) The vapor pressure of water at 55°C = 118 mmHg or $118(133.3) = 15730 \text{ N/m}^2 = p_A$. The relative humidity = $\bar{p}_A(100)/p_A = 4660(100)/15730 = 29.6\%$.

(e) Dew point. From point D proceed at constant humidity to the saturation curve at point E, at which the dew point temperature is 31.5°C.

(f) Humid volume. At 55°C, the specific volume of dry air is $0.93 \text{ m}^3/\text{kg}$ dry air. The humid volume of saturated air = $1.10 \text{ m}^3/\text{kg}$ dry air. Interpolating for 26.1% humidity,

$$v_H = 0.93 + (1.10 - 0.93)(0.261) = 0.974 \text{ m}^3/\text{kg dry air}$$

(g) Humid heat, Eq. (7.10):

$$C_S = C_B + Y' C_A = 1005 + 0.030(1884) = 1061.5 \text{ J (for wet air) / (kg dry air) } \cdot \text{K}$$

(h) Enthalpy. At 55°C, the enthalpy of dry air is $56000 \text{ J/kg dry air}$; that for saturated air is $352000 \text{ N} \cdot \text{m/kg dry air}$. Interpolating for 26.1% humidity gives

$$H' = 56000 + (352000 - 56000)(0.261) = 133300 \text{ J/kg dry air}$$

Alternatively, Eq. (7.13) or Table 7.1:

$$\begin{aligned} H' &= C_S(t_G - t_0) + Y'\lambda_0 = (1005 + 1884Y')t_G + 2502300Y' \\ &= [1005 + 1884(0.030)]55 + 2502300(0.030) = 133.4 \text{ kJ/kg dry air} \end{aligned}$$

As another alternative line DF is drawn parallel to the adjacent adiabatic-saturation curves. At F , the enthalpy is 134 kJ/kg dry air, or nearly the same as at D .

Illustration 7.7 If 100 m^3 of the moist air of Illustration 7.6 is heated to 110°C , how much heat is required?

SOLUTION After heating the mixture will be at point G of Fig. 7.6. The mass of dry air $= W_B = 100/v_H = 100/0.974 = 102.7 \text{ kg}$. Eq. (7.11): $Q = W_B C_S \Delta t = 102.7(1061.5)(110 - 55) = 6.00 \times 10^6 \text{ J}$.

Adiabatic-Saturation Curves

Consider the operation indicated schematically in Fig. 7.7. Here the entering gas is contacted with liquid, for example, in a spray, and as a result of diffusion and heat transfer between gas and liquid the gas leaves at conditions of humidity and temperature different from those at the entrance. The operation is adiabatic inasmuch as no heat is gained or lost to the surroundings. A mass balance for substance A gives

$$L' = G_S'(Y_2' - Y_1') \quad (7.14)$$

An enthalpy balance is

$$G_S' H_1' + L' H_L = G_S' H_2' \quad (7.15)$$

therefore

$$H_1' + (Y_2' - Y_1') H_L = H_2' \quad (7.16)$$

This can be expanded by the definition of H' given in Eq. (7.13),

$$C_{S1}(t_{G1} - t_0) + Y_1' \lambda_0 + (Y_2' - Y_1') C_{A,L}(t_L - t_0) = C_{S2}(t_{G2} - t_0) + Y_2' \lambda_0 \quad (7.17)$$

In the special case where the leaving gas-vapor mixture is saturated, and therefore at conditions t_{as} , Y'_{as} , H'_{as} , and the liquid enters at t_{as} , the gas is humidified by evaporation of liquid and cooled. Equation (7.17) becomes, on expansion of the humid-heat terms,

$$\begin{aligned} C_B(t_{G1} - t_0) + Y_1' C_A(t_{G1} - t_0) + Y_1' \lambda_0 + (Y'_{as} - Y_1') C_{A,L}(t_{as} - t_0) \\ = C_B(t_{as} - t_0) + Y'_{as} C_A(t_{as} - t_0) + Y'_{as} \lambda_0 \end{aligned} \quad (7.18)$$

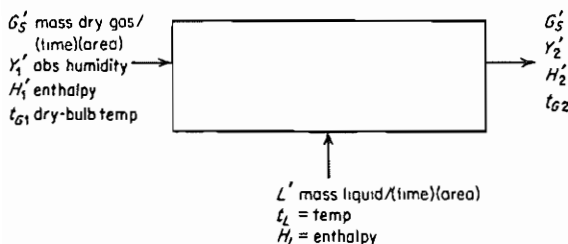


Figure 7.7 Adiabatic gas-liquid contact.

Heat of vaporization = Sensible Heat

By subtracting $Y_1' C_A t_{as}$ from both sides and simplifying this becomes

$$(C_B + Y_1' C_A)(t_{G1} - t_{as}) = C_{S1}(t_{G1} - t_{as}) \\ = (Y_{as}' - Y_1')[C_A(t_{as} - t_0) + \lambda_0 - C_{A,L}(t_{as} - t_0)] \quad (7.19)$$

Reference to Fig. 7.3 shows the quantity in brackets to be equal to λ_{as} . Consequently,

$$C_{S1}(t_{G1} - t_{as}) = (Y_{as}' - Y_1')\lambda_{as} \quad (7.20)$$

or $t_{G1} - t_{as} = (Y_{as}' - Y_1') \frac{\lambda_{as}}{C_{S1}} \quad (7.21)$

adiabatic saturation line *latent heat* *-Adiabatic*

This is the equation of a curve on the psychrometric chart, the "adiabatic-saturation curve"† which passes through the points (Y_{as}', t_{as}) on the 100 percent saturation curve and (Y_1', t_{G1}) . Since the humid heat C_{S1} contains the term Y_1' , the curve is not straight but slightly concave upward. For any vapor-gas mixture there is an *adiabatic-saturation temperature* t_{as} such that if contacted with liquid at t_{as} , the gas will become humidified and cooled. If sufficient contact time is available, the gas will become saturated at (Y_{as}', t_{as}) but otherwise will leave unsaturated at (Y_2', t_{G2}) , a point on the adiabatic-saturation curve for the initial mixture. Eventually, as Eq. (7.20) indicates, the sensible heat given up by the gas in cooling equals the latent heat required to evaporate the added vapor.

The psychrometric chart (Fig. 7.5) for air-water contains a family of adiabatic-saturation curves, as previously noted. Each point on the curve represents a mixture whose adiabatic-saturation temperature is at the intersection of the curve with the 100 percent humidity curve.

Illustration 7.8 Air at 83°C, $Y' = 0.030$ kg water/kg dry air, 1 std atm is contacted with water at the adiabatic-saturation temperature and is thereby humidified and cooled to 90% saturation. What are the final temperature and humidity of the air?

SOLUTION The point representing the original air is located on the psychrometric chart (Fig. 7.5a). The adiabatic-saturation curve through the point reaches the 100% saturation curve at 40°C, the adiabatic-saturation temperature. This is the water temperature. On this curve, 90% saturation occurs at 41.5°C, $Y' = 0.0485$ kg water/kg air, the outlet-air conditions.

Wet-Bulb Temperature

The wet-bulb temperature is the steady-state temperature reached by a small amount of liquid evaporating into a large amount of unsaturated vapor-gas mixture. Under properly controlled conditions it can be used to measure the humidity of the mixture. For this purpose a thermometer whose bulb has been

† The adiabatic-saturation curve is nearly one of constant enthalpy per unit mass of dry gas. As Eq. (7.16) indicates, H_{as}' differs from H_1' by the enthalpy of the evaporated liquid at its entering temperature t_{as} , but this difference is usually unimportant.

covered with a wick kept wet with the liquid is immersed in a rapidly moving stream of the gas mixture. The temperature indicated by this thermometer will ultimately reach a value lower than the dry-bulb temperature of the gas if the latter is unsaturated, and from a knowledge of this value the humidity is computed.

Consider a drop of liquid immersed in a rapidly moving stream of unsaturated vapor-gas mixture. If the liquid is initially at a temperature higher than the gas dew point, the vapor pressure of the liquid will be higher at the drop surface than the partial pressure of vapor in the gas, and the liquid will evaporate and diffuse into the gas. The latent heat required for the evaporation will at first be supplied at the expense of the sensible heat of the liquid drop, which will then cool down. As soon as the liquid temperature is reduced below the dry-bulb temperature of the gas, heat will flow from the gas to the liquid, at an increasing rate as the temperature difference becomes larger. Eventually the rate of heat transfer from the gas to the liquid will equal the rate of heat requirement for the evaporation, and the temperature of the liquid will remain constant at some low value, the wet-bulb temperature t_w . The mechanism of the wet-bulb process is essentially the same as that governing the adiabatic saturation, except that in the case of the former the humidity of the gas is assumed not to change during the process.

Refer to Fig. 7.8, sketched in the manner of the film theory, where a drop of liquid is shown already at the steady-state conditions and the mass of gas is so large as it passes the drop that its humidity is not measurably affected by the evaporation. Since both heat and mass transfer occur simultaneously, Eq. (3.71)

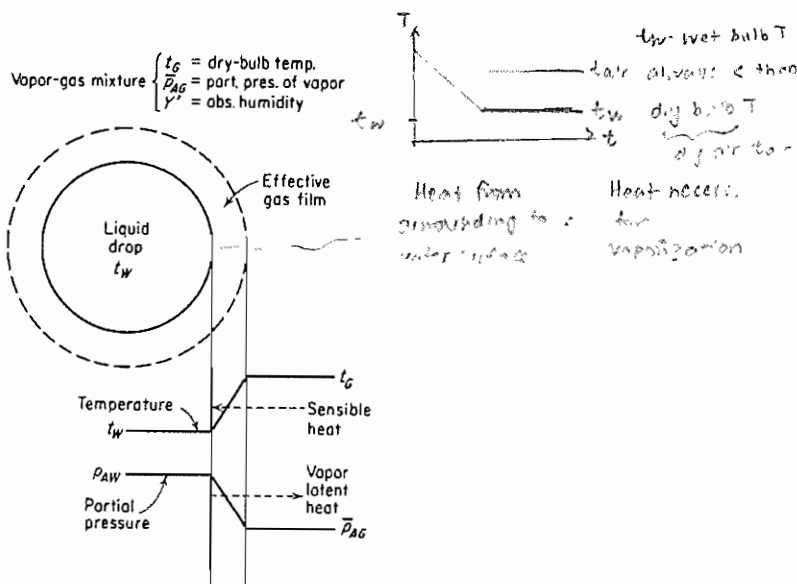


Figure 7.8 The wet-bulb temperature.

applies with $q_r = 0$ since no heat passes through the gas-liquid interface, and $N_B = 0$. Therefore†

$$\text{heat transfer } q_s = \frac{N_A M_A C_A}{1 - e^{-N_A M_A C_A / h_G}} (t_G - t_w) \approx h_G (t_G - t_w) \quad (7.22)$$

and the approximation of the right-hand side is usually satisfactory since ordinarily the rate of mass transfer is small. Further,

$$\text{mass transfer } N_A = F \ln \frac{1 - p_{A,w}/p_t}{1 - \bar{p}_{A,G}/p_t} \approx k_G (\bar{p}_{A,G} - p_{A,w}) \quad (7.23)$$

where the approximation on the right is usually satisfactory since N_A is small [the form of Eq. (7.23) reflects the fact that N_A is negative if q_s is taken to be positive]. $p_{A,w}$ is the vapor pressure of A at t_w . Substituting Eqs. (7.22) and (7.23) into Eq. (3.71) with N_B and q_r equal to zero, we get

$$h_G (t_G - t_w) + \lambda_w M_A k_G (\bar{p}_{A,G} - p_{A,w}) = 0 \quad (7.24)$$

where λ_w is the latent heat at the wet-bulb temperature per unit of mass. From this,

$$t_G - t_w = \frac{\lambda_w M_A k_G (p_{A,w} - p_{A,G})}{h_G} = \frac{\lambda_w M_B \bar{p}_{B,M} k_G (Y'_w - Y')}{h_G} \quad (7.25)$$

where $\bar{p}_{B,M}$ is the average partial pressure of the gas. Since (Table 3.1) $M_B \bar{p}_{B,M} k_G = k_Y$, Eq. (7.25) becomes

$$\text{ratio w/w h.t.} \quad t_G - t_w = \frac{\lambda_w (Y'_w - Y')}{h_G / k_Y} \quad (7.26)$$

which is the form of the relationship commonly used. The quantity $t_G - t_w$ is the wet-bulb depression.

In order to use Eq. (7.26) for determination of Y , it is necessary to have at hand appropriate values of h_G / k_Y , the psychrometric ratio.‡ Values of h_G and k_Y can be estimated independently for the particular shape of the wetted surface by correlations like those of Table 3.3, using the heat-mass-transfer analogy if necessary. Alternatively, experimental values of the ratio can be employed. Henry and Epstein [10] have critically examined the data and methods of

† For very careful measurements, the possibility of the liquid surface's receiving heat by radiation from either the gas itself or from the surroundings must also be considered. Assuming that the source of radiation is at temperature t_G , we have

$$q_r = (h_G + h_R)(t_G - t_w)$$

where the radiative heat transfer is described by an equivalent convection-type coefficient h_R . In wet-bulb thermometry, the effect of radiation can be minimized by using radiation shields and maintaining a high velocity of gas to keep h_G relatively high (at least 5 to 6 m/s in the case of air-water-vapor mixtures at ordinary temperatures). The relative size of h_G and h_R in any case can be estimated by standard methods [13]. It is necessary to observe the additional precaution of feeding the wick surrounding the thermometer bulb with an adequate supply of liquid preadjusted as nearly as practicable to the wet-bulb temperature.

‡ The quantity $h_G / k_Y C_S$ is also sometimes termed the psychrometric ratio.

measurement and have produced some measurements of their own. For flow of gases past cylinders, such as wet-bulb thermometers, and past single spheres, the results for 18 vapor-gas systems are well correlated by

$$\frac{h_G}{k_Y C_S} = \left(\frac{Sc}{Pr} \right)^{0.567} = Le^{0.567} \quad (7.27)$$

for rates of flow which are turbulent, independent of the Reynolds number. A large range of Lewis numbers Le , 0.335 to 7.2, is made possible by using not only surfaces wetted with evaporating liquids but also cylinders and spheres cast from volatile solids, which provide large Schmidt numbers.

Vapor-air systems are the most important. For dilute mixtures, where $C_S = C_B$, and with Pr for air taken as 0.707, Eq. (7.27) becomes (SI units)†

$$\frac{h_G}{k_Y} = 1223 Sc^{0.567} \quad (7.28)$$

For the system *air-water vapor*, for which Dropkin's [3] measurements are generally conceded to be the most authoritative, a thorough analysis [23, 24] led to the value $h_G/k_Y = 950 \text{ N} \cdot \text{m/kg} \cdot \text{K}$,† which is recommended for this system. It agrees closely with Eq. (7.28).

It will be noted that Eq. (7.26) is identical with Eq. (7.21) for the adiabatic-saturation temperature, but with replacement of C_{S1} by h_G/k_Y . These are nearly equal for air-water vapor at moderate humidities, and for many practical purposes the adiabatic-saturation curves of Fig. 7.5 can be used instead of Eq. (7.26). This is *not* the case for most other systems.

The Lewis relation We have seen that for the system air-water vapor, h_G/k_Y is approximately equal to C_S , or, approximately, $h_G/k_Y C_S = 1$. This is the so-called *Lewis relation* (after W. K. Lewis). Not only does it lead to near equality of the wet-bulb and adiabatic-saturation temperatures (as in the case of air-water vapor) but also to other simplifications to be developed later. It can be shown, through consideration of Eqs. (3.31) and (3.33), with $J_A = N_A$ and equality of the eddy diffusivities E_D and E_H , that the Lewis relation will be followed only if the thermal and molecular diffusivities are identical, or if $Sc = Pr$, or $Le = 1$. This is, of course, the conclusion also reached from the empirical equation (7.27). Le is essentially unity for air-water vapor but not for most other systems.

Illustration 7.9 For an air-water-vapor mixture of dry-bulb temperature 65°C , a wet-bulb temperature 35°C was determined under conditions such that the radiation coefficient can be considered negligible. The total pressure was 1 std atm. Compute the humidity of the air.

SOLUTION At $t_w = 35^\circ\text{C}$, $\lambda_w = 2419.300 \text{ J/kg}$, and $Y_w = 0.0365 \text{ kg H}_2\text{O/kg dry air}$

† With h_G/k_Y expressed as $\text{Btu/lb} \cdot ^\circ\text{F}$, the coefficient of Eq. (7.28) for air mixtures becomes 0.292 and for air-water vapor $h_G/k_Y = 0.227$.

(Fig. 7.5a); $h_G/k_Y = 950 \text{ J/kg} \cdot \text{K}$, $t_G = 65^\circ\text{C}$. Eq. (7.26):

$$65 - 35 = \frac{2419300(0.0365 - Y')}{950}$$

$$Y' = 0.0247 \text{ kg H}_2\text{O/kg air}$$

Alternatively as an approximation, the adiabatic-saturation curve for $t_{as} = 35^\circ\text{C}$ in Fig. 7.5a is followed to a dry-bulb temperature 65°C , where Y' is read as $0.0238 \text{ kg H}_2\text{O/kg air}$.

Illustration 7.10 Estimate the wet-bulb and adiabatic-saturation temperatures for a toluene-air mixture of 60°C dry-bulb temperature, $Y' = 0.050 \text{ kg vapor/kg air}$, 1 std atm.

SOLUTION *Wet-bulb temperature* $t_G = 60^\circ\text{C}$, $Y' = 0.050 \text{ kg toluene/kg air}$. $D_{AB} = 0.92 \times 10^{-5} \text{ m}^2/\text{s}$ at 59°C , 1 std atm. At 60°C , ρ for air = 1.060 kg/m^3 and $\mu = 1.95 \times 10^{-5} \text{ kg/m} \cdot \text{s}$.

Sc should be calculated for mean conditions between those of the gas-vapor mixture and the wet-bulb saturation conditions. However, for the dilute mixture considered here, the bulk-gas value of Sc is satisfactory and is essentially independent of temperature

$$Sc = \frac{\mu}{\rho D_{AB}} = \frac{1.95 \times 10^{-5}}{1.060(0.92 \times 10^{-5})} = 2.00$$

Eq. (7.28): $h_G/k_Y = 1223(2.00)^{0.567} = 1812 \text{ J/kg} \cdot \text{K}$ (observed value = 1842). Eq. (7.26):

$$60 - t_w = \frac{\lambda_w}{1812} (Y'_w - 0.050)$$

Solution for t_w is by trial and error. Try $t_w = 35^\circ\text{C}$. $p_{A,w} = 46.2 \text{ mmHg}$, $Y'_w = [46.2/(760 - 46.2)] 92/29 = 0.2056$, $\lambda_w = (96.6 \text{ cal/gm})(4187) = 404460 \text{ J/kg}$. The equation provides $t_w = 25.3^\circ\text{C}$ instead of the 35°C assumed. Upon repeated trials, t_w is computed to be 31.8°C . **Ans.**

Adiabatic-saturation temperature $t_{G1} = 60^\circ\text{C}$, $Y'_1 = 0.05$, C for toluene vapor = $1256 \text{ J/kg} \cdot \text{K}$. $C_{S1} = 1005 + 1256(0.05) = 1067.8 \text{ J/kg} \cdot \text{K}$. Eq. (7.21):

$$60 - t_{as} = (Y'_{as} - 0.05) \frac{\lambda_{as}}{1067.8}$$

In the same fashion as the wet-bulb temperature, t_{as} is calculated by trial and found to be 25.7°C . **Ans.**

GAS-LIQUID CONTACT OPERATIONS

Direct contact of a gas with a pure liquid may have any of several purposes:

1. Adiabatic operations.

- Cooling a liquid.* The cooling occurs by transfer of sensible heat and also by evaporation. The principal application is cooling of water by contact with atmospheric air (water cooling).
- Cooling a hot gas.* Direct contact provides a nonfouling heat exchanger which is very effective, providing the presence of some of the vapor of the liquid is not objectionable.
- Humidifying a gas.* This can be used for controlling the moisture content of air for drying, for example.

- d. *Dehumidifying a gas.* Contact of a warm vapor-gas mixture with a cold liquid results in condensation of the vapor. There are applications in air conditioning, recovery of solvent vapors from gases used in drying, and the like.
2. Nonadiabatic operations.
 - a. *Evaporative cooling.* A liquid or gas inside a pipe is cooled by water flowing in a film about the outside, the latter in turn being cooled by direct contact with air.
 - b. *Dehumidifying a gas.* A gas-vapor mixture is brought into contact with refrigerated pipes, and the vapor condenses upon the pipes.

Although operations of this sort are simple in the sense that mass transfer is confined to the gas phase (there can be no mass transfer within the pure liquid), they are nevertheless complex owing to the large heat effects which accompany evaporation or condensation.

ADIABATIC OPERATIONS

These are usually carried out in some sort of packed tower, frequently with countercurrent flow of gas and liquid. General relationships will be developed first, to be particularized for specific operations.

Fundamental Relationships

Refer to Fig. 7.9, which shows a tower of unit cross-sectional area. A mass balance for substance A over the lower part of the tower (envelope I) is

$$L' - L_1 = G'_S(Y' - Y'_1) \quad (7.29)$$

$$\text{or} \quad dL' = G'_S dY' \quad (7.30)$$

Similarly, an enthalpy balance is

$$L'H_L + G'_S H'_1 = L_1 H_{L1} + G'_S H' \quad (7.31)$$

These can be applied to the entire tower by putting subscript 2 on the unnumbered terms.

The rate relationships are fairly complex and will be developed in the manner of Olander [20]. Refer to Fig. 7.10, which represents a section of the tower of differential height dZ and shows the liquid and gas flowing side by side, separated by the gas-liquid interface. The changes in temperature, humidity, etc., are all differential over this section.

The interfacial surface of the section is dS . If the specific interfacial surface per packed volume is a (not the same as the packing surface a_p), since the volume of packing per unit cross section is dZ , then $dS = a dZ$. If the packing is incompletely wetted by the liquid, the surface for mass transfer a_M , which is the liquid-gas interface, will be smaller than that for heat transfer a_H , since heat

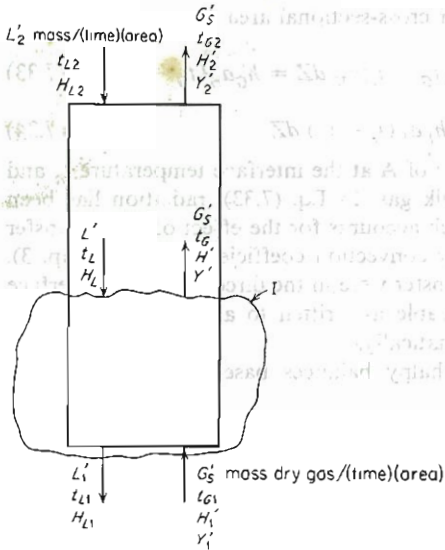


Figure 7.9 Continuous countercurrent adiabatic gas-liquid contact.

transfer may also occur between the packing and the fluids. Note that a_M corresponds to a_v of Chap. 6. The transfer rates are then:

Mass, as mass rate per tower cross-sectional area:

$$N_A M_A a_M dZ = -G'_S dY' = M_A F_G \left(\ln \frac{1 - \bar{p}_{A,i}/p_i}{1 - p_{A,G}/p_i} \right) a_M dZ \quad (7.32)$$

Handwritten notes:
 - $N_A M_A a_M dZ$: mass transfer rate
 - $-G'_S dY'$: mass transfer rate
 - $M_A F_G$: mass transfer coefficient factor
 - $\ln \frac{1 - \bar{p}_{A,i}/p_i}{1 - p_{A,G}/p_i}$: change in humidity
 - $a_M dZ$: volume
 - dZ : height of non-transferring element

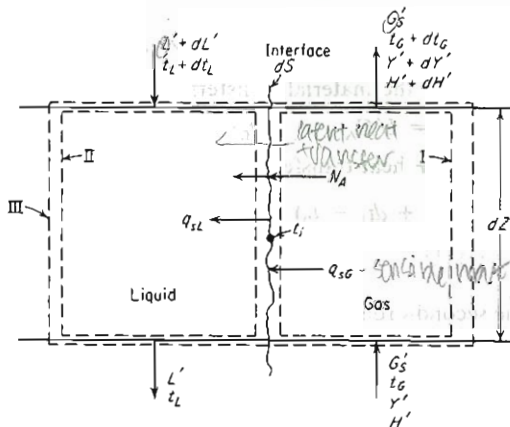


Figure 7.10 Differential section of a packed tower.

Sensible heat, as energy rate per tower cross-sectional area:

$$\text{Gas: } \underbrace{q_{SG} a_H dZ}_{\text{heat flow}} = \frac{N_A M_A C_A}{1 - e^{-N_A M_A C_A / h_G}} (t_G - t_i) a_H dZ = \overset{\text{heat transfer}}{h'_G a_H (t_G - t_i) dZ} \quad (7.33)$$

$$\text{Liquid: } \underbrace{q_{SL} a_H dZ}_{\text{volume flow}} = h_L a_H (t_i - t_L) dZ \overset{\text{heat transfer}}{\text{in the liquid}} \quad (7.34)$$

In Eq. (7.32), $\bar{p}_{A,i}$ is the vapor pressure of A at the interface temperature t_i , and $\bar{p}_{A,G}$ is the partial pressure in the bulk gas. In Eq. (7.33), radiation has been neglected, and the coefficient h'_G , which accounts for the effect of mass transfer on heat transfer, replaces the ordinary convection coefficient h_G (see Chap. 3). The rate equations are written as if transfer were in the direction gas to interface to liquid, but they are directly applicable as written to all situations; correct signs for the fluxes will develop automatically.

We now require a series of enthalpy balances based on the envelopes sketched in Fig. 7.10.

Envelope I:

Rate enthalpy in = $G'_S H'$

$$\text{Rate enthalpy out} = G'_S (H' + dH') - (G'_S dY') [C_A (t_G - t_0) + \lambda_0] \quad (7.35)$$

The second term is the enthalpy of the transferred vapor [recall that N_A and $G'_S dY'$ have opposite signs in Eq. (7.32)].

Rate in - rate out = heat-transfer rate

$$G'_S H' - G'_S (H' + dH') + (G'_S dY') [C_A (t_G - t_0) + \lambda_0] = h'_G a_H (t_G - t_i) dZ \quad (7.36)$$

If dH' , obtained by differentiation of Eq. (7.13), is substituted, this reduces to

$$- G'_S C_S dt_G = h'_G a_H (t_G - t_i) dZ \quad (7.37)$$

Envelope II:

Rate enthalpy in = $(L' + dL') C_{A,L} (t_L + dt_L - t_0) + (-G'_S dY') C_{A,L} (t_i - t_0)$

Here the second term is the enthalpy of the material transferred, now a liquid.

Rate enthalpy out = $L' C_{A,L} (t_L - t_0)$

Rate out = rate in + heat-transfer rate

$$L' C_{A,L} (t_L - t_0) = (L' + dL') C_{A,L} (t_L + dt_L - t_0) - (G'_S dY') C_{A,L} (t_i - t_0) + h_L a_H (t_i - t_L) dZ \quad (7.38)$$

If Eq. (7.30) is substituted and the second-order differential $dY' dt_L$ ignored, this becomes

$$L' C_{A,L} dt_L = (G'_S C_{A,L} dY' - h_L a_H dZ) (t_i - t_L) \quad (7.39)$$

Envelope III:

$$\text{Rate enthalpy in} = G'_S H' + (L' + dL') C_{A,L} (t_L + dt_L - t_0)$$

$$\text{Rate enthalpy out} = L' C_{A,L} (t_L - t_0) + G'_S (H' + dH')$$

Rate in = rate out (adiabatic operation)

$$G'_S H' + (L' + dL') C_{A,L} (t_L + dt_L - t_0) = L' C_{A,L} (t_L - t_0) + G'_S (H' + dH') \quad (7.40)$$

Substitutions of Eq. (7.30) and the differential of Eq. (7.13) for dH' are made, and the term $dH' dt_L$ is ignored, whereupon this becomes

$$L'_j C_{A,L} dt_L = G'_S \{ C_S dt_G + [C_A(t_G - t_0) - C_{A,L}(t_L - t_0) + \lambda_0] dY' \} \quad (7.41)$$

Handwritten notes:
 $L'_j C_{A,L} dt_L$ is constant everywhere.
 $C_A + Y' C_B$ is temperature.
 λ_0 is latent heat.
 dY' is change in humidity level.
 G'_S is total heat transfer.

These will now be applied to the adiabatic operations.

Water Cooling with Air

This is without question the most important of the operations. Water, warmed by passage through heat exchangers, condensers, and the like, is cooled by contact with atmospheric air for reuse. The latent heat of water is so large that only a small amount of evaporation produces large cooling effects. Since the rate of mass transfer is usually small, the temperature level is generally fairly low, and the Lewis relation applies reasonably well for the air-water system, the relationships of the previous section can be greatly simplified by making reasonable approximations.

Thus, if the sensible-heat terms of Eq. (7.41) are ignored in comparison with the latent heat, we have

$$L' C_{A,L} dt_L = G'_S C_S dt_G + G'_S \lambda_0 dY' \approx G'_S dH \quad (7.42)$$

Here the last term on the right ignores the Y' which appears in the definition of C_S . Integrating, on the further assumption that L' is essentially constant (little evaporation), gives

$$L' C_{A,L} (t_{L2} - t_{L1}) = G'_S (H_2' - H_1') \quad (7.43)$$

This enthalpy balance can be represented graphically by plotting the gas enthalpy H' against t_L , as in Fig. 7.11. The line ON on the chart represents Eq. (7.43), and it passes through the points representing the terminal conditions for the two fluids. Insofar as $L_2 - L_1$ is small in comparison with L' , the line is straight and of slope $L' C_{A,L} / G'_S$. The equilibrium curve in the figure is plotted for conditions of the gas at the gas-liquid interface, i.e., the enthalpy of saturated gas at each temperature.

If the mass-transfer rate is small, as it usually is, Eq. (7.32) can be written as

$$G'_S dY' = k_Y a_M (Y_i' - Y') dZ \quad (7.44)$$

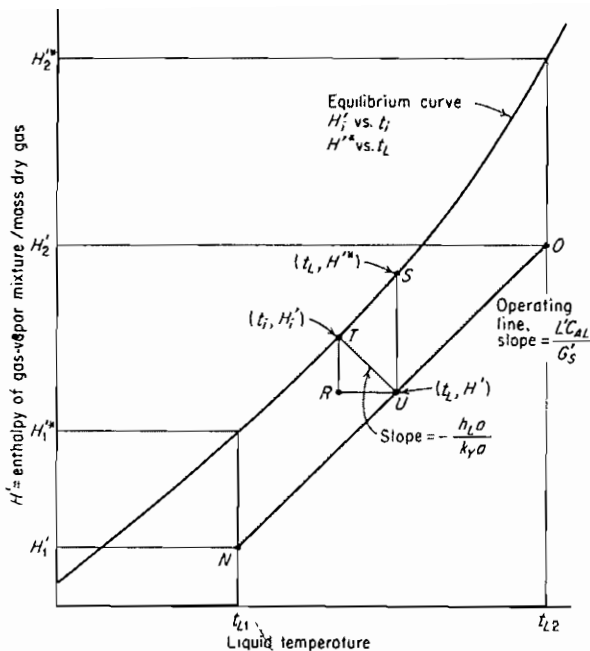


Figure 7.11 Operating diagram for a water cooler.

and Eq. (7.37) as

$$G'_S C_S dt_G = h_G a_H (t_i - t_G) dZ \quad (7.45)$$

When the sensible heat of the transferred vapor is ignored, Eq. (7.39) becomes

$$L' C_{A,L} dt_L = h_L a_H (t_L - t_i) dZ \quad (7.46)$$

Substituting Eqs. (7.44) and (7.45) into (7.42) gives

$$G'_S dH' = h_G a_H (t_i - t_G) dZ + \lambda_0 k_Y a_M (Y'_i - Y') dZ \quad (7.47)$$

If $h_G a_H / C_S k_Y a_M = r$, this becomes

$$G'_S dH' = k_Y a_M [(C_S r t_i + \lambda_0 Y'_i) - (C_S r t_G + \lambda_0 Y')] dZ \quad (7.48)$$

For the special case where $r = 1$ [16, 17] the terms in parentheses are gas enthalpies. The restriction that $r = 1$ requires $Le = 1$ (air-water), and $a_M = a_H = a$ (the latter will be true only for thoroughly irrigated tower filling; even for air-water contacting, values of r as high as 2 have been observed with low liquid rates [9]). With these understood, Eq. (7.48) is

$$G'_S dH' = k_Y a (H'_i - H') dZ \quad (7.49)$$

which is remarkable in that the mass-transfer coefficient is used with an

enthalpy driving force. Combining Eqs. (7.42), (7.46), and (7.49) then provides

$$G'_S dH' = k_y a (H'_i - H') dZ = h_L a (t_L - t_i) dZ \quad (7.50)$$

At a position in the apparatus corresponding to point *U* on the operating line (Fig. 7.11), point *T* represents the interface conditions and the distance *TR* the enthalpy driving force $H'_i - H'$ within the gas phase. By making constructions like the triangle *RTU* at several places along the operating line, corresponding H'_i and H' values can be obtained. Equation (7.50) then provides, assuming $k_y a$ is constant,

$$\int_{H'_i}^{H'_2} \frac{dH'}{H'_i - H'} = \frac{k_y a}{G'_S} \int_0^Z dZ = \frac{k_y a Z}{G'_S} \quad (7.51)$$

The integral can be evaluated graphically and the packed height *Z* computed. The enthalpy integral of Eq. (7.51) is sometimes given another interpretation. Thus,

$$\int_{H'_i}^{H'_2} \frac{dH'}{H'_i - H'} = \frac{H'_2 - H'_1}{(H'_i - H')_{av}} = N_{IG} \quad (7.52)$$

where the middle part of the equation is the number of times the average driving force divides into the enthalpy change. This is a measure of the difficulty of enthalpy transfer, called the *number of gas-enthalpy transfer units* N_{IG} . Consequently,

$$Z = H_{IG} N_{IG} \quad (7.53)$$

where the height of a gas-enthalpy transfer unit $= H_{IG} = G'_S / k_y a$. H_{IG} is frequently preferred over $k_y a$ as a measure of packing performance since it is less dependent upon rates of flow and has the simple dimension of length.

As discussed in Chap. 5, an overall driving force representing the enthalpy difference for the bulk phases but expressed in terms of H' can be used, such as the vertical distance *SU* (Fig. 7.11). This requires a corresponding overall coefficient and leads to overall numbers and heights of transfer units:†

$$N_{IOG} = \int_{H'_1}^{H'_2} \frac{dH'}{H'^* - H'} = \frac{K_y a Z}{G'_S} = \frac{Z}{H_{IOG}} \quad (7.54) \quad (=7.51)$$

The use of Eq. (7.53) is satisfactory (see Chap. 5) only if the equilibrium enthalpy curve of Fig. 7.11 is straight, which is not strictly so, or if $h_L a$ is infinite, so that the interface temperature equals the bulk-liquid temperature. Although the few data available indicate that $h_L a$ is usually quite large (see, for

† The water-cooling-tower industry frequently uses Eq. (7.54) in another form:

$$\frac{K_y a Z}{L'} = \int_{t_{L1}}^{t_{L2}} \frac{dt_L}{H'^* - H'}$$

which results from combining Eqs. (7.42) and (7.54) and setting $C_{A,L}$ for water = 1.

example, Illustration 6.7), there are uncertainties owing to the fact that many have been taken under conditions such that $h_{GAH}/C_S k_Y a_M = r$ was *not* unity even though assumed to be so. In any case, it frequently happens that, for cooling-tower packings, only $K_Y a$ or H_{IOG} , and not the individual phase coefficients, are available.

Just as with concentrations (Chap. 5), an operating line on the enthalpy coordinates of Fig. 7.11 which anywhere touches the equilibrium curve results in a zero driving force and consequently an infinite interfacial surface, or infinite height Z , to accomplish a given temperature change in the liquid. This condition would then represent the limiting ratio of L'/G'_S permissible. It is also clear that point N , for example, will be below the equilibrium curve so long as the entering-air enthalpy H'_1 is less than the saturation enthalpy H'_1^* for air at t_{L1} . Since the enthalpy H' is for most practical purposes only a function of the adiabatic-saturation temperature (or, for air-water, the wet-bulb temperature), the entering-air wet-bulb temperature must be below t_{L1} *but its dry-bulb temperature need not be*. For this reason, it is perfectly possible to cool water to a value of t_{L1} less than the entering-air dry-bulb temperature t_{G1} . It is also possible to operate a cooler with entering air saturated, so long as its temperature is less than t_{L1} . The difference between the exit-liquid temperature and the entering-air wet-bulb temperature, $t_{L1} - t_{w1}$, called the *wet-bulb temperature approach*, is then a measure of the driving force available for diffusion at the lower end of the equipment. In the design of cooling towers, this is ordinarily specified to be from 2.5 to 5°C, with t_{w1} set at the "5 percent wet-bulb temperature" (the wet-bulb temperature which is exceeded only 5 percent of the time on the average during the summer months).

Makeup fresh water in recirculating water systems must be added to replace losses from entrainment (drift, or windage), evaporation losses, and blowdown. Windage losses can be estimated as 0.1 to 0.3 percent of the recirculation rate for induced-draft towers. If makeup water introduces dissolved salts (hardness) which will otherwise accumulate, a small amount of water is deliberately discarded (blowdown) to keep the salt concentration at some predetermined level. A calculation is demonstrated in Illustration 7.11. Chlorine treatment of the water to control algae and slime and addition of chromate-phosphate mixtures to inhibit corrosion have been common in the past, but restrictions on discharges to the environment by the blowdown have led to the use of nonchromate inhibitors. Many other practical details are available [14, 15].

The use of overall mass-transfer coefficients does not distinguish between convective and evaporative cooling of the liquid and will not permit computation of the humidity or dry-bulb temperature of the exit air. The air will ordinarily be very nearly saturated, and for purposes of estimating makeup requirements it may be so assumed. The temperature-humidity history of the air as it passes through the tower can be estimated by a graphical method on the $H't_L$ diagram (Fig. 7.11) if $h_L a$ and $k_Y a$ are known [17] but approach of the gas to saturation is very critical to the computations, and it is recommended instead

that these be done by the methods outlined later (see page 255) which make no assumptions. Cooling towers for systems other than air-water ($Le \neq 1$) or when $a_M \neq a_H$ must also be treated by the general methods discussed later. Some cooling towers use a cross flow of air and water, for which methods of computation are also available [11, 22, 27].

Illustration 7.11 A plant requires 15 kg/s (1984 lb/min) of cooling water to flow through its distillation-equipment condensers, thereby removing 270 W (55 270 Btu/min) from the condensers. The water will leave the condensers at 45°C. It is planned to cool the water for reuse by contact with air in an induced-draft cooling tower. The design conditions are as follows: entering air at 30°C dry-bulb, 24°C wet-bulb temperature, water to be cooled to within 5°C of the inlet-air wet-bulb temperature, i.e., to 29°C; an air/water-vapor ratio of 1.5 times the minimum. Makeup water will come from a well at 10°C, hardness 500 ppm dissolved solids. The circulating water is not to contain more than 2000 ppm hardness. For the packing to be used, $K_Y a$ is expected to be $0.90 \text{ kg/m}^3 \cdot \text{s} \cdot \Delta Y$ (202 lb/ft² · h · ΔY), for a liquid rate at least $2.7 \text{ kg/m}^2 \cdot \text{s}$ and a gas rate $2.0 \text{ kg/m}^2 \cdot \text{s}$ (1991 and 1474 lb/ft² · h, respectively). Compute the dimensions of the packed section and the makeup-water requirement.

SOLUTION Refer to Fig. 7.12, which represents the flowsheet of the operation. The entering air humidity and enthalpy are taken from Fig. 7.5a. The operating diagram, Fig. 7.13, contains the saturated-air-enthalpy curve, and on this plot is point N representing the condition at the bottom of the tower ($t_{L1} = 29^\circ\text{C}$, $H'_1 = 72\,000 \text{ N} \cdot \text{m/kg dry air}$). The operating line will pass through N and end at $t_{L2} = 45^\circ\text{C}$. For the minimum value of G'_S , the operating line will have the least slope which causes it to touch the equilibrium curve and will consequently pass through point O , where $H'_2 = 209\,500 \text{ N} \cdot \text{m/kg dry air}$. The slope of the line $O'N$ is therefore

$$\frac{L' C_{A,L}}{G'_{S,\min}} = \frac{15(4187)}{G'_{S,\min}} = \frac{209\,500 - 72\,000}{45 - 29}$$

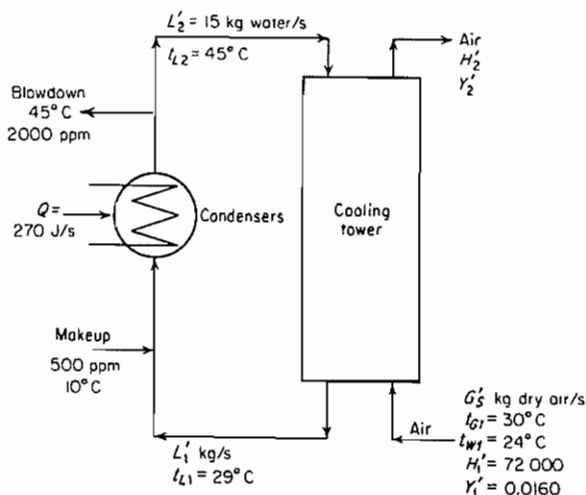


Figure 7.12 Flowsheet for Illustration 7.11.

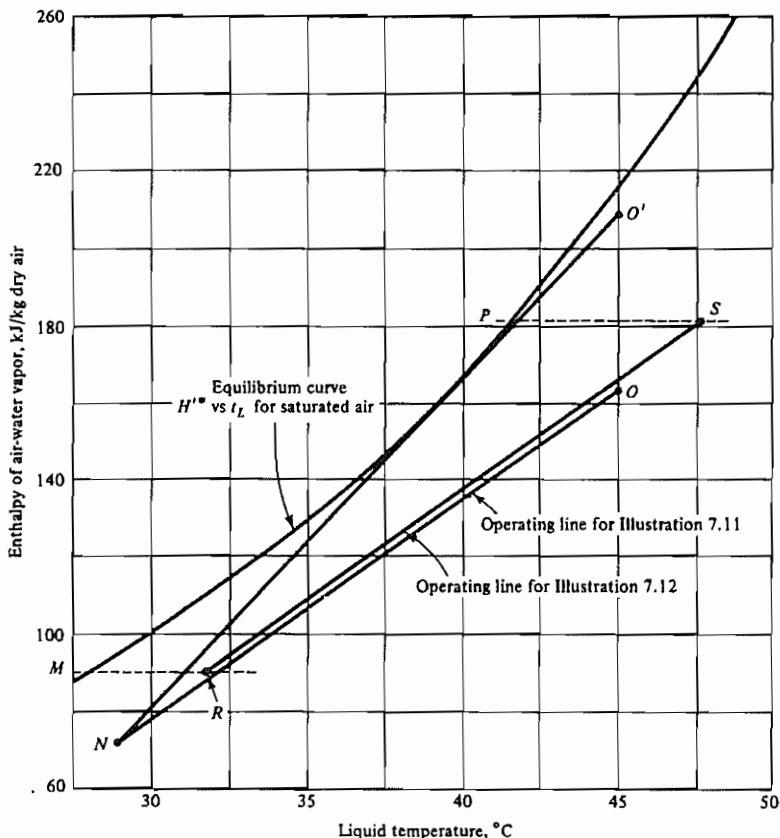


Figure 7.13 Solutions to Illustrations 7.11 and 7.12.

whence $G'_{S, \min} = 7.31 \text{ kg dry air/s}$. For a gas rate of 1.5 times the minimum, $G'_S = 1.5(7.31) = 10.97 \text{ kg dry air/s}$. Therefore

$$\frac{H'_2 - 72\,000}{45 - 29} = \frac{15(4187)}{10.97}$$

and $H'_2 = 163\,600 \text{ N} \cdot \text{m/kg dry air}$, plotted at point O . The operating line is therefore line ON . For a liquid rate at least $2.7 \text{ kg/m}^2 \cdot \text{s}$, the tower cross section would be $15/2.7 = 5.56 \text{ m}^2$. For a gas rate of at least $2.0 \text{ kg/m}^2 \cdot \text{s}$, the cross section will be $10.97/2.0 = 5.50 \text{ m}^2$. The latter will therefore be used, since the liquid rate will then exceed the minimum to ensure that $K_Y a = 0.90$.

Basis: 1 m^2 cross section, $G'_S = 2.0 \text{ kg/m}^2 \cdot \text{s}$. The driving force $H'^* - H'$ is obtained at

frequent intervals of t_L from Fig. 7.13 as follows

$t_L, ^\circ\text{C}$	$\chi H''$ (equilibrium curve),	H' (operating line),	10^5
	J/kg	J/kg	$H'' - H'$
29	100 000	72 000	3.571
32.5	114 000	92 000	4.545
35	129 800	106 500	4.292
37.5	147 000	121 000	3.846
40	166 800	135 500	3.195
42.5	191 000	149 500	2.410
45	216 000	163 500	1.905

The data of the last two columns are plotted against each other, H' as abscissa, and the area under the curve is 3.25. From Eq. (7.54)

$$3.25 = \frac{K_Y a Z}{G'_S} = \frac{0.90 Z}{2.0}$$

$$z = 7.22 \text{ m (23.7 ft) packed height} \quad \text{Ans.}$$

Note: In this case $N_{tOG} = 3.25$, and $H_{tOG} = G'_S / K_Y a = 2.0 / 0.90 = 2.22 \text{ m}$.

Makeup-water requirement For present purposes, define

E = evaporation rate, kg/h

W = windage loss, kg/h

B = blowdown rate, kg/h

M = makeup rate, kg/h

x_C = weight fraction hardness in circulated water

x_M = wt fraction hardness in makeup water

For continuous makeup and continuous blowdown, a total material balance is

$$M = B + E + W$$

and a hardness balance is

$$M x_M = (B + W) x_C$$

Elimination of M results in

$$B = E \frac{x_M}{x_C - x_M} - W$$

Assuming the outlet air ($H' = 163\,500 \text{ N} \cdot \text{m/kg}$) is essentially saturated, $Y'_2 = 0.0475$. The approximate rate of evaporation is then

$$E = 2.0(5.50)(0.0475 - 0.0160) = 0.3465 \text{ kg/s}$$

The windage loss is estimated as 0.2 percent of the circulation rate,

$$W = 0.002(15) = 0.03 \text{ kg/s}$$

Since the weight fractions x_C and x_M are proportional to the corresponding ppm values, the blowdown rate is

$$B = 0.3465 \frac{500}{2000 - 500} - 0.03 = 0.0855 \text{ kg/s}$$

The makeup rate is then estimated to be

$$M = B + E + W = 0.0855 + 0.3465 + 0.03 = 0.462 \text{ kg/s (3670 lb/h) Ans.}$$

Illustration 7.12 In the cooler of Illustration 7.11, to what temperature would the water be cooled if, after the tower was built and operated at the design L' and G'_S values, the entering air entered at dry-bulb temperature $t_{G1} = 32^\circ\text{C}$, wet-bulb temperature $t_{w1} = 28^\circ\text{C}$?

SOLUTION For the new conditions, $H'_1 = 90\,000 \text{ J/kg}$, and the new operating line must originate on the broken line at M in Fig. 7.13. Since in all likelihood the heat load on the plant condensers, which is ultimately transferred to the air, will remain the same, the change in air enthalpy will be the same as in Illustration 7.11:

$$H'_2 - 90\,000 = 163\,600 - 72\,000$$

$$H_2 = 181\,600 \text{ J/kg}$$

The new operating line must therefore end on the broken line at P in Fig. 7.13, and because of the same ratio L'/G'_S as in Illustration 7.11 it must be parallel to the original line NO . Since at the same flow rates as in Illustration 7.11 the value of H_{IOG} remains the same, for the same packing depth N_{IOG} remains at 3.25. The new operating line RS is therefore located by trial so that $N_{IOG} = 3.25$. The temperature at $R = t_{L1}$ is 31.7°C , which is the temperature to which the water will be cooled. **Ans.**

Dehumidification of Air-Water Vapor

If a warm vapor-gas mixture is contacted with cold liquid so that the humidity of the gas is greater than that at the gas-liquid interface, vapor will diffuse toward the liquid and the gas will be dehumidified. In addition, sensible heat can be transferred as a result of temperature differences within the system. For air-water-vapor mixtures ($Le = 1$) contacted with cold water, the methods of water cooling apply with only obvious modification. The operating line on the gas-enthalpy-liquid-temperature graph will be above the equilibrium curve, the driving force is $H' - H'^*$, and Eq. (7.54) can be used with this driving force. For all other systems, for which $Le \neq 1$, the general methods below must be used.

30/5/97

Recirculating Liquid-Gas Humidification-Cooling

This is a special case where the liquid enters the equipment at the adiabatic-saturation temperature of the entering gas. This can be achieved by continuously reintroducing the exit liquid to the contactor immediately, without addition or removal of heat on the way, as in Fig. 7.14. The development which follows applies to any liquid-gas system, regardless of the Lewis number. In such a system, the temperature of the entire liquid will fall to, and remain at, the adiabatic-saturation temperature. The gas will be cooled and humidified, following along the path of the adiabatic-saturation curve on the psychrometric chart which passes through the entering-gas conditions. Depending upon the degree of contact, the gas will approach more or less closely equilibrium with the liquid, or its adiabatic-saturation conditions. This supposes that the makeup liquid enters

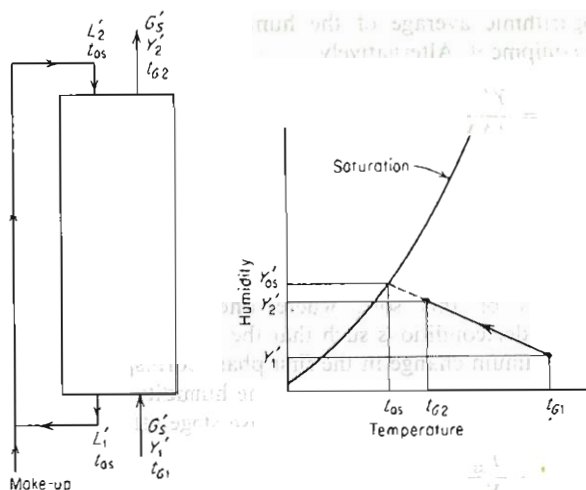


Figure 7.14 Recirculating liquid, gas humidification-cooling.

at the adiabatic-saturation temperature also, but for most purposes the quantity of evaporation is so small relative to the total liquid circulation that minor deviations from this temperature for the makeup liquid can be ignored.

As has been shown previously, the enthalpy of the gas is practically a function only of its adiabatic-saturation temperature, which remains constant throughout the operation. The enthalpy of the liquid at constant temperature is also constant, so that an operating "line" on a plot such as Fig. 7.11 would be merely a single point on the equilibrium curve. This diagram therefore cannot be used for design purposes. The temperature and humidity changes, which lie entirely within the gas phase, can be used, however, and these are shown schematically in Fig. 7.14. If mass transfer is used as a basis for design, Eq. (7.44) becomes

$$G'_S dY = k_Y a (Y'_{as} - Y') dZ \quad (7.55)$$

$$\int_{Y_1}^{Y_2} \frac{dY'}{Y'_{as} - Y'} = \frac{k_Y a}{G'_S} \int_0^Z dZ \quad (7.56)$$

and since Y'_{as} is constant,

$$\ln \frac{Y'_{as} - Y_1}{Y'_{as} - Y_2} = \frac{k_Y a Z}{G'_S} \quad (7.57)$$

Equation (7.57) can be used directly, or it can be rearranged by solving for G'_S and multiplying each side by $Y'_2 - Y'_1$ or its equivalent,

$$G'_S (Y'_2 - Y'_1) = \frac{k_Y a Z [(Y'_{as} - Y'_1) - (Y'_{as} - Y'_2)]}{\ln [(Y'_{as} - Y'_1) / (Y'_{as} - Y'_2)]} = k_Y a Z (\Delta Y')_{av} \quad (7.58)$$

where $(\Delta Y')_{av}$ is the logarithmic average of the humidity-difference driving forces at the ends of the equipment. Alternatively,

$$N_{IG} = \frac{Y'_2 - Y'_1}{(\Delta Y')_{av}} = \ln \frac{Y'_{as} - Y'_1}{Y'_{as} - Y'_2} \quad (7.59)$$

and

$$H_{IG} = \frac{G'_S}{k_Y a} = \frac{Z}{N_{IG}} \quad (7.60)$$

where N_{IG} is the number of gas-phase transfer units and H_{IG} the corresponding height of a transfer unit.[†]

In contacting operations of this sort, where one phase approaches equilibrium with the other under conditions such that the characteristics of the latter do not change, the maximum change in the first phase corresponds to the operation of one theoretical stage (see Chap. 5). Since the humidity in adiabatic equilibrium with the liquid is Y'_{as} , the Murphree gas-phase stage efficiency is

$$E_{MG} = \frac{Y'_2 - Y'_1}{Y'_{as} - Y'_1} = 1 - \frac{Y'_{as} - Y'_2}{Y'_{as} - Y'_1} = 1 - e^{-k_Y a Z / G'_S} = 1 - e^{-N_{IG}} \quad (7.61)$$

If heat transfer is used as the basis for design, similar treatment of Eq. (7.45) leads to

$$G'_S C_{S1} (t_{G1} - t_{G2}) = \frac{h_G a Z [(t_{G1} - t_{as}) - (t_{G2} - t_{as})]}{\ln [(t_{G1} - t_{as}) / (t_{G2} - t_{as})]} = h_G a Z (\Delta t)_{av} \quad (7.62)$$

where $h_G a$ is the volumetric-heat-transfer coefficient of sensible-heat transfer between the bulk of the gas and the liquid surface.

Illustration 7.13 A horizontal spray chamber (Fig. 7.19) with recirculated water is used for adiabatic humidification and cooling of air. The active part of the chamber is 2 m long and has cross section of 2 m². With an air rate 3.5 m³/s at dry-bulb temperature 65.0°C, $Y' = 0.0170$ kg water/kg dry air, the air is cooled and humidified to a dry-bulb temperature 42.0°C. If a duplicate spray chamber operated in the same manner were to be added in series with the existing chamber, what outlet conditions could be expected for the air?

SOLUTION For the existing chamber, the adiabatic-saturation line on Fig. 7.5 for $t_{G1} = 65.0^\circ\text{C}$, $Y'_1 = 0.0170$ shows $t_{as} = 32.0^\circ\text{C}$, $Y'_{as} = 0.0309$, and at $t_{G2} = 42.0^\circ\text{C}$, $Y'_2 = 0.0265$. From Eq. (7.57) with $Z = 2$ m:

$$\ln \frac{0.0309 - 0.0170}{0.0309 - 0.0265} = \frac{k_Y a (2)}{G'_S}$$

$$\frac{k_Y a}{G'_S} = 0.575 \text{ m}^{-1}$$

[†] To be entirely consistent with the definition of Chap. 8, Eq. (7.60) should read $H_{IG} = G'_S / k_Y a (1 - y_A)$. The value of $1 - y_A$ in the present application, however, is ordinarily very close to unity.

For the extended chamber, with $Z = 4$ m, $k_Y a / G'_S$ and t_{as} will remain the same. Therefore, Eq. (7.57):

$$\ln \frac{0.0309 - 0.0170}{0.0309 - Y'_2} = 0.575(4)$$

$$Y'_2 = 0.0295$$

With the same adiabatic-saturation curve as before, $t_{G2} = 34.0^\circ\text{C}$.

General Methods

For all other countercurrent operations, and even for those discussed above when the approximations are not appropriate or when $\text{Le} \neq 1$, we must return to the equations developed earlier. Equating the right-hand sides of Eqs. (7.39) and (7.41) provides

$$t_i = t_L + \frac{G'_S \{ C_S(dt_G/dZ) + [C_{A,G} - C_{A,L}t_L + (C_{A,L} - C_A)t_0 + \lambda_0](dY'/dZ) \}}{G'_S C_{A,L}(dY'/dZ) - h_L a_H} \quad (7.63)$$

The humidity gradient in this expression is obtained from Eq. (7.32):

$$\frac{dY'}{dZ} = - \frac{M_A F_G a_M}{G'_S} \ln \frac{1 - \bar{p}_{A,i}/p_t}{1 - \bar{p}_{A,G}/p_t} = - \frac{M_A F_G a_M}{G'_S} \ln \frac{Y' + M_A/M_B}{Y'_i + M_A/M_B} \quad (7.64)$$

$$\approx - \frac{M_A k_G a_M}{G'_S} (\bar{p}_{A,G} - \bar{p}_{A,i}) \approx - \frac{k_Y a_M}{G'_S} (Y' - Y'_i) \quad (7.64a)$$

where the approximations of Eq. (7.64a) are suitable for low vapor concentrations. The temperature gradient is taken from Eq. (7.37):

$$\frac{dt_G}{dZ} = - \frac{h'_G a_H (t_G - t_i)}{G'_S C_S} \approx - \frac{h_G a_H (t_G - t_i)}{G'_S C_S} \quad (7.65)$$

where $h_G a_H$ rather than $h'_G a_H$ may be used at low transfer rates. Unless a_H and a_M are separately known (which is not usual), there will be difficulty in evaluating h'_G exactly. Here we must assume $a_H = a_M$:

$$h'_G a_H = \frac{N_A M_A C_A a_H}{1 - e^{-N_A M_A C_A a_H / h_G a_H}} \approx \frac{N_A M_A C_A a_M}{1 - e^{-N_A M_A C_A a_M / h_G a_H}} \quad (7.66)$$

$$\approx - \frac{G'_S C_A (dY'/dZ)}{1 - e^{G'_S C_A (dY'/dZ) / h_G a_H}}$$

The effect of the approximation is not normally important.

Equations (7.64) and (7.65) are integrated numerically, using a procedure outlined in Illustration 7.14. Extensive trial and error is required, since t_i from Eq. (7.63) is necessary before $\bar{p}_{A,i}$ (or Y'_i) can be computed. The t_i - $\bar{p}_{A,i}$ relation is that of the vapor-pressure curve. If at any point in the course of the

calculations $\bar{p}_{A,G}$ (or Y') at a given t_G calculates to be larger than the corresponding saturation vapor concentration, a fog may form in the gas phase, in which case the entire analysis is invalid and new conditions at the terminals of the tower must be chosen.

Illustration 7.14 A producer gas, 65% N_2 , 35% CO , initially dry at 1 std atm and $315^\circ C$, flowing at a rate of $5 \text{ m}^3/\text{s}$ ($10\,600 \text{ ft}^3/\text{min}$), is to be cooled to $27^\circ C$ by countercurrent contact with water entering at $18^\circ C$. A tower packed with 50-mm ceramic Raschig rings will be used, with $L'_2/G'_2 = 2.0$. Specify the diameter of a suitable tower and the packed height.

SOLUTION For the entering gas, $Y'_1 = 0$, $M_B = 28.0$, $p_1 = 1 \text{ std atm}$, $t_{G1} = 315^\circ C$, whence

$$\rho_{G1} = \frac{28.0}{22.41} \frac{273}{273 + 315} = 0.580 \text{ kg/m}^3$$

therefore

$$\text{Gas in} = 5(0.580) = 2.90 \text{ kg/s}$$

At the top of the tower, $t_{L2} = 18^\circ C$, $\rho_{L2} = 1000 \text{ kg/m}^3$, $\mu_{L2} = 1.056 \times 10^{-3} \text{ kg/m} \cdot \text{s}$, and $t_{G2} = 27^\circ C$. Since the outlet gas is likely to be nearly saturated ($Y' = 0.024$), estimate $Y'_2 = 0.022$, to be checked later. The outlet gas rate $= 2.90(1.022) = 2.964 \text{ kg/s}$.

$$M_{av} = \frac{1.022}{1/28.0 + 0.022/18.02} = 27.7 \text{ kg/kmol}$$

$$\rho_{G2} = \frac{27.7}{22.41} \frac{273}{273 + 27} = 1.125 \text{ kg/m}^3$$

$$\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L - \rho_G} \right)^{1/2} = 2 \left(\frac{1.125}{1000 - 1.125} \right)^{1/2} = 0.0671$$

Fig. 6.34: at a gas-pressure drop of $400 \text{ (N/m}^2\text{)/m}$, the ordinate is $0.073 = G'^2 C_f \mu_L^{0.1} / \rho_G (\rho_L - \rho_G) g_c$. Table 6.3: $C_f = 65$; $J = 1$, $g_c = 1$, whence $G'_2 = 1.583 \text{ kg/m}^2 \cdot \text{s}$ (tentative).

The tower cross section is $2.964/1.583 = 1.87 \text{ m}^2$ (tentative). The corresponding diameter is 1.54 , say 1.50 m , for which the cross section is $\pi(1.50)^2/4 = 1.767 \text{ m}^2$ (final). $G'_2 = 2.90/1.767 = 1.641 \text{ kg/m}^2 \cdot \text{s}$ (final). $G'_2 = 2.964/1.767 = 1.677 \text{ kg/m}^2 \cdot \text{s}$ (final); $L'_2 = 2(1.677) = 3.354 \text{ kg/m}^2 \cdot \text{s}$.

Calculations will be started at the bottom, for which L'_1 and t_{L1} must be known. An overall water balance [Eq. (7.29)] is

$$3.354 - L'_1 = 1.641(0.022 - 0)$$

$$L'_1 = 3.319 \text{ kg/m}^2 \cdot \text{s}$$

The heat capacities of CO and N_2 are $C_B = 1089$, and that of water vapor is $C_A = 1884 \text{ J/kg} \cdot \text{K}$. $C_{S1} = 1089$, $C_{S2} = 1089 + 1884(0.022) = 1130 \text{ N} \cdot \text{m}/(\text{kg dry gas}) \cdot \text{K}$. For convenience use as base temperature $t_0 = 18^\circ C$, for which $\lambda_0 = 2.46 \times 10^6 \text{ J/kg}$. $C_{AL} = 4187 \text{ J/kg} \cdot \text{K}$. An overall enthalpy balance [Eq. (7.32)]:

$$3.354(4187)(18 - 18) + 1.641(1089)(315 - 18) = 3.319(4187) \\ \times (t_{L1} - 18) + 1.641[1130(27 - 18) + 0.022(2.46 \times 10^6)]$$

$t_{L1} = 49.2^\circ C$, for which $\mu_L = 0.557 \times 10^{-3} \text{ kg/m} \cdot \text{s}$, $\rho_L = 989 \text{ kg/m}^3$, thermal conductivity $= 0.64 \text{ W/m} \cdot \text{K}$, Prandtl number $Pr_L = 3.77$. For the entering gas, the viscosity is $0.0288 \times 10^{-3} \text{ kg/m} \cdot \text{s}$, the diffusivity of water vapor $0.8089 \times 10^{-4} \text{ m}^2/\text{s}$, $Sc_G = 0.614$, and $Pr_G = 0.74$.

With the data of Chap. 6 and the methods of Illustration 6.7, we obtain $a = a_{vw} = 53.1 \text{ m}^2/\text{m}^3$, $F_{Ga} = 0.0736 \text{ kmol/m}^3 \cdot \text{s}$, $h_{Ga} = 4440$, and $h_{La} = 350\,500 \text{ W/m}^3 \cdot \text{K}$.

At the bottom, t_i will be estimated by trial and error. After several trials, assume $t_i = 50.3^\circ C$. $\bar{p}_{A,i} = 93.9 \text{ mmHg}/760 = 0.1235 \text{ std atm}$. Eq. (7.64):

$$\frac{dY'}{dZ} = - \frac{18.02(0.0736)}{1.641} \ln \frac{1 - 0.1235}{1 - 0} = 0.1066 \text{ (kg H}_2\text{O/kg dry gas)/m}$$

Eq. (7.66)

$$h_G a = \frac{-1.641(1884)(0.1066)}{1 - \exp[1.641(1844)(0.1066)/4440]} = 4280 \text{ W/m}^2 \cdot \text{K}$$

Eq. (7.65)

$$\frac{dt_G}{dZ} = \frac{-4280(315 - 50.3)}{1.641(1089)} = -630^\circ\text{C/m}$$

For use in Eq. (7.63):

$$C_A t_G - C_{A,L} t_L + (C_{A,L} - C_A) t_0 + \lambda_0 = 1884(315) - 4187(49.2) \\ + (4187 - 1884)(18) + 2.46 \times 10^6 = 2.889 \times 10^6 \text{ J/kg}$$

Eq. (7.63):

$$t_i = 49.2 + \frac{1.641[1089(-630) + (2.889 \times 10^6)(0.1066)]}{1.641(4187)(0.1066) - 350500} = 50.3^\circ\text{C} \quad (\text{check})$$

A suitably small increment in gas temperature is now chosen, and with the computed gradients assumed constant over a small range, the conditions at the end of the increment are computed. For example, take $\Delta t_G = -30^\circ\text{C}$.

$$\Delta Z = \frac{\Delta t_G}{dt_G/dZ} = \frac{-30}{-630} = 0.0476 \text{ m}$$

$$t_G = t_G(\text{at } Z = 0) + \Delta t_G = 315 - 30 = 285^\circ\text{C}$$

$$Y' = Y'(\text{at } Z = 0) + \frac{dY'}{dZ} \Delta Z = 0 + 0.1066(0.0476)$$

$$= 5.074 \times 10^{-3} \text{ kg H}_2\text{O/kg dry gas}$$

$$\bar{P}_{A,G} = \frac{Y'}{Y' + M_A/M_B} = \frac{5.074 \times 10^{-3}}{5.074 \times 10^{-3} + 18/28} = 7.83 \times 10^{-3} \text{ std atm}$$

$$C_S = 1089 + 1884(5.074 \times 10^{-3}) = 1099 \text{ J/kg} \cdot \text{K}$$

L' is calculated by a water balance [Eq. (7.29)] over the increment:

$$L' - 3.319 = 1.641(5.074 \times 10^{-3} - 0)$$

$$L' = 3.327 \text{ kg/m}^2 \cdot \text{s}$$

t_L is calculated by an enthalpy balance [Eq. (7.31)]:

$$3.327(4187)(t_L - 18) + 1.641(1089)(315 - 18) = 3.319(4187)(49.2 - 18) \\ + 1.641[1099(285 - 18) + (5.074 \times 10^{-3})(2.46 \times 10^6)]$$

therefore

$$t_L = 47.1^\circ\text{C}$$

New gradients dY'/dZ and dt_G/dZ at this level in the tower and another interval are then computed in the same manner. The process is repeated until the gas temperature falls to 27°C . The intervals of Δt_G chosen must be small as the gas approaches saturation. The entire computation is readily adapted to a digital computer. The computed $t_G - Y'$ results are shown in Fig. 7.15, along with the saturation humidity curve. At $t_G = 27^\circ\text{C}$, the value of Y_2' was calculated to be 0.0222, which is considered sufficiently close to the value assumed earlier, 0.022. The sum of the ΔZ 's = $Z = 1.54 \text{ m}$. This relatively small packed depth emphasizes the effectiveness of this type of operation.

The direct contact of a vapor-laden gas with cold liquid of the same composition as the vapor can be useful as a means of vapor recovery. For this, a flowsheet of the sort shown in Fig. 7.16 can be employed and the tower designed

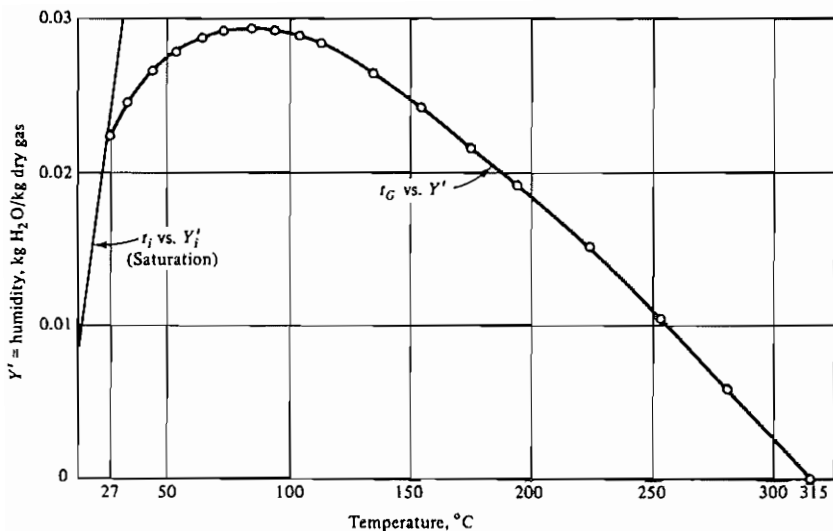


Figure 7.15 Temperatures and humidities for Illustration 7.14.

through Eqs. (7.63) to (7.69). The method is direct, whereas alternative methods of vapor recovery require additional operations for their completion. For example, absorption of the vapor into a solvent must be followed by distillation or stripping to recover the solvent and obtain the recovered solute; adsorption onto a solid similarly requires additional steps. With gas-vapor mixtures other than air-water vapor, however, the molecular diffusivity is likely to be less than the thermal diffusivity, so that $Le = Sc/Pr = \alpha/D_{AB}$ may exceed 1.0. Heat transfer is then faster than mass transfer, the path of the gas on a psychrometric chart tends to enter the supersaturation region above the saturation curve, and fog may result if suitable nucleation conditions are present [25, 26]. The difficulty is

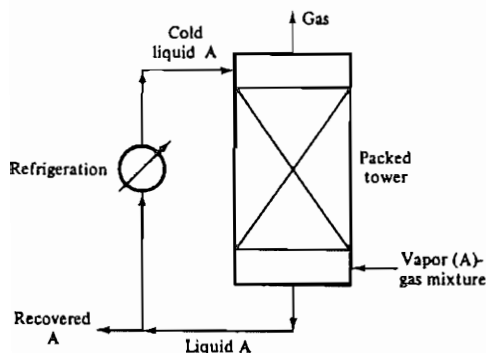


Figure 7.16 Recovery of vapor from a gas mixture by direct contact with a cold liquid.

alleviated if the entering vapor-gas mixture is sufficiently superheated. An example is given in Prob. 7.11.

Equipment

Any of the gas-liquid contact devices described in Chap. 6 are applicable to the operations described here, and conventional packed and tray towers are very effective in these services.

Water-cooling towers Air and water are low-cost substances, and where large volumes must be handled, as in many water-cooling operations, equipment of low initial cost and low operating cost is essential. The framework and internal packing are frequently of redwood, a material which is very durable in continuous contact with water, or Douglas fir. Impregnation of the wood under pressure with fungicides such as coal-tar creosote, pentachlorophenols, acid copper chromate, and the like, is commonplace. Siding for the towers is commonly redwood, asbestos cement, glass-reinforced polyester plastic, and the like. Towers have been built entirely of plastic. The internal packing ("fill") is usually a modified form of hurdle (see Chap. 6), horizontal slats arranged staggered with alternate tiers at right angles. Plastic packing may be polypropylene, molded in a grid or other form [5]. A great many arrangements are used [12, 15]. The void space is very large, usually greater than 90 percent, so that the gas-pressure drop will be as low as possible. The air-water interfacial surface consequently includes not only that of the liquid films which wet the slats (or other packing) but also the surface of the droplets which fall as rain from each tier of packing to the next.

The common arrangements are shown schematically in Fig. 7.17. Of the *natural-circulation towers* (Fig. 7.17*a* and *b*) the atmospheric towers depend on prevailing winds for air movement. The natural-draft design [6, 8] ensures more positive air movement even in calm weather by depending upon the displacement of the warm air inside the tower by the cooler outside air. Fairly tall chimneys are then required. Both these tower types must be relatively tall in order to operate at a small wet-bulb-temperature approach. Natural-draft equipment is used commonly in the southwestern United States and in the Middle East, where the humidity is usually low, in parts of Europe where air temperatures are generally low, and with increasing frequency everywhere as energy for fan power becomes more costly.

Mechanical-draft towers may be of the forced-draft type (Fig. 7.17*c*), where the air is blown into the tower by a fan at the bottom. These are particularly subject to recirculation of the hot, humid discharged air into the fan intake owing to the low discharge velocity, which materially reduces the tower effectiveness. Induced draft, with the fan at the top, avoids this and also permits more uniform internal distribution of air. The arrangements of Fig. 7.17*d* and *e* are most commonly used, and a more detailed drawing is shown in Fig. 7.18. Liquid rates are ordinarily in the range $L' = 0.7$ to $3.5 \text{ kg/m}^2 \cdot \text{s}$ (500 to 2500

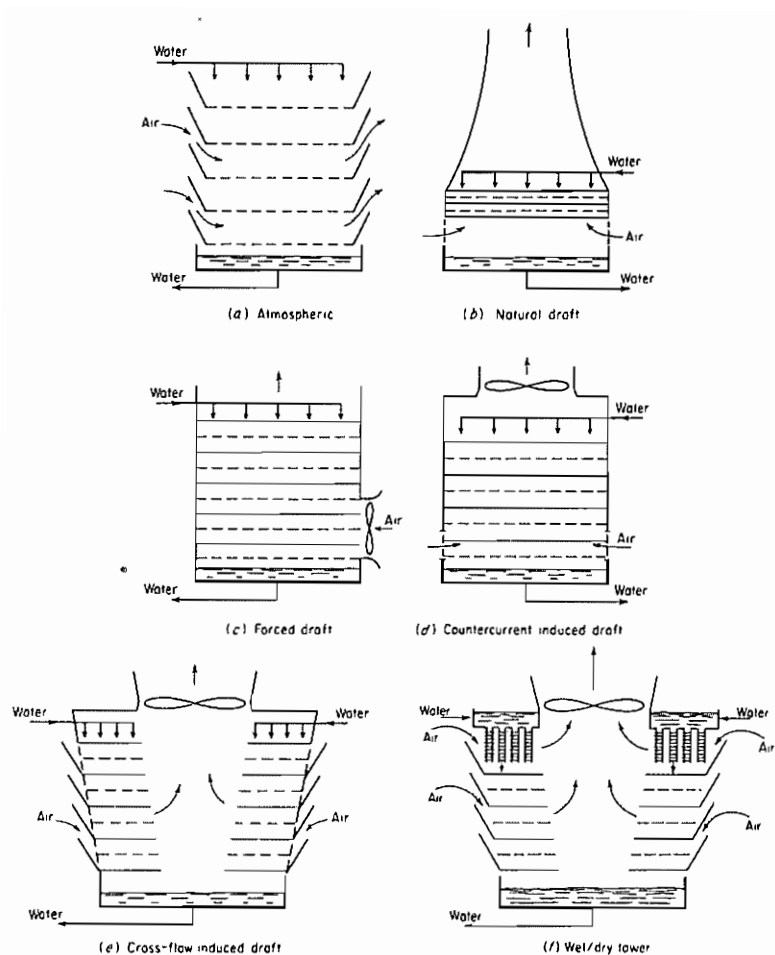


Figure 7.17 Cooling-tower arrangements.

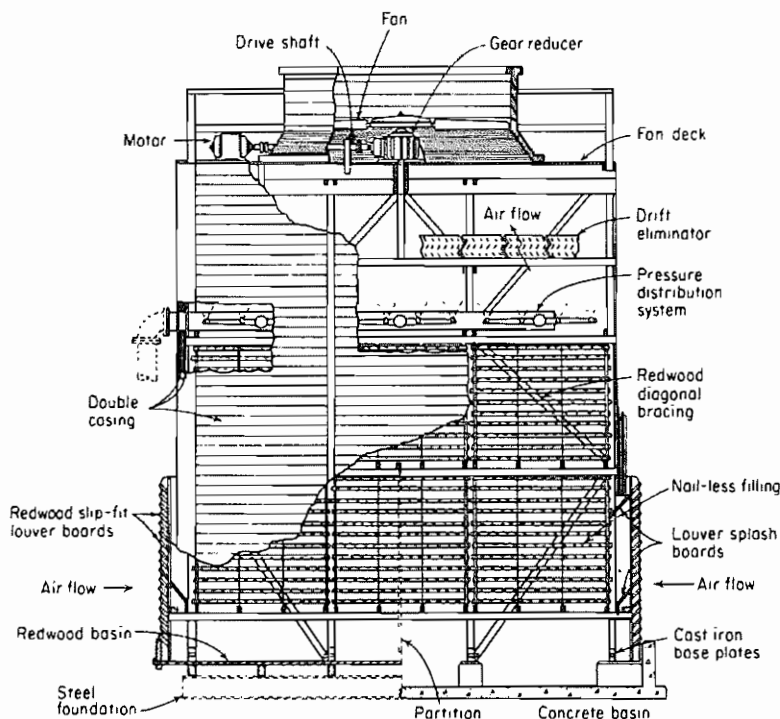


Figure 7.18 Induced-draft cooling tower. (The Marley Co., Inc.)

$\text{lb/ft}^2 \cdot \text{h}$), and superficial air rates are of the order of $G'_S = 1.6$ to $2.8 \text{ kg/m}^2 \cdot \text{s}$ (1200 to $2100 \text{ lb/ft}^2 \cdot \text{h}$), whereupon the air pressure drop is ordinarily less than 250 N/m^2 (25 mmHg). If fogging is excessive, finned-type heat exchangers can be used to evaporate the fog by heat from the hot water to be cooled, as in Fig. 7.17f.

Mass-transfer rates The correlations of mass-transfer coefficients for the standard packings discussed in Chap. 6 are suitable for the operations discussed here (see particularly Illustration 6.7). Additional data for humidification with Berl saddles [9]; Intalox saddles, and Pall rings [18†] are available. Data for some of the special tower fillings generally used for water-cooling towers are available in texts specializing in this type of equipment [15, 19].

† The data for $\text{H}_2\text{O}-\text{H}_2$ and $\text{H}_2\text{O}-\text{CO}_2$ should be used with caution since $H-L$ diagrams were used in interpreting the data despite the fact that Le is substantially different from unity for these systems.

Tray towers These are very effective but are not commonly used in humidification, dehumidification, or gas-cooling operations for reasons of cost and relatively high pressure drop, except under special circumstances. Design considerations are available [1, 4].

Spray chambers These are essentially horizontal spray towers and may be arranged as in Fig. 7.19. They are frequently used for adiabatic humidification-cooling operations with recirculating liquid. With large liquid drops, gas rates up to roughly 0.8 to $1.2 \text{ kg/m}^2 \cdot \text{s}$ (600 to $900 \text{ lb/ft}^2 \cdot \text{h}$) are possible, but in any case entrainment eliminators are necessary.

Heat-transfer surfaces at the inlet and outlet provide for preheating and afterheating of the air, so that processes of the type shown in Fig. 7.20 can be carried out. If large humidity changes by this method are required, preheating the air to unusually high temperatures is necessary, however. As an alternative, the spray water can be heated above the adiabatic-saturation temperature to which it will tend to come by direct injection of steam or by heating coils. Dehumidification can be practiced by cooling the water before spraying or by using refrigerating coils directly in the spray chamber. Operations of this sort cannot be followed with assurance on the enthalpy-temperature diagrams described earlier owing to the departure from strictly countercurrent-flow conditions which prevail. When an adequate spray density is maintained, it can be assumed that three banks of sprays in series will bring the gas to substantial equilibrium with the incoming spray liquid.

For comfort air conditioning, many compact devices are provided with a variety of these facilities, and automatic controls are available.

Spray ponds These are sometimes used for water cooling where close approach to the air wet-bulb temperature is not required. Spray ponds are essentially

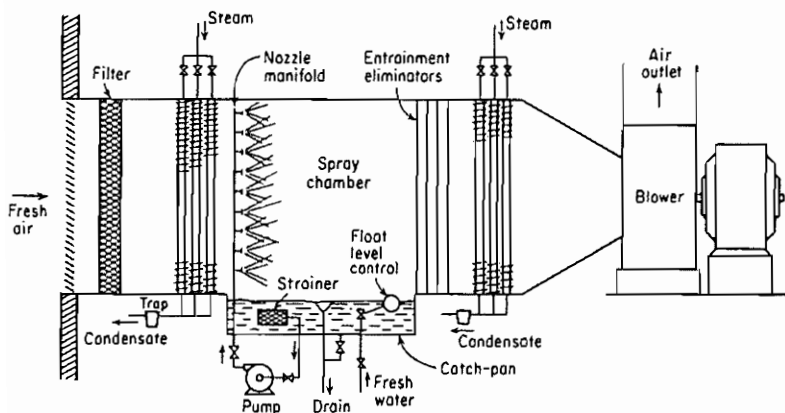


Figure 7.19 Schematic arrangement of a spray chamber.

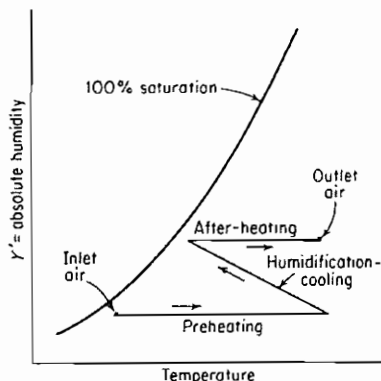


Figure 7.20 A simple conditioning process.

fountains, where the water is sprayed upward into the air and allowed to fall back into a collection basin. They are subject to high windage losses of water.

Dehumidification by other methods Adsorption, e.g., methods using activated silica gel, alumina, or molecular sieves as adsorbents (see Chap. 11), and washing gases with water solutions containing dissolved substances which appreciably lower the partial pressure of the water (see Chap. 8) are other commonly used dehumidification processes, particularly when very dry gases are required.

NONADIABATIC OPERATION: EVAPORATIVE COOLING

In evaporative cooling, a fluid is cooled while it flows through a tube. Water flows in a film or spray about the outside of the tube, and air is blown past the water to carry away the heat removed from the tube-side fluid. Advantage is taken of the large heat-transfer rate resulting when the spray water is evaporated into the airstream. A schematic arrangement is shown in Fig. 7.21a. The tube-side fluid usually flows through a bank of tubes in parallel, as shown in Fig. 7.21b and c. Since the water is recirculated from top to bottom of the heat exchanger, the temperature t_{L2} at which it enters is the same as that at which it leaves, t_{L1} . While the water temperature does not remain constant as it passes through the device, it does not vary greatly from the terminal value.

Figure 7.22 shows the temperature and gas-enthalpy profiles through a typical section of the exchanger. In the present analysis [24] the heat-transfer system will be divided at the bulk-water temperature t_L , corresponding to a saturation gas enthalpy H_1^* . The overall heat-transfer coefficient U_o based on the outside tube surface, from tube fluid to bulk water, is then given by

$$\frac{1}{U_o} = \frac{d_o}{d_i h_T} + \frac{d_o}{d_{av}} \frac{z_m}{k_m} + \frac{1}{h_L} \quad (7.67)$$

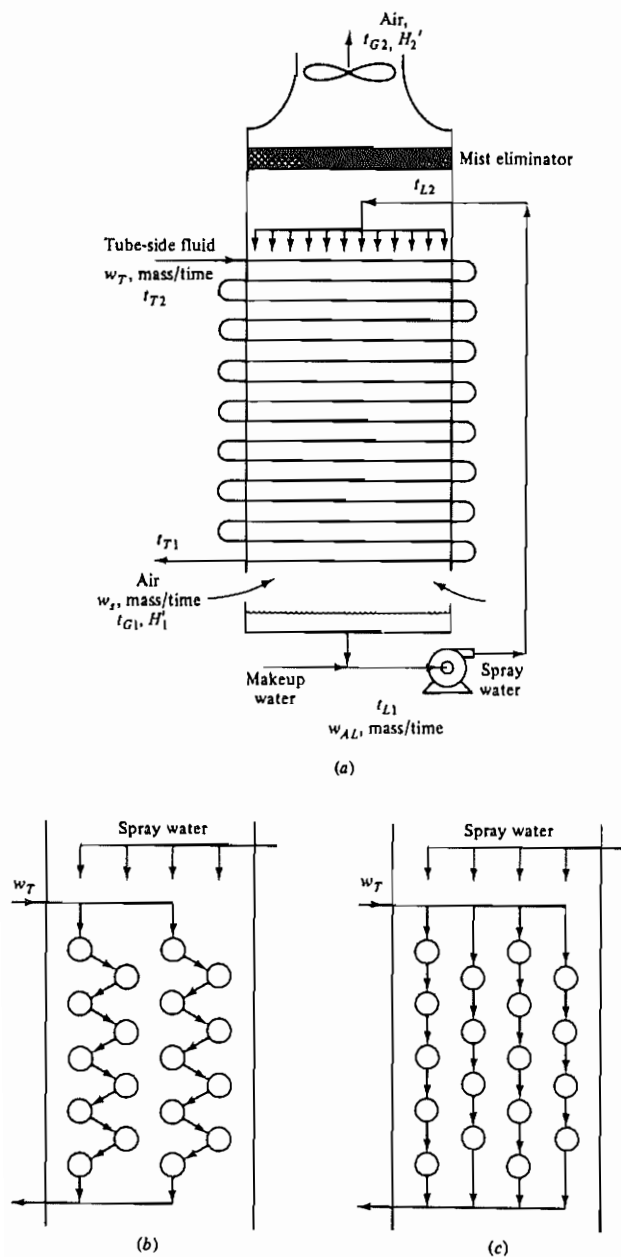


Figure 7.21 Evaporative cooler: (a) schematic arrangement, (b) and (c) arrangements of tube-side fluid flow.

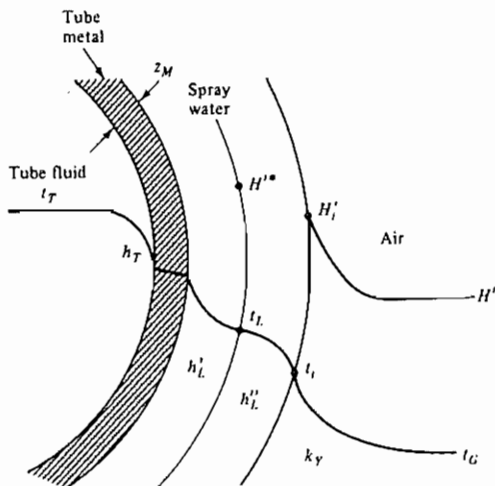


Figure 7.22 Temperature and enthalpy gradients, evaporative cooler.

The overall coefficient K_Y for use with gas enthalpies, from $H_1'^*$ to H' , is

$$\frac{1}{K_Y} = \frac{1}{k_Y} + \frac{m}{h_L''} \quad (7.68)$$

where m is the slope of a chord such as chord TS on Fig. 7.11:

$$m = \frac{H_1'^* - H_1'}{t_L - t_1}$$

Since the variation in t_L is small, for all practical purposes m can be taken as constant,

$$m = \frac{dH_1'^*}{dt_L} \quad (7.69)$$

If A_o is the outside surface of all the tubes, $A_o x$ is the area from the bottom to the level where the bulk-water temperature is t_L . Here x is the fraction of the heat-transfer surface to that level. For a differential portion of the exchanger, the heat loss by the tube-side fluid is

$$w_T C_T dt_L = U_o A_o dx (t_T - t_L)$$

whence
$$\frac{dt_T}{dx} = \frac{U_o A_o}{w_T C_T} (t_T - t_L) \quad (7.70)$$

The heat lost by the cooling water is

$$w_{A,L} C_{A,L} dt_L = K_Y A_o dx (H_1'^* - H') - U_o A_o dx (t_T - t_L)$$

or
$$\frac{dt_L}{dx} = \frac{K_Y A_o}{w_{A,L} C_{A,L}} (H_1'^* - H') - \frac{U_o A_o}{w_{A,L} C_{A,L}} (t_T - t_L) \quad (7.71)$$

The heat gained by the air is

$$w_S dH' = K_Y A_o dx (H'^* - H')$$

$$\text{or} \quad \frac{dH'}{dx} = \frac{K_Y A_o}{w_S} (H'^* - H') \quad (7.72)$$

Subtraction of Eq. (7.71) from Eq. (7.70) results in

$$\frac{d(t_T - t_L)}{dx} + \alpha_1(t_T - t_L) + \beta_1(H'^* - H') = 0 \quad (7.73)$$

Multiplying Eq. (7.71) by m and then subtracting Eq. (7.72) provides

$$\frac{d(H'^* - H')}{dx} + \alpha_2(t_T - t_L) + \beta_2(H'^* - H') = 0 \quad (7.74)$$

$$\text{where} \quad \alpha_1 = - \left(\frac{U_o A_o}{w_T C_T} + \frac{U_o A_o}{w_{A,L} C_{A,L}} \right) \quad \alpha_2 = \frac{m U_o A_o}{w_T C_T} \quad (7.75)$$

$$\text{and} \quad \beta_1 = \frac{K_Y A_o}{W_{A,L} C_{A,L}} \quad \beta_2 = - \left(\frac{m K_Y A_o}{w_{A,L} C_{A,L}} - \frac{K_Y A_o}{w_S} \right) \quad (7.76)$$

Integration of Eqs. (7.73) and (7.74), with U_o and K_Y assumed constant, yields

$$t_T - t_L = M_1 e^{r_1 x} + M_2 e^{r_2 x} \quad (7.77)$$

$$H'^* - H' = N_1 e^{r_1 x} + N_2 e^{r_2 x} \quad (7.78)$$

For use with average driving forces, these yield

$$Q = w_T C_T (t_{T2} - t_{T1}) = w_S (H'_2 - H'_1) = U_o A_o (t_T - t_L)_{av} = K_Y A_o (H'^* - H')_{av} \quad (7.79)$$

$$\text{where} \quad (t_T - t_L)_{av} = \frac{M_1}{r_1} (e^{r_1} - 1) + \frac{M_2}{r_2} (e^{r_2} - 1) \quad (7.80)$$

$$(H'^* - H')_{av} = \frac{N_1}{r_1} (e^{r_1} - 1) + \frac{N_2}{r_2} (e^{r_2} - 1) \quad (7.81)$$

In these expressions, r_1 and r_2 are the roots of the quadratic equation

$$r^2 + (\alpha_1 + \beta_2)r + (\alpha_1 \beta_2 - \alpha_2 \beta_1) = 0 \quad (7.82)$$

$$\text{and} \quad N_j = \frac{M_j(r_j + \alpha_1)}{\beta_1} \quad j = 1, 2 \quad (7.83)$$

These expressions permit the design of the evaporative cooler provided values of the various transfer coefficients are available. The available data [24] are meager despite the widespread use of these devices.

For h_T use standard correlation for sensible-heat changes or vapor condensation [13] for flow inside tubes.

For tubes 19.05 mm OD ($\frac{3}{4}$ in) on 38-mm (1.5-in) center, triangular pitch (Fig. 7.21b and c), $\Gamma/d_o = 1.36$ to 3.0 kg $H_2O/m^2 \cdot s$ (1000 to 2200 lb/ft² · h), $G'_{S, \min} = 0.68$ to 5 kg/m² · s (500 to 3700 lb/ft² · h), water 15 to 70°C, in

kilograms, meters, newtons, seconds, and kelvins and t_L in degrees Celsius†

$$h_L' = (982 + 15.58 t_L) \left(\frac{\Gamma}{d_o} \right)^{1/3} \quad \text{W/m}^2 \cdot \text{K} \quad (7.84)$$

$$h_L'' = 11\,360 \text{ W/m}^2 \cdot \text{K} \quad (7.85)$$

$$k_Y = 0.0493 [G_{S, \min}' (1 + Y_{av}')^{0.905}] \quad \text{kg/m}^2 \cdot \text{s} \cdot \Delta Y' \quad (7.86)$$

For other tube diameters, since the general relationship is expected to be

$$\text{Sh} = \text{const Re}^n \text{Sc}^{n'}$$

k_Y can be expected to vary as $G^n d_o^{n-1}$, or in this case as $d_o^{-0.095}$

Illustration 7.15 An evaporative cooler of the sort shown in Fig. 7.21a and b is to be used to cool oil. The oil, flowing at a rate 4.0 kg/s in the tubes, will enter at 95°C. Its average properties are density = 800 kg/m³, viscosity = 0.005 kg/m · s, thermal conductivity = 0.1436 W/m · K, and heat capacity = 2010 J/kg · K.

The cooler consists of a rectangular vertical shell, 0.75 m wide, fitted with 400 Admiralty Metal tubes, 19.05 mm OD, 1.65 mm wall thickness, 3.75 m long, on equilateral-triangular centers 38 mm apart. The tubes are arranged in horizontal rows of 20 tubes through which the oil flows in parallel, in stacks 10 tubes tall, as in Fig. 7.21b. Cooling water will be recycled at the rate 10 kg/s. The design air rate is to be 2.3 kg dry air/s, entering at 30°C dry-bulb temperature, standard atmospheric pressure, humidity 0.01 kg H₂O/kg dry air.

Estimate the temperature to which the oil will be cooled.

SOLUTION The minimum area for airflow is on a plane through the diameters of tubes in a horizontal row. The free area is

$$[0.75 - 20(0.01905)](3.75) = 1.384 \text{ m}^2$$

$$w_s = 2.3 \text{ kg/s} \quad G_{S, \min}' = \frac{2.3}{1.384} = 1.662 \text{ kg/m}^2 \cdot \text{s}$$

$Y_1' = 0.01 \text{ kg H}_2\text{O/kg dry air}$. Estimate $Y_2' = 0.06$. $Y_{av}' = (0.01 + 0.06)/2 = 0.035$. Eq. (7.86):

$$k_Y = 0.0493 [1.662(1 + 0.035)]^{0.905} = 0.05826 \text{ kg/m}^2 \cdot \text{s} \cdot \Delta Y'$$

Fig. 7.5:

$$H_1'(\text{satd at } t_w = 19.5^\circ\text{C}) = 56\,000 \text{ J/kg}$$

Outside surface of tubes = $A_o = 400\pi(0.01905)(3.75) = 89.8 \text{ m}^2$. The cooling water is distributed over 40 tubes. Since the tubes are staggered,

$$\Gamma = \frac{10}{40(2)(3.75)} = 0.0333 \text{ kg/m} \cdot \text{s} \quad \Gamma/d_o = \frac{0.0333}{0.01905} = 1.750 \text{ kg/m}^2 \cdot \text{s}$$

Tentatively take $av \ t_L \approx t_{L1} = t_{L2} = 28^\circ\text{C}$. Eq. (7.84):

$$h_L' = [982 + 15.58(28)](1.750)^{1/3} = 1709 \text{ W/m}^2 \cdot \text{K}$$

† For units of Btu, hours, feet, pounds mass, and degrees Fahrenheit Eqs. (7.84) to (7.86) become

$$h_L' = 13.72(1 + 0.0123 t_L) \left(\frac{\Gamma}{d_o} \right)^{1/3} \quad \text{Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F}$$

$$h_L'' = 2000 \text{ Btu/ft}^2 \cdot \text{h} \cdot ^\circ\text{F}$$

$$k_Y = 0.0924 [G_{S, \min}' (1 + Y_{av}')^{0.905}] \quad \text{lb/ft}^2 \cdot \text{h} \cdot \Delta Y'$$

Eq. (7.85):

$$h_L'' = 11\,360 \text{ W/m}^2 \cdot \text{K}$$

Fig. 7.5:

$$m = \frac{dH''}{dt_L} \text{ at } 28^\circ\text{C} = 5000 \text{ J/kg} \cdot \text{K}$$

Eq. (7.68):

$$\frac{1}{K_T} = \frac{1}{0.05826} + \frac{5000}{11\,360}$$

Tube ID = $d_i = [19.05 - 2(1.65)]/1000 = 0.01575 \text{ m}$. Inside cross section, each tube = $\pi(0.01575)^2/4 = 1.948 \times 10^{-4} \text{ m}^2$. G_T' = tube-fluid mass velocity = $4.0/20 (1.948 \times 10^{-4}) = 1026.7 \text{ kg/m}^2 \cdot \text{s}$;

$$\text{Re}_T = \frac{d_i G_T'}{\mu_T} = \frac{0.01575(1026.7)}{0.005} = 3234 \quad \text{Pr}_T = \frac{C_T \mu_T}{k_T} = \frac{2010(0.005)}{0.1436} = 70.0$$

From a standard correlation (fig. 10-8, p. 10-14, "The Chemical Engineers' Handbook," 5th ed.), $h_T = 364 \text{ W/m}^2 \cdot \text{K}$.

$$d_{av} \text{ for tubes} = \frac{0.01905 + 0.01575}{2} = 0.01740 \text{ m} \quad k_m = 112.5 \text{ W/m} \cdot \text{K}$$

Eq. (7.67):

$$\frac{1}{U_o} = \frac{0.01905}{0.01575(364)} + \frac{0.01905}{0.01740} \frac{0.00165}{112.5} + \frac{1}{1709}$$

$$U_o = 255 \text{ W/m}^2 \cdot \text{K}$$

$w_T = 4.0 \text{ kg/s}$, $w_{A,L} = 10 \text{ kg/s}$, $w_S = 2.3 \text{ kg/s}$; $C_T = 2010$, $C_{A,L} = 4187 \text{ N} \cdot \text{m/kg} \cdot \text{K}$. Equations (7.75) and (7.76) then yield

$$\alpha_1 = -3.393 \quad \alpha_2 = 18\,794 \quad \beta_1 = 12.08 \times 10^{-5} \quad \beta_2 = 1.402$$

With these values, the roots of Eq. (7.82) are

$$r_1 = 3.8272 \quad r_2 = -1.8362$$

Since t_L is unknown, proceed as follows. Eq. (7.83):

$$N_1 = \frac{M_1(3.8272 - 3.393)}{12.08 \times 10^{-5}} = 3594 M_1 \quad (7.87)$$

$$N_2 = \frac{M_2(-1.8362 - 3.393)}{12.08 \times 10^{-5}} = -43\,288 M_2 \quad (7.88)$$

From Eq. (7.77) at $x = 1$ (top)

$$95 - t_{L2} = M_1 e^{3.8272} + M_2 e^{-1.8362} = 45.934 M_1 + 0.15942 M_2 \quad (7.89)$$

From Eq. (7.78) at $x = 0$ (bottom)

$$H_1'' - 56\,000 = N_1 + N_2 \quad (7.90)$$

Eq. (7.80):

$$\begin{aligned} (t_T - t_L)_{av} &= \frac{M_1}{3.8272} (e^{3.8272} - 1) + \frac{M_2}{-1.8362} (e^{-1.8362} - 1) \\ &= 11.741 M_1 + 0.4578 M_2 \end{aligned} \quad (7.91)$$

Eq. (7.81):

$$(H'' - H')_{av} = 11.741 N_1 + 0.4578 N_2 \quad (7.92)$$

Equation (7.79), with Eqs. (7.91) and (7.92) for the average temperature and enthalpy differences, gives

$$U_o A_o (t_T - t_L)_{av} = K_Y A_o (H'^* - H')_{av}$$

$$255(89.8)(11.741M_1 + 0.4578M_2) = 0.0568(89.8)(11.741N_1 + 0.4578N_2) \quad (7.93)$$

Elimination of the M 's and the N 's by solving Eqs. (7.87) to (7.90) and (7.93) simultaneously (with $t_{L2} = t_{L1}$) results in

$$H_1'^* = 106\,560 - 532.2t_{L1}$$

This is solved simultaneously with the saturated-enthalpy curve of Fig. 7.5, whence $t_{L1} = 28^\circ\text{C}$, thus checking the value assumed at the start. $H_1'^* = 91\,660$ J/kg. Equations (7.87) to (7.90) then provide $M_1 = 1.461$, $M_2 = -0.7204$, and from Eq. (7.91) $(t_T - t_L)_{av} = 16.83^\circ\text{C}$. Eq. (7.79):

$$Q = 4.0(2010)(95 - t_{T1}) = 255(89.8)(16.83) = 2.3(H_2' - 56\,000)$$

$$t_{T1} = 47.0^\circ\text{C} = \text{the outlet temperature} \quad \text{Ans.}$$

$$H_2' = 223\,500 \text{ J/kg}$$

If the outlet air is essentially saturated, this corresponds to $Y' = 0.0689$ and the average $Y' = (0.01 + 0.0689)/2 = 0.0395$, which is considered sufficiently close to the 0.035 assumed earlier.

Note: It is estimated that in the absence of the spray water, all other quantities as above, the oil would be cooled to 81°C .

NOTATION FOR CHAPTER 7

Any consistent set of units may be used, except as noted.

a	specific interfacial surface, based on volume of packed section, L^2/L^3
a_H	specific interfacial surface for heat transfer, L^2/L^3
a_M	specific interfacial surface for mass transfer, L^2/L^3
A_o	outside surface of tubes, L^2
c	molar density, mole/L^3
C	heat capacity of a gas or vapor, unless otherwise indicated, at constant pressure, FL/MT
C_S	humid heat, heat capacity of a vapor-gas mixture per unit mass of dry gas content, FL/MT
C_T	heat capacity of tube-side fluid, FL/MT
d_{av}	mean diameter of a tube, L
d_i	inside diameter of a tube, L
d_o	outside diameter of a tube, L
D_{AB}	molecular diffusivity of vapor A in mixture with gas B, L^2/Θ
E_D	eddy diffusivity of mass, L^2/Θ
E_H	eddy diffusivity of heat, L^2/Θ
E_{MG}	Murphree gas-phase stage efficiency, fractional
F	mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta$
G	molar mass velocity, $\text{mole}/\text{L}^2\Theta$
G_S'	superficial mass velocity of dry gas, $\text{M}/\text{L}^2\Theta$
h	convective heat-transfer coefficient, $\text{FL}/\text{L}^2\Theta$
h'	convective heat-transfer coefficient corrected for mass transfer, $\text{FL}/\text{L}^2\Theta$
h_L', h_L''	water heat-transfer coefficients, evaporative cooler, $\text{FL}/\text{L}^2\Theta$
h_R	radiation heat-transfer coefficient in convection form, $\text{FL}/\text{L}^2\Theta$
H	enthalpy, FL/M
H'	enthalpy of a vapor-gas mixture per unit mass of dry gas, FL/M

H_{IG}	height of a gas transfer unit, L
H_{IOG}	overall height of a gas-enthalpy transfer unit, L
J_A	flux of mass for no bulk flow, mole/L ² Θ
k	thermal conductivity, FL ² /L ² TΘ
k_G	gas-phase mass-transfer coefficient, mole/L ² Θ(F/L ³)
k_Y	gas-phase mass-transfer coefficient, M/L ² Θ(M/M)
K_Y	overall gas-phase mass-transfer coefficient, M/L ² Θ(M/M)
ln	natural logarithm
log	common logarithm
L'	superficial mass velocity of liquid, M/L ² Θ
Le	Lewis number, Sc/Pr, dimensionless
m	slope of a chord of the saturated-enthalpy curve on a $H'^* - t_L$ diagram; dH'^*/dt_L , FL/MT
M	mol wt, M/mole
M_1, M_2	quantities defined by Eq. (7.83)
N	mass-transfer flux, mole/L ² Θ
N_{IG}	number of gas transfer units, dimensionless
N_{IOG}	number of overall gas transfer units, dimensionless
N_1, N_2	quantities defined by Eq. (7.83)
p	vapor pressure, F/L ²
$\bar{p}$	partial pressure, F/L ²
$\bar{p}_B, \mu$	mean partial pressure of nondiffusing gas, F/L ²
Pr	Prandtl number, $C \mu / k$, dimensionless
q_s	sensible-heat-transfer flux, FL/L ² Θ
q_T	total-heat-transfer flux, FL/L ² Θ
Q	evaporative-cooler heat load, FL/Θ; gain in enthalpy, FL/M
r	$h_G a_H / C_S k_Y a_M$ [not in Eq. (7.82)], dimensionless
r_1, r_2	roots of Eq. (7.82)
R	universal gas constant, FL/mole K
S	interfacial surface, L ²
Sc	Schmidt number, $\mu / \rho D_{AB}$, dimensionless
Sh	Sherwood number, $k_Y d_o / M_B c D_{AB}$, dimensionless
t	temperature, T
t_{as}	adiabatic saturation temperature, T
t_{DP}	dew-point temperature, T
t_W	wet-bulb temperature, T
t_0	reference temperature, T
T	absolute temperature, T
U	internal energy, FL/M
U_o	overall heat-transfer coefficient based on outside tube surface, FL/L ² TΘ
v	molal specific volume, L ³ /mole
v_H	humid volume, volume vapor-gas mixture/mass dry gas, L ³ /M
w	mass rate, M/Θ
w_S	mass rate of dry gas, M/Θ
W_B	mass of dry gas, M
x	fraction of heat-transfer surface traversed in the direction of airflow, dimensionless
Y'	absolute humidity, mass vapor/mass dry gas, M/M
z_m	metal thickness, L
Z	length or height of active part of equipment, L
α	thermal diffusivity, L ² /Θ
α_1, α_2	quantities defined by Eq. (7.75)
β_1, β_2	quantities defined by Eq. (7.76)
Γ	mass rate of spray water per tube per tube length (Fig. 7.21c); use twice the tube length for Fig. 7.21b, M/LΘ

Δ	difference
λ	latent heat of vaporization, FL/M
λ'	molal latent heat of vaporization, FL/mole
μ	viscosity, M/L θ
ρ	density, M/L ³

Subscripts

0	reference condition
1, 2	positions 1, 2
as	adiabatic saturation
av	average
A	substance A, the vapor
bp	boiling point
B	substance B, the gas
G	pertaining to the gas
i	interface
L	pertaining to the liquid
m	metal
min	minimum
nbp	normal boiling point
o	overall; outside
r	reference substance
s	saturation
T	tube-side fluid
W	wet-bulb temperature

Superscript

*	at saturation
---	---------------

REFERENCES

1. Barrett, E. C., and S. G. Dunn: *Ind. Eng. Chem. Process Des. Dev.*, **13**, 353 (1974).
2. "Cooling Towers," American Institute of Chemical Engineers, New York, 1972.
3. Dropkin, D.: *Cornell Univ. Eng. Expt. Stn. Bull.*, **23**, 1936; **26**, 1939.
4. Fair, J. R.: *Chem. Eng.*, **81**, 91 (June 12, 1972).
5. Fuller, A. L., A. L. Kohl, and E. Butcher: *Chem. Eng. Prog.*, **53**, 501 (1957).
6. Furzer, I. A.: *Ind. Eng. Chem. Process Des. Dev.*, **7**, 558, 561 (1968).
7. Grosvenor, W. M.: *Trans. AIChE*, **1**, 184 (1908).
8. Hayashi, Y., E. Hirai, and M. Okubo: *Heat Transfer Jap. Res.*, **2**(2), 1 (1973).
9. Hensel, S. L., and R. E. Treybal: *Chem. Eng. Prog.*, **48**, 362 (1952).
10. Henry, H. C., and N. Epstein: *Can. J. Chem. Eng.*, **48**, 595, 602, 609 (1970).
11. Inazume, H., and S. Kageyama: *Chem. Eng. Sci.*, **30**, 717 (1975).
12. Kelly, N. W., and L. K. Swenson: *Chem. Eng. Prog.*, **52**, 263 (1956).
13. Kreith, F.: "Principles of Heat Transfer," International Textbook, Scranton, Pa., 1958.
14. Kuehnmister, A. M.: *Chem. Eng.*, **78**, 112 (May 31, 1971).
15. McKelvey, K. K., and M. Brooke: "The Industrial Cooling Tower," Van Nostrand, Princeton, N.J., 1959.
16. Merkel, F.: *Ver. Dtsch. Forschungsarb.*, **275** (1925).
17. Mickley, H. S.: *Chem. Eng. Prog.*, **45**, 739 (1949).
18. Nemunaitis, R. R., and J. S. Eckert: *Chem. Eng. Prog.*, **71** (8), 60 (1975).
19. Norman, W. S.: "Absorption, Distillation, and Cooling Towers," Wiley, New York, 1961.

20. Olander, D. R.: *AIChE J.*, **6**, 346 (1960); *Ind. Eng. Chem.*, **53**, 121 (1961).
21. Othmer, D. F., et al.: *Ind. Eng. Chem.*, **32**, 841 (1940); **34**, 952 (1942).
22. Park, J. E., and J. M. Vance: *Chem. Eng. Prog.*, **67** (7), 55 (1971).
23. Parker, R. O.: D. Eng. Sci. dissertation, New York University, 1959; The Psychrometric Ratio for the System Air-Water Vapor, *1st Interam. Congr. Chem. Eng., San Juan, 1961*.
24. Parker, R. O., and R. E. Treybal: *Chem. Eng. Prog. Symp. Ser.*, **57**(32), 138 (1960).
25. Schrodt, J. T.: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 20 (1972).
26. Steinmeyer, D. E.: *Chem. Eng. Prog.*, **68**(7), 64 (1972).
27. Wnek, W. J., and R. H. Snow: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 343 (1972).

PROBLEMS

7.1 Prepare a logarithmic reference-substance plot of the vapor pressure of acetone over a temperature range of 10°C to its critical temperature, 235°C, with water as reference substance. With the help of the plot, determine (a) the vapor pressure of acetone at 65°C, (b) the temperature at which acetone has a vapor pressure of 500 torr, and (c) the latent heat of vaporization of acetone at 40°C (accepted value = 536.1 kJ · mol⁻¹).

7.2 A mixture of nitrogen and acetone vapor at 800 mmHg total pressure, 25°C, has a percentage saturation of 80%. Calculate (a) the absolute molal humidity, (b) the absolute humidity, kg acetone/kg nitrogen, (c) the partial pressure of acetone, (d) the relative humidity, (e) the volume percent acetone, and (f) the dew point.

7.3 In a plant for the recovery of acetone which has been used as a solvent, the acetone is evaporated into a stream of nitrogen gas. A mixture of acetone vapor and nitrogen flows through a duct, 0.3 by 0.3 m cross section. The pressure and temperature at one point in the duct are 800 mmHg, 40°C, and at this point the average velocity is 3.0 m/s. A wet-bulb thermometer (wick wet with acetone) indicates a temperature at this point of 27°C. Calculate the kilograms of acetone per second carried by the duct.

Ans.: 0.194 kg/s.

7.4 Would you expect the wet-bulb temperature of hydrogen-water vapor mixtures to be equal to, greater than, or less than the adiabatic-saturation temperature? Take the pressure as 1 std atm. Diffusivity for H₂O-H₂ = 7.5×10^{-5} m²/s at 0°C, 1 std atm. Heat capacity of H₂ = 14 650 N · m/kg · K, thermal conductivity = 0.173 W/m · K.

7.5 An air-water-vapor mixture, 1 std atm, 180°C, flows in a duct (wall temperature = 180°C) at 3 m/s average velocity. A wet-bulb temperature, measured with an ordinary, unshielded thermometer covered with a wetted wick (9.5 mm outside diameter) and inserted in the duct at right angles to the duct axis, is 52°C. Under these conditions, the adiabatic-saturation curves of the psychrometric chart do not approximate wet-bulb lines, radiation to the wet bulb and the effect of mass transfer on heat transfer are not negligible, and the k -type (rather than F) mass-transfer coefficients should not be used.

(a) Make the best estimate you can of the humidity of the air, taking these matters into consideration.

(b) Compute the humidity using the ordinary wet-bulb equation (7.26) with the usual h_G/k_Y , and compare.

7.6 Prepare a psychrometric chart for the mixture acetone-nitrogen at a pressure of 800 mmHg over the ranges -15 to 60°C, $Y' = 0$ to 3 kg vapor/kg dry gas. Include the following curves, all plotted against temperature: (a) 100, 75, 50, and 25% humidity; (b) dry and saturated humid volumes; (c) enthalpy of dry and saturated mixtures expressed as N · m/kg dry gas, referred to liquid acetone and nitrogen gas at -15°C; (d) wet-bulb curves for $t_w = 25^\circ\text{C}$; (e) adiabatic-saturation curves $t_{as} = 25$ and 40°C.

7.7 A drier requires 1.50 m³/s (3178 ft³/min) of air at 65°C, 20% humidity. This is to be prepared

from air at 27°C dry-bulb, 18°C wet-bulb temperatures by direct injection of steam into the airstream followed by passage of the air over steam-heated finned tubes. The available steam is saturated at 35 000 N/m² (5.08 lb_f/in²) gauge. Compute the kilograms of steam per second required (a) for direct injection and (b) for the heat exchanger.

Ans.: (a) 0.0463 kg/s; (b) 0.0281 kg/s.

7.8 For the purpose of carrying out a catalytic reaction of acetone vapor with another reagent, an acetone-nitrogen mixture will be produced by passing nitrogen upward through a tower packed with 38-mm (1.5-in) Berl saddles, irrigated with liquid acetone recirculated from the bottom to the top of the tower, with addition of makeup acetone as needed. The process is to be one of adiabatic gas humidification, as in Fig. 7.14. The N₂ will be preheated to the necessary temperature and introduced at the rate 1.5 kg/s (1106 lb/h). The desired ultimate humidity is 0.6 kg acetone/kg N₂, and the circulated liquid acetone is to be at 25°C, the adiabatic-saturation temperature. The surface tension of acetone at 25°C is 23.2 dyn/cm.

(a) To what temperature should the N₂ be preheated?

(b) Specify the liquid-circulation rate and the dimensions of a suitable packed section of the tower.

Ans.: 7.1 kg/s; 1.5 m diam, 1.1 m depth.

(c) Another possible process would involve heating the recirculating liquid, admitting the gas at ordinary temperatures. Are there advantages to this alternative? What portion of this chapter could be used for the design of the tower?

7.9 A recently installed induced-draft cooling tower was guaranteed by the manufacturer to cool 2000 U.S. gal/min (0.1262 m³/s) of 43°C water to 30°C when the available air has a wet-bulb temperature 24°C. A test on the tower, when operated at full fan capacity, provided the following data:

Inlet water, 0.1262 m³/s, 46.0°C

Outlet water, 25.6°C

Inlet air, 24.0°C dry-bulb, 15.6°C wet-bulb temperature

Outlet air, 37.6°C, essentially saturated

(a) What is the fan capacity, m³/s?

(b) Can the tower be expected to meet the guarantee conditions? Note that to do so, N_{OG} in the test must at least equal the guarantee value if H_{OG} is unchanged.

7.10 It is desired to dehumidify 1.2 m³/s (2540 ft³/min) of air, available at 38.0°C dry-bulb, 30.0°C wet-bulb temperatures, to a wet-bulb temperature of 15.0°C in a countercurrent tower using water chilled to 10.0°C. The packing will be 50-mm (2-in) Raschig rings. To keep entrainment at a minimum, G'_S will be 1.25 kg air/m² · s (922 lb/ft² · h), and a liquid rate of 1.5 times the minimum will be used.

(a) Specify the cross section and height of the packed portion of the tower.

Ans.: $Z = 1.013$ m.

(b) What will be the temperature of the outlet water?

7.11 A new process involves a product wet with methanol, and it is planned to dry the product by a continuous evaporation of the methanol into a stream of hot carbon dioxide (CO₂ rather than air in order to avoid an explosive mixture). One of the schemes being considered is that of Fig. 7.16. The tower will be packed with 25-mm ceramic Raschig rings. The CO₂-methanol mixture from the drier, at the rate 0.70 kg/(m² tower cross section) · s (516 lb/ft² · h) total flow, 1 std atm, 80°C, will contain 10% methanol vapor by volume. The refrigerated liquid methanol will enter the tower at -15°C at a rate of 4.75 kg/m² · s (3503 lb/ft² · h). The methanol of the effluent gas is to be reduced to 2.0% by volume.

(a) Calculate the value of the coefficients $F_G a$, $h_G a$, and $h_L a$ at the top of the tower. For this purpose assume that the outlet gas is at -9°C. Take the surface tension of methanol as 25 dyn/cm.

Ans.: $F_{Ga} = 0.0617$ kmol/m³ · s, $h_{Ga} = 2126$, $h_{La} = 32\,900$ W/m³ · K.

(b) Compute the required depth of packing. Take the base temperature for enthalpies as -15°C , at which temperature the latent heat of vaporization of methanol is 1197.5 kJ/kg . Take $C_{A,L} = 2470$, $C_A = 1341$, $C_B = 846 \text{ J/kg} \cdot \text{K}$. In order to simplify the computations, assume that the interfacial area, void space, gas viscosity, Sc_G , and Pr_G all remain constant. Then F_G varies as $G/G^{0.36}$. Since $h_L a$ is very large, assume also that $t_i = t_L$. For present purposes, ignore any solubility of CO_2 in methanol.

Ans.: 0.64 m.

It is necessary to consider the equilibrium characteristics of gas-liquid systems. A very brief discussion of such matters was presented in Chap. 2, but more will be required here.

CHAPTER

EIGHT

GAS ABSORPTION

If a quantity of a single gas and a relatively nonvolatile liquid are brought to equilibrium in the manner described in Figure 2, the resulting concentration of dissolved gas in the liquid is said to be the gas solubility at the prevailing temperature and pressure. At fixed temperature, the solubility of a gas in a liquid increases with pressure in the manner, for example, of curve A, Fig. 2.1, which shows the solubility of ammonia in water at 30°C.

Different gases and different liquid solvents, of course, exhibit different solubilities. Ordinarily, the solubility is determined experimentally for each system. If the equilibrium pressure of a gas at a given liquid concentration is high, as in the case of curve B, Fig. 2.1, the gas is said to be relatively insoluble in the liquid, while if it is low, as for curve A, the solubility is said to be high. But it is also a relative matter.

Gas absorption is an operation in which a gas mixture is contacted with a liquid for the purposes of preferentially dissolving one or more components of the gas and to provide a solution of them in the liquid. For example, the gas from by-product coke ovens is washed with water to remove ammonia and again with an oil to remove benzene and toluene vapors. Objectionable hydrogen sulfide is removed from such a gas or from naturally occurring hydrocarbon gases by washing with various alkaline solutions in which it is absorbed. Valuable solvent vapors carried by a gas stream can be recovered for reuse by washing the gas with an appropriate solvent for the vapors. Such operations require mass transfer of a substance from the gas stream to the liquid. When mass transfer occurs in the opposite direction, i.e., from the liquid to the gas, the operation is called desorption, or stripping. For example, the benzene and toluene are removed from the absorption oil mentioned above by contacting the liquid solution with steam, whereupon the vapors enter the gas stream and are carried away, and the absorption oil can be used again. Since the principles of both absorption and desorption are basically the same, we can study both operations at the same time.

Ordinarily, these operations are used only for solute recovery or solute removal. Separation of solutes from each other to any important extent requires the fractionation techniques of distillation.

EQUILIBRIUM SOLUBILITY OF GASES IN LIQUIDS

The rate at which a gaseous constituent of a mixture will dissolve in an absorbent liquid depends upon the departure from equilibrium which exists, and

therefore it is necessary to consider the equilibrium characteristics of gas-liquid systems. A very brief discussion of such matters was presented in Chap. 5, but some elaboration will be required here.

Two-Component Systems

If a quantity of a single gas and a relatively nonvolatile liquid are brought to equilibrium in the manner described in Chap. 5, the resulting concentration of dissolved gas in the liquid is said to be the *gas solubility* at the prevailing temperature and pressure. At fixed temperature, the solubility concentration will increase with pressure in the manner, for example, of curve *A*, Fig. 8.1, which shows the solubility of ammonia in water at 30°C.

Different gases and liquids yield separate solubility curves, which must ordinarily be determined experimentally for each system. If the equilibrium pressure of a gas at a given liquid concentration is high, as in the case of curve *B* (Fig. 8.1), the gas is said to be relatively insoluble in the liquid, while if it is low, as for curve *C*, the solubility is said to be high. But these are relative matters

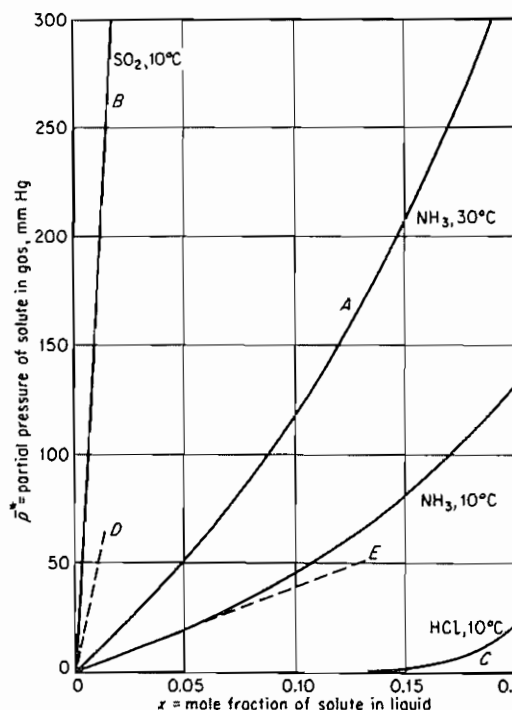


Figure 8.1 Solubilities of gases in water.

only, for it is possible to produce any ultimate gas concentration in the liquid if sufficient pressure is applied, so long as the liquefied form of the gas is completely soluble in the liquid.

The solubility of any gas is influenced by the temperature, in a manner described by van't Hoff's law of mobile equilibrium: if the temperature of a system at equilibrium is raised, that change will occur which will absorb heat. Usually, but not always, the solution of a gas results in an evolution of heat, and it follows that in most cases the solubility of a gas decreases with increasing temperature. As an example, curve *A* (Fig. 8.1) for ammonia in water at 30°C lies above the corresponding curve for 10°C. At the boiling point of the solvent, provided its vapor pressure is less than that of the gas or vapor solute, the gas solubility will be zero. On the other hand, the solubility of many of the low-molecular-weight gases such as hydrogen, oxygen, nitrogen, methane, and others in water increases with increased temperature above about 100°C and therefore at pressures above atmospheric [29, 35]. This phenomenon can be usefully exploited, e.g., in operations such as certain ore-leaching operations where oxygen-bearing solutions are required. Quantitative treatment of such equilibria is beyond the scope of this book, but excellent treatments are available [27, 34].

Multicomponent Systems

If a mixture of gases is brought into contact with a liquid, under certain conditions the equilibrium solubilities of each gas will be independent of the others, provided, however, that the equilibrium is described in terms of the *partial pressures* in the gas mixture. If all but one of the components of the gas are substantially insoluble, their concentrations in the liquid will be so small that they cannot influence the solubility of the relatively soluble component, and the generalization applies. For example, curve *A* (Fig. 8.1) will also describe the solubility of ammonia in water when the ammonia is diluted with air, since air is so insoluble in water, provided that the ordinate of the plot is considered as the partial pressure of ammonia in the gas mixture. This is most fortunate, since the amount of experimental work in gathering useful solubility data is thereby considerably reduced. If several components of the mixture are appreciably soluble, the generalization will be applicable only if the solute gases are indifferent to the nature of the liquid, which will be the case only for ideal solutions. For example, a mixture of propane and butane gases will dissolve in a nonvolatile paraffin oil independently since the solutions that result are substantially ideal. On the other hand, the solubility of ammonia in water can be expected to be influenced by the presence of methylamine, since the resulting solutions of these gases are not ideal. The solubility of a gas will also be influenced by the presence of a nonvolatile solute in the liquid, such as a salt in water solution, when such solutions are nonideal.

Ideal Liquid Solutions

When the liquid phase can be considered ideal, we can compute the equilibrium partial pressure of a gas from the solution without resort to experimental determination.

There are four significant characteristics of ideal solutions, all interrelated:

1. The average intermolecular forces of attraction and repulsion in the solution are unchanged on mixing the constituents.
2. The volume of the solution varies linearly with composition.
3. There is neither absorption nor evolution of heat in mixing the constituents.
For gases dissolving in liquids, however, this criterion should not include the heat of condensation of the gas to the liquid state.
4. The total vapor pressure of the solution varies linearly with composition expressed as mole fractions.

In reality there are no ideal solutions, and actual mixtures only approach ideality as a limit. Ideality would require that the molecules of the constituents be similar in size, structure, and chemical nature, and the nearest approach to such a condition is perhaps exemplified by solutions of optical isomers of organic compounds. Practically, however, many solutions are so nearly ideal that for engineering purposes they can be so considered. Adjacent or nearly adjacent members of a homologous series of organic compounds particularly fall in this category. So, for example, solutions of benzene in toluene, of ethyl and propyl alcohols, or of the paraffin hydrocarbon gases in paraffin oils can ordinarily be considered as ideal solutions.

When the gas mixture in equilibrium with an ideal liquid solution also follows the ideal-gas law, the partial pressure $\bar{p}^*$ of a solute gas A equals the product of its vapor pressure p at the same temperature and its mole fraction in the solution x . This is *Raoult's law*.

$$\bar{p}^* = px \quad (8.1)$$

(The asterisk is used to indicate equilibrium.) The nature of the solvent liquid does not enter into consideration except insofar as it establishes the ideality of the solution, and it follows that the solubility of a particular gas in ideal solution in any solvent is always the same.

Illustration 8.1 After long contact with a hydrocarbon oil and establishment of equilibrium, a gas mixture has the following composition at $2 \times 10^5 \text{ N/m}^2$ total pressure, 24°C : methane 60%, ethane 20%, propane 8%, *n*-butane 6%, *n*-pentane 6%. Calculate the composition of the equilibrium solution.

SOLUTION The equilibrium partial pressure $\bar{p}^*$ of each constituent in the gas is its volume fraction multiplied by the total pressure. These and the vapor pressures p of the constituents at 24°C are tabulated below. The prevailing pressure is above the critical value for methane, and at this low total pressure its solubility may be considered negligible. For each constituent its mole fraction in the liquid is calculated by Eq. (8.1), $\bar{p}^*/p$, and the last column of the table lists

these as the answers to the problem. The remaining liquid, $1 - 0.261 = 0.739$ mole fraction, is the solvent oil.

Component	Equilibrium partial pressure $\bar{p}^*$, $\text{N/m}^2 \times 10^{-5}$	Vapor pressure p at 24°C , $\text{N/m}^2 \times 10^{-5}$	Mole fraction in the liquid $x = \bar{p}^*/p$
Methane	1.20 = 0.6(2)		
Ethane	0.4	42.05	0.0095
Propane	0.16	8.96	0.018
<i>n</i> -Butane	0.12	2.36	0.051
<i>n</i> -Pentane	0.12	0.66	0.182
Total			0.261

For total pressures in excess of those for which the ideal-gas law applies, Raoult's law can frequently be used with fugacities substituted for the pressure terms [27, 34].

Nonideal Liquid Solutions

For liquid solutions which are not ideal, Eq. (8.1) will give highly incorrect results. Line *D* (Fig. 8.1), for example, is the calculated partial pressure of ammonia in equilibrium with water solutions at 10°C , assuming Raoult's law to be applicable, and it clearly does not represent the data. On the other hand, the straight line *E* is seen to represent the 10°C ammonia-water data very well up to mole fractions of 0.06 in the liquid. The equation of such a line is

$$y^* = \frac{\bar{p}^*}{p_t} = mx \quad (8.2)$$

where m is a constant. This is *Henry's law*, † and it is seen to be applicable with different values of m for each of the gases in the figure over at least a modest liquid-concentration range. Failure to follow Henry's law over wide concentration ranges may be the result of chemical interaction with the liquid or electrolytic dissociation, as is the case with ammonia-water, or nonideality in the gas phase. Most gases can be expected to follow the law to equilibrium pressures to about $5 \times 10^5 \text{ N/m}^2$ (≈ 5 std atm), although if solubility is low, as with hydrogen-water, it may be obeyed to pressures as high as 34 N/m^2 [35]. Gases of the vapor type (which are below their critical temperature) will generally

† Equation (8.2) is frequently written as $y^* = Kx$, but here we wish to distinguish between the Henry-law constant and mass-transfer coefficients. For conditions under which Eq. (8.2) is inapplicable, it is frequently used in empirical fashion to describe experimental data, but the value of m will then be expected to vary with temperature, pressure, and concentration and should be listed as a function of these variables.

follow the law up to pressures of approximately 50 percent of the saturation value at the prevailing temperature provided no chemical action occurs in the liquid. In any case m must be established experimentally.

The advantages of straight-line plotting for interpolating and extrapolating experimental data are, of course, very great, and an empirical method of wide utility is an extension [32] of the reference-substance vapor-pressure plot described in Chap. 7. As an example of this, Fig. 8.2 shows the data for ammonia-water solutions, covering a wide range of concentrations and temperatures. The coordinates are logarithmic. The abscissa is marked with the vapor pressure of a convenient reference substance, in this case water, and the ordinate is the equilibrium partial pressure of the solute gas. Points are plotted where the corresponding temperatures for the vapor pressure of the reference substance and the partial pressure of solute are identical. For example, at 32.2°C the vapor pressure of water is 36 mmHg, the partial pressure of ammonia for a 10 mole percent solution is 130 mmHg, and these pressures locate point *A* on the figure. The lines for constant liquid composition are straight with few exceptions. A temperature scale can later be substituted for the reference vapor-pressure scale, using the steam tables where water is the reference substance. A similar plot,

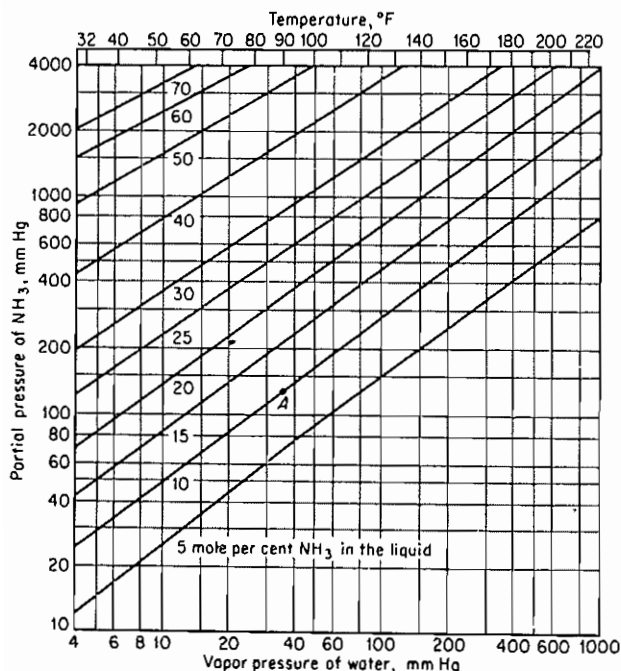


Figure 8.2 Reference-substance plot for gas solubility. The system ammonia-water, water as reference.

with water as reference substance, can also be drawn for the partial pressures of water over these solutions. An important property of such plots is their ability to provide enthalpy data. For substance (solute or solvent) J, let the slope of the line for concentration x_J be m'_{Jr} . Then

$$\bar{H}_J = C_J(t - t_0) + \lambda_{J0} - m'_{Jr}\lambda_{r,t} \quad (8.3)$$

$\bar{H}_J$ = partial enthalpy of component J in solution

C_J = molar heat capacity of gaseous J

λ = appropriate latent heat of vaporization

λ_{J0} is taken as zero if J is a gas at temperature t_0 . The enthalpy of the solution relative to pure components at t_0 is then

$$H_L = \sum_J \bar{H}_J x_J \quad (8.4)$$

Choice of Solvent for Absorption

If the principal purpose of the absorption operation is to produce a specific solution, as in the manufacture of hydrochloric acid, for example, the solvent is specified by the nature of the product. If the principal purpose is to remove some constituent from the gas, some choice is frequently possible. Water is, of course, the cheapest and most plentiful solvent, but the following properties are important considerations:

1. *Gas solubility.* The gas solubility should be high, thus increasing the rate of absorption and decreasing the quantity of solvent required. Generally solvents of a chemical nature similar to that of the solute to be absorbed will provide good solubility. Thus hydrocarbon oils, and not water, are used to remove benzene from coke-oven gas. For cases where the solutions formed are ideal, the solubility of the gas is the same in terms of mole fractions for all solvents. But it is greater in terms of weight fractions for solvents of low molecular weight, and smaller weights of such solvents, as measured in pounds, need to be used. A chemical reaction of solvent with the solute will frequently result in very high gas solubility, but if the solvent is to be recovered for reuse, the reaction must be reversible. For example, hydrogen sulfide can be removed from gas mixtures using ethanolamine solutions since the sulfide is readily absorbed at low temperatures and easily stripped at high temperatures. Caustic soda absorbs hydrogen sulfide excellently but will not release it in a stripping operation.
2. *Volatility.* The solvent should have a low vapor pressure since the gas leaving an absorption operation is ordinarily saturated with the solvent and much may thereby be lost. If necessary, a second, less volatile liquid can be used to recover the evaporated portion of the first, as in Fig. 8.3. This is sometimes done, for example, in the case of hydrocarbon absorbers, where a relatively

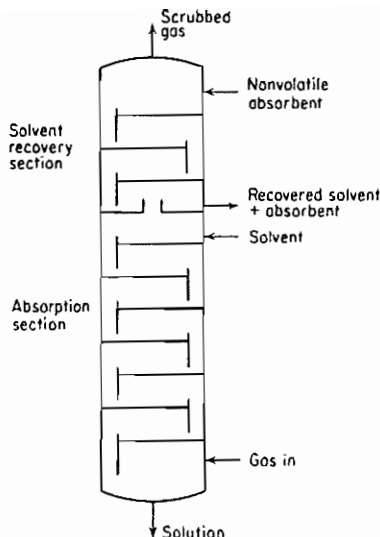


Figure 8.3 Tray absorber with volatile-solvent recovery section.

volatile solvent oil is used in the principal portion of the absorber because of the superior solubility characteristics and the volatilized solvent is recovered from the gas by a nonvolatile oil. Similarly, hydrogen sulfide can be absorbed by a water solution of sodium phenolate, but the desulfurized gas is further washed with water to recover the evaporated phenol.

3. *Corrosiveness*. The materials of construction required for the equipment should not be unusual or expensive.
4. *Cost*. The solvent should be inexpensive, so that losses are not costly, and should be readily available.
5. *Viscosity*. Low viscosity is preferred for reasons of rapid absorption rates, improved flooding characteristics in absorption towers, low pressure drops on pumping, and good heat-transfer characteristics.
6. *Miscellaneous*. The solvent if possible should be nontoxic, nonflammable, and chemically stable and should have a low freezing point.

ONE COMPONENT TRANSFERRED; MATERIAL BALANCES

The basic expressions for material balances and their graphical interpretation were presented for any mass-transfer operation in Chap. 5. Here they are adapted to the problems of gas absorption and stripping.

Countercurrent Flow

Figure 8.4 shows a countercurrent tower which may be either a packed or spray tower, filled with bubble-cap trays or of any internal construction to bring about

25/7/97

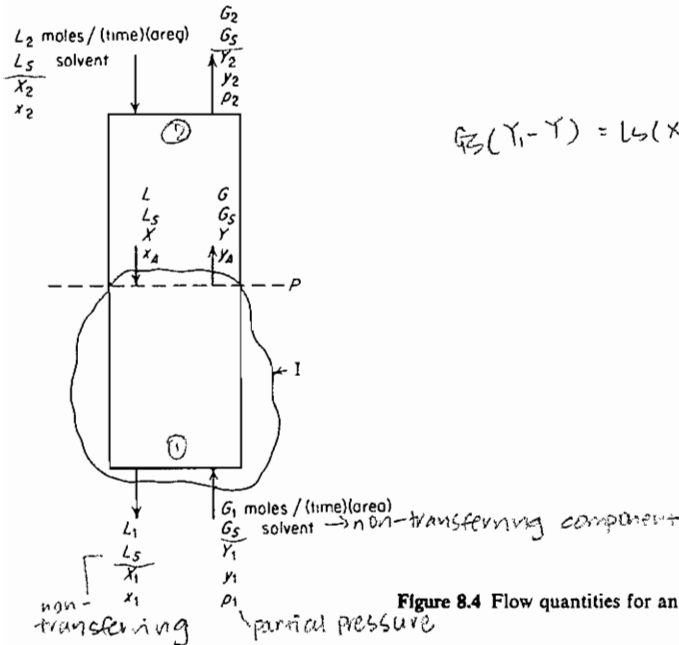


Figure 8.4 Flow quantities for an absorber or stripper.

liquid-gas contact. The gas stream at any point in the tower consists of G total mol/(area of tower cross section) (time), made up of diffusing solute A of mole fraction y , partial pressure $\bar{p}$, or mole ratio Y , and nondiffusing, essentially insoluble gas G_S mol/(area) (time). The relationship between these is

$$Y = \frac{y}{1 - y} = \frac{\bar{p}}{p_t - \bar{p}} \quad (8.5)$$

$$G_S = G(1 - y) = \frac{G}{1 + Y} \quad (8.6)$$

Similarly the liquid stream consists of L total mol/(area) (time), containing x mole fraction soluble gas, or mole ratio X , and essentially nonvolatile solvent L_S mol/(area) (time).

$$X = \frac{x}{1 - x} \quad (8.7)$$

$$L_S = L(1 - x) = \frac{L}{1 + X} \quad (8.8)$$

Since the solvent gas and solvent liquid are essentially unchanged in quantity as they pass through the tower, it is convenient to express the material balance in terms of these. A solute balance about the lower part of the tower

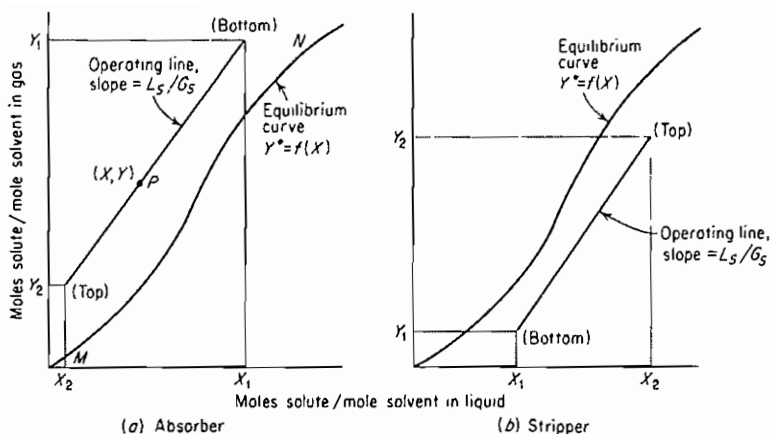


Figure 8.5 Operating lines for absorber and stripper.

(envelope I) is†

$$G_S(Y_1 - Y) = L_S(X_1 - X) \quad (8.9)$$

This is the equation of a straight line (the operating line) on X, Y coordinates, of slope L_S/G_S , which passes through (X_1, Y_1) . Substitution of X_2 and Y_2 for X and Y shows the line to pass through (X_2, Y_2) , as on Fig. 8.5a for an absorber. This line indicates the relation between the liquid and gas concentration at any level in the tower, as at point P .

The equilibrium-solubility data for the solute gas in the solvent liquid can also be plotted in terms of these concentration units on the same diagram, as curve MN , for example. Each point on this curve represents the gas concentration in equilibrium with the corresponding liquid at its local concentration and temperature. For an absorber (mass transfer from gas to liquid) the operating line always lies above the equilibrium-solubility curve, while for a stripper (mass transfer from liquid to gas) the line is always below, as in Fig. 8.5b.

The operating line is straight only when plotted in terms of the mole-ratio units. In terms of mole fractions or partial pressures the line is curved, as in Fig. 8.6 for an absorber. The equation of the line is then

$$G_S \left(\frac{y_1}{1-y_1} - \frac{y}{1-y} \right) = G_S \left(\frac{\bar{p}_1}{p_1 - \bar{p}_1} - \frac{\bar{p}}{p_1 - \bar{p}} \right) = L_S \left(\frac{x_1}{1-x_1} - \frac{x}{1-x} \right) \quad (8.10)$$

† Equations (8.9) and (8.10) and the corresponding Figs. 8.4 to 8.8 are written as for packed towers, with 1 indicating the streams at the bottom and 2 those at the top. For tray towers, where tray numbers are used as subscripts, as in Fig. 8.12, the same equations apply, but changes in subscripts must be made. Thus Eq. (8.9) becomes, when applied to an entire tray tower,

$$G_S(Y_{N_r+1} - Y_1) = L_S(X_{N_r} - X_0)$$

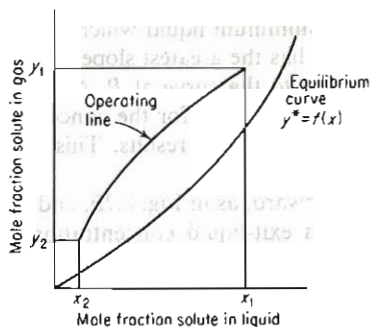


Figure 8.6 Operating line in mole fractions.

The total pressure p_t at any point can ordinarily be considered constant throughout the tower for this purpose.

Minimum Liquid-Gas Ratio for Absorbers

In the design of absorbers, the quantity of gas to be treated G or G_s , the terminal concentrations Y_1 and Y_2 , and the composition of the entering liquid X_2 are ordinarily fixed by process requirements, but the quantity of liquid to be used is subject to choice. Refer to Fig. 8.7a. The operating line must pass through point D and must end at the ordinate Y_1 . If such a quantity of liquid is used to give operating line DE , the exit liquid will have the composition X_1 . If less liquid is used, the exit-liquid composition will clearly be greater, as at point F , but since the driving forces for diffusion are less, the absorption is more difficult. The time of contact between gas and liquid must then be greater, and

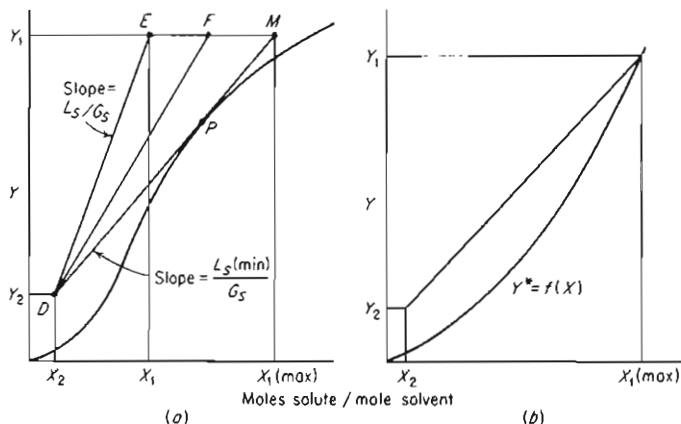


Figure 8.7 Minimum liquid-gas ratio, absorption.

the absorber must be correspondingly taller. The minimum liquid which can be used corresponds to the operating line DM , which has the greatest slope for any line touching the equilibrium curve and is tangent to the curve at P . At P the diffusional driving force is zero, the required time of contact for the concentration change desired is infinite, and an infinitely tall tower results. This then represents the limiting liquid-gas ratio.

The equilibrium curve is frequently concave upward, as in Fig. 8.7*b*, and the minimum liquid-gas ratio then corresponds to an exit-liquid concentration in equilibrium with the entering gas.

These principles also apply to strippers, where an operating line which anywhere touches the equilibrium curve represents a maximum ratio of liquid to gas and a maximum exit-gas concentration.

Cocurrent Flow

When gas and liquid flow cocurrently, as in Fig. 8.8, the operating line has a negative slope $-L_S/G_S$. There is no limit on this ratio, but an infinitely tall tower would produce an exit liquid and gas in equilibrium, as at (X_e, Y_e) . Cocurrent flow may be used when an exceptionally tall tower is built in two sections, as in Fig. 8.9, with the second section operated in cocurrent flow to save on the large-diameter gas pipe connecting the two. It may also be used if the gas to be dissolved in the liquid is a pure substance, where there is no advantage to countercurrent operation, or if a rapid, irreversible chemical reaction with the dissolved solute occurs in the liquid, where only the equivalent of one theoretical stage is required.

Illustration 8.2 A coal gas is to be freed of its light oil by scrubbing with wash oil as an absorbent and the light oil recovered by stripping the resulting solution with steam. The circumstances are as follows.

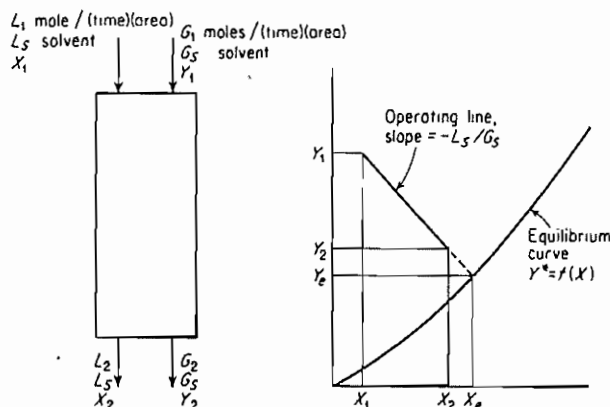


Figure 8.8 Cocurrent absorber.

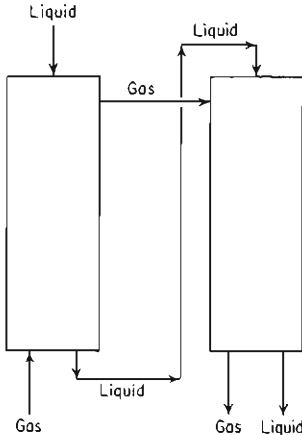


Figure 8.9 Countercurrent-ocurrent arrangement for very tall towers.

Absorber Gas in, $0.250 \text{ m}^3/\text{s}$ ($31\,800 \text{ ft}^3/\text{h}$) at 26°C , $p_i = 1.07 \times 10^5 \text{ N/m}^2$ (803 mmHg), containing 2.0% by volume of light oil vapors. The light oil will be assumed to be entirely benzene, and a 95% removal is required. The wash oil is to enter at 26°C , containing 0.005 mole fraction benzene, and has an av mol wt 260. An oil circulation rate of 1.5 times the minimum is to be used. Wash oil–benzene solutions are ideal. The temperature will be constant at 26°C .

Stripper The solution from the absorber is to be heated to 120°C and will enter the stripper at 1 std atm pressure. Stripping steam will be at standard atmospheric pressure, superheated to 122°C . The debenzolized oil, 0.005 mole fraction benzene, is to be cooled to 26°C and returned to the absorber. A steam rate of 1.5 times the minimum is to be used. The temperature will be constant at 122°C .

Compute the oil-circulation rate and the steam rate required.

SOLUTION

Absorber Basis: 1 s. Define L , L_S , G , and G_S in terms of kmol/s.

$$G_1 = 0.250 \frac{273}{273 + 26} \frac{1.07 \times 10^5}{1.0133 \times 10^5} \frac{1}{22.41} = 0.01075 \text{ kmol/s}$$

$$y_1 = 0.02 \quad Y_1 = \frac{0.02}{1 - 0.02} = 0.0204 \text{ kmol benzene/kmol dry gas}$$

$$G_S = 0.01075 (1 - 0.02) = 0.01051 \text{ kmol dry gas/s}$$

For 95% removal of benzene,

$$Y_2 = 0.05 (0.0204) = 0.00102 \text{ kmol benzene/kmol dry gas}$$

$$x_2 = 0.005 \quad X_2 = \frac{0.005}{1 - 0.005} = 0.00503 \text{ kmol benzene/kmol oil}$$

At 26°C , the vapor pressure of benzene is $p = 100 \text{ mmHg} = 13\,330 \text{ N/m}^2$. Equation (8.1) for ideal solutions: $\bar{p}^* = 13\,330x$.

$$y^* = \bar{p}^*/p_i \quad p_i = 1.07 \times 10^5 \text{ N/m}^2 \quad Y^* = \frac{y^*}{1 - y^*} \quad X = \frac{x}{1 - x}$$

Substitution in Eq. (8.1) yields

$$\frac{Y^*}{1 + Y^*} = 0.125 \frac{X^*}{1 + X^*}$$

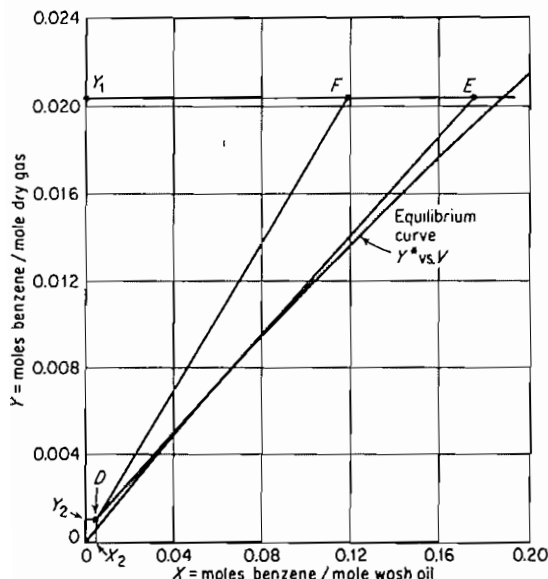


Figure 8.10 Solution to Illustration 8.2, absorption.

which is the equilibrium-curve equation for the absorber, plotted in Fig. 8.10. Operating lines originate at point D in this figure. For the minimum oil rate, line DE is drawn as the line of maximum slope which touches the equilibrium curve (tangent to the curve). At $Y_1 = 0.0204$, $X_1 = 0.176$ kmol benzene/kmol wash oil (point E).

$$\min L_S = \frac{G_S(Y_1 - Y_2)}{X_1 - X_2} = \frac{0.01051(0.0204 - 0.00102)}{0.176 - 0.00503} = 1.190 \times 10^{-3} \text{ kmol oil/s}$$

For 1.5 times the minimum, $L_S = 1.5(1.190 \times 10^{-3}) = 1.787 \times 10^{-3}$ kmol/s.

$$X_1 = \frac{G_S(Y_1 - Y_2)}{L_S} + X_2 = \frac{0.01051(0.0204 - 0.00102)}{1.787 \times 10^{-3}} + 0.00503 \\ = 0.1190 \text{ kmol benzene/kmol oil}$$

The operating line is DF.

Stripper At 122°C , the vapor pressure of benzene is $2400 \text{ mmHg} = 319.9 \text{ kN/m}^2$. The equilibrium curve for the stripper is therefore

$$\frac{Y^*}{1 + Y^*} = \frac{319.9}{101.33} \frac{X}{1 + X} = \frac{3.16X}{1 + X}$$

which is drawn in Fig. 8.11. For the stripper, $X_2 = 0.1190$, $X_1 = 0.00503$ kmol benzene/kmol oil, $Y_1 = 0$ kmol benzene/kmol steam. For the minimum steam rate, line MN is drawn tangent to the equilibrium curve, and at N the value of $Y_2 = 0.45$ kmol benzene/kmol steam.

$$\min G_S = \frac{L_S(X_2 - X_1)}{Y_2 - Y_1} = \frac{(1.787 \times 10^{-3})(0.1190 - 0.00503)}{0.45 - 0} \\ = 4.526 \times 10^{-4} \text{ kmol steam/s}$$

For 1.5 times the minimum, the steam rate is $1.5(4.526 \times 10^{-4}) = 6.79 \times 10^{-4}$ kmol/s, corresponding to line MP.

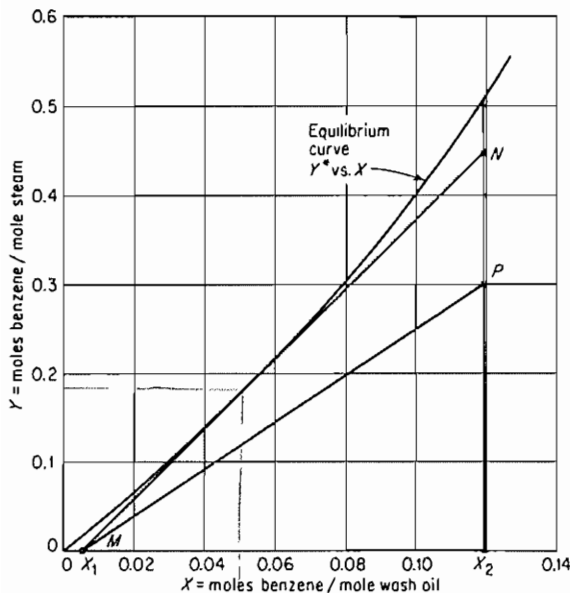


Figure 8.11 Solution to Illustration 8.2, stripping.

COUNTERCURRENT MULTISTAGE OPERATION; ONE COMPONENT TRANSFERRED

Tray towers and similar devices bring about stepwise contact of the liquid and gas and are therefore countercurrent multistage cascades. On each tray of a sieve-tray tower, for example, the gas and liquid are brought into intimate contact and separated, somewhat in the manner of Fig. 5.14, and the tray thus constitutes a stage. Few of the tray devices described in Chap. 6 actually provide the parallel flow on each tray as shown in Fig. 5.14. Nevertheless it is convenient to use the latter as an arbitrary standard for design and for measurement of performance of actual trays regardless of their method of operation. For this purpose a *theoretical*, or *ideal*, tray is defined as one where the average composition of all the gas leaving the tray is in equilibrium with the average composition of all the liquid leaving the tray.

The number of ideal trays required to bring about a given change in composition of the liquid or the gas, for either absorbers or strippers, can then be determined graphically in the manner of Fig. 5.15. This is illustrated for an absorber in Fig. 8.12, where the liquid and gas compositions corresponding to each tray are marked on the operating diagram. Ideal tray 1, for example, brings about a change in liquid composition from X_0 to X_1 and of gas composition

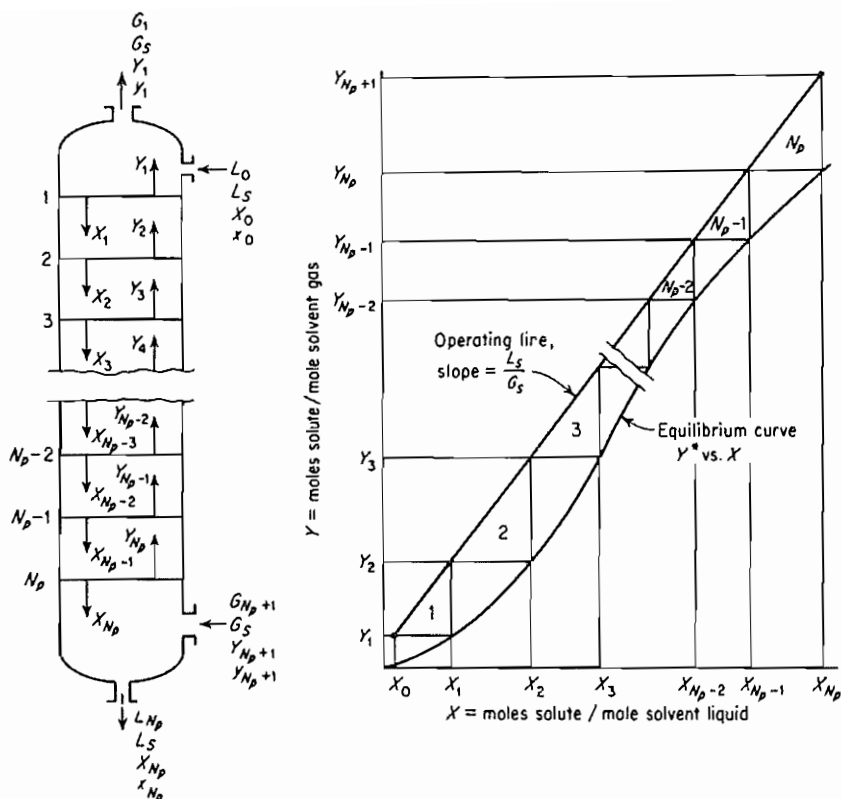


Figure 8.12 Tray absorber.

from Y_2 to Y_1 . The step marked 1 on the operating diagram therefore represents this ideal tray. The nearer the operating line to the equilibrium curve, the more steps will be required, and should the two curves touch at any point corresponding to a minimum L_s/G_s ratio, the number of steps will be infinite. The steps can equally be constructed on diagrams plotted in terms of any concentration units, such as mole fractions or partial pressures. The construction for strippers is the same, except, of course, that the operating line lies below the equilibrium curve.

It is usually convenient, for tray towers, to define the flow rates L and G simply as mol/h, rather than to base them on unit tower cross section.

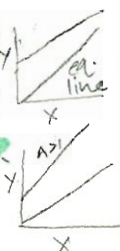
Dilute Gas Mixtures

Where both operating line and equilibrium curve can be considered straight, the number of ideal trays can be determined without recourse to graphical methods.

This will frequently be the case for relatively dilute gas and liquid mixtures. Henry's law [Eq. (8.2)] often applies to dilute solutions, for example. If the quantity of gas absorbed is small, the total flow of liquid entering and leaving the absorber remains substantially constant, $L_0 \approx L_N \approx L$ total mol/(area) (time), and similarly the total flow of gas is substantially constant at G total mol/(area) (time). An operating line plotted in terms of mole fractions will then be substantially straight. For such cases, the Kremser equations (5.50) to (5.57) and Fig. 5.16 apply.† Small variations in A from one end of the tower to the other due to changing L/G as a result of absorption or stripping or to change in gas solubility with concentration or temperature can be roughly allowed for by using the geometric average of the values of A at top and bottom [17]. For large variations, either more elaborate corrections [13, 20, 33] for A , graphical computations, or tray-to-tray numerical calculations as developed below must be used.

The Absorption Factor A

The absorption factor $A = L/mG$ is the ratio of the slope of the operating line to that of the equilibrium curve. For values of A less than unity, corresponding to convergence of the operating line and equilibrium curve for the lower end of the absorber, Fig. 5.16 indicates clearly that the fractional absorption of solute is definitely limited, even for infinite theoretical trays. On the other hand, for values of A greater than unity, any degree of absorption is possible if sufficient trays are provided. For a fixed degree of absorption from a fixed amount of gas, as A increases beyond unity, the absorbed solute is dissolved in more and more liquid and becomes therefore less valuable. At the same time, the number of trays decreases, so that the equipment cost decreases. From these opposing cost tendencies it follows that in all such cases there will be a value of A , or of L/G , for which the most economical absorption results. This should be obtained generally by computing the total costs for several values of A and observing the minimum. As a rule of thumb for purposes of rapid estimates, it has been



† These become, in terms of mole fractions,

$$\text{Absorption: } \frac{y_{N_p+1} - y_1}{y_{N_p+1} - mx_0} = \frac{A^{N_p+1} - A}{A^{N_p+1} - 1} \quad (5.54a)$$

$$N_p = \frac{\log \left[\frac{y_{N_p+1} - mx_0}{y_1 - mx_0} \left(1 - \frac{1}{A} \right) + \frac{1}{A} \right]}{\log A} \quad (5.55a)$$

$$\text{Stripping: } \frac{x_0 - x_{N_p}}{x_0 - y_{N_p+1}/m} = \frac{S^{N_p+1} - S}{S^{N_p+1} - 1} \quad (5.50a)$$

$$N_p = \frac{\log \left[\frac{x_0 - y_{N_p+1}/m}{x_{N_p} - y_{N_p+1}/m} \left(1 - \frac{1}{S} \right) + \frac{1}{S} \right]}{\log S} \quad (5.51a)$$

where $A = L/mG$, and $S = mG/L$. if $A < 1 \Rightarrow$ absorption limited

1/G - slope of operating line

frequently found [8] that the most economical A will be in the range from 1.25 to 2.0.

The reciprocal of the absorption factor is called the stripping factor S .

Illustration 8.3 Determine the number of theoretical trays required for the absorber and the stripper of Illustration 8.2.

SOLUTION

Absorber Since the tower will be a tray device, the following notation changes will be made (compare Figs. 8.4 and 8.12):

L_1 is changed to L_{N_p}	G_1 is changed to G_{N_p+1}	x_1 is changed to x_{N_p}
L_2	G_2	x_2
X_1	Y_1	y_1
X_2	Y_2	y_2

The operating diagram is established in Illustration 8.2 and replotted in Fig. 8.13, where the theoretical trays are stepped off. Between 7 and 8 (approximately 7.6) theoretical trays are

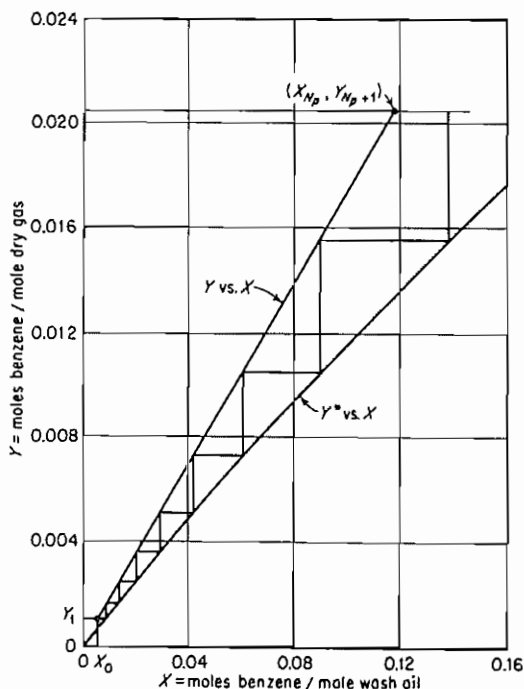


Figure 8.13 Illustration 8.3, the absorber.

required. Alternatively, the number of theoretical trays can be computed analytically.

$$\begin{aligned}
 y_{N_p+1} &= 0.02 & y_1 &= \frac{0.00102}{1 + 0.00102} = 0.00102 \\
 x_0 &= 0.005 & m &= y^*/x = 0.125 \\
 L_{N_p} &= L_S(1 + X_{N_p}) = (1.787 \times 10^{-3})(1 + 0.1190) = 2.00 \times 10^{-3} \text{ kmol/s} \\
 A_{N_p} &= \frac{L_{N_p}}{mG_{N_p}} \approx \frac{L_{N_p}}{mG_{N_p+1}} = \frac{2.00 \times 10^{-3}}{0.125 (0.01075)} = 1.488 \\
 L_0 &= L_S(1 + X_0) = (1.787 \times 10^{-3})(1 + 0.00503) = 1.796 \times 10^{-3} \text{ kmol/s} \\
 G_1 &= G_S(1 + Y_1) = 0.01051 (1 + 0.00102) = 0.01052 \text{ kmol/s} \\
 A_1 &= \frac{L_1}{mG_1} \approx \frac{L_0}{mG_1} = \frac{1.796 \times 10^{-3}}{0.125 (0.01052)} = 1.366 \\
 A &= [1.488 (1.366)]^{0.5} = 1.424 \\
 \frac{y_1 - mx_0}{y_{N_p+1} - mx_0} &= \frac{0.00102 - 0.125 (0.005)}{0.02 - 0.125 (0.005)} = 0.0204
 \end{aligned}$$

From Fig. 5.16 or Eq. (5.55) $N_p = 7.7$ equilibrium stages.

Stripper The trays were determined graphically in the same manner as for the absorber and found to be 6.7. Figure 5.16 for this case gave 6.0 trays owing to the relative nonconstancy of the stripping factor, $1/A_{N_p} = S_{N_p} = 1.197$, $1/A_1 = S_1 = 1.561$. The graphical method should be used.

Nonisothermal Operation

Many absorbers and strippers deal with dilute gas mixtures and liquids, and it is frequently satisfactory in these cases to assume that the operation is isothermal. But actually absorption operations are usually exothermic, and when large quantities of solute gas are absorbed to form concentrated solutions, the temperature effects cannot be ignored. If by absorption the temperature of the liquid is raised to a considerable extent, the equilibrium solubility of the solute will be appreciably reduced and the capacity of the absorber decreased (or else much larger flow rates of liquid will be required). If the heat evolved is excessive, cooling coils can be installed in the absorber or the liquid can be removed at intervals, cooled, and returned to the absorber. For stripping, an endothermic action, the temperature tends to fall.

Consider the tray tower of Fig. 8.14. If Q_T is the heat removed per unit time from the entire tower by any means whatsoever, an enthalpy balance for the entire tower is

$$L_0 H_{L0} + G_{N_p+1} H_{G, N_p+1} = L_{N_p} H_{L, N_p} + G_1 H_{G1} + Q_T \quad (8.11)$$

where H represents in each case the molal enthalpy of the stream at its particular concentration and condition. It is convenient to refer all enthalpies to the condition of pure liquid solvent, pure diluent (or solvent) gas, and pure solute at some base temperature t_0 , with each substance assigned zero enthalpy

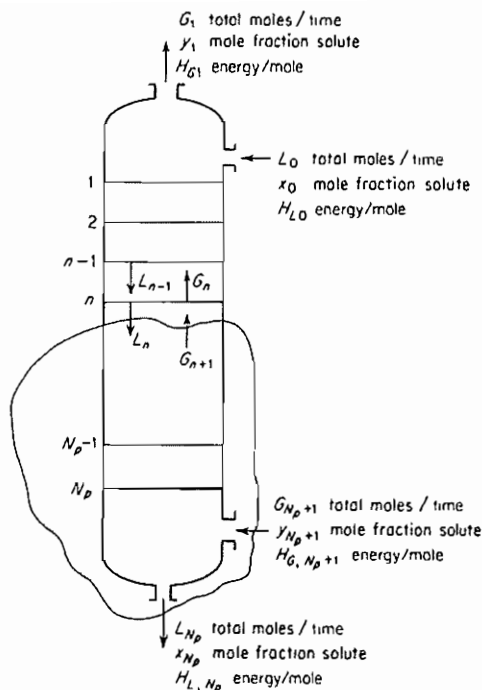


Figure 8.14 Nonisothermal operation.

for its normal state of aggregation at t_0 and 1 atm pressure. Thus, the molal enthalpy of a liquid solution, temperature t_L , composition x mole fraction solute, can be obtained either from Eq. (8.4) or, depending on the nature of the data available, from

$$H_L = C_L(t_L - t_0) + \Delta H_S \quad (8.12)$$

where the first term on the right represents the sensible heat and the second the molal enthalpy of mixing, or integral heat of solution, at the prevailing concentration and at the base temperature t_0 , *per mole of solution*. If heat is evolved on mixing, ΔH_S will be a negative quantity. If the absorbed solute is a gas at t_0 , 1 std atm, the gas enthalpy will include only sensible heat. If the absorbed solute is a liquid at the reference conditions, as in the case of many vapors, the enthalpy of the gas stream must also include the latent heat of vaporization of the solute vapor (see Chap. 7). For ideal solutions, ΔH_S for mixing liquids is zero, and the enthalpy of the solution is the sum of the enthalpies of the separate, unmixed constituents. If the ideal liquid solution is formed from a gaseous solute, the heat evolved is the latent heat of condensation of the absorbed solute.

For *adiabatic* operation, Q_T of Eq. (8.11) is zero and the temperature of the streams leaving an absorber will generally be higher than the entering temperatures owing to the heat of solution. The rise in temperature causes a decrease in solute solubility, which in turn results in a larger minimum L/G and a larger number of trays than for isothermal absorption. The design of such absorbers may be done numerically, calculating tray by tray from the bottom to the top. The principle of an ideal tray, that the effluent streams from the tray are in equilibrium both with respect to composition and temperature, is utilized for each tray. Thus, total and solute balances up to tray n , as shown by the envelope, Fig. 8.14, are

$$L_n + G_{N_p+1} = L_{N_p} + G_{n+1} \quad (8.13)$$

$$L_n x_n + G_{N_p+1} y_{N_p+1} = L_{N_p} x_{N_p} + G_{n+1} y_{n+1} \quad (8.14)$$

from which L_n and x_n are computed. An enthalpy balance is

$$L_n H_{L,n} + G_{N_p+1} H_{G,N_p+1} = L_{N_p} H_{L,N_p} + G_{n+1} H_{G,n+1} \quad (8.15)$$

from which the temperature of stream L_n can be obtained. Stream G_n is then at the same temperature as L_n and in composition equilibrium with it. Equations (8.13) to (8.15) are then applied to tray $n-1$, and so forth. To get started, since usually only the temperatures of the entering streams L_0 and G_{N_p+1} are known, it is usually necessary to estimate the temperature t_1 of the gas G_1 (which is the same as the top tray temperature), and use Eq. (8.11) to compute the temperature of the liquid leaving at the bottom of the tower. The estimate is checked when the calculations reach the top tray, and if necessary the entire computation is repeated. The method is best illustrated by an example.

Illustration 8.4 One kilomole per unit time of a gas consisting of 75% methane, CH_4 , and 25% n -pentane vapor, $n\text{-C}_5\text{H}_{12}$, 27°C , 1 std atm, is to be scrubbed with 2 kmol/unit time of a nonvolatile paraffin oil, mol wt 200, heat capacity $1.884 \text{ kJ/kg} \cdot \text{K}$, entering the absorber free of pentane at 35°C . Compute the number of ideal trays for adiabatic absorption of 98% of the pentane. Neglect solubility of CH_4 in the oil, and assume operation to be at 1 std atm (neglect pressure drop for flow through the trays). The pentane forms ideal solutions with the paraffin oil.

SOLUTION Heat capacities over the range of temperatures to be encountered are

$$\text{CH}_4 = 35.59 \text{ kJ/kmol} \cdot \text{K} \quad n\text{-C}_5\text{H}_{12} = \begin{cases} 119.75 \text{ kJ/kmol} \cdot \text{K} & \text{as vapor} \\ 177.53 \text{ kJ/kmol} \cdot \text{K} & \text{as liquid} \end{cases}$$

The latent heat of vaporization of $n\text{-C}_5\text{H}_{12}$ at $0^\circ\text{C} = 27\,820 \text{ kJ/kmol}$. Use a base temperature $t_0 = 0^\circ\text{C}$. Enthalpies referred to 0°C , liquid pentane, liquid paraffin oil, and gaseous methane, are then

$$\begin{aligned} H_L &= (1-x)(1.884)(200)(t_L - 0) + x(177.53)(t_L - 0) \\ &= t_L(376.8 - 199.3x) \text{ kJ/kmol liquid solution} \\ H_G &= (1-y)(35.59)(t_G - 0) + y[119.75(t_G - 0) + 27\,820] \\ &= t_G(35.59 + 84.16y) + 27\,820y \text{ kJ/kmol gas-vapor mixture} \end{aligned}$$

For solutions which follow Raoult's law, Eqs. (8.1) and (8.2) provide

$$y^* = \frac{p_x}{p_i} = mx$$

where p = vapor pressure of n -pentane. Thus the vapor pressure of n -C₅H₁₂ is 400 mmHg (53.32 kN/m²) at 18.5°C, whence $m = p/p_i = 400/760 = 0.530$. Similarly,

$t, ^\circ\text{C}$	20	25	30	35	40	43
m	0.575	0.69	0.81	0.95	1.10	1.25

It is convenient to prepare a graph of these for interpolation.

Basis: unit time. From the given data,

$$G_{N_p+1} = 1 \text{ kmol} \quad y_{N_p+1} = 0.25$$

$$H_{G, N_p+1} = 27[35.59 + 84.16 (0.25)] + 27 \ 820 (0.25) = 8484 \text{ kJ/kmol}$$

$$L_0 = 2 \text{ kmol} \quad x_0 = 0 \quad H_{L0} = 35 (376.8) = 13 \ 188 \text{ kJ/kmol}$$

$$n\text{-C}_5\text{H}_{12} \text{ absorbed} = 0.98 (0.25) = 0.245 \text{ kmol}$$

$$n\text{-C}_5\text{H}_{12} \text{ in } G_1 = 0.25 - 0.245 = 0.005 \text{ kmol}$$

$$G_1 = 0.75 + 0.005 = 0.755 \text{ kmol} \quad \text{Required } y_1 = \frac{0.005}{0.755} = 0.00662$$

$$L_{N_p} = 2 + 0.245 = 2.245 \text{ kmol} \quad x_{N_p} = \frac{0.245}{2.245} = 0.1091$$

Assume $t_1 = 35.6^\circ\text{C}$ (to be checked later).

$$H_{G1} = 35.6[35.59 + 84.16 (0.00662)] + 27 \ 820 (0.00662) = 1471 \text{ kJ/kmol}$$

Eq. (8.11) with $Q_T = 0$:

$$2(13 \ 188) + 8484 = L_{N_p} H_{L, N_p} + 0.755 (1471)$$

$$L_{N_p} H_{L, N_p} = 37 \ 749 \text{ kJ}$$

$$H_{L, N_p} = \frac{37 \ 749}{2.245} = t_{N_p} [376.8 - 199.3 (0.1091)]$$

$$t_{N_p} = 42.3^\circ\text{C}$$

$$m_{N_p} = 1.21 \quad y_{N_p} = m_{N_p} x_{N_p} = 1.21 (0.1091) = 0.1330$$

$$G_{N_p} = \frac{0.75}{1 - 0.1330} = 0.865 \text{ kmol}$$

$$H_{G, N_p} = 42.3[35.59 + 84.16 (0.1330)] + 27 \ 820 (0.1330) = 5679 \text{ kJ/kmol}$$

Eq. (8.13) with $n = N_p - 1$:

$$L_{N_p-1} + G_{N_p+1} = L_{N_p} + G_{N_p}$$

$$L_{N_p-1} + 1 = 2.245 + 0.865$$

$$L_{N_p-1} = 2.110 \text{ kmol}$$

Eq. (8.14) with $n = N_p - 1$:

$$2.110 x_{N_p-1} + 0.25 = 0.245 + 0.865 (0.1330)$$

$$x_{N_p-1} = 0.0521$$

Eq. (8.15) with $n = N_p - 1$:

$$2.110H_{L, N_p-1} + 8484 = 3374 + 0.865(5679)$$

$$H_{L, N_p-1} = 14\,302 \text{ kJ/kmol}$$

$$14\,302 = t_{N_p-1} [376.8 - 199.3(0.0521)]$$

$$t_{N_p-1} = 39.0^\circ\text{C}$$

The computations are continued upward through the tower in this manner until the gas composition falls at least to $y = 0.00662$. The results are:

$n = \text{tray no.}$	$t_n, ^\circ\text{C}$	x_n	y_n
$N_p = 4$	42.3	0.1091	0.1320
$N_p - 1 = 3$	39.0	0.0521	0.0568
$N_p - 2 = 2$	36.8	0.0184	0.01875
$N_p - 3 = 1$	35.5	0.00463	0.00450

Figure 8.15 shows the calculated gas composition and tray temperature plotted against tray number. The required $y_1 = 0.00662$ occurs at about 3.75 ideal trays, and the temperature on the top tray is essentially that assumed. Had this not been so, a new assumed value of t_1 and a new

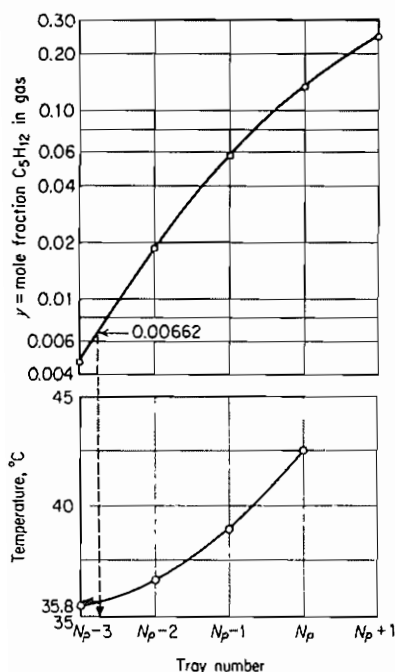


Figure 8.15 Solution to Illustration 8.4.

calculation would have been required. An integral number of trays will require a slightly greater (for $N_p = 3$) or less (for $N_p = 4$) liquid flow, but since a tray efficiency must still be applied to obtain the number of real trays, the nonintegral number is ordinarily accepted.

A graphical solution using an enthalpy-concentration diagram (see Chap. 9) is also possible [43]. When the temperature rise for the liquid is large or for concentrated gases entering and small values of L/G , the rate of convergence of the calculated and assumed t_1 is very slow and the computation is best done on a digital computer. There may be a temperature maximum at some tray other than the bottom one. If the number of trays is fixed, the outlet-gas composition and top-tray temperature must both be found by trial and error. Cases when the carrier gas is absorbed and the solvent evaporates [3, 4] are considered to be multicomponent systems (which see).

Real Trays and Tray Efficiency

Methods for estimating the Murphree tray efficiency corrected for entrainment E_{MGE} for sieve trays are discussed in detail in Chap. 6. For a given absorber or stripper, these permit estimation of the tray efficiency as a function of fluid compositions and temperature as they vary from one end of the tower to the other. Usually it is sufficient to make such computations at only three or four locations and then proceed as in Fig. 8.16. The broken line is drawn between equilibrium curve and operating line at a fractional vertical distance from the operating line equal to the prevailing Murphree gas efficiency. Thus the value of E_{MGE} for the bottom tray is the ratio of the lengths of lines, AB/AC . Since the broken line then represents the real effluent compositions from the trays, it is used instead of the equilibrium curve to complete the tray construction, which now provides the number of real trays.

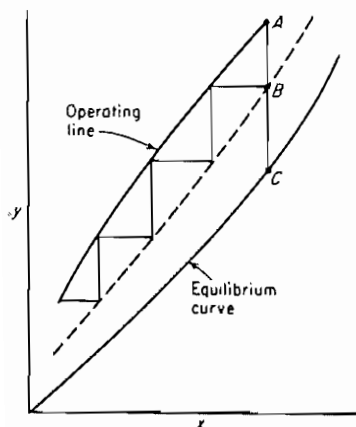


Figure 8.16 Use of Murphree efficiencies for an absorber.

When the Murphree efficiency is constant for all trays, and under conditions such that the operating line and equilibrium curves are straight (Henry's law, isothermal operation, dilute solutions), the overall tray efficiency can be computed and the number of real trays can be determined analytically:

$$E_O = \frac{\text{equilibrium trays}}{\text{real trays}} = \frac{\log[1 + E_{MGE}(1/A - 1)]}{\log(1/A)} \quad (8.16)$$

For rough estimates, Fig. 6.24 is useful for both bubble-cap and sieve trays.

Illustration 8.5 A process for making small amounts of hydrogen by cracking ammonia is being considered, and residual, uncracked ammonia is to be removed from the resulting gas. The gas will consist of H_2 and N_2 in the molar ratio 3:1, containing 3% NH_3 by volume, at a pressure of 2 bars and temperature of $30^\circ C$.

There is available a sieve-tray tower, 0.75 m diameter, containing 14 cross-flow trays at 0.5 m tray spacing. On each tray, the downflow weir is 0.53 m long and extends 60 mm above the tray floor. The perforations are 4.75 mm in diameter, arranged in triangular pitch on 12.5-mm centers, punched in sheet metal 2.0 mm thick. Assume isothermal operation at $30^\circ C$. Estimate the capacity of the tower to remove ammonia from the gas by scrubbing with water.

SOLUTION In this case "capacity" will be taken to mean obtaining a low NH_3 content of the effluent gas at reasonable gas rates.

From the geometry of the tray arrangement, and in the notation of Chap. 6, the following are readily calculated:

$$T = \text{tower diameter} = 0.75 \text{ m} \quad A_t = \text{tower cross section} = 0.4418 \text{ m}^2$$

$$A_d = \text{downspout cross section} = 0.04043 \text{ m}^2 \quad A_n = A_t - A_d = 0.4014 \text{ m}^2$$

$$A_o = \text{perforation area} = 0.0393 \text{ m}^2 \quad Z = \text{distance between downspouts} = 0.5307 \text{ m}$$

$$W = \text{weir length} = 0.53 \text{ m} \quad h_w = \text{weir height} = 0.060 \text{ m}$$

$$t = \text{tray spacing} = 0.5 \text{ m} \quad z = \text{average flow width} = \frac{T + W}{2} = 0.64 \text{ m}$$

Use a weir crest $h_1 = 0.040$ m. Eq. (6.34): $W_{eff} = 0.4354$ m. Eq. (6.32): $q = 1.839$ (0.4354) $(0.04)^{3/2} = 6.406 \times 10^{-3} \text{ m}^3/\text{s}$ liquid flow. *Note:* this is a recommended rate because it produces a liquid depth on the tray of 10 cm.

$$\text{Liquid density} = \rho_L = 996 \text{ kg/m}^3$$

$$\text{Liquid rate} = 6.38 \text{ kg/s} = 0.354 \text{ kmol/s}$$

$$\begin{aligned} \text{Av mol wt gas in} &= 0.03 (17.03) + 0.97 (0.25) (28.02) + 0.97 (0.75) (2.02) \\ &= 8.78 \text{ kg/kmol} \end{aligned}$$

$$\rho_G = \frac{8.78}{22.41} \frac{2.0}{0.986} \frac{273}{273 + 30} = 0.716 \text{ kg/m}^3$$

The flooding gas rate depends upon the liquid rate chosen. With the preceding values, the flooding gas rate is found by simultaneous solution of Eqs. (6.29) and (6.30) by trial and error. For these solutions, $\sigma = 68 \times 10^{-3} \text{ N/m}$ surface tension. As a final trial, use a gas mass rate = 0.893 kg/s.

$$\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = \frac{6.38}{0.893} \left(\frac{0.716}{996} \right)^{0.5} = 0.1916$$

Table 6.2: $\alpha = 0.04893$, $\beta = 0.0302$. Eq. (6.30): $C_F = 0.0834$. Eq. (6.29): $V_F = 3.11 \text{ m/s}$ based on A_n . Therefore flooding gas mass rate = 3.11 (0.4014) (0.716) = 0.893 kg/s (check). Use 80%

of the flooding value. $V = 0.8 (3.11) = 2.49$ m/s based on A_n . Gas mass rate = $0.80 (0.893) = 0.7144$ kg/s = 0.0814 kmol/s.

The methods of Chap. 6 provide (Chap. 6 notation): $V_o = 2.49 (0.4014)/0.0393 = 25.43$ m/s, $l = 0.002$ m, $d_o = 0.00475$ m, $C_o = 1.09 (0.00475/0.002)^{0.25} = 1.353$, $l = 0.002$ m, $\mu_G = 1.13$ kg/m · s, $Re_o = d_o V_o \rho_G / \mu_G = 7654$, $f = 0.0082$, $h_D = 0.0413$ m, $V_a = VA_n/A_a = 3.33$ m/s, $h_L = 0.0216$ m, $h_R = 0.0862$ m, $h_G = 0.1491$ m, $h_2 = 3.84 \times 10^{-3}$ m, $h_3 = 0.1529$ m.

Since $h_W + h_1 + h_3 = 0.2529$ m is essentially equal to $l/2$, flooding will not occur. Flow rates are reasonable. $V/V_F = 0.8$, and $(L'/G') (\rho_G/\rho_L)^{0.5} = 0.239$. Fig. 6.17: E = fractional entrainment = 0.009 .

At the prevailing conditions, the methods of Chap. 2 provide $D_{NH_3(\text{mean})}$ through the N_2-H_2 mixture = $D_G = 2.296 \times 10^{-5}$ m²/s. The gas viscosity is estimated to be 1.122×10^{-5} kg/m · s, whence $Sc_G = \mu_G/\rho_G D_G = 0.683$. The diffusivity of NH_3 in dilute water solutions is 2.421×10^{-9} = D_L at 30°C.

For dilute solutions, NH_3-H_2O follows Henry's law, and at 30°C $m = y^*/x = 0.850$. The absorption factor $A = L/mG = 0.354/0.850 (0.0814) = 5.116$. Note: "Optimum" values of A such as 1.25 to 2.0 cited earlier apply to new designs but not necessarily to existing towers.

$$\begin{array}{ll} \text{Eq. (6.61): } N_{IG} = 1.987 & \text{Eq. (6.64): } \theta_L = 1.145 \text{ s} \\ \text{Eq. (6.62): } N_{IL} = 1.691 & \text{Eq. (6.52): } N_{IOG} = 1.616 \\ \text{Eq. (6.51): } E_{OG} = 0.801 & \text{Eq. (6.63): } D_E = 0.01344 \text{ m}^2/\text{s} \\ \text{Eq. (6.59): } Pe = 18.30 & \text{Eq. (6.58): } \eta = 0.1552 \\ \text{Eq. (6.57): } E_{MG} = 0.86 & \text{Eq. (6.60): } E_{MGE} = 0.853 \end{array}$$

Since the quantity of gas absorbed is relatively small with respect to the total flow and the liquid solutions are dilute, the Murphree efficiency will be taken as constant for all trays. For the dilute solutions encountered here, the operating line in terms of mole-fraction concentrations is essentially straight. Consequently, Eq. (8.16):

$$E_O = \frac{\log[1 + 0.853 (1/5.116 - 1)]}{\log(1/5.116)} = 0.710$$

(Note: Fig. 6.24, with abscissa = 1.306×10^{-5} for these data, shows $E_O \approx 0.7$ for absorption of NH_3 in water, the point plotted as +.) Eq. (8.16): $N_p = 14 (0.710) = 9.94$ theoretical trays. Fig. 5.16 is off-scale. Therefore, the Kremser equation Eq. (5.54a), gives

$$\frac{y_{N_p+1} - y_1}{y_{N_p+1} - mx_0} = \frac{y_{N_p+1} - y_1}{y_{N_p+1}} = \frac{0.03 - y_1}{0.03} = \frac{(5.116)^{10.94} - 5.116}{(5.116)^{10.94} - 1}$$

$$y_1 = 2.17 \times 10^{-9} \text{ mole fraction } NH_3 \text{ in effluent Ans.}$$

CONTINUOUS-CONTACT EQUIPMENT

Countercurrent packed and spray towers operate in a different manner from plate towers in that the fluids are in contact continuously in their path through the tower, rather than intermittently. Thus, in a packed tower the liquid and gas compositions change continuously with height of packing. Every point on an operating line therefore represents conditions found somewhere in the tower, whereas for tray towers, only the isolated points on the operating line corresponding to trays have real significance.

Height Equivalent to an Equilibrium Stage (Theoretical Plate)

A simple method for designing packed towers, introduced many years ago, ignores the differences between stagewise and continuous contact. In this method the number of theoretical trays or plates required for a given change in concentration is computed by the methods of the previous section. This is then multiplied by a quantity, the height equivalent to a theoretical tray or plate (**HETP**) to give the required height of packing to do the same job. The **HETP** must be an experimentally determined quantity characteristic for each packing. Unfortunately it is found that the **HETP** varies, not only with the type and size of the packing but also very strongly with flow rates of each fluid and for every system with concentration as well, so that an enormous amount of experimental data would have to be accumulated to permit utilization of the method. The difficulty lies in the failure to account for the fundamentally different action of tray and packed towers, and the method has now largely been abandoned.

Absorption of One Component

Consider a packed tower of unit cross section, as in Fig. 8.17. The total effective interfacial surface for mass transfer, as a result of spreading of the liquid in a film over the packing, is S per unit tower cross section. This is conveniently described as the product of a specific interfacial surface, surface per volume of

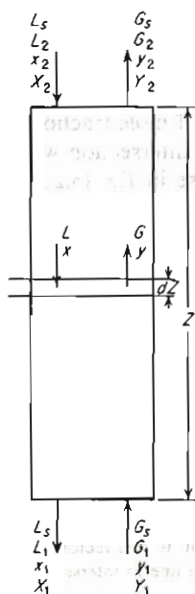


Figure 8.17 Packed tower.

packing, by the packed volume Z volume per unit tower cross section, or height (the quantity a is the a_A of Chap. 6). In the differential volume dZ , the interface surface is

$$dS = a \, dZ \quad (8.17)$$

The quantity of solute A in the gas passing the differential section of the tower under consideration is Gy mol/(area) (time), and the rate of mass transfer is therefore $d(Gy)$ mol A/(differential volume) (time). Since $N_B = 0$ and $N_A/(N_A + N_B) = 1.0$, application of Eq. (5.20) provides

$$N_A = \frac{\text{moles A absorbed}}{(\text{interfacial surface}) (\text{time})} = \frac{d(Gy)}{a \, dZ} = F_G \ln \frac{1 - y_i}{1 - y} \quad (8.18)$$

Both G and y vary from one end of the tower to the other, but G_S , the solvent gas which is essentially insoluble, does not. Therefore,

$$d(Gy) = d\left(\frac{G_S y}{1 - y}\right) = \frac{G_S \, dy}{(1 - y)^2} = \frac{G \, dy}{1 - y} \quad (8.19)$$

Substituting in Eq. (8.18), rearranging, and integrating give

$$Z = \int_0^Z dZ = \int_{y_2}^{y_1} \frac{G \, dy}{F_G a (1 - y) \ln[(1 - y_i)/(1 - y)]} \quad (8.20)$$

The value of y_i can be found by the methods of Chap. 5, using Eq. (5.21) with $N_A/\Sigma N = 1$:

$$\frac{1 - y_i}{1 - y} = \left(\frac{1 - x}{1 - x_i}\right)^{F_L/F_G} = \left(\frac{1 - x}{1 - x_i}\right)^{F_L a / F_G a} \quad (8.21)$$

For any value of (x, y) on the operating curve plotted in terms of mole fractions, a curve of x_i vs. y_i from Eq. (8.21) is plotted to determine the intersection with the equilibrium curve. This provides the local y and y_i for use in Eq. (8.20).† Equation (8.20) can then be integrated graphically after plotting the integrand as ordinate vs. y as abscissa.

However, it is more customary to proceed as follows [6]. Since

$$y - y_i = (1 - y_i) - (1 - y) \quad (8.22)$$

the numerator and denominator of the integral of Eq. (8.20) can be multiplied respectively by the right- and left-hand sides of Eq. (8.22) to provide

$$Z = \int_{y_2}^{y_1} \frac{G(1 - y)_i \, dy}{F_G a (1 - y)(y - y_i)} \quad (8.23)$$

† As demonstrated in Illustration 8.6, for moderately dilute solutions it is satisfactory to determine y_i by a line of slope $-k_x a / k_y a$ drawn from (x, y) on the operating line to intersection with the equilibrium curve, in accordance with Eq. (5.2).

where $(1 - y)_{iM}$ is the logarithmic mean of $1 - y_i$ and $1 - y$. When we define a height of a gas transfer unit H_{IG} as

$$H_{IG} = \frac{G}{F_G a} = \frac{G}{k_y a (1 - y)_{iM}} = \frac{G}{k_G a p_i (1 - y)_{iM}} \quad (8.24)$$

Eq. (8.23) becomes

$$Z = \int_{y_2}^{y_1} H_{IG} \frac{(1 - y)_{iM} dy}{(1 - y)(y - y_i)} \approx H_{IG} \int_{y_2}^{y_1} \frac{(1 - y)_{iM} dy}{(1 - y)(y - y_i)} = H_{IG} N_{IG} \quad (8.25)$$

Here advantage is taken of the fact that the ratio $G/F_G a = H_{IG}$ is very much more constant than either G or $F_G a$, and in many cases may be considered constant within the accuracy of the available data. In the integral of Eq. (8.25) containing only the y terms, if we disregard the ratio $(1 - y)_{iM}/(1 - y)$, the remainder is seen to be the number of times the average $y - y_i$ divides into the change of gas concentration $y_1 - y_2$. As in Chap. 7, this is a measure of the difficulty of the absorption, and the integral is called the *number of gas transfer units* N_{IG} . H_{IG} is then the packing height providing one gas transfer unit. Equation (8.25) can be further simplified by substituting the arithmetic average for the logarithmic average $(1 - y)_{iM}$ [9]

$$(1 - y)_{iM} = \frac{(1 - y_i) - (1 - y)}{\ln[(1 - y_i)/(1 - y)]} \approx \frac{(1 - y_i) + (1 - y)}{2} \quad (8.26)$$

which involves little error. N_{IG} then becomes

$$N_{IG} = \int_{y_2}^{y_1} \left(\frac{dy}{y - y_i} \right) + \frac{1}{2} \ln \frac{1 - y_2}{1 - y_1} \quad (8.27)$$

which makes for simpler graphical integration. A plot of $1/(y - y_i)$ vs. y for the graphical integration of Eq. (8.27) often covers awkwardly large ranges of the ordinate. This can be avoided [37] by replacing dy by its equal $y d \ln y$, so that

$$N_{IG} = 2.303 \int_{\log y_2}^{\log y_1} \frac{y}{y - y_i} d \log y + 1.152 \log \frac{1 - y_2}{1 - y_1} \quad (8.28)$$

For dilute solutions, the second term on the right of Eqs. (8.27) and (8.28) is negligible, $F_G a \approx k_y a$, and y_i can be obtained by plotting a line of slope $-k_x a/k_y a$ from points (x, y) on the operating line to intersection with the equilibrium curve.

The above relationships all have their counterparts in terms of liquid concentrations, derived in exactly the same way

$$Z = \int_{x_2}^{x_1} \frac{L dx}{F_L a (1-x) \ln [(1-x)/(1-x_i)]} = \int_{x_2}^{x_1} \frac{L(1-x)_{iM} dx}{F_L a (1-x)(x_i-x)} \quad (8.29)$$

$$Z = \int_{x_2}^{x_1} H_{iL} \frac{(1-x)_{iM} dx}{(1-x)(x_i-x)} \approx H_{iL} \int_{x_2}^{x_1} \frac{(1-x)_{iM} dx}{(1-x)(x_i-x)} = H_{iL} N_{iL} \quad (8.30)$$

$$H_{iL} = \frac{L}{F_L a} = \frac{L}{k_x a (1-x)_{iM}} = \frac{L}{k_L a c (1-x)_{iM}} \quad (8.31)$$

$$N_{iL} = \int_{x_2}^{x_1} \frac{dx}{x_i - x} + \frac{1}{2} \ln \frac{1-x_1}{1-x_2} \quad (8.32)$$

where H_{iL} = height of liquid transfer unit

N_{iL} = number of liquid transfer units

$(1-x)_{iM}$ = logarithmic mean of $1-x$ and $1-x_i$
Either set of equations leads to the same value of Z .

Strippers The same relationships apply as for absorption. The *driving forces* $y-y_i$ and x_i-x which appear in the above equations are then negative, but since for strippers $x_2 > x_1$ and $y_2 > y_1$, the result is a positive Z as before.

Illustration 8.6 The absorber of Illustrations 8.2 and 8.3 is to be a packed tower, 470 mm diameter, filled with 38-mm (1.5-in) Berl saddles. The circumstances are as follows:

Gas Benzene content: in, $y_1 = 0.02$ mole fraction, $Y_1 = 0.0204$ mol/mol dry gas.

Out:

$$y_2 = 0.00102 \quad Y_2 = 0.00102$$

Nonabsorbed gas:

Av mol wt = 11.0

Rate in = $0.250 \text{ m}^3/\text{s} = 0.01075 \text{ kmol/s}$ 0.01051 kmol/s nonbenzene

Temperature = 26°C pressure $p_1 = 1.07 \times 10^5 \text{ N/m}^2$ (803 mmHg)

Viscosity = $10^{-3} \text{ kg/m} \cdot \text{s}$ (0.010 cP) $D_{AG} = 1.30 \times 10^{-5} \text{ m}^2/\text{s}$

Liquid Benzene content in, $x_2 = 0.005$ mol fraction, $X_2 = 0.00503$ mol benzene/mol oil.

Out:

$$x_1 = 0.1063 \quad X_1 = 0.1190$$

Benzene-free oil:

Mol wt = 270 viscosity = $2 \times 10^{-3} \text{ kg/m} \cdot \text{s}$ (2.0 cp)

Density = 840 kg/m^3 rate = $1.787 \times 10^{-3} \text{ kmol/s}$

Temperature = 26°C $D_{A,L} = 4.77 \times 10^{-10} \text{ m}^2/\text{s}$

Surface tension = 0.03 N/m^2 $m = y^*/x = 0.1250$

Compute the depth of packing required.

SOLUTION To plot the operating line, use Eq. (8.9) to calculate X and Y values (or read from Fig. 8.10) and convert to $x = X/(1 + X)$, $y = Y/(1 + Y)$. Thus,

$$0.01051 (0.0204 - Y) = (1.787 \times 10^{-3}) (0.1190 - X)$$

X	x	Y	y
0.00503	0.00502	0.00102	0.00102
0.02	0.01961	0.00357	0.00356
0.04	0.0385	0.00697	0.00692
0.06	0.0566	0.01036	0.01025
0.08	0.0741	0.01376	0.01356
0.10	0.0909	0.01714	0.01685
0.1190	0.1063	0.0204	0.0200

Values of y and x (the operating line) are plotted along with the equilibrium line, $y = 0.1250x$, in Fig. 8.18. Although the curvature in this case is not great, the operating line is not straight on mole-fraction coordinates.

The cross-sectional area of the absorber $= \pi(0.47)^2/4 = 0.1746 \text{ m}^2$. At the bottom,

$$L' = \frac{(1.787 \times 10^{-3})(260) + 0.1190(1.787 \times 10^{-3})(78)}{0.1746} = 2.75 \text{ kg/m}^2 \cdot \text{s}$$

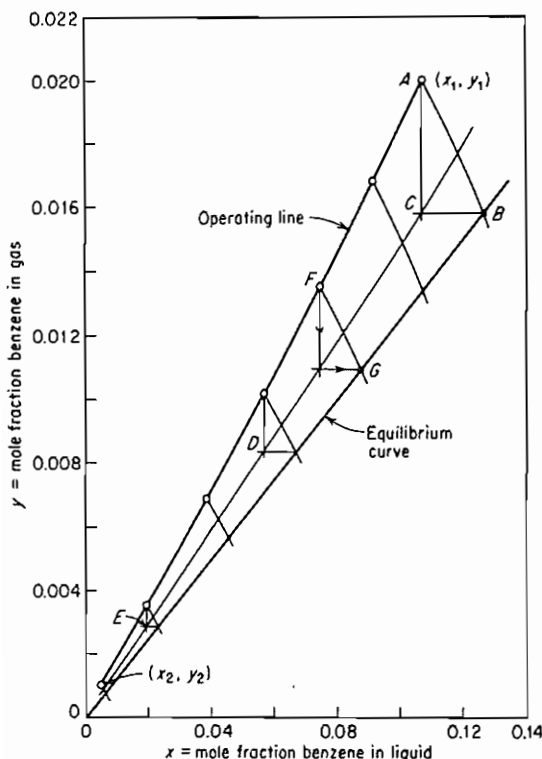


Figure 8.18 Solution to Illustration 8.6.

Similarly at the top, $L' = 2.66 \text{ kg/m}^2 \cdot \text{s}$, $Av L' = 2.71 \text{ kg/m}^2 \cdot \text{s}$. In similar fashion for the gas, at the bottom $G' = 0.761 \text{ kg/m}^2 \cdot \text{s}$; at the top $G' = 0.670 \text{ kg/m}^2 \cdot \text{s}$, $Av G' = 0.716 \text{ kg/m}^2 \cdot \text{s}$.

The flow quantities change so little from one end of the tower to the other that the average values can be used to compute the mass-transfer coefficients, which can then be taken as constant. The circumstances are precisely those of Illustration 6.6, where it was found that

$$F_G a = 0.0719 \text{ kmol/m}^3 \cdot \text{s} \quad F_L a = 0.01377 \text{ kmol/m}^3 \cdot \text{s}$$

Interface compositions corresponding to points on the operating line of Fig. 8.18 are determined with Eq. (8.21). For example, at $(x = 0.1063, y = 0.0200)$ on the operating line at point *A*, Fig. 8.18, Eq. (8.21) becomes

$$\frac{1 - y_i}{1 - 0.0200} = \left(\frac{1 - 0.1063}{1 - x_i} \right)^{0.01377/0.0719}$$

This is plotted (y_i vs. x_i) as curve *AB* on Fig. 8.18 to intersect the equilibrium curve at *B*, where $y_i = 0.01580$.

Note: In this case, since the concentrations are low, an essentially correct value of y_i can be obtained by the following less tedious calculation. Table 3.1 shows

$$k_y a = \frac{F_G a}{p_{B,M}/p_i} = \frac{F_G a}{(1 - y)_{i,M}} \quad k_x a = \frac{F_L a}{(1 - x)_{i,M}}$$

At point *A* on the operating line, $1 - y = 1 - 0.02 = 0.98$, $1 - x = 1 - 0.1063 = 0.8937$. As an approximation, these can be taken as $(1 - y)_{i,M}$ and $(1 - x)_{i,M}$, respectively. Then

$$k_y a = \frac{0.0719}{0.98} = 0.0734 \text{ kmol/m}^3 \cdot \text{s} \cdot (\text{mole fraction})$$

$$k_x a = \frac{0.01377}{0.8937} = 0.01541 \text{ kmol/m}^3 \cdot \text{s} \cdot (\text{mole fraction})$$

In accordance with Eq. (5.2), $-k_x a/k_y a = -0.01541/0.0734 = -0.210$, and if a straight line of this slope is drawn from point *A*, it intersects the equilibrium curve at $y_i = 0.01585$. Curve *AB*, in other words, is nearly a straight line of slope -0.210 . The error by this method becomes even less at lower concentrations.

Where the F 's must be used, it is sufficient to proceed as follows. In Fig. 8.18, draw the vertical line *AC* to intersect with the horizontal line *CB*, thus locating the intersection at *C*. Repeat the calculation at two other locations, such as at *D* and *E*, and draw the curve *EDC*. Interfacial concentrations G corresponding to any point *F* on the operating line can then be obtained by the construction shown on the figure leading to the point *G*.

In a similar manner the tabulated values of y_i were determined

y	y_i	$\frac{y}{y - y_i}$	$\log y$
0.00102	0.000784	4.32	$-2.9999 = \log y_2$
0.00356	0.00285	5.02	-2.4486
0.00692	0.00562	5.39	-2.1599
0.01025	0.00830	5.26	-1.9893
0.01356	0.01090	5.10	-1.8687
0.01685	0.01337	4.84	-1.7734
0.0200	0.01580	4.76	$-1.6990 = \log y_1$

Since the mass-transfer coefficients are essentially constant, Eq. (8.28) will be used to determine N_{iG} . From the data of the above tabulation, the curve for graphical integration of Eq. (8.28) is plotted in Fig. 8.19. The area under the curve (to the zero ordinate) is 6.556. Then

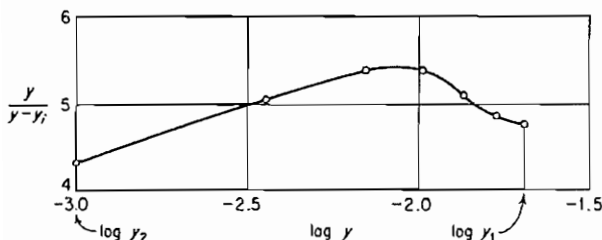


Figure 8.19 Graphical integration for Illustration 8.6.

by Eq. (8.28),

$$N_{IG} = 2.303 (6.556) + 1.152 \log \frac{1 - 0.00102}{1 - 0.0200} = 15.11$$

The average gas rate is

$$G = \frac{0.01075 + 0.01051 / (1 - 0.00102)}{2(0.1746)} = 0.0609 \text{ kmol/m}^2 \cdot \text{s}$$

Eq. (8.24):

$$H_{IG} = \frac{G}{F_{Ga}} = \frac{0.0609}{0.0719} = 0.847 \text{ m}$$

$$Z = H_{IG} N_{IG} = 0.847 (15.11) = 12.8 \text{ m packed depth} \quad \text{Ans}$$

Note: The circumstances are such that, for this problem, the simpler computations of Illustration 8.7 are entirely adequate. The method used above, however, is suitable for solutions of any concentration and where the equilibrium curve is not straight. If flow rates vary appreciably from top to bottom of the tower, so that mass-transfer coefficients vary, this is easily allowed for in determining y_i . In such cases, Eq. (8.20) would be used to compute Z .

Overall Coefficients and Transfer Units ($m = dy_i/dx_i = \text{const}$)

For cases where the equilibrium distribution curve is straight and the ratio of mass-transfer coefficients is constant, it was shown in Chap. 5 that overall mass-transfer coefficients are convenient. The expressions for the height of packing can then be written

$$Z = N_{IOG} H_{IOG} \quad \left. \begin{array}{l} (1-y)_{im} \\ \text{wg mean of } 1-y \text{ and } 1-y_i \end{array} \right\} \text{point of HETP} \quad (8.33)$$

$$N_{IOG} = \int_{y_2}^{y_1} \frac{(1-y)_{*M} dy}{(1-y)(y-y_*)} \quad (8.34)$$

$$N_{IOG} = \int_{y_2}^{y_1} \frac{dy}{y-y_*} + \frac{1}{2} \ln \frac{1-y_2}{1-y_1} \quad (8.35)$$

$$N_{IOG} = \int_{Y_2}^{Y_1} \frac{dY}{Y-Y_*} + \frac{1}{2} \ln \frac{1+Y_2}{1+Y_1} \quad (8.36)$$

$$H_{IOG} = \frac{G}{F_{OG} a} = \frac{G}{K_y a (1-y)_{*M}} = \frac{G}{K_G a p_i (1-y)_{*M}} \quad (8.37)$$

Here y^* (or Y^*) is the solute concentration in the gas corresponding to equilibrium with the bulk liquid concentration x (or X), so that $y - y^*$ (or $Y - Y^*$) is simply the vertical distance between operating line and equilibrium curve. $(1 - y)_{*M}$ is the logarithmic average of $1 - y$ and $1 - y^*$. These methods are convenient since interfacial concentrations need not be obtained, and Eq. (8.36) is especially convenient since the operating line on X, Y coordinates is straight. N_{IOG} is the number of overall gas transfer units, H_{IOG} the height of an overall gas transfer unit.

Equations (8.33) to (8.37) are usually used when the principal mass-transfer resistance resides within the gas. For cases where the principal mass-transfer resistance lies within the liquid, it is more convenient to use

$$Z = N_{IOL} H_{IOL} \quad (8.38)$$

$$N_{IOL} = \int_{x_2}^{x_1} \frac{(1 - x)_{*M} dx}{(1 - x)(x^* - x)} \quad (8.39)$$

$$N_{IOL} = \int_{x_2}^{x_1} \frac{dx}{x^* - x} + \frac{1}{2} \ln \frac{1 - x_1}{1 - x_2} \quad (8.40)$$

$$N_{IOL} = \int_{X_2}^{X_1} \frac{dX}{X^* - X} + \frac{1}{2} \ln \frac{1 + X_1}{1 + X_2} \quad (8.41)$$

$$H_{IOL} = \frac{L}{F_{OL} a} = \frac{L}{K_x a (1 - x)_{*M}} = \frac{L}{K_L a c (1 - x)_{*M}} \quad (8.42)$$

overall liq. transfer unit $\uparrow$ *log mean of 1-y and (1-y*)*

Dilute Solutions

The computation of the number of transfer units for dilute mixtures can be greatly simplified. When the gas mixture is dilute, for example, the second term of the definition of N_{IOG} [Eq. (8.35)] becomes entirely negligible and can be discarded

$$N_{IOG} = \int_{y_2}^{y_1} \frac{dy}{y - y^*} \quad (8.43)$$

If the equilibrium curve in terms of mole fractions is linear over the range of compositions x_1 to x_2 , then

$$y^* = mx + r \quad (8.44)$$

If the solutions are dilute, the operating line can be considered as a straight line as well

$$y = \frac{L}{G}(x - x_2) + y_2 \quad (8.45)$$

so that the driving force $y - y^*$ is then linear in x

$$y - y^* = qx + s \quad (8.46)$$

where q , r , and s are constants. Therefore Eq. (8.43) becomes

$$N_{IOG} = \frac{L}{G} \int_{x_2}^{x_1} \frac{dx}{qx + s} = \frac{L}{Gq} \ln \frac{(y - y^*)_1}{(y - y^*)_2} = \frac{y_1 - y_2}{\ln[(y - y^*)_1 / (y - y^*)_2]} \quad (8.47)$$

$$N_{IOG} = \frac{y_1 - y_2}{(y - y^*)_M} \quad \text{total change} / \text{avg. driving force} \quad (8.48)$$

where $(y - y^*)_M$ is the logarithmic average of the concentration differences at the ends of the tower. This equation is sometimes used in the familiar rate form obtained by substituting the definition of N_{IOG}

$$G(y_1 - y_2) = K_G a Z p_i (y - y^*)_M \quad (8.49)$$

Dilute Solutions, Henry's Law

If Henry's law applies [r of Eq. (8.44) = 0], by elimination of x between Eqs. (8.44) and (8.45) and substitution of y^* in Eq. (8.43) there results for absorbers [9]

$$N_{IOG} = \frac{\ln \left[\frac{y_1 - mx_2}{y_2 - mx_2} \left(1 - \frac{1}{A} \right) + \frac{1}{A} \right]}{1 - 1/A} \quad (8.50)$$

where $A = L/mG$, as before. For strippers, the corresponding expression in terms of N_{IOL} is similar

$$N_{IOL} = \frac{\ln \left[\frac{x_2 - y_1/m}{x_1 - y_1/m} (1 - A) + A \right]}{1 - A} \quad (8.51)$$

These are shown in convenient graphical form in Fig. 8.20.

Graphical Construction for Transfer Units [2]

Equation (8.48) demonstrates that one overall gas transfer unit results when the change in gas composition equals the average overall driving force causing the change. Consider now the operating diagram of Fig. 8.21, where line KB has been drawn so as to be everywhere vertically halfway between the operating line and equilibrium curve. The step CFD , which corresponds to one transfer unit, has been constructed by drawing the horizontal line CEF so that line $CE = EF$ and continuing vertically to D . $y_G - y_H$ can be considered as the average driving force for the change in gas composition $y_D - y_F$ corresponding to this step. Since $GE = EH$, and if the operating line can be considered straight, $DF = 2(GE) = GH$, the step CFD corresponds to one transfer unit. In similar fashion the other transfer units were stepped off ($JK = KL$, etc.). For computing N_{IOL} ,

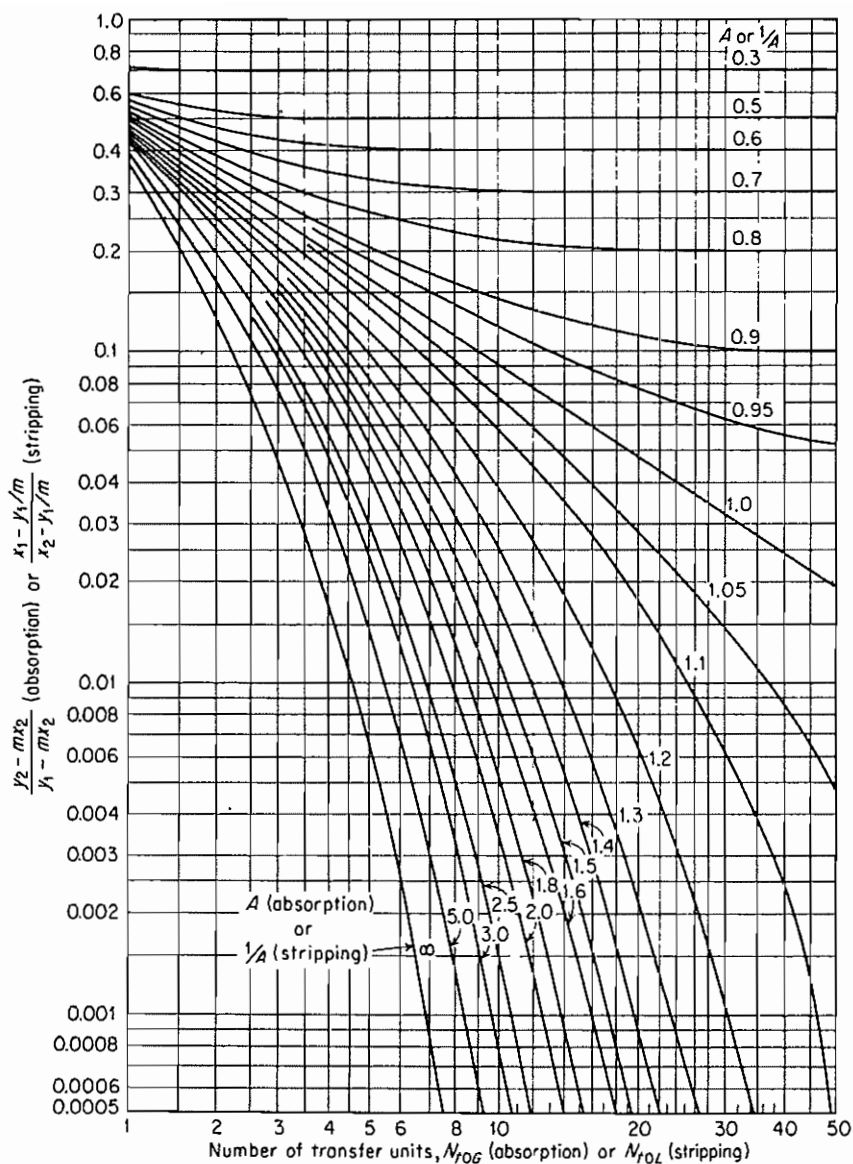


Figure 8.20 Number of transfer units for absorbers or strippers with constant absorption or stripping factor.

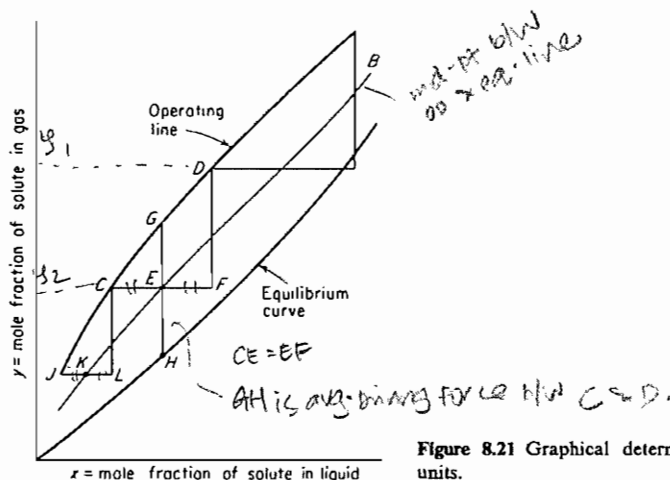


Figure 8.21 Graphical determination of transfer units.

the line KB would be drawn horizontally halfway between equilibrium curve and operating line and would bisect the vertical portions of the steps.

Overall Heights of Transfer Units

When overall numbers of transfer units are appropriate, the overall heights of transfer units can be synthesized from those for the individual phases through the relationships developed in Chap. 5. Thus, Eq. (5.25a), with $m' = m'' = m = \text{const.}$ can be written

$$\frac{G}{F_{OG}a} = \frac{G(1-y)_{iM}}{F_G a (1-y)_{*M}} + \frac{mG}{L} \frac{L}{F_L a} \frac{(1-x)_{iM}}{(1-y)_{*M}} \quad (8.52)$$

whence, by definition of the heights of transfer units,

$$H_{IOG} = H_{IG} \frac{(1-y)_{iM}}{(1-y)_{*M}} + \frac{mG}{L} H_{IL} \frac{(1-x)_{iM}}{(1-y)_{*M}} \quad (8.53)$$

If the mass-transfer resistance is essentially all in the gas, $y_i \approx y^*$, and

$$H_{IOG} = H_{IG} + \frac{mG}{L} H_{IL} \frac{(1-x)_{iM}}{(1-y)_{*M}} \quad (8.54)$$

and, for dilute solutions, the concentration ratio of the last equation can be dropped. In similar fashion, Eq. (5.26a) yields

$$H_{IOL} = H_{IL} \frac{(1-x)_{iM}}{(1-x)_{*M}} + \frac{L}{mG} H_{IG} \frac{(1-y)_{iM}}{(1-x)_{*M}} \quad (8.55)$$

and if the mass-transfer resistance is essentially all in the liquid,

$$H_{IOL} = H_{IL} + \frac{L}{mG} H_{IG} \frac{(1-y)_{iM}}{(1-x)_{*M}} \quad (8.56)$$

The concentration ratio of the last equation can be dropped for dilute solutions. Data for the individual phase coefficients are summarized for standard packings in Chap. 6.

Illustration 8.7 Repeat the computation of Illustration 8.6 using the simplified procedures for dilute mixtures.

SOLUTION

Number of transfer units (a) Use Eq. (8.48). $y_1 = 0.02$, $x_1 = 0.1063$, $y_1^* = mx_1 = 0.125$ (0.1063) = 0.01329, $(y - y^*)_1 = 0.02 - 0.01329 = 0.00671$. $y_2 = 0.00102$, $x_2 = 0.0050$, $y_2^* = mx_2 = 0.125$ (0.005) = 0.000625, $(y - y^*)_2 = 0.00102 - 0.000625 = 0.000395$.

$$(y - y^*)_M = \frac{0.00671 - 0.000395}{\ln(0.00671/0.000395)} = 0.00223$$

$$N_{tOG} = \frac{0.02 - 0.00102}{0.00223} = 8.51 \quad \text{Ans.}$$

(b) Eq. (8.50) or Fig. 8.20: $\text{av } A = 1.424$ (Illustration 8.3).

$$\frac{y_2 - mx_2}{y_1 - mx_2} = \frac{0.00102 - 0.125(0.005)}{0.02 - 0.125(0.005)} = 0.0204$$

From either Eq. (8.50) or Fig. 8.20, $N_{tOG} = 9.16$. Ans.

(c) The graphical construction for transfer units is shown in Fig. 8.22. The mole-ratio coordinates are satisfactory for dilute mixtures (even though Fig. 8.21 is drawn for mole fractions), and the operating line and equilibrium curve were redrawn from Fig. 8.10. The line BD was drawn everywhere vertically midway between operating line and equilibrium curve and

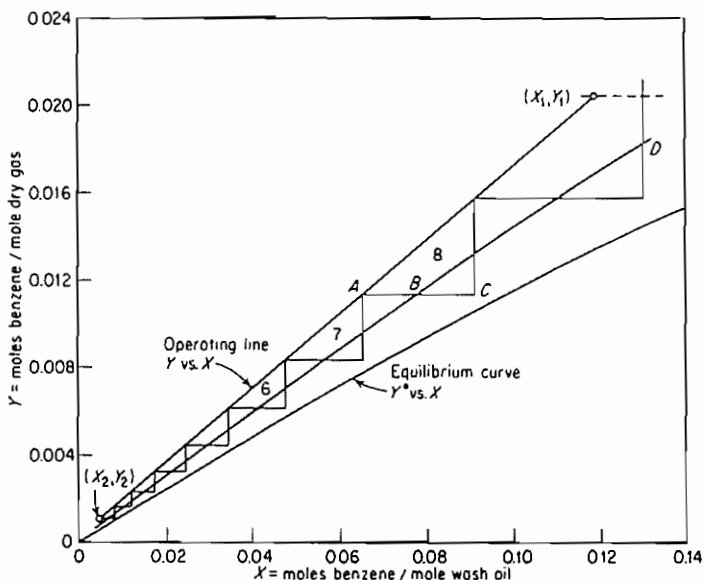


Figure 8.22 Solution to Illustration 8.7.

the transfer-unit steps were constructed by making the horizontal line segments such as AB and BC equal. The number of transfer units required is 8.7. **Ans.**

(d) Eq. (8.36). From Fig. 8.10, for each value of X , values of Y are read from the operating line DF and Y^* from the equilibrium curve (or they can be computed from the equations of these lines).

X	Y	Y^*	$\frac{1}{Y - Y^*}$
$X_2 = 0.00502$	0.00102	0.000625	2531
0.02	0.00357	0.00245	893
0.04	0.00697	0.00483	467
0.06	0.01036	0.00712	309
0.08	0.01376	0.00935	227
0.10	0.01714	0.01149	177.0
$X_1 = 0.1190$	0.0204	0.01347	144.3

The integral of Eq. (8.36) can be evaluated graphically from a plot of $1/(Y - Y^*)$ as ordinate vs. Y as abscissa or, following the suggestion of Eq. (8.28), from a plot of $Y/(Y - Y^*)$ as ordinate vs. $\log Y$ as abscissa. The integral term is then 8.63. Then

$$N_{OG} = 8.63 + \frac{1}{2} \ln \frac{1 + 0.00102}{1 + 0.0204} = 8.62 \quad \text{Ans.}$$

Height of a transfer unit Since the solutions are dilute, Eq. (8.54) becomes

$$H_{OG} = H_{IG} + \frac{mG}{L} H_{IL} + H_{IG} + \frac{H_{IL}}{A}$$

From Illustration 6.4, $F_G a = 0.0719 \text{ kmol/m}^3 \cdot \text{s}$, and $F_L a = 0.01377 \text{ kmol/m}^3 \cdot \text{s}$. $\text{Av } G = 0.0609 \text{ kmol/m}^2 \cdot \text{s}$ (Illustration 8.6). $\text{Av } L = (1.787 \times 10^{-3}) [(1 + 0.1190) + (1 + 0.00503)] / 2(0.1746) = 0.01087 \text{ kmol/m}^2 \cdot \text{s}$. Eq. (8.24):

$$H_{IG} = \frac{G}{F_G a} = \frac{0.0609}{0.0719} = 0.847 \text{ m}$$

Eq. (8.31):

$$H_{IL} = \frac{L}{F_L a} = \frac{0.01087}{0.01377} = 0.789$$

$$H_{OG} = 0.847 + \frac{0.789}{1.424} = 1.401 \text{ m}$$

$$Z = H_{OG} N_{OG} = 1.401 (9.16) = 12.83 \text{ m} \quad \text{Ans.}$$

Nonisothermal Operation

All the relationships thus far developed for packed towers are correct for either isothermal or nonisothermal operation, but for the nonisothermal case they do not tell the entire story. During absorption, release of energy at the interface due to latent heat and heat of solution raises the interface temperature above that of the bulk liquid. This changes physical properties, mass-transfer coefficients, and equilibrium concentrations. More subtle side effects associated with volume changes and contributions of thermal diffusion can enter the problem [4, 5, 11, 42]. As a practical matter, where the energy release is large, as in absorption of

HCl or nitrogen oxides into water, unless vigorous attempts to remove the heat are made, the temperature rise limits the ultimate concentration of the product solution [23].

For relatively dilute solutions, a conservative approximation can be made by assuming that all the heat evolved on absorption is taken up by the liquid, thus neglecting the temperature rise of the gas. This results in a liquid temperature higher than that which is likely, leading to a tower taller than need be. A somewhat more correct but still approximate approach neglects the thermal- and mass-transfer resistances of the liquid [38]. The method which follows allows for these resistances and also for the simultaneous mass transfer of solvent, which in the warm parts of the tower evaporates and in the cooler parts recondenses [41]. It has been experimentally confirmed [36].

Adiabatic Absorption and Stripping

Because of the possibility of substantial temperature effects, allowance must be made for evaporation and condensation of the liquid solvent. For this problem, therefore, the components are defined as follows:

A = principal transferred solute, present in gas and liquid phases

B = carrier gas, not dissolving, present only in gas

C = principal solvent of liquid, which can evaporate and condense; present in gas and liquid phases

Refer to Fig. 8.23, a schematic representation of a differential section of the packed tower [41]. For convenience, the phases are shown separated by a vertical surface. Enthalpies are referred to the pure unmixed components in their normal state of aggregation at some convenient temperature t_0 and the tower pressure. Liquid enthalpies are given by Eq. (8.12), expressed per mole of solution

$$H_L = C_L(t_L - t_0) + \Delta H_S \quad (8.12)$$

$$dH_L = C_L dt_L + d\Delta H_S \quad (8.57)$$

Gas enthalpies are expressed per mole of carrier gas B

$$H_G = C_B(t_G - t_0) + Y_A[C_A(t_G - t_0) + \lambda_{A0}] + Y_C[C_C(t_G - t_0) + \lambda_{C0}] \quad (8.58)$$

$$dH_G = C_B dt_G + Y_A C_A dt_G + [C_A(t_G - t_0) + \lambda_{A0}] dY_A + Y_C C_C dt_G + [C_C(t_G - t_0) + \lambda_{C0}] dY_C \quad (8.59)$$

If solute A is a gas at t_0 , λ_{A0} is taken as zero.

Transfer rates The fluxes of mass and heat shown in Fig. 8.23 are taken as positive in the direction gas to liquid, negative in the opposite direction. The relationships developed below are correct for any combination of them. Consider the gas phase: solute A and solvent vapor C may transfer, but $N_B = 0$.

$$N_A a dZ = R_A F_{GA} \left(\ln \frac{R_A - y_{A,i}}{R_A - y_A} \right) a dZ = -G_B dY_A \quad (8.60)$$

$$\text{and} \quad N_C a dZ = R_C F_{GC} \left(\ln \frac{R_C - y_{C,i}}{R_C - y_C} \right) a dZ = -G_B dY_C \quad (8.61)$$

where $R_A = N_A/(N_A + N_C)$, $R_C = N_C/(N_A + N_C)$. Since the fluxes N_A and N_C may have either sign, the R 's can be greater or less than 1.0, positive or negative. However $R_A + R_C = 1.0$. The heat-transfer rate for the gas is

$$q_G a dZ = h'_G a (t_G - t_i) dZ \quad (8.62)$$

where h'_G is the convection coefficient corrected for mass transfer. Similarly for the liquid,

$$N_A a dZ = R_A F_{L,i} \left(\ln \frac{R_A - x_A}{R_A - x_{A,i}} \right) a dZ \quad (8.63)$$

$$N_C a dZ = R_C F_{L,i} \left(\ln \frac{R_C - x_C}{R_C - x_{C,i}} \right) a dZ \quad (8.64)$$

$$q_L a dZ = h_L a (t_i - t_L) dZ \quad (8.65)$$

Mass balance Refer to envelope III, Fig. 8.23.

$$L + dL + G = L + G + dG$$

$$dL = dG$$

Since

$$G = G_B(1 + Y_A + Y_C)$$

then

$$dL = G_B(dY_A + dY_C) \quad (8.66)$$

Enthalpy balances The balance for the gas employs envelope I, Fig. 8.23, which extends to, but does not include, the interface

$$\text{Rate enthalpy in} = G_B H'_G \quad (8.67)$$

$$\text{Rate enthalpy out} = G_B(H'_G + dH'_G) - G_B dY_A [C_A(t_G - t_0) + \lambda_{A0}]$$

$$- G_B dY_C [C_C(t_G - t_0) + \lambda_{C0}] \quad (8.68)$$

The last two terms are the enthalpies of the transferred mass. At steady state,

$$\text{Rate enthalpy in} - \text{rate enthalpy out} = \text{rate of heat transfer}$$

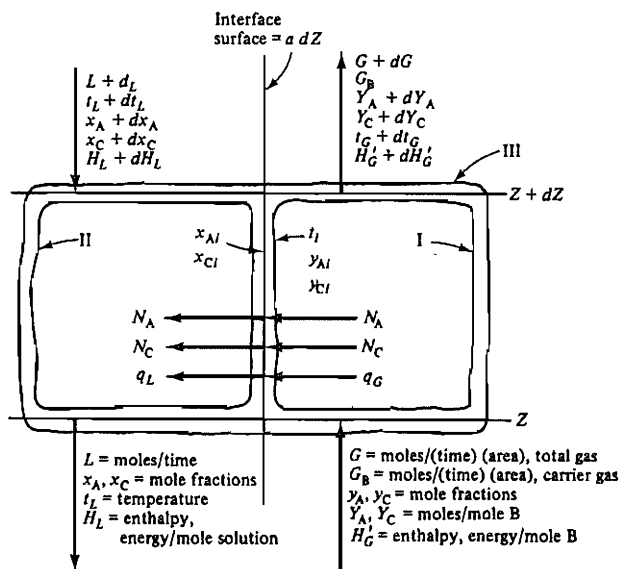


Figure 8.23 Differential section of a packed absorber or stripper.

Substitution of Eqs. (8.62), (8.67), and (8.68), with Eq. (8.59) for H_G , then results in

$$- G_B(C_B + Y_A C_A + Y_C C_C) dt_G = h'_G a(t_G - t_i) dZ \quad (8.69)$$

Similarly for the liquid. Envelope II extends to, but does not include, the interface

$$\text{Rate enthalpy in} = (L + dL)(H_L + dH_L) + (N_A \bar{H}_{A,i} + N_C \bar{H}_{C,i}) a dZ \quad (8.70)$$

$\bar{H}_{J,i}$ is the partial enthalpy of a component J ($= A, C$) in solution at temperature t_i , concentration $x_{J,i}$, and the last term is the enthalpy of the substance transferred from the gas

$$\text{Rate enthalpy out} = LH_L \quad (8.71)$$

At steady state,

$$\text{Rate enthalpy out} = \text{rate enthalpy in} + \text{rate of heat transfer}$$

Substituting Eqs. (8.12), (8.57), (8.60), (8.61), (8.65), (8.66), (8.70), and (8.71) and dropping the second-order differential $dH dL$ results in

$$L(C_L dt_L + d\Delta H_S) = G_B \left\{ -[C_L(t_L - t_0) + \Delta H_S](dY_A + dY_C) + \bar{H}_{A,i} dY_A + H_{C,i} dY_C \right\} - h_L a(t_i - t_L) dZ \quad (8.72)$$

Finally, an enthalpy balance using envelope III is

$$\text{Rate enthalpy in} = G_B H'_G + (H_L + dH_L)(L + dL) \quad (8.73)$$

$$\text{Rate enthalpy out} = LH_L + G_B(H'_G + dH'_G) \quad (8.74)$$

At steady state,

$$\text{Rate enthalpy in} = \text{rate enthalpy out}$$

Substituting Eqs. (8.12), (8.57), (8.59), (8.66), (8.73), and (8.74) and ignoring $dH_L dL$ produces

$$L(C_L dt_L + d\Delta H_S) = G_B \{ C_B dt_G + Y_A C_A dt_G + [C_A(t_G - t_0) + \lambda_{A0}] dY_A + Y_C C_C dt_G + [C_C(t_G - t_0) + \lambda_{C0}] dY_C - (dY_A + dY_C)[C_L(t_L - t_0) + \Delta H_S] \} \quad (8.75)$$

Interface conditions Equating the right sides of Eqs. (8.72) and (8.75) provides the interface temperature

$$t_i = t_L + \frac{G_B}{h_L a} \left\{ [\bar{H}_{A,i} - C_A(t_G - t_0) - \lambda_{A0}] \frac{dY_A}{dZ} + [\bar{H}_{C,i} - C_C(t_G - t_0) - \lambda_{C0}] \frac{dY_C}{dZ} - (C_B + Y_A C_A + Y_C C_C) \frac{dt_G}{dZ} \right\} \quad (8.76)$$

The gradients for this expression are provided by Eqs. (8.60), (8.61), and (8.69):

$$\frac{dY_A}{dZ} = - \frac{R_A F_{G,A} a}{G_B} \ln \frac{R_A - y_{A,i}}{R_A - y_A} \quad (8.77)$$

$$\frac{dY_C}{dZ} = - \frac{R_C F_{G,C} a}{G_B} \ln \frac{R_C - y_{C,i}}{R_C - y_C} \quad (8.78)$$

$$\frac{dt_G}{dZ} = - \frac{h'_G a(t_G - t_i)}{G_B(C_B + Y_A C_A + Y_C C_C)} \quad (8.79)$$

The interface conditions are obtained from Eqs. (8.60) and (8.63) for A

$$y_{A,i} = R_A - (R_A - y_A) \left(\frac{R_A - x_A}{R_A - x_{A,i}} \right)^{F_{L,A}/F_{G,A}} \quad (8.80)$$

Similarly for C

$$y_{C,i} = R_C - (R_C - y_C) \left(\frac{R_C - x_C}{R_C - x_{C,i}} \right)^{F_{L,C}/F_{G,C}} \quad (8.81)$$

These are solved simultaneously with their respective equilibrium-distribution curves, as described in Chap. 5 (see Illustration 5.1). The calculation is by trial: R_A is assumed, whence $R_C = 1 - R_A$, and the correct R 's are those for which $x_{A,i} + x_{C,i} = 1.0$.

The partial enthalpies of Eq. (8.76) can be obtained in a number of ways, e.g., from activity-coefficient data [12] or from integral heats of solution by the method of tangent intercepts [22, 40]. Since, however, the equilibrium partial pressures of components A and C from these solutions are required over a range of temperatures in any event, the simplest procedure is that described earlier in connection with Fig. 8.2 and Eq. (8.3). For ideal solutions, $\bar{H}_{j,i}$ is the enthalpy of pure liquid J at t_i for all $x_{j,i}$.

Mass-transfer coefficients must be those associated with the individual liquid and gas phases: overall coefficients will not serve. The correlations of Chap. 6 will provide these for Berl saddles and Raschig rings, and additional data are summarized in Ref. 14. The k -type coefficients are not suitable because in some cases the transfer is opposite the concentration gradient, but k 's can be converted into F 's (Table 3.1). Similarly data in the form of H_i 's can be converted into Fa [Eqs. (8.24) and (8.31)]. Heat-transfer coefficients can be estimated, if not otherwise available, through the heat-mass-transfer analogy. For the gas, the correction for mass transfer [Eq. (3.70)] provides

$$h'_{Ga} = \frac{-G_B(C_A dY_A/dZ + C_C dY_C/dZ)}{1 - \exp[G_B(C_A dY_A/dZ + C_C dY_C/dZ)/h_{GA}]} \quad (8.82)$$

A corresponding correction for the liquid, which would involve changing ΔH_S with concentration, is not available, but h_L is usually sufficiently large to make a correction unimportant. Mean diffusivities for three-component gas mixtures are estimated through application of Eq. (2.35), which reduces to

$$D_{A,m} = \frac{R_A - y_a}{R_A \left(\frac{y_B}{D_{AB}} + \frac{y_A + y_C}{D_{AC}} \right) - \frac{y_A}{D_{AC}}} \quad (8.83)$$

$$D_{C,m} = \frac{R_C - y_C}{R_C \left(\frac{y_B}{D_{CB}} + \frac{y_A + y_C}{D_{AC}} \right) - \frac{y_C}{D_{AC}}} \quad (8.84)$$

In the design of a tower, the cross-sectional area and hence the mass velocities of gas and liquid are established through pressure-drop considerations (Chap. 6). Assuming that entering flow rates, compositions, and temperatures, pressure of absorption, and percentage absorption (or stripping) of one component are specified, the packed height is then fixed. The problem is then to estimate the packed height and the conditions (temperature, composition) of the outlet streams. Fairly extensive trial and error is required, for which the relations outlined above can best be solved with a digital computer [15, 36]. Computer calculations can also be adapted to the problem of multicomponent mixtures [15]. The procedure for the three-component case is outlined in the following example [41].

Illustration 8.8 A gas consisting of 41.6% ammonia (A), 58.4% air (B) at 20°C, 1 std atm, flowing at the rate 0.0339 kmol/m²·s (25 lb mol/ft²·h), is to be scrubbed countercurrently with water (C) entering at 20°C at a rate 0.271 kmol/m²·s (200 lb mol/ft²·h), to remove 99% of the ammonia. The adiabatic absorber is to be packed with 38-mm ceramic Berl saddles. Estimate the packed height.

SOLUTION At 20°C, $C_A = 36\,390$, $C_B = 29\,100$, $C_C = 33\,960$ J/kmol·K. $\lambda_C = 44.24 \times 10^6$ J/kmol. These and other data are obtained from "The Chemical Engineers' Handbook" [31].

Enthalpy base = NH_3 gas, H_2O liquid, air at 1 std atm, $t_0 = 20^\circ\text{C}$. $\lambda_{\text{AO}} = 0$, $\lambda_{\text{CO}} = 44.24 \times 10^6$ J/kmol.

Gas in:

$$G_B = 0.0339(0.584) = 0.01980 \text{ kmol air/m}^2 \cdot \text{s}$$

$$y_A = 0.416 \quad Y_A = \frac{0.416}{1 - 0.416} = 0.7123 \text{ kmol NH}_3/\text{kmol air}$$

$$y_C = Y_C = 0 \quad H'_G = 0$$

Liquid in:

$$L = 0.271 \text{ kmol/m}^2 \cdot \text{s} \quad x_A = 0 \quad x_C = 1.0 \quad H_L = 0$$

Gas out:

$$Y_A = 0.7123(1 - 0.99) = 0.007123 \text{ kmol NH}_3/\text{kmol air}$$

Assume outlet gas $t_G = 23.9^\circ\text{C}$ (to be checked)

Assume $y_C = 0.0293$ (saturation) (to be checked)

$$y_C = \frac{Y_C}{Y_C + 0.007123 + 1} \quad Y_C = 0.0304 \text{ kmol H}_2\text{O}/\text{kmol air}$$

Eq. (8.58):

$$H'_G = 1463 \text{ kJ/kmol air}$$

Liquid out:

$$\text{H}_2\text{O content} = 0.271 - 0.0304(0.01980) = 0.2704 \text{ kmol/m}^2 \cdot \text{s}$$

$$\text{NH}_3 \text{ content} = 0.01980(0.7123 - 0.007123) = 0.01396 \text{ kmol/m}^2 \cdot \text{s}$$

$$L = 0.2704 + 0.01396 = 0.2844 \text{ kmol/m}^2 \cdot \text{s}$$

$$x_A = \frac{0.01396}{0.2844} = 0.0491 \quad x_C = 0.9509$$

$$\text{At } x_A = 0.0491, t_0 = 20^\circ\text{C}, \Delta H_S = -1709.6 \text{ kJ/kmol solution [24]}$$

Conditions at the bottom of the tower Enthalpy balance, entire tower:

Enthalpy gas in + enthalpy liquid in = enthalpy gas out + enthalpy liquid out

$$0 + 0 = 0.01980(1463000) + 0.2844[C_L(t_L - 20) - 1709600]$$

The value of C_L , obtained by trial at $(t_L + 20)/2$, equals $75481 \text{ J/kmol} \cdot \text{K}$, whence $t_L = 41.3^\circ\text{C}$.

For the gas, $M_{G,av} = 24.02$, $\rho_G = 0.999 \text{ kg/m}^3$, $\mu_G = 1.517 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, k (thermal conductivity) = $0.0261 \text{ W/m} \cdot \text{K}$, $C_p = 1336 \text{ J/kg} \cdot \text{K}$; $D_{AB} = 2.297 \times 10^{-5} \text{ m}^2/\text{s}$, $D_{AC} = 3.084 \times 10^{-5} \text{ m}^2/\text{s}$, $D_{CB} = 2.488 \times 10^{-5} \text{ m}^2/\text{s}$, $\text{Pr}_G = 0.725$.

For the liquid, $M_{L,av} = 17.97$, $\rho_L = 953.1 \text{ kg/m}^3$, $\mu_L = 6.408 \times 10^{-4} \text{ kg/m} \cdot \text{s}$, $D_{AL} = 3.317 \times 10^{-9} \text{ m}^2/\text{s}$, k (thermal conductivity) = $0.4777 \text{ W/m} \cdot \text{K}$; $\text{Sc}_L = 202$, $\text{Pr}_L = 5.72$.

The mass velocities at the bottom are $G' = GM_{G,av} = 0.0339(24.02) = 0.8142 \text{ kg/m}^2 \cdot \text{s}$, and $L' = LM_{L,av} = 0.2844(17.97) = 5.11 \text{ kg/m}^2 \cdot \text{s}$. From the data of Chap. 6, $d_S = 0.0472 \text{ m}$, $a = 57.57 \text{ m}^2/\text{m}^3$, $\phi_L = 0.054$, $\epsilon = 0.75$, and $\epsilon_{Lo} = 0.696$ (Chap. 6 notation).

Eq. (6.72): $k_L = 3.616 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$. Table 3.1: $F_L = k_L c$, $c = \rho_L/M_L = 953.1/17.97 = 53.04 \text{ kmol/m}^3$. Therefore $F_L = 0.01918 \text{ kmol/m}^2 \cdot \text{s}$. The heat-mass-transfer analogy used with Eq. (6.72) provides

$$\frac{h_L d_S}{k} = 25.1 \left(\frac{d_S L'}{\mu_L} \right)^{0.45} \text{Pr}_L^{0.5}$$

whence, with $k = 0.4777 \text{ W/m} \cdot \text{K}$, $h_L = 8762 \text{ W/m}^2 \cdot \text{K}$. The heat-transfer analog of Eq. (6.69) is

$$\frac{h_G}{C_p G'} \text{Pr}_G^{2/3} = 1.195 \left[\frac{d_S G'}{\mu_G (1 - \epsilon_{Lo})} \right]^{-0.36}$$

from which $h_G = 62.45 \text{ W/m}^2 \cdot \text{K}$.

To obtain the gas mass-transfer coefficients, the mean diffusivities of A and C, and hence R_A and R_C , must be known. Assume $R_A = 1.4$, whence $R_C = 1 - 1.4 = -0.4$. With $y_A = 0.416$, $y_B = 0.584$, $y_C = 0$, Eqs. (8.83) and (8.84) yield $D_{A,m} = 2.387 \times 10^{-5} \text{ m}^2/\text{s}$ and $D_{C,m} = 2.710 \times 10^{-5} \text{ m}^2/\text{s}$. Therefore $Sc_{G,A} = \mu_G/\rho_G D_{A,m} = 0.636$, $Sc_{G,C} = 0.560$. Eq. (6.69): $F_{G,A} = 2.13 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$, $F_{G,C} = 2.32 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s}$. Eqs. (8.80) and (8.81) then become

$$y_{A,i} = 1.4 - (1.4 - 0.416) \left(\frac{1.4 - 0.0491}{1.4 - x_{A,i}} \right)^{0.01918/2.13 \times 10^{-3}} \quad (8.85)$$

$$y_{C,i} = -0.4 - (-0.4 - 0) \left(\frac{-0.4 - 0.9509}{-0.4 - x_{C,i}} \right)^{0.01918/2.32 \times 10^{-3}} \quad (8.86)$$

These are to be plotted to obtain the intersections with their respective distribution curves at temperature t_i . Equilibrium data for NH_3 and H_2O vapors in contact with solutions are plotted in Figs. 8.2 and 8.24, with water as reference substance.

After one trial, assume $t_i = 42.7^\circ\text{C}$, to be checked. The data of Figs. 8.2 and 8.24 are used to draw the equilibrium curves of Fig. 8.25, and Eqs. (8.85) and (8.86) are plotted on their respective portions. Similarly the counterparts of these equations for $R_A = 1.3$, $R_C = -0.7$ and $R_A = 1.5$, $R_C = -0.5$ are plotted. By interpolation, the intersections of the curves with the equilibrium lines satisfy $x_{A,i} + x_{C,i} = 1.0$ for $R_A = 1.38$, $R_C = -0.38$, and $x_{A,i} = 0.0786$, $y_{A,i} = 0.210$, and $x_{C,i} = 0.9214$, $y_{C,i} = 0.0740$.

Eq. (8.77): $dY_A/dZ = -1.706 \text{ (kmol H}_2\text{O/kmol air)/m}$. Eq. (8.78): $dY_C/dZ = 0.4626 \text{ (kmol H}_2\text{O/kmol air)/m}$. Eq. (8.82): $h_G a = 4154 \text{ W/m}^3 \cdot \text{K}$. Eq. (8.79): $dt_G/dZ = 86.5 \text{ K/m}$.

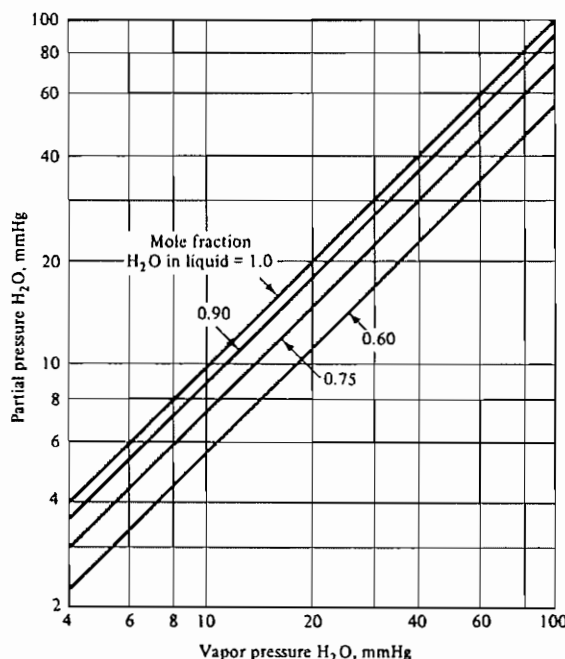


Figure 8.24 Equal-temperature reference-substance graph for partial pressures of water over aqueous ammonia solutions. Reference substance is water.

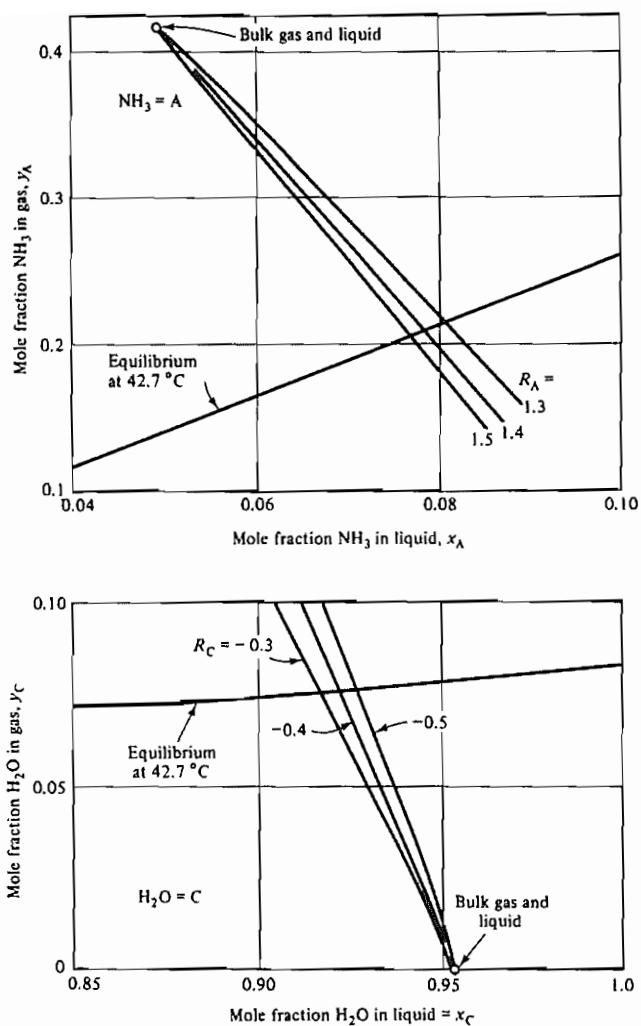


Figure 8.25 Determination of interface conditions, Illustration 8.8.

When the curves of Figs. 8.2 and 8.24 are interpolated for concentrations $x_{A,i}$ and $x_{C,i}$, the slopes are $m'_{A,i} = 0.771$ and $m'_{C,i} = 1.02$. At 42.7°C , $\lambda_{\text{H}_2\text{O}} = \lambda_{\text{C}} = 43.33 \times 10^6 \text{ J/kmol}$. Then Eq. (8.3) for A:

$$\bar{H}_{A,i} = 36\,390(42.7 - 20) + 0 - 0.771(43.33 \times 10^6) = -32.58 \times 10^6 \text{ J/kmol}$$

Similarly for C: $H_{C,i} = 0.784 \times 10^6 \text{ J/kmol}$. Eq. (8.76): $t_i = 42.5^\circ\text{C}$, which is sufficiently close to the 42.7°C assumed earlier.

An interval of ΔY_A up the tower Take a small increment in ΔY_A . For a desk calculator, $\Delta Y_A = -0.05$ is suitable (for a digital computer, -0.01 would be more appropriate). Then $\Delta Z = \Delta Y_A / (dY_A/dZ) = -0.05 / (-1.706) = 0.0293$ m. At this level,

$$Y_{A, \text{next}} = Y_A + \Delta Y_A = 0.7123 - 0.05 = 0.6623 \text{ kmol/kmol air}$$

$$Y_{C, \text{next}} = Y_C + \Delta Y_C = Y_C + (dY_C/dZ) \Delta Z = 0 + 0.4626(0.0293) = 0.01355 \text{ kmol H}_2\text{O/kmol air}$$

$$t_{G, \text{next}} = t_G + (dt_G/dZ) \Delta Z = 20 + 86.5(0.0293) = 22.53^\circ\text{C}$$

$$L_{\text{next}} = L + G_B(\Delta Y_A + \Delta Y_C) = 0.2837 \text{ kmol/m}^2 \cdot \text{s}$$

$$x_{A, \text{next}} = \frac{G_B \Delta Y_A + Lx_A}{L_{\text{next}}} = 0.0457 \text{ mole fraction NH}_3$$

$$H'_{G, \text{next}} = 7.33 \times 10^5 \text{ J/kmol air}$$

$$H_{L, \text{next}} = (LH_L)_{\text{top}} + \frac{G_B(H'_{G, \text{next}} - H'_{G, \text{top}})}{L_{\text{next}}} = -4.868 \times 10^5 \text{ J/kmol}$$

The previous computations can now be repeated at this level, leading ultimately to a new ΔY_A . The calculations are continued until the specified gas outlet composition is reached, whereupon the assumed outlet gas temperature and water concentration can be checked. The latter are adjusted, as necessary, and the entire computation repeated until a suitable check is obtained. The packed depth required is then the sum of the final ΔZ 's. The computations started above lead to $Z = 1.58$ m, $t_G = 23.05^\circ\text{C}$. The final results are shown in Fig. 8.26 [26]. Water is stripped in the lower part of the tower and reabsorbed in the upper part.

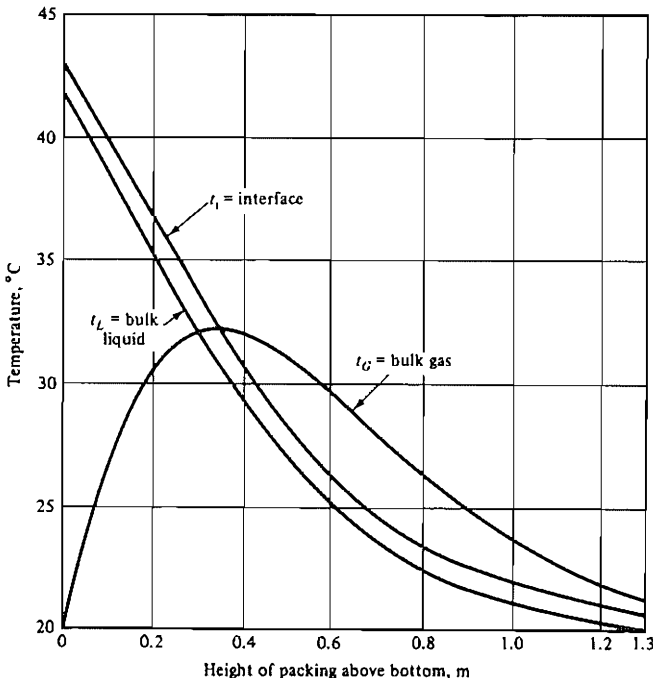


Figure 8.26 Temperature profiles, Illustration 8.8.

In the course of such calculations, it may develop that the direction of mass transfer of the solvent liquid, especially as a vapor in the gas phase, is *against* the concentration gradient. This is entirely possible (see Chap. 2) and is one of the reasons why *F*-type (rather than *k*-type) mass-transfer coefficients are essential. The bulk-gas temperature must never be less than the dew point, since this will lead to fog formation for which the computations do not allow. A larger liquid rate, with consequently lower liquid temperatures, will result in less solvent vaporization and less likelihood of fog formation.

MULTICOMPONENT SYSTEMS

Except for the consideration of solvent vaporization in the above discussion of adiabatic packed towers, it has thus far been assumed that only one component of the gas stream has an appreciable solubility. When the gas contains several soluble components or the liquid several volatile ones for stripping, some modifications are needed. The almost complete lack of solubility data for multicomponent systems, except where ideal solutions are formed in the liquid phase and the solubilities of the various components are therefore mutually independent, unfortunately makes estimates of even the ordinary cases very difficult. However, some of the more important industrial applications fall in the ideal-solution category, e.g., the absorption of hydrocarbons from gas mixtures in nonvolatile hydrocarbon oils as in the recovery of natural gasoline.

In principle, the problem for tray towers should be capable of solution by the same tray-to-tray methods used in Illustration 8.4 for one component absorbed, through Eqs. (8.11) to (8.15). These expressions are indeed valid. If, as with one component absorbed, computations are to be started at the bottom of the tower, the outlet liquid temperature and composition must be known. This, as before, requires that the outlet gas temperature be estimated, but in addition, in order to complete the enthalpy balance of Eq. (8.11), also requires that the complete outlet gas composition with respect to each component be estimated at the start. Herein lies the difficulty.

The quantities which are ordinarily fixed before an absorber design is started are the following:

1. Rate of flow, composition, and temperature of entering gas
2. Composition and temperature of entering liquid (but not flow rate)
3. Pressure of operation
4. Heat gain or loss (even if set at zero, as for adiabatic operation)

Under these circumstances it can be shown [30] that the principal variables still remaining are:

1. The liquid flow rate (or liquid/gas ratio)
2. The number of ideal trays
3. The fractional absorption of any one component

Any two of these last but not all three can be arbitrarily fixed for a given design. When two have been specified, the third is automatically fixed, as is the extent of absorption of all substances not already specified and the outlet-stream temperatures. For example, if the liquid rate and number of equilibrium trays are specified, the extent of absorption of each substance of the gas is automatically fixed and cannot be arbitrarily chosen. Or if the liquid rate and extent of absorption of one substance are specified, the number of equilibrium trays and the extent of absorption of all components are automatically fixed and cannot be chosen.

As a result, for the tray-to-tray calculations suggested above, not only must the outlet gas temperature be guessed but also the complete outlet gas composition, all to be checked at the end of the calculation. This becomes so hopeless a trial-and-error procedure that it cannot be done practically without some guidance. This is provided through an approximate procedure, either that offered by the Kremser equations, which apply only for constant absorption factor, or through some procedure allowing for variation of the absorption factor with tray number. To establish the latter, we first need an exact expression for the absorber with varying absorption factor. This was first derived by Horton and Franklin [21] as outlined below.

Refer to Fig. 8.27, which shows a multitray absorber or stripper. Since all components can transfer between gas and liquid, there may be no substance which passes through at constant rate in the gas, for example. It is convenient, therefore, to define all gas compositions in terms of the entering gas and similarly all liquid compositions in terms of the entering liquid. Thus, for any component in the liquid leaving any tray n ,

$$X'_n = \frac{\text{moles component in } L_n/\text{time}}{L_0} = \frac{x_n L_n}{L_0}$$

and for any component in the gas G_n ,

$$Y'_n = \frac{\text{moles component in } G_n/\text{time}}{G_{N_p+1}} = \frac{y_n G_n}{G_{N_p+1}}$$

where x and y are the usual mole fractions.

The equations which follow can all be written separately for each component. Consider the tower of Fig. 8.27 to be an absorber. A material balance for any component about equilibrium tray n is

$$L_0(X'_n - X'_{n-1}) = G_{N_p+1}(Y'_{n+1} - Y'_n) \quad (8.87)$$

The equilibrium relation for the equilibrium tray is†

$$y'_n = m_n x'_n \quad (8.88)$$

or, in terms of the new concentrations,

$$Y'_n \frac{G_{N_p+1}}{G_n} = m_n X'_n \frac{L_0}{L_n} \quad (8.89)$$

† The ratio y/x at equilibrium is usually written as K , but here we use m to distinguish this from the mass-transfer coefficients.

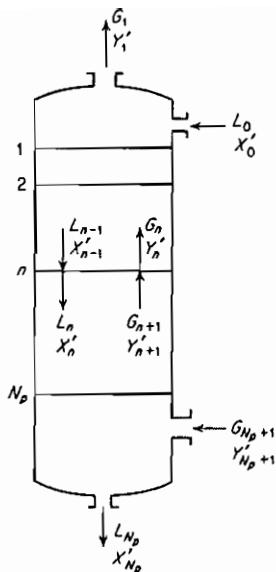


Figure 8.27 Multicomponent absorber or stripper.

Similarly, for tray $n - 1$,

$$Y'_{n-1} \frac{G_{N_p+1}}{G_{n-1}} = m_{n-1} X'_{n-1} \frac{L_0}{L_{n-1}} \quad (8.90)$$

Solving Eqs. (8.89) and (8.90) for the X 's, substituting in Eq. (8.87), and rearranging, we have

$$Y'_n = \frac{Y'_{n+1} + A_{n-1} Y'_{n-1}}{1 + A_n} \quad (8.91)$$

where $A_n = L_n/m_n G_n$ and $A_{n-1} = L_{n-1}/m_{n-1} G_{n-1}$ are the component absorption factors on the two trays.

If the absorber contained only one tray ($n = 1$), Eq. (8.91) would read

$$Y'_1 = \frac{Y'_2 + A_0 Y'_0}{1 + A_1} \quad (8.92)$$

From Eq. (8.90)

$$Y'_0 = m_0 X'_0 \frac{L_0}{L_0} \frac{G_0}{G_{N_p+1}} = \frac{m_0 X'_0 G_0}{G_{N_p+1}} \quad (8.93)$$

and

$$A_0 Y'_0 = \frac{L_0}{m_0 G_0} \frac{m_0 X'_0 G_0}{G_{N_p+1}} = \frac{L_0 X'_0}{G_{N_p+1}} \quad (8.94)$$

Substituting this into Eq. (8.92) provides

$$Y'_1 = \frac{Y'_2 + L_0 X'_0 / G_{N_p+1}}{1 + A_1} \quad (8.95)$$

If the absorber contained two trays, Eq. (8.91) with $n = 2$ would become

$$Y'_2 = \frac{Y'_3 + A_1 Y'_1}{1 + A_2} \quad (8.96)$$

Substituting Y'_1 from Eq. (8.95) and rearranging gives

$$Y'_2 = \frac{(A_1 + 1)Y'_1 + A_1 L_0 X'_0 / G_{N_p+1}}{A_1 A_2 + A_2 + 1} \quad (8.97)$$

Similarly, for a three-tray absorber,

$$Y'_3 = \frac{(A_1 A_2 + A_2 + 1)Y'_2 + A_1 A_2 L_0 X'_0 / G_{N_p+1}}{A_1 A_2 A_3 + A_2 A_3 + A_3 + 1} \quad (8.98)$$

and for N_p trays,

$$Y'_{N_p} = \frac{(A_1 A_2 A_3 \cdots A_{N_p-1} + A_2 A_3 \cdots A_{N_p-1} + \cdots + A_{N_p-1} + 1) \times Y'_{N_p+1} + A_1 A_2 \cdots A_{N_p-1} L_0 X'_0 / G_{N_p+1}}{A_1 A_2 A_3 \cdots A_{N_p} + A_2 A_3 \cdots A_{N_p} + \cdots + A_{N_p} + 1} \quad (8.99)$$

In order to eliminate Y'_{N_p} , which is inside the absorber, a component material balance about the entire absorber,

$$L_0(X'_{N_p} - X'_0) = G_{N_p+1}(Y'_{N_p+1} - Y'_1) \quad (8.100)$$

and Eq. (8.89) for $n = N_p$,

$$Y'_{N_p} = m_{N_p} X'_{N_p} \frac{L_0}{L_{N_p}} \frac{G_{N_p}}{G_{N_p+1}} = \frac{L_0 X'_{N_p}}{A_{N_p} G_{N_p+1}} \quad (8.101)$$

are solved simultaneously to eliminate X'_{N_p} , the result substituted for Y'_{N_p} in Eq. (8.101), whence rearrangement yields

$$\frac{Y'_{N_p+1} - Y'_1}{Y'_{N_p+1}} = \frac{A_1 A_2 A_3 \cdots A_{N_p} + A_2 A_3 \cdots A_{N_p} + \cdots + A_{N_p}}{A_1 A_2 A_3 \cdots A_{N_p} + A_2 A_3 \cdots A_{N_p} + \cdots + A_{N_p} + 1} - \frac{L_0 X'_0}{G_{N_p+1} Y'_{N_p+1}} \frac{A_2 A_3 A_4 \cdots A_{N_p} + A_3 A_4 \cdots A_{N_p} + \cdots + A_{N_p} + 1}{A_1 A_2 A_3 \cdots A_{N_p} + A_2 A_3 \cdots A_{N_p} + \cdots + A_{N_p} + 1} \quad (8.102)$$

Equation (8.102) is an expression for the fractional absorption of any component, exact because it is based only upon material balances and the condition of equilibrium which defines an ideal tray.

A similar expression for strippers is

$$\frac{X'_0 - X'_{N_p}}{X'_0} = \frac{S_1 S_2 \cdots S_{N_p} + S_1 S_2 \cdots S_{N_p-1} + \cdots + S_1}{S_1 S_2 \cdots S_{N_p} + S_1 S_2 \cdots S_{N_p-1} + \cdots + S_1 + 1} - \frac{G_{N_p+1} Y'_{N_p+1}}{L_0 X'_0} \frac{S_1 S_2 \cdots S_{N_p-1} + S_1 S_2 \cdots S_{N_p-2} + \cdots + S_1 + 1}{S_1 S_2 \cdots S_{N_p} + S_1 S_2 \cdots S_{N_p-1} + \cdots + S_1 + 1} \quad (8.103)$$

In order to use Eqs. (8.102) or (8.103), the L/G ratio for each tray and the tray temperatures (which determine the m 's) are required to compute the A 's or the S 's. If the liquids are not ideal, m for any component on any tray additionally depends upon the complete liquid composition on the tray. The same is true for the gas compositions under conditions for which the gas solutions are not ideal. The equations are practically useful, therefore, only for ideal solutions.

As an approximation [21], the gas rate G_n for tray n of an absorber can be estimated on the assumption that the fractional absorption is the same for each

tray:

$$\frac{G_n}{G_{n+1}} \approx \left(\frac{G_1}{G_{N_p+1}} \right)^{1/N_p} \quad (8.104)$$

or

$$G_n \approx G_{N_p+1} \left(\frac{G_1}{G_{N_p+1}} \right)^{(N_p+1-n)/N_p} \quad (8.105)$$

The liquid rate L_n can then be obtained by material balance to the end of the tower. Similarly in strippers,

$$L_n \approx L_0 \left(\frac{L_{N_p}}{L_0} \right)^{n/N_p} \quad (8.106)$$

where G_n is determined by material balance. If the molar latent heats and heat capacities are all alike for all components, and if no heat of solution is evolved, the temperature rise on absorption is roughly proportional to the amount of absorption, so that, approximately,

$$\frac{G_{N_p+1} - G_{n+1}}{G_{N_p+1} - G_1} \approx \frac{t_{N_p} - t_n}{t_{N_p} - t_0} \quad (8.107)$$

and similarly for stripping,

$$\frac{L_0 - L_n}{L_0 - L_{N_p}} \approx \frac{t_0 - t_n}{t_0 - t_{N_p}} \quad (8.108)$$

In order further to simplify the computations, Edmister [13] has written the Horton-Franklin equations in terms of average or "effective" absorption and stripping factors instead of the A 's and S 's for each tray. Thus, for absorption, Eq. (8.102) becomes

$$\frac{Y'_{N_p+1} - Y'_1}{Y'_{N_p+1}} = \left(1 - \frac{L_0 X'_0}{A' G_{N_p+1} Y'_{N_p+1}} \right) \frac{A_E^{N_p+1} - A_E}{A_E^{N_p+1} - 1} \quad (8.109)$$

For a two-tray absorber, it develops that

$$A' = \frac{A_{N_p}(A_1 + 1)}{A_{N_p} + 1} \quad (8.110)$$

and

$$A_E = [A_{N_p}(A_1 + 1) + 0.25]^{0.5} - 0.5 \quad (8.111)$$

These are exact, but it is found that Eqs. (8.104) to (8.111) apply reasonably well for any number of trays, provided that unusual temperature profiles (such as a maximum temperature on an intermediate tray) do not develop. If $X'_0 = 0$, Eq. (8.109) is the Kremser equation. It is also convenient to note that $(A_E^{N_p+1} - A_E)/(A_E^{N_p+1} - 1)$ is the Kremser function [Eq. (5.54)], and the ordinate of Fig. 5.16 is 1 minus this function.

Similarly for stripping,

$$\frac{X'_0 - X'_{N_p}}{X'_0} = \left(1 - \frac{G_{N_p+1} Y'_{N_p+1}}{S' L_0 X'_0} \right) \frac{S_E^{N_p+1} - S_E}{S_E^{N_p+1} - 1} \quad (8.112)$$

$$\text{with} \quad S' = \frac{S_1(S_{N_p} + 1)}{S_1 + 1} \quad (8.113)$$

$$\text{and} \quad S_E = [S_1(S_{N_p} + 1) + 0.25]^{0.5} - 0.5 \quad (8.114)$$

Equation (8.112) becomes the Kremser equation if $Y'_{N_p+1} = 0$.

Components present only in the entering gas will be absorbed, and those present only in the entering liquid will be stripped. If a component is present in both streams, the terms in the parentheses of the right-hand sides of Eqs. (8.109) and (8.112) are computed and the equation is used which provides the positive quantity.

Equations (8.109) and (8.112) can be used to determine the number of equilibrium trays required to absorb or strip a component to a specified extent and to estimate the extent of absorption or stripping of all other components. This then provides a basis for using the exact equations of Horton and Franklin, Eqs. (8.102) and (8.103). These latter can be used only for an integral number of equilibrium trays. A change of L_0/G_{N_p+1} may be necessary to meet the specifications exactly with such an integral number; alternatively the nearest larger integral number of trays may be accepted.

Illustration 8.9 A gas analyzing 70 mol % CH_4 , 15% C_2H_6 , 10% $n\text{-C}_3\text{H}_8$, and 5% $n\text{-C}_4\text{H}_{10}$, at 25°C , 2 std atm, is to be scrubbed in an adiabatic tray absorber with a liquid containing 1 mol % $n\text{-C}_4\text{H}_{10}$, 99% nonvolatile hydrocarbon oil, at 25°C , using 3.5 mol liquid/mol entering gas. The pressure is to be 2 std atm, and at least 70% of the C_3H_8 of the entering gas is to be absorbed. The solubility of CH_4 in the liquid will be considered negligible, and the other components form ideal solutions. Estimate the number of ideal trays required and the composition of the effluent streams.

SOLUTION In what follows, CH_4 , C_2H_6 , C_3H_8 , and C_4H_{10} will be identified as C_1 , C_2 , C_3 , and C_4 , respectively. Physical properties are:

Component	Average specific heat at $0\text{--}40^\circ\text{C}$, kJ/kmol $\cdot$ K		Latent heat of vaporization at 0°C , kJ/kmol	$m = \frac{y^*}{x} \dagger$		
	Gas	Liquid		25°C	27.5°C	30°C
C_1	35.59					
C_2	53.22	105.1	10 032	13.25	13.6	14.1
C_3	76.04	116.4	16 580	4.10	4.33	4.52
C_4	102.4	138.6	22 530	1.19	1.28	1.37
Oil		37.7				

$\dagger$ Values of m from C. L. Depriester, *Chem. Eng. Prog. Symp. Ser.*, 49(7), 1 (1953); "The Chemical Engineers' Handbook," 5th ed., pp. 13-12 and 13-13.

Basis: 1 kmol entering gas; $t_0 = 0^\circ\text{C}$; enthalpies are referred to gaseous C_1 and other components liquid at 0°C . Gas in: $G_{N_p+1} = 1$ kmol, $t_{N_p+1} = 25^\circ\text{C}$.

Component	$Y'_{N_p+1} = y_{N_p+1}$	Enthalpy, kJ/kmol = H_{G, N_p+1} per component	$H_{G, N_p+1} y_{N_p+1}$
C_1	0.70	$35.59(25 - 0) = 889.75$	622.8
C_2	0.15	$53.22(25 - 0) + 10\,032 = 16\,363$	1704.5
C_3	0.10	$76.04(25 - 0) + 16\,580 = 18\,480$	1848.0
C_4	0.05	$102.4(25 - 0) + 22\,530 = 25\,090$	1254.5
	1.00		5429.8 = H_{G, N_p+1}

Liquid in: $L_0 = 3.5$ kmol $t = 25^\circ\text{C}$

Component	$X'_0 = x_0$	$L_0 X'_0$	Enthalpy, kJ/kmol = H_{L_0} per component	$H_{L_0} L_0 X'_0$
C_4	0.01	0.035	$138.6(25 - 0) = 3465$	121.28
Oil	0.99	3.465	$377(25 - 0) = 9425$	32\,658
	1.00	3.50		32\,780 = $H_{L_0} L_0$

Preliminary calculations The total absorption is estimated to be 0.15 kmol, and an average temperature 25°C is assumed. A rough value of L/G at the top is then $3.5/(1 - 0.15) = 4.12$, and at the bottom $(3.5 + 0.15)/1.0 = 3.65$, with an average of 3.90. The number of equilibrium trays is fixed by the specified C_3 absorption. For C_3 , m at $25^\circ\text{C} = 4.10$, rough $A = 3.90/4.10 = 0.951$. Equation (8.109) with fractional absorption = 0.7 and $X_0 = 0$ gives

$$0.7 = \frac{0.951^{N_p+1} - 0.951}{0.951^{N_p+1} - 1}$$

This can be solved for N_p directly. Alternatively, since Fig. 5.16 has as its ordinate $(1 - \text{Kremser function}) = 0.3$, then $A = 0.951$ and N_p can be read. Either way, $N_p = 2.53$. Since the Horton-Franklin equation will be used later, an integral number of trays $N_p = 3$ is chosen. Fig. 5.16 with $N_p = 3$, $A = 0.951$, then shows an ordinate of 0.27, or 0.73 fraction C_3 absorbed.

$$\frac{Y'_{N_p+1} - Y'_1}{Y'_{N_p+1}} = \frac{0.10 - Y'_1}{0.1} = 0.73$$

$$Y'_1 = 0.027 \text{ kmol } C_3 \text{ in } G_1$$

For C_2 , $m = 13.25$ at 25°C , rough $A = 3.90/13.25 = 0.294$. At low A 's, Fig. 5.16 shows the ordinate to be $1 - A$, and the fractional absorption is 0.294.

$$\frac{Y'_{N_p+1} - Y'_1}{Y'_{N_p+1}} = \frac{0.15 - Y'_1}{0.15} = 0.294$$

$$Y'_1 = 0.1059 \text{ kmol } C_2 \text{ in } G_1$$

C_4 is present both in the entering liquid and gas. At 25°C , $m = 1.19$, rough $A = 3.90/1.19 = 3.28$, rough $S = 1/3.28 = 0.305$. Then

$$1 - \frac{L_0 X'_0}{A' G_{N_p+1} Y'_{N_p+1}} = 1 - \frac{3.5(0.01)}{3.28(1.0)(0.05)} = +0.7866$$

$$1 - \frac{G_{N_p+1} Y'_{N_p+1}}{S' L_0 X'_0} = 1 - \frac{1(0.05)}{0.305(3.5)(0.01)} = -3.684$$

The C_4 will therefore be absorbed. Eq. (8.109):

$$\frac{0.05 - Y'_1}{0.05} = 0.7866 \frac{3.28^4 - 3.28}{3.28^4 - 1}$$

$$Y'_1 = 0.01145 \text{ kmol } C_4 \text{ in } G_1$$

[Alternatively, at $N_p = 3$, $A = 3.28$, Fig. 5.16 provides an ordinate of 0.0198, whence $(0.05 - Y'_1)/0.05 = 0.7866(1 - 0.0198)$, and $Y'_1 = 0.01145$.]

Edmister method Estimate the top-tray temperature $t_1 = 26^\circ\text{C}$ (to be checked). The results of the preliminary calculations then give the following approximate results:

G_1

Component	Y'_1	Enthalpy, kJ/kmol = H_{G1} per component	$H_{G1} Y'_1$
C_1	0.70	$35.59(26 - 0) = 925$	647.5
C_2	0.1059	$53.22(26 - 0) + 10\,032 = 11\,420$	1209.4
C_3	0.027	$76.04(26 - 0) + 16\,580 = 18\,560$	501.12
C_4	0.01145	$102.4(26 - 0) + 22\,530 = 25\,190$	288.4
	$\frac{0.8444}{G_1} =$		$\frac{2646}{H_{G1} G_1}$

L_3

Component	$L_3 x_3$, kmol	Enthalpy, kJ/kmol = H_{L3} per component	$H_{L3} L_3 x_3$
C_2	$0.15 - 0.1059 = 0.0441$	$105(t_3 - 0)$	$4.64 t_3$
C_3	$0.01 - 0.027 = 0.073$	$116.4(t_3 - 0)$	$8.50 t_3$
C_4	$0.035 + 0.05 - 0.01148 = 0.0736$	$138.6(t_3 - 0)$	$10.2 t_3$
Oil	$\frac{3.465}{L_3} =$	$377(t_3 - 0)$	$\frac{1306.3 t_3}{1329 t_3}$
	$L_3 = 3.656$		

Overall enthalpy balance, Eq. (8.11) with $Q_T = 0$, $G_{N_p+1} = 1.0$, $N_p = 3$:

$$32\,780 + 5429.8 = 1329 t_3 + 2646$$

$$t_3 = 26.8^\circ\text{C}$$

Equation (8.109) can now be used to obtain a second approximation of the effluent composition. Eq. (8.105) with $n = 2$:

$$G_2 = 1.0 \left(\frac{0.8444}{1.0} \right)^{(3+1-2)/3} = 0.8934$$

Eq. (8.13) with $n = 1$:

$$L_1 + 1.0 = 3.656 + 0.8934$$

$$L_1 = 3.549 \quad \frac{L_1}{G_1} = \frac{3.549}{0.8444} = 4.203$$

Eq. (8.105) with $n = 3$:

$$G_3 = 1.0 \left(\frac{0.8444}{1.0} \right)^{(3+1-3)/3} = 0.945$$

$$\frac{L_3}{G_3} = \frac{3.656}{0.945} = 3.869$$

For C_4 :

$$m \text{ at } 26^\circ\text{C} = 1.225 \quad A_1 = \frac{4.203}{1.225} = 3.43$$

$$m \text{ at } 26.8^\circ\text{C} = 1.250 \quad A_3 = \frac{3.869}{1.250} = 3.095 = A_N$$

Eq. (8.110):

$$A' = 3.348$$

Eq. (8.111):

$$A_E = 3.236$$

Eq. (8.109):

$$Y'_1 = 0.01024 \text{ kmol } C_4 \text{ in } G_1$$

$$C_4 \text{ in } L_3 = 0.035 + 0.05 - 0.01024 = 0.07476 \text{ kmol} = L_3 x_3$$

In similar fashion the Y'_i 's for the other components, together with the outlet gas and outlet liquid enthalpies, can be computed:

Component	Y'_i	$H_{G1} Y'_i, 26^\circ\text{C}$	$L_3 x_3$	$H_3 L_3 x_3$
C_1	0.700	647.5	0	0
C_2	0.1070	1222	0.0430	4.519 t_3
C_3	0.0274	508.5	0.0726	8.45 t_3
C_4	0.01024	288.4	0.07476	10.36 t_3
Oil			3.465	1306.3 t_3
	$0.8446 = G_1$	$2666.4 = H_{G1} G_1$	$3.655 = L_3$	$1329.6 t_3 = H_{L3} L_3$

Eq. (8.11) shows $t_3 = 26.7^\circ\text{C}$, sufficiently close to that of the first estimate (26.8°C) for us to proceed with Eq. (8.102).

Horton-Franklin method Equations (8.105), (8.107), and (8.13), used as above with $G_1 = 0.8446$, now show

Tray n	G_n	L_n	L_n/G_n	$t_n, ^\circ\text{C}$
1	0.8446	3.549	4.202	26.0
2	0.8935	3.600	4.029	26.1
3	0.9453	3.655	3.866	26.7

The temperatures permit tabulation of m 's and the L_n/G_n 's permit calculation of A for each component on each tray. Equation (8.102) then gives Y'_i , and a material balance gives the moles of each component in L_3 :

Component	A_1	A_2	A_3	Y'_i [Eq. (8.102)]	$L_3 x_3$
C_1				0.700	0
C_2	0.314	0.301	0.286	0.1072	0.0428
C_3	1.002	0.959	0.910	0.0273	0.0727
C_4	3.430	3.290	3.093	0.01117	0.0738
Oil					3.465
				$0.8457 = G_1$	$3.654 = L_3$

An enthalpy balance again shows $t_3 = 26.7^\circ\text{C}$.

For most purposes, the preceding results will be reasonably satisfactory. But they are correct only if the A values for each component on each tray are correct. These can be checked by first verifying the L/G ratios. For the components found in both liquid and gas

$$A_n = \frac{L_n}{m_n G_n} = \frac{L_n}{(y_n/x_n) G_n} \quad (8.115)$$

or
$$G_n y_n = \frac{L_n x_n}{A_n} \quad (8.116)$$

If there are G_S mol/time of nonabsorbed gas, then

$$G_n = G_S + \sum \frac{L_n x_n}{A_n} \quad (8.117)$$

and Eq. (8.13) will provide $L_n x_n$, whose sum is L_n . The L_n/G_n ratios can then be compared with those used previously.

Illustration 8.9, continued For tray 3, $G_S = 0.7$ kmol C_1 . For C_2 , $G_3 y_3 = L_3 x_3 / A_3 = 0.0428 / 0.286 = 0.1497$, and Eq. (8.14) with $n = 2$ provides $L_2 x_2 = 0.0428 + 0.1497 - 0.15 = 0.0425$. Similarly,

Component	$G_3 y_3 = L_3 x_3 / A_3$	$L_2 x_2$
C_1	0.700	0
C_2	0.1497	0.0425
C_3	0.0799	0.0526
C_4	0.0239	0.0477
Oil		3.465
	<u>0.9535 = G_3</u>	<u>3.6078 = L_2</u>

Therefore

$$\frac{L_2}{G_3} = \frac{3.654}{0.9535} = 3.832 \text{ (3.866 used previously)}$$

Similarly L_2/G_2 and L_1/G_1 can be computed and compared with values used previously.

The procedure then to be followed is to repeat the use of Eq. (8.102) with the new L/G ratios to provide A 's, repeating the check of L/G and use of Eq. (8.102) until agreement is reached. This still leaves the temperatures to be checked. For an assumed top-tray temperature, an overall tower enthalpy balance provides t_N . With the compositions and flow rates for each stream as last determined, individual tray enthalpy balances provide the temperature of each tray. These are repeated until the assumed t_1 agrees with that calculated. New m 's and A 's are then used with Eq. (8.102) and the entire procedure repeated until a completely consistent set of L/G 's and temperatures are obtained, at which time the problem is solved. The procedure is very tedious, and methods for reducing the work, including the use of high-speed computers, have been given much study [16, 18, 19, 39].

Use of Reflux—Reboiled Absorbers

Refer again to the preceding illustration. Considering only the three substances absorbed, the wet gas contained these roughly in the proportions $C_2H_6 : C_3H_8 : C_4H_{10} = 50 : 33 : 17$. The rich oil leaving the absorber contained the same substances in the approximate ratio $21 : 36 : 37$. Despite its low solubility, the proportion of absorbed gas which is ethane is high, owing to the relatively high proportion of this substance in the original wet gas. If it is desired to reduce the relative proportions of ethane in the rich oil, the oil just as it leaves the tower must be in contact with a gas relatively leaner in ethane and richer in propane and butane. Such a gas can be obtained by heating a part of the rich oil, which will then evolve a gas of the required low ethane content. This evolved gas can then be returned to an extension of the absorber below the inlet of the wet gas, where in rising past the oil it will strip out the ethane (and also any methane which may have been absorbed as well). The ethane not absorbed will now, of course, leave with the lean gas. The heat exchanger where a portion of the rich oil is heated is called a *reboiler*, and the arrangement of Fig. 8.28 can be used. Calculation methods for multicomponents are available [20, 25].

A stream returned to a cascade of stages, as represented by the trays in the absorber, for the purposes of obtaining an enrichment beyond that obtained by countercurrent contact with the feed to the cascade, is called *reflux*. This principle is used extensively in distillation, liquid extraction, and adsorption, but it is applicable to any countercurrent enrichment operation.

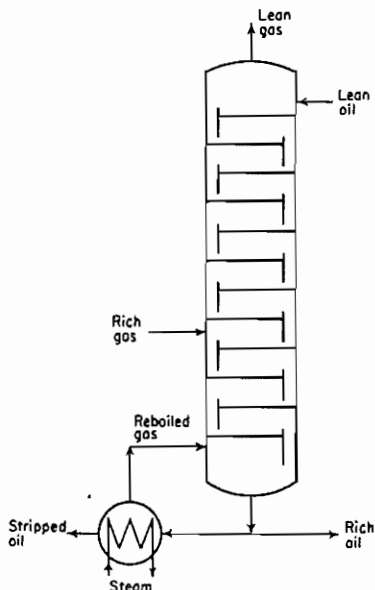


Figure 8.28 Reboiled absorber.

ABSORPTION WITH CHEMICAL REACTION

Many industrial absorption processes are accompanied by chemical reaction. Reaction in the liquid of the absorbed component with a reagent in the absorbing liquid is especially common. Sometimes the reagent and product of reaction are both soluble, as in the absorption of carbon dioxide into an aqueous solution of the ethanolamines or other alkaline solutions. In contrast, furnace gases containing sulfur dioxide can be contacted with slurries of insoluble ground limestone in water, to form insoluble calcium sulfite. There are many other important examples, for which Ref. 28 should be consulted. Reaction of the absorbed solute and a reagent accomplishes two things favorable to the absorption rate: (1) destruction of the absorbed solute as it forms a compound reduces the equilibrium partial pressure of the solute, consequently increasing the concentration difference between the bulk gas and the interface; and the absorption rate is also increased; (2) the liquid-phase mass-transfer coefficient is increased in magnitude, which also contributes to increased absorption rates. These effects have been given extensive theoretical analysis [1, 10, 38] but relatively little experimental verification.

NOTATION FOR CHAPTER 8

Any consistent set of units may be used.

a	specific interface surface, area/packed volume L^2/L^3
A	absorption factor, L/mG , dimensionless
A'	effective absorption factor [Eq. (8.110)], dimensionless
A_E	effective absorption factor [Eq. (8.111)], dimensionless
A, B, C	components A, B, C
c	concentration, moles/volume, mole/L^3
C_J	heat capacity of component J as a gas, $FL/\text{mole } T$
C_L	heat capacity of a liquid, $FL/\text{mole } T$
C_p	heat capacity of a gas at constant pressure, $FL/\text{mole } T$
d_p	equivalent diameter of packing, L
D	diffusivity, L^2/Θ
D_{IJ}	diffusivity of component I in a binary gas IJ mixture, L^2/Θ
D_{Im}	mean diffusivity of I in a multicomponent mixture, L^2/Θ
D_L	liquid diffusivity, L^2/Θ
e	2.7183
E_{MGE}	Murphree gas tray efficiency corrected for entrainment, fractional
E_O	overall tray efficiency, fractional
$F_{G,J}$	gas-phase mass-transfer coefficient for component J, $\text{mole}/L^2\Theta$
$F_{L,J}$	liquid-phase mass-transfer coefficient for component J, $\text{mole}/L^2\Theta$
F_O	overall mass-transfer coefficient, $\text{mole}/L^2\Theta$
G	total gas rate for tray towers, mole/Θ ; for packed towers, superficial molar mass velocity, $\text{mole}/L^2\Theta$
G_J	gas molar mass velocity of component J, $\text{mole}/L^2\Theta$
G'	gas mass velocity $M/L^2\Theta$
h	heat-transfer coefficient, $FL/L^2\Theta T$
h'	heat-transfer coefficient corrected for mass transfer, $FL/L^2\Theta T$

H_G	molar enthalpy of a gas referred to pure substance at temperature t_0 , FL/mole
H'_G	molar enthalpy of a gas per mole of carrier gas, referred to pure substance at temperature t_0 , FL/mole
$\bar{H}_{J,t}$	partial enthalpy of component J in solution at concentration x_J , temperature t_J , FL/mole
H_L	enthalpy of a liquid solution referred to pure substance at temperature t_0 , FL/mole
H_i	height of a transfer unit, L
H_{iO}	height of an overall transfer unit, L
ΔH_S	integral heat of solution per mole of solution, FL/mole
k	thermal conductivity, FL/LT θ
k_G	gas mass-transfer coefficient, mole/L ² θ (F/L ²)
k_L	liquid mass-transfer coefficient, mole/L ² θ (mole/L ³)
k_x	liquid mass-transfer coefficient, mole/L ² θ (mole fraction)
k_y	gas mass-transfer coefficient, mole/L ² θ (mole fraction)
K	overall mass-transfer coefficient (units indicated by subscripts, as for k 's)
$\ln$	natural logarithm
$\log$	common logarithm
L	total molar liquid rate, for tray towers, mole/ θ
L'	total liquid mass rate (tray towers), M/ θ ; total mass rate per unit tower cross section (packed towers), M/L ² θ
L_S	solvent mass rate (tray towers), M/ θ ; total mass rate per unit tower cross section (packed towers), M/L ² θ
m	slope of the equilibrium curve, dy^*/dx ; equilibrium distribution ratio, y^*/x , dimensionless
$m'_{J,r}$	slope of a logarithmic reference-substance plot at concentration x_J , $(\log \bar{p}_J^*)/(\log p_r)$
M	molecular weight, M/mole
N_J	mass-transfer flux of component J, mole/L ² θ
N_p	number of equilibrium trays
N_t	number of transfer units
N_{iO}	number of overall transfer units
p	vapor pressure of a pure substance, F/L ²
$\bar{p}$	partial pressure, F/L ²
p_t	total pressure, F/L ²
Pr	Prandtl number, $C_p \mu / k$, dimensionless
q	flux of heat transfer, FL/L ² θ ; a constant
Q_T	rate of total heat removal, FL/ θ
r	const
R_J	$N_J / \Sigma N$, dimensionless
s	const
S	stripping factor, mG/L , dimensionless
S'	effective stripping factor [Eq. (8.113)], dimensionless
S_E	effective stripping factor [Eq. (8.114)], dimensionless
Sc	Schmidt number, $\mu / \rho D$, dimensionless
t	temperature, T
x	concentration in the liquid, mole fraction, mole/mole
X	concentration in the liquid, mole/mole solvent, mole/mole
X'	concentration in the liquid, mole/mole entering liquid, mole/mole
y	concentration in the gas, mole fraction, mole/mole
Y	concentration in the gas, mole/mole carrier gas, mole/mole
Y'	concentration in the gas, mole/mole entering gas, mole/mole
Z	height of packing, L
Δ	difference
ϵ	volume fraction voids, dry packing, L ³ /L ³
ϵ_o	volume fraction voids, irrigated packing, L ³ /L ³
λ_{JO}	molar latent heat of vaporization of J at temperature t_0 , FL/mole

$\lambda_{r,i}$	molar latent heat of vaporization of reference substance at temperature i , FL/mole
μ	viscosity, M/L Θ
ρ	density, M/L ³
φ_i	fractional total liquid holdup, volume liquid/volume packed space, L ³ /L ³

Subscripts

av	average
A, B, C, J	components A, B, C, J
G	gas
i	interface
L	liquid
M	logarithmic mean
n	effluent from tray n
N _p	effluent from tray N _p
O	overall
r	reference substance
0	liquid entering top tray
	reference temperature for enthalpy
1	bottom of a packed tower; effluent from tray 1 (tray tower)
2	top of a packed tower; effluent from tray 2 (tray tower)

Superscript

*	in equilibrium with the other phase
---	-------------------------------------

REFERENCES

1. Astarita, G.: "Mass Transfer with Chemical Reaction," Elsevier, Amsterdam, 1966.
2. Baker, T. C.: *Ind. Eng. Chem.*, **27**, 977 (1935).
3. Bourne, J. R., and G. C. Coggan: *Chem. Eng. Sci.*, **24**, 196 (1969).
4. Bourne, J. R., U. von Stockar, and G. C. Coggan: *Ind. Eng. Chem. Process Des. Dev.*, **13**, 115, 124 (1974).
5. Chiang, S. H., and H. L. Toor: *AIChE J.*, **10**, 398 (1964).
6. Chilton, T. H., and A. P. Colburn: *Ind. Eng. Chem.*, **27**, 255 (1935).
7. Coggan, G. C., and J. R. Bourne: *Trans. Inst. Chem. Eng. Lond.*, **47**, T96, T160 (1969).
8. Colburn, A. P.: *Trans. AIChE*, **35**, 211 (1939).
9. Colburn, A. P.: *Ind. Eng. Chem.*, **33**, 111 (1941).
10. Danckwerts, P. V.: "Gas-Liquid Reactions," McGraw-Hill Book Company, New York, 1970.
11. De Lancey, G. B., and S. H. Chiang: *Ind. Eng. Chem. Fundam.*, **9**, 138 (1970).
12. Denbigh, K.: "Principles of Chemical Equilibrium," Cambridge University Press, London, 1961.
13. Edmister, W. C.: *Ind. Eng. Chem.*, **35**, 837 (1943).
14. Fair, J. R., in R. H. Perry and C. H. Chilton (eds.), "The Chemical Engineers' Handbook," 5th ed., pp. 18-32 to 18-39, McGraw-Hill Book Company, New York, 1973.
15. Feintuch, H. M., and R. E. Treybal: *Ind. Eng. Chem. Process Des. Dev.*, in press.
16. Friday, J. R., and B. D. Smith: *AIChE J.*, **10**, 698 (1964).
17. Hachmuth, K. H.: *Chem. Eng. Prog.*, **45**, 716 (1949); **47**, 523, 621 (1951).
18. Holland, C. D.: "Multicomponent Distillation," Prentice-Hall, Englewood Cliffs, N. J., 1963.
19. Holland, C. D.: "Fundamentals and Modeling of Separation Processes," Prentice-Hall, Englewood Cliffs, N.J., 1975.
20. Holland, C. D., G. P. Pendon, and S. E. Galloway: *Hydrocarbon Process.*, **54**(1), 101 (1975).
21. Horton, G., and W. B. Franklin: *Ind. Eng. Chem.*, **32**, 1384 (1940).
22. Hougen, O. A., Watson, K. M., and Ragatz, R. A.: "Chemical Process Principles, Part One," 2d ed., Wiley, New York, 1954.

23. Hulsworth, C. E.: *Chem. Eng. Prog.*, **69**(2), 50 (1973).
24. "International Critical Tables," vol. V, p. 213, McGraw-Hill Book Company, New York, 1929.
25. Ishii, Y., and F. D. Otto: *Can. J. Chem. Eng.*, **51**, 601 (1973).
26. Khurani, M. K.: private communication, Dec. 13, 1969.
27. King, M. B.: "Phase Equilibrium in Mixtures," Pergamon, London, 1969.
28. Kohl, A., and F. Riesenfeld: "Gas Purification," 2d ed., Gulf Publishing, Houston, 1974.
29. Koyashi, R., and D. L. Katz: *Ind. Eng. Chem.*, **45**, 440, 446 (1953).
30. Kwauk, M.: *AIChE J.*, **2**, 240 (1956).
31. Liley, P. E., and W. R. Gambill, in R. H. Perry and C. H. Chilton (eds.), "The Chemical Engineers' Handbook," 5th ed., sec. 3, McGraw-Hill Book Company, New York, 1973.
32. Othmer, D. F., et al.: *Ind. Eng. Chem.*, **36**, 858 (1944); **51**, 89 (1959).
33. Owens, W. R., and R. N. Maddox: *Ind. Eng. Chem.*, **60**(12), 14 (1968).
34. Prausnitz, J. M.: "Molecular Thermodynamics of Fluid Phase Equilibria," Prentice-Hall, Englewood Cliffs, N.J., 1969.
35. Pray, H. A., C. C. Schweickert, and B. H. Minnich: *Ind. Eng. Chem.*, **44**, 1146 (1952).
36. Raal, J. D., and M. K. Khurani: *Can. J. Chem. Eng.*, **51**, 162 (1973); *Process Technol. Int.*, **18**(6-7), 267 (1973).
37. Rackett, H. G.: *Chem. Eng.*, **71**, 108 (Dec. 21, 1964).
38. Sherwood, T. K., R. L. Pigford, and C. R. Wilke: "Mass Transfer," McGraw-Hill Book Company, New York, 1975.
39. Smith, B. D.: "Design of Equilibrium Stage Processes," McGraw-Hill Book Company, New York, 1963.
40. Smith, J. M., and H. C. Van Ness: "Introduction to Chemical Engineering Thermodynamics," 3d ed., McGraw-Hill Book Company, New York, 1975.
41. Treybal, R. E.: *Ind. Eng. Chem.*, **61**(7), 36 (1969).
42. Verma, S. L., and G. B. de Lancey: *AIChE J.*, **21**, 96 (1975).
43. Wankat, C. P.: *Ind. Eng. Chem. Process Des. Dev.*, **11**, 302 (1972).

PROBLEMS

8.1 The equilibrium partial pressures of carbon dioxide over aqueous solutions of monoethanolamine (30 wt%) [Mason and Dodge, *Trans. AIChE*, **32**, 27 (1936)] are:

$\frac{\text{mol CO}_2}{\text{mol solution}}$	Partial pressure CO ₂ , mmHg		
	25°C	50°C	75°C
0.050			65
0.052		7.5	93.5
0.054		13.6	142.6
0.056		25.0	245
0.058	5.6	47.1	600
0.060	12.8	96.0	
0.062	29.0	259	
0.064	56.0		
0.066	98.7		
0.068	155		
0.070	232		

A plant manufacturing dry ice will burn coke in air to produce a flue gas which, when cleaned and cooled, will contain 15% CO₂, 6% O₂, 79.0% N₂. The gas will be blown into a sieve-tray-tower scrubber at 1.2 std atm, 25°C, to be scrubbed with a 30% ethanolamine solution entering at 25°C. The scrubbing liquid, which is recycled from a stripper, will contain 0.058 mol CO₂/mol solution.

The gas leaving the scrubber is to contain 2% CO₂. Assume isothermal operation.

(a) Determine the minimum liquid/gas ratio, mol/mol.

(b) Determine the number of kilograms of solution to enter the absorber per cubic meter of entering gas, for an L/G ratio of 1.2 times the minimum.

Ans.: 18.66.

(c) Determine the number of theoretical trays for the conditions of part (b).

Ans.: 2.5.

(d) The viscosity of the solution is 6.0 cP; sp gr = 1.012. Estimate the average m and the overall tray efficiency to be expected. How many real trays are required?

8.2 (a) Determine the ideal trays for the absorption of Prob. 8.1, assuming adiabatic operation. Use 25.6 kg absorbent solution/m³ gas in, which will be about 1.2 times the minimum. The heat of solution of CO₂ is 1675 kJ evolved/kg CO₂ absorbed (720 Btu/lb), referred to gaseous CO₂ and liquid solution. The specific heat of the solution is 3.433 kJ/(kg solution) · K (0.82 Btu/lb · °F).

Ans.: 2.6.

(b) Suppose the absorber planned for isothermal operation (L/G and theoretical trays of Prob. 8.1) were operated adiabatically. What concentration of CO₂ in the exit gas could be expected? [Note that this normally requires trial-and-error determination of both the top-tray temperature and effluent gas concentration. However, study of the calculations of part (a) should indicate that in this case the top-tray temperature need not be determined by trial.]

Ans.: 4.59%.

8.3 Derive Eq. (8.16). *Hint:* Start with the definition of E_{MGE} and locate a *pseudo-equilibrium line*, which would be used together with the operating line for graphically constructing steps representing real trays. Then use the Kremser equation (5.55a) with the pseudo-equilibrium line, by moving the origin of the Y, X diagram to the intercept of the pseudo-equilibrium line with the Y axis.

8.4 Carbon disulfide, CS₂, used as a solvent in a chemical plant, is evaporated from the product in a drier into an inert gas (essentially N₂) in order to avoid an explosion hazard. The vapor-N₂ mixture is to be scrubbed with an absorbent hydrocarbon oil, which will be subsequently steam-stripped to recover the CS₂. The CS₂-N₂ mixture has a partial pressure of CS₂ equal to 50 mmHg at 24°C (75°F) and is to be blown into the absorber at essentially standard atmospheric pressure at the expected rate of 0.40 m³/s (50 000 ft³/h). The vapor content of the gas is to be reduced to 0.5%. The absorption oil has av mol wt 180, viscosity 2 cP, and sp gr 0.81 at 24°C. The oil enters the absorber essentially stripped of all CS₂, and solutions of oil and CS₂, while not actually ideal, follow Raoult's law [see Ewell et al.: *Ind. Eng. Chem.*, 36, 871 (1944)]. The vapor pressure of CS₂ at 24°C = 346 mmHg. Assume isothermal operation.

(a) Determine the minimum liquid/gas ratio.

(b) For a liquid/gas ratio of 1.5 times the minimum, determine the kilograms of oil to enter the absorber per hour.

(c) Determine the number of theoretical trays required, both graphically and analytically.

(d) For a conventional sieve-tray tower, estimate the overall tray efficiency to be expected and the number of real trays required.

8.5 Design a suitable sieve tray for the absorber of Prob. 8.4 and compute its hydraulics (which may be considered constant for all the trays) and the number of real trays required, graphically and through Eq. (8.16). Take the surface tension as 30 dyn/cm.

8.6 Determine the number of equilibrium trays for the absorber of Prob. 8.4, assuming adiabatic operation. Use a liquid rate of 2.27 kg/s, which is about 1.5 times the minimum. The specific heats are:

	kJ/kmol · K	Btu/lb mol · °F
Oil	362.2	86.5
CS ₂ , liquid	76.2	18.2
CS ₂ , vapor	46.89	11.2

The latent heat of vaporization of CS₂ is 27 910 kJ/kmol at 24°C (12 000 Btu/lb mol).

Ans.: 3.6.

8.7 Starting with Eq. (8.34), replace the y 's by the equivalent Y 's and derive Eq. (8.36).

8.8 With the help of the Kremser equation and Eq. (8.50), derive the relation between N_p and N_{IOG} for constant absorption factor. Establish the conditions when $N_p = N_{IOG}$.

8.9 Design a tower packed with 50-mm (2-in) ceramic Raschig rings for the carbon disulfide scrubber of Prob. 8.4. Assume isothermal operation and use a liquid/gas ratio of 1.5 times the minimum and a gas-pressure drop not exceeding $327 \text{ (N/m}^2\text{)/m}$ ($0.4 \text{ inH}_2\text{O/ft}$) of packing. The liquid surface tension = 30 dyn/cm . A procedure follows.

(a) Determine the tower diameter.

Ans.: 0.725 m.

(b) Using average (top and bottom) flow rates and fluid properties, compute the mass-transfer coefficients $F_G a$ and $F_L a$ and heights of transfer units H_{IG} , H_{IL} , and H_{IOG} .

(c) Compute N_{IOG} and with H_{IG} , the packing height.

Ans.: 9.61, 4.73 m.

(d) Compute N_{IOG} through the following methods and the corresponding packing height for each: Eq. (8.36), Eq. (8.48), and Fig. 8.20.

(e) Compare the gas-pressure drop for the full depth of packing with that for all the trays of Prob. 8.5. At a power cost of 15 cents per kilowatt-hour and a blower-motor efficiency of 50%, calculate the annual (350-day) difference in power cost for the two towers.

8.10 For dilute mixtures and cases when Henry's law applies, prove that the number of overall transfer units for cocurrent gas absorption in packed towers is given by

$$N_{IOG} = \frac{A}{A+1} \ln \frac{y_1 - mx_1}{y_2 - mx_2}$$

where subscript 1 indicates the top (where gas and liquid enter) and subscript 2 indicates the bottom of the tower.

8.11 Benzene vapor in a coke-oven gas is scrubbed from the gas with wash oil in a countercurrent packed scrubber. The resulting benzene-wash-oil solution is then heated to 125°C and stripped in a tray tower, using steam as the stripping medium. The stripped wash oil is then cooled and recycled to the absorber. Some data relative to the operation follow:

Absorption:

Benzene in entering gas = 1 mol %

Pressure of absorber = 800 mmHg

Oil circulation rate = $2 \text{ m}^3/1000 \text{ m}^3$ gas at STP ($15 \text{ U.S. gal/1000 ft}^3$)

Oil sp gr = 0.88 mol wt = 260

Henry's-law constant $m = \frac{y^*}{x}$ for benzene-wash-oil = $\begin{cases} 0.095 & \text{at } 25^\circ\text{C} \\ 0.130 & \text{at } 27^\circ\text{C} \end{cases}$

N_{IOG} = 5 overall gas transfer units

Stripping:

Pressure = 1 std atm steam = 1 std atm, 125°C

Henry's-law constant for benzene $m = \frac{y^*}{x} = 3.08 \text{ at } 125^\circ\text{C}$

Number of equilibrium stages = 5

(a) In the winter, it is possible to cool the recycled stripped oil to 20°C , at which temperature the absorber then operates. Under these conditions 72.0 kg steam is used in the stripper per 1000 m^3 gas at STP (4.5 lb/1000 ft^3). Calculate the percent benzene recovery from the coke-oven gas in the winter.

Ans.: 92.6%.

(b) In the summer it is impossible to cool the recycled wash oil to lower than 27°C with the available cooling water. It has been suggested that the steam rate to the stripper can be increased so as to obtain the same percentage benzene recovery as in the winter. Assuming that the absorber then operates at 27°C, with the same oil rate, and that N_{IOG} and equilibrium stages remain the same, what steam rate in the summer would be required?

(c) If the oil rate cannot be increased but the steam rate in the summer is increased by 20% over the winter value, what summer recovery of benzene can be expected?

Ans.: 86.2%.

8.12 It is desired to reduce the ammonia content of 0.0472 m³/s (26.7°C, 1 std atm) (6000 ft³/h) of an ammonia-air mixture from 5.0 to 0.04% by volume by water scrubbing. There is available a 0.305-m (1-ft)-diameter tower packed with 25-mm Berl saddles to a depth of 3.66 m (12 ft). Is the tower satisfactory, and if so, what water rate should be used? At 26.7°C, ammonia-water solutions follow Henry's law up to 5 mol% ammonia in the liquid, and $m = 1.414$.

8.13 A tower, 0.6 m diameter, packed with 50-mm ceramic Raschig rings to a depth of 1.2 m, is to be used for producing a solution of oxygen in water for certain pollution-control operations. The packed space will be pressurized at 5 std atm abs with pure oxygen gas from a gas cylinder. There is to be no gas outlet, and gas will enter from the cylinder only to maintain pressure as oxygen is dissolved. Water will flow down the packing continuously at 1.50 kg/s. The temperature is to be 25°C. Assuming that the entering water is oxygen-free, what concentration of oxygen can be expected in the water effluent?

Ans.: 1.108×10^{-4} mole fraction.

Data: at 25°C, the diffusivity of O₂ in water = 2.5×10^{-5} cm²/s, viscosity of water = 0.894 cP, and the solubility of O₂ in H₂O follows Henry's law: $\bar{p} = 4.38 \times 10^4 x$, where $\bar{p}$ = equilibrium partial pressure, atm, and x = mole fraction O₂ in the liquid. Neglect water-vapor content of the gas.

8.14 A system for recovering methanol from a solid product wet with methanol involves evaporation of the methanol into a stream of inert gas, essentially nitrogen. In order to recover the methanol from the nitrogen, an absorption scheme involving washing the gas countercurrently with water in a packed tower is being considered. The resulting methanol-water solution is then to be distilled to recover the methanol. Note: An alternative scheme is to wash the gas with refrigerated methanol, whence no distillation is required (see Prob. 7.11). A still different approach is to adsorb the methanol onto activated carbon (see Chap. 11).

The absorption tower will be filled with 38-mm (1.5-in) ceramic Raschig rings. Use a gas-pressure drop not to exceed 400 N/m² per meter of packed depth. Tower shells are available in 25-mm-diameter increments.

Partial pressures of methanol and water over aqueous solutions of methanol are available in "The Chemical Engineers' Handbook," 5th ed., p. 3-68. Plot these in the manner of Fig. 8.2, using water as reference substance.

(a) Assume isothermal operation at 26.7°C (80°F). Gas in: 0.70 m³/s (90 000 ft³/h) at 1 std atm, 26.7°C; partial pressure methanol = 100 mmHg. $Sc_G = 0.783$ (MeOH-N₂). Gas out: methanol partial pressure = 15 mmHg. Liquid in: water, methanol-free, 26.7°C, 0.15 kg/s (1190 lb/h); neglect evaporation of water. Calculate the depth of packing required.

(b) Assume adiabatic operation. Gas in: 0.70 m³/s at 1 std atm, 26.7°C, partial pressure methanol = 100 mmHg, partial pressure water = 21.4 mmHg. Gas out: methanol partial pressure = 15 mmHg. Liquid in: water, methanol-free, 26.7°C, 1.14 kg/s (9000 lb/h). Assume that the gas leaves at $t_G = 30^\circ\text{C}$, with $\bar{p}_{H_2O} = 26$ mmHg. Do not neglect evaporation of water. Calculate the following for the bottom of the tower: t_L , t_i , dt_G/dZ , dY_A/dZ , dY_C/dZ . For $\Delta Y_A = -0.02$, compute the next value of Z , Y_A , Y_S , t_G , and t_L .

Additional data: heat of solution ["International Critical Tables," vol. IV, p. 159]:

x_A , mole fraction MeOH in liquid	0.5	0.10	0.15	0.20	0.25	0.30
ΔH_S , integral heat of solution, kJ/kmol soln at 20°C	-341.7	-597.8	-767.6	-872.3	-914.1	-916.4

Use a base temperature $t_0 = 20^\circ\text{C}$. The latent heat of vaporization of methanol at $20^\circ\text{C} = 1163$ kJ/kg. Heat capacities of liquid solutions: see "Chemical Engineers' Handbook," 5th ed., p. 3-136. Molar heat capacities of gases: N_2 , 29.141; MeOH , 52.337; H_2O , 33.579 kJ/kmol $\cdot$ K.

8.15 A gas containing 88% CH_4 , 4% C_2H_6 , 5% $n\text{-C}_3\text{H}_8$, and 3% $n\text{-C}_4\text{H}_{10}$ is to be scrubbed isothermally at 37.8°C (100°F), 5 std atm pressure, in a tower containing the equivalent of eight equilibrium trays, 80% of the C_3H_8 is to be removed. The lean oil will contain 0.5 mol% of C_4H_{10} but none of the other gaseous constituents. The rest of the oil is nonvolatile. What quantity of lean oil, mol/mol wet gas, should be used, and what will be the composition of rich oil and scrubbed gas? The Henry's-law constants m are

CH_4	C_2H_6	$n\text{-C}_3\text{H}_8$	$n\text{-C}_4\text{H}_{10}$
32	6.7	2.4	0.74

8.16 An absorber of four theoretical trays, to operate adiabatically at 1.034 kN/m² (150 lb_f/in²), is fed with 1 mol per unit time each of liquid and gas, each entering at 32.2°C (90°F), as follows:

Component	Liquid, mole fraction	Gas, mole fraction
CH_4		0.70
C_2H_6		0.12
$n\text{-C}_3\text{H}_8$		0.08
$n\text{-C}_4\text{H}_{10}$	0.02	0.06
$n\text{-C}_5\text{H}_{12}$	0.01	0.04
Nonvolatile oil	0.97	

Estimate the composition and rate (mol/time) of the exit gas.

Data: Henry's-law constants, enthalpies at 32.2°C relative to saturated liquid at -129°C (-200°F), and molar heat capacities are given in Table 8.1.

Table 8.1 Data for Prob. 8.16

Component	m	32.2°C (90°F)		m		C_p , kJ/kmol $\cdot$ K (Btu/lb mol $\cdot$ °F)	
		$H_G^\dagger$ kJ/kmol $\times 10^{-3}$ (Btu/lb mol)	H_L kJ/kmol $\times 10^{-3}$ (Btu/lb mol)	37.8°C (100°F)	43°C (110°F)	Gas	Liquid
CH_4	16.5	12.91 (5510)	9.77 (4200)	17.0	17.8	37.7 (9.0)	50.2 (12.01)
C_2H_6	3.40	22.56 (9700)	15.58 (6700)	3.80	4.03	62.8 (15.0)	83.7 (20.0)
$n\text{-C}_3\text{H}_8$	1.16	31.05 (13 350)	16.86 (7250)	1.30	1.44	79.6 (19.0)	129.8 (31.0)
$n\text{-C}_4\text{H}_{10}$	0.35	41.05 (17 650)	20.70 (8900)	0.41	0.47	96.3 (23.0)	159.1 (38.0)
$n\text{-C}_5\text{H}_{12}$	0.123	50.94 (21 400)	24.66 (10 600)	0.140	0.165	117.2 (28.0)	184.2 (44.0)
Oil							376.8 (90.0)

$\dagger$ Data from DePriester, *Chem. Eng. Progr. Symp. Ser.*, 49(7) 1 (1953); "Chemical Engineers' Handbook," 5th ed., pp. 13-12 and 13-13; Maxwell, "Data Book on Hydrocarbons," Van Nostrand, Princeton, N.J., 1950.

8.17 In this chapter it was shown that there is an optimum, or most economical, value of the absorption factor. There are also optimum values for:

- (a) Outlet gas concentration when there are valuable solutes
- (b) The solute concentration in the liquid entering the absorber where the absorbent liquid is recirculated through a stripper or fractionator for solute recovery
- (c) The tray spacing in a tray tower
- (d) The gas pressure drop per unit height in a packed tower

For each item explain why there is an optimum value.

DISTILLATION

Distillation is a method of separating the components of a solution which depends upon the distribution of the substances between a gas and a liquid phase, applied to cases where all components are present in both phases. Instead of introducing a new substance into the mixture in order to provide the second phase, as is done in gas absorption or desorption, the new phase is created from the original solution by vaporization or condensation.

In order to make clear the distinction between distillation and the other operations, let us cite a few specific examples. In the separation of a solution of common salt and water, the water can be completely vaporized from the solution without removal of salt since the latter is for all practical purposes quite nonvolatile at the prevailing conditions. This is the operation of evaporation. Distillation, on the other hand, is concerned with the separation of solutions where all the components are appreciably volatile. In this category, consider the separation of the components of a liquid solution of ammonia and water. By contacting the ammonia-water solution with air, which is essentially insoluble in the liquid, the ammonia can be stripped or desorbed by processes which were discussed in Chap. 8, but the ammonia is then mixed with water vapor and air and is not obtained in pure form. On the other hand, by application of heat, we can partially vaporize the solution and thereby create a gas phase consisting of nothing but water and ammonia. And since the gas will be richer in ammonia than the residual liquid, a certain amount of separation will have resulted. By appropriate manipulation of the phases or by repeated vaporizations and condensations it is then ordinarily possible to make as complete a separation as may be desired, recovering both components of the mixture in as pure a state as we wish.

The advantages of such a separation method are clear. In distillation the new phase differs from the original by its heat content, but heat is readily added or removed, although of course the cost of doing this must inevitably be considered. Absorption or desorption operations, on the other hand, which depend upon the introduction of a foreign substance, provide us with a new solution which in turn may have to be separated by one of the diffusional operations unless it happens that the new solution is useful directly.

There are in turn certain limitations to distillation as a separation process. In absorption or similar operations, where it has been agreed to introduce a foreign substance to provide a new phase for distribution purposes, we can ordinarily choose from a great variety of solvents in order to provide the greatest possible separation effect. For example, since water is ineffectual in absorbing hydrocarbon gases from a gas mixture, we choose instead a hydrocarbon oil which provides a high solubility. But in distillation there is no such choice. The gas which can be created from a liquid by application of heat inevitably consists only of the components constituting the liquid. Since the gas is therefore chemically very similar to the liquid, the change in composition resulting from the distribution of the components between the two phases is ordinarily not very great. Indeed, in some cases the change in composition is so small that the process becomes impractical; it may even happen that there is no change in composition whatsoever.

Nevertheless the direct separation, which is ordinarily possible by distillation, into pure products requiring no further processing has made this perhaps the most important of all the mass-transfer operations.

VAPOR-LIQUID EQUILIBRIA

The successful application of distillation methods depends greatly upon an understanding of the equilibria existing between the vapor and liquid phases of the mixtures encountered. A brief review of these is therefore essential. The emphasis here will be on binary mixtures.

Pressure-Temperature-Concentration Phase Diagram

Let us first consider binary mixtures which we shall term *ordinary*; by this is meant that the liquid components dissolve in all proportions to form homogeneous solutions which are not necessarily ideal and that no complications of maximum or minimum boiling points occur. We shall consider component A of the binary mixture A-B as the more volatile, i.e., the vapor pressure of pure A at any temperature is higher than the vapor pressure of pure B. The vapor-liquid equilibrium for each pure substance of the mixture is of course its vapor-pressure-temperature relationship, as indicated in Fig. 7.1. For binary mixtures an additional variable, concentration, must likewise be considered. Mole fractions are the most convenient concentration terms to use, and throughout this discus-

sion x will be the mole fraction of the more volatile substance A in the liquid and y^* the corresponding equilibrium mole fraction of A in the vapor.

Complete graphical representation of the equilibria requires a three-dimensional diagram [29, 47], as in Fig. 9.1. The curve marked p_A is the vapor-pressure curve of A, lying entirely in the nearest composition plane at $x = 1.0$. The curve extends from its critical point C_A to its triple point T_A , but the complications of the solid phase which do not enter into distillation operations will not be considered. Similarly curve p_B is the vapor pressure of pure B, in the far plane at $x = 0$. The liquid and vapor regions at compositions between $x = 0$ and 1.0 are separated by a double surface which extends smoothly from p_A to p_B . The shape of this double surface is most readily studied by considering sections at constant pressure and constant temperature, examples of which are shown in the figure.

Constant-Pressure Equilibria

Consider first a typical section at constant pressure (Fig. 9.2a). The intersection of the double surface of Fig. 9.1 with the constant-pressure plane produces a looped curve without maxima or minima extending from the boiling point of pure B to that of pure A at the pressure in question. The upper curve provides the temperature-vapor composition (t - y^*) relationship, the lower that of the temperature-liquid composition (t - x). Liquid and vapor mixtures at equilibrium are at the same temperature and pressure throughout, so that horizontal *tie lines*

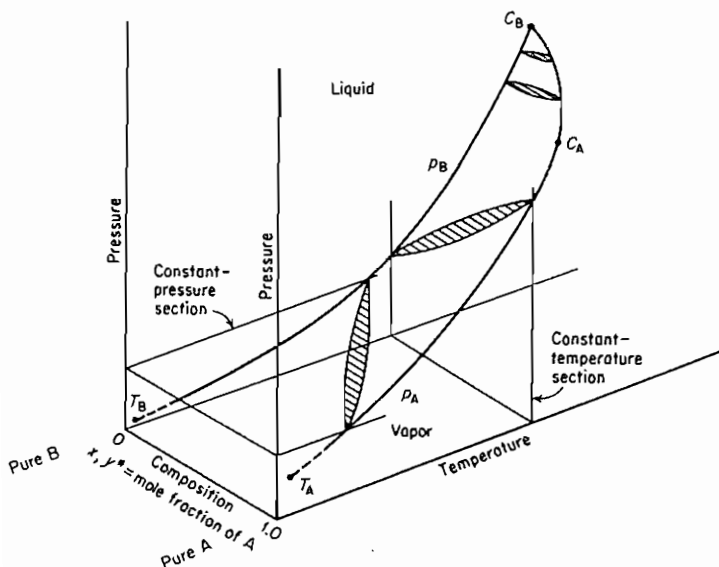


Figure 9.1 Binary vapor-liquid equilibria.

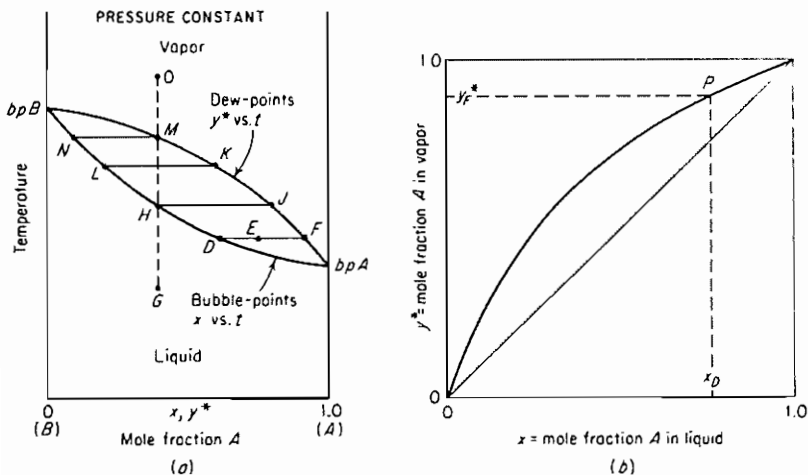


Figure 9.2 Constant-pressure vapor-liquid equilibria.

such as line DF join equilibrium mixtures at D and F . There are an infinite number of such tie lines for this diagram. A mixture on the lower curve, as at point D , is a saturated liquid; a mixture on the upper curve, as at F , is a saturated vapor. A mixture at E is a two-phase mixture, consisting of a liquid phase of composition at D and a vapor phase of composition at F in such proportions that the average composition of the entire mixture is represented by E . The relative amounts of the equilibrium phases are related to the segments of the tie line,

$$\frac{\text{Moles of } D}{\text{Moles of } F} = \frac{\text{line } EF}{\text{line } DE} \quad (9.1)$$

Consider a solution at G in a closed container which can be kept at constant pressure by moving a piston. The solution is entirely liquid. If it is heated, the first bubble of vapor forms at H and has the composition at J , richer in the more volatile substance, and hence the lower curve is called the *bubble-point-temperature curve*. As more of the mixture is vaporized, more of the vapor forms at the expense of the liquid, giving rise, for example, to liquid L and its equilibrium vapor K , although the composition of the entire mass is still the original as at G . The last drop of liquid vaporizes at M and has the composition at N . Superheating the mixture follows the path MO . The mixture has vaporized over a temperature range from H to M , unlike the single vaporization temperature of a pure substance. Thus, the term *boiling point* for a solution ordinarily has no meaning since vaporization occurs over a temperature range, i.e., from the bubble point to the dew point. If the mixture at O is cooled, all the phenomena reappear in reverse order. Condensation, for example, starts at M , whence the upper curve is termed the dew-point curve, and continues to H .

If a solution like that at H is boiled in an open vessel, on the other hand, with the vapors escaping into the atmosphere, since the vapor is richer in the more volatile substance, the liquid residue must therefore become leaner. The temperature and composition of the saturated residual liquid therefore move along the lower curve toward N as the distillation proceeds.

The vapor-liquid equilibrium compositions can also be shown on a distribution diagram (x vs. y^*) as in Fig. 9.2*b*. Point P on the diagram represents the tie line DF , for example. Since the vapor is richer in the more volatile substance, the curve lies above the 45° diagonal line, which has been drawn in for comparison.

Relative Volatility

The greater the distance between the equilibrium curve and the diagonal of Fig. 9.2*b*, the greater the difference in liquid and vapor compositions and the easier the separation by distillation. One numerical measure of this is called the *separation factor* or, particularly in the case of distillation, the *relative volatility* α . This is the ratio of the concentration ratio of A and B in one phase to that in the other and is a measure of the separability,

$$\alpha = \frac{y^*/(1-y^*)}{x/(1-x)} = \frac{y^*(1-x)}{x(1-y^*)} \quad (9.2)$$

The value of α will ordinarily change as x varies from 0 to 1.0. If $y^* = x$ (except at $x = 0$ or 1), $\alpha = 1.0$ and no separation is possible. The larger the value of α above unity, the greater the degree of separability.

Increased Pressures

At higher pressures the sections at constant pressure will of course intersect the double surface of Fig. 9.1 at increased temperatures. The intersections can be projected onto a single plane, as in Fig. 9.3*a*. It should be noted that not only do the looped curves occur at higher temperatures, but they also usually become narrower. This is readily seen from the corresponding distribution curves of Fig. 9.3*b*. The relative volatilities and hence the separability therefore usually become less at higher pressures. As the critical pressure of one component is exceeded, there is no longer a distinction between vapor and liquid for that component, and for mixtures the looped curves are therefore shorter, as at pressures above p_{13} , the critical pressure for A in the figure. Distillation separations can be made only in the region where a looped curve exists.

For particular systems, the critical pressure of the less volatile substance may be reached before that of the more volatile, and it is also possible that the double surface of Fig. 9.1 will extend at intermediate compositions to a small extent beyond the critical pressures of either substance.

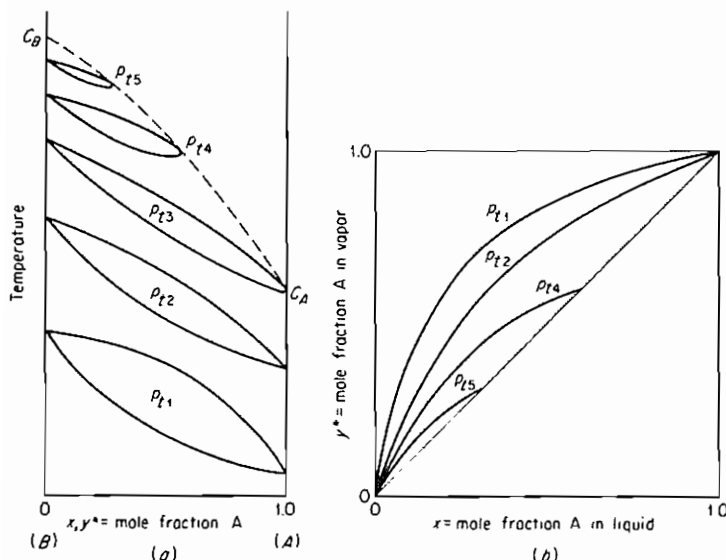


Figure 9.3 Vapor-liquid equilibria at increased pressures.

Constant-Temperature Equilibria

A typical constant-temperature section of the three-dimensional phase diagram is shown in Fig. 9.4. The intersection of the constant-temperature plane with the double surface of Fig. 9.1 provides the two curves which extend without maxima or minima from the vapor pressure of pure B to that of pure A. As before, there are an infinite number of horizontal tie lines, such as TV , which join an

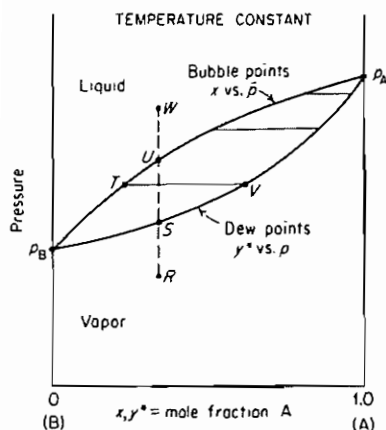


Figure 9.4 Constant-temperature vapor-liquid equilibria.

equilibrium vapor as at V to its corresponding liquid at T . A solution in a closed container at W is entirely a liquid, and if the pressure is reduced at constant temperature, the first bubble of vapor forms at U , complete vaporization occurs at S , and further reduction in pressure results in a superheated vapor as at R .

Vapor-liquid equilibrium data, except in the special situations of ideal and regular solutions, must be determined experimentally. Descriptions of experimental methods [18], extensive bibliographies [67], and lists of data [7, 22] are available.

Ideal Solutions—Raoult's Law

Before studying the characteristics of mixtures which deviate markedly from those just described, let us consider the equilibria for the limiting case of mixtures whose vapors and liquids are ideal. The nature of ideal solutions and the types of mixtures which approach ideality were discussed in Chap. 8.

For an ideal solution, the equilibrium partial pressure $\bar{p}^*$ of a constituent at a fixed temperature equals the product of its vapor pressure p when pure at this temperature and its mole fraction in the liquid. This is Raoult's law.

$$\bar{p}_A^* = p_A x \quad \bar{p}_B^* = p_B(1 - x) \quad (9.3)$$

If the vapor phase is also ideal,

$$p_t = \bar{p}_A^* + \bar{p}_B^* = p_A x + p_B(1 - x) \quad (9.4)$$

and the total as well as the partial pressures are linear in x at a fixed temperature. These relationships are shown graphically in Fig. 9.5. The equilibrium vapor composition can then be computed at this temperature. For example, the value of y^* at point D on the figure equals the ratio of the

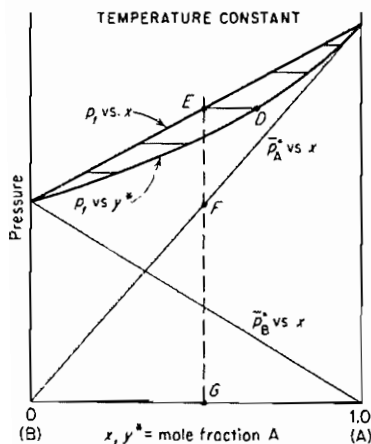


Figure 9.5 Ideal solutions.

distances FG to EG ,

$$y^* = \frac{\bar{p}_A^*}{p_i} = \frac{p_A x}{p_i} \quad (9.5)$$

$$1 - y^* = \frac{\bar{p}_B^*}{p_i} = \frac{p_B(1 - x)}{p_i} \quad (9.6)$$

The relative volatility α is, by substitution in Eq. (9.2),

$$\alpha = \frac{p_A}{p_B} \quad (9.7)$$

For ideal solutions, it is then possible to compute the entire vapor-liquid equilibria from the vapor pressures of the pure substances. For pressures too high to apply the ideal-gas law, fugacities are used instead of pressures [29, 42]. It is also possible to compute the equilibria for the special class of solutions known as *regular* [21]. For all other mixtures, however, it is necessary to obtain the data experimentally.

Illustration 9.1 Compute the vapor-liquid equilibria at constant pressure of 1 std atm for mixtures of *n*-heptane with *n*-octane, which may be expected to form ideal solutions.

SOLUTION The boiling points at 1 std atm of the substances are *n*-heptane (A), 98.4°C and *n*-octane (B), 125.6°C. Computations are therefore made between these temperatures. For example, at 110°C, $p_A = 1050$ mmHg, $p_B = 484$ mmHg, $p_i = 760$ mmHg. Eq. (9.4):

$$x = \frac{p_i - p_B}{p_A - p_B} = \frac{760 - 484}{1050 - 484} = 0.487 \text{ mole fraction heptane in liquid}$$

Eq. (9.5):

$$y^* = \frac{p_A x}{p_i} = \frac{1050(0.487)}{760} = 0.674 \text{ mole fraction heptane in vapor}$$

Eq. (9.7):

$$\alpha = \frac{p_A}{p_B} = \frac{1050}{484} = 2.17$$

In similar fashion, the data of the following table can be computed:

$t, ^\circ\text{C}$	p_A, mmHg	p_B, mmHg	x	y^*	α
98.4	760	333	1.0	1.0	2.28
105	940	417	0.655	0.810	2.25
110	1050	484	0.487	0.674	2.17
115	1200	561	0.312	0.492	2.14
120	1350	650	0.1571	0.279	2.08
125.6	1540	760	0	0	2.02

Curves of the type of Fig. 9.2 can now be plotted. Note that although the vapor pressures of the pure substances vary considerably with temperature, α for ideal solutions does not. In this case, an average of the computed α 's is 2.16, and substituting this in Eq. (9.2), rearranged,

$$y^* = \frac{\alpha x}{1 + x(\alpha - 1)} = \frac{2.16x}{1 + 1.16x}$$

provides an expression which for many purposes is a satisfactory empirical relation between y^* and x for this system at 1 atm.

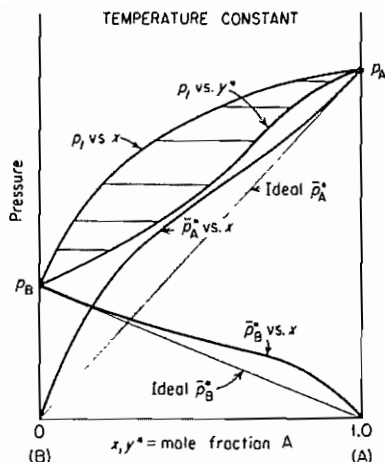


Figure 9.6 Positive deviations from ideality.

Positive Deviations from Ideality

A mixture whose total pressure is greater than that computed for ideality [Eq. (9.4)] is said to show positive deviations from Raoult's law. Most mixtures fall into this category. In these cases the partial pressures of each component are larger than the ideal, as shown in Fig. 9.6.† It should be noted that as the concentration for each component approaches unity mole fraction, the partial pressures for that substance approach ideality tangentially. Raoult's law, in other words, is nearly applicable to the substance present in very large concentrations. This is the case for all substances except where association within the vapor or electrolytic dissociation within the liquid occurs.

The distribution diagram (x vs. y^*) for systems of this type appears much the same as that of Fig. 9.2b.

Minimum-boiling mixtures—azeotropes When the positive deviations from ideality are sufficiently large and the vapor pressures of the two components are not too far apart, the total-pressure curves at constant temperature may rise through a maximum at some concentration, as in Fig. 9.7a. Such a mixture is said to form an *azeotrope*, or *constant-boiling mixture*. The significance of this is more readily seen by study of the constant-pressure section (Fig. 9.7b or c). The liquid- and vapor-composition curves are tangent at point L , the point of

† The ratio of the actual equilibrium partial pressure of a component $\bar{p}^*$ to the ideal value px is the activity coefficient referred to the pure substance: $\gamma = \bar{p}^*/px$. Since γ is greater than unity in these cases and $\log \gamma$ is therefore positive, the deviations are termed positive deviations from ideality. A very extensive science of the treatment of nonideal solutions through activity coefficients has been developed by which, from a very small number of data, all the vapor-liquid equilibria of a system can be predicted [42].

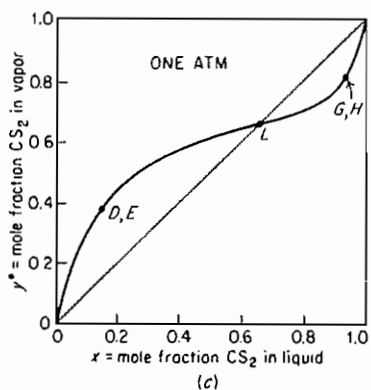
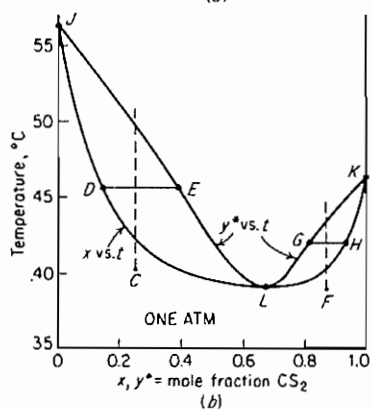
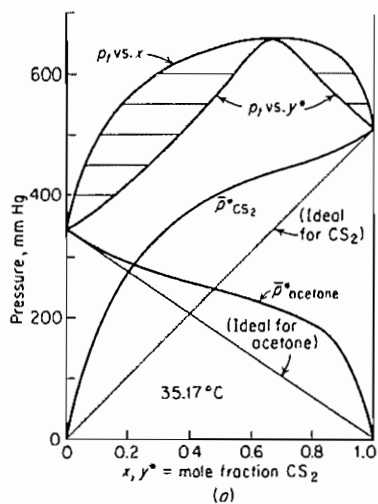


Figure 9.7 Minimum-boiling azeotropism in the system carbon disulfide–acetone: (a) at constant temperature; (b) and (c) at constant pressure.

azeotropism at this pressure, which represents the minimum-boiling temperature for this system. For all mixtures of composition less than L , such as those at C , the equilibrium vapor (E) is richer in the more volatile component than the liquid (D). For all mixtures richer than L , however, such as at F , the equilibrium vapor (G) is less rich in the more volatile substance than the liquid (H). A mixture of composition L gives rise to a vapor of composition identical with the liquid, and it consequently boils at constant temperature and without change in composition. If solutions either at D or H are boiled in an open vessel with continuous escape of the vapors, the temperature and composition of the residual liquids in each case move along the lower curve away from point L (toward K for a liquid at H , and toward J for one at D).

Solutions like these cannot be completely separated by ordinary distillation methods at this pressure, since at the azeotropic composition $y^* = x$ and $\alpha = 1.0$.† The azeotropic composition as well as its boiling point changes with pressure. In some cases, changing the pressure may eliminate azeotropism from the system.

Azeotropic mixtures of this sort are very common, and thousands have been recorded [25]. One of the most important is the ethanol-water azeotrope which at 1 atm occurs at 89.4 mole percent ethanol and 78.2°C. Azeotropism disappears in this system at pressures below 70 mmHg.

Partial liquid miscibility Some substances exhibit such large positive deviations from ideality that they do not dissolve completely in the liquid state, e.g., isobutanol-water (Fig. 9.8). The curves through points C and E represent the solubility limits of the constituents at relatively low temperatures. Mixtures of composition and temperature represented by points within the central area, such as point D , form two liquid phases at equilibrium at C and E , and line CE is a liquid tie line. Mixtures in the regions on either side of the solubility limits such as at F are homogeneous liquids. The solubility ordinarily increases with increased temperature, and the central area consequently decreases in width. If the pressure were high enough for vaporization not to occur, the liquid-solubility curves would continue along the broken extensions as shown. At the prevailing pressure, however, vaporization occurs before this can happen, giving rise to the branched vapor-liquid equilibrium curves. For homogeneous liquids such as that at F , the vapor-liquid equilibrium phenomena are normal, and such a mixture boils initially at H to give the first bubble of vapor of composition J . The same is true of any solution richer than M , except that here the vapor is leaner in the more volatile component. Any two-phase liquid mixture within the composition range from K to M will boil at the temperature of the line KM , and all these give rise to the same vapor of composition L . A liquid mixture of average composition L , which produces a vapor of the same composition, is sometimes called a

† For compositions to the right of L (Fig. 9.7) α as usually computed is less than unity, and the reciprocal of α is then ordinarily used.

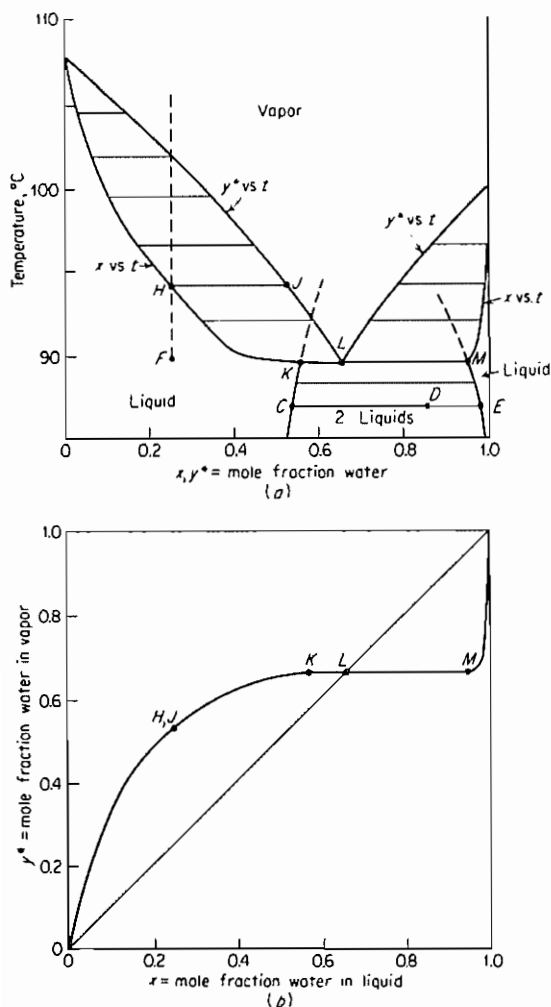


Figure 9.8 Isobutanol-water at 1 std atm.

heteroazeotrope. The corresponding distribution diagram with the tie line HJ , solubility limits at the bubble point K and M , and the azeotropic point L is shown in Fig. 9.8*b*.

In relatively few instances the azeotropic composition lies outside the limits of solubility, as in the systems methyl ethyl ketone–water and phenol–water. In others, especially when the components have a very large difference in their boiling points, no azeotrope can form, as for ammonia–toluene and carbon dioxide–water.

Insoluble liquids; steam distillation The mutual solubility of some liquids is so small that they can be considered substantially insoluble: points *K* and *M* (Fig. 9.8) are then for all practical purposes on the vertical axes of these diagrams. This is the case for a mixture such as a hydrocarbon and water, for example. If the liquids are completely insoluble, the vapor pressure of either component cannot be influenced by the presence of the other and each exerts its true vapor pressure at the prevailing temperature. When the sum of the separate vapor pressures equals the total pressure, the mixture boils, and the vapor composition is readily computed, assuming the applicability of the simple gas law,

$$p_A + p_B = p_t \quad (9.8)$$

$$y^* = \frac{p_A}{p_t} \quad (9.9)$$

So long as two liquid phases are present, the mixture will boil at the same temperature and produce a vapor of constant composition.

Illustration 9.2 A mixture containing 50 g water and 50 g ethylaniline, which can be assumed to be essentially insoluble, is boiled at standard atmospheric pressure. Describe the phenomena that occur.

SOLUTION Since the liquids are insoluble, each exerts its own vapor pressure, and when the sum of these equals 760 mmHg, the mixture boils.

<i>t</i> , °C	<i>p_A</i> (water), mmHg	<i>p_B</i> (ethylaniline), mmHg	<i>p_t</i> = <i>p_A</i> + <i>p_B</i> , mmHg
38.5	51.1	1	52.1
64.4	199.7	5	205
80.6	363.9	10	374
96.0	657.6	20	678
99.15	737.2	22.8	760
113.2	1225	40	1265
204		760	

The mixture boils at 99.15°C.

$$y^* = \frac{p_A}{p_t} = \frac{737.2}{760} = 0.97 \text{ mole fraction water}$$

$$1 - y^* = \frac{p_B}{p_t} = \frac{22.8}{760} = 0.03 \text{ mole fraction ethylaniline}$$

The original mixture contained 50/18.02 = 2.78 g mol water and 50/121.1 = 0.412 g mol ethylaniline. The mixture will continue to boil at 99.15°C, with an equilibrium vapor of the indicated composition, until all the water has evaporated together with 2.78(0.03/0.97) = 0.086 g mol of the ethylaniline. The temperature will then rise to 204°C, and the equilibrium vapor will be pure ethylaniline.

Note that by this method of distillation with steam, so long as liquid water is present, the high-boiling organic liquid can be made to vaporize at a temperature much lower than its normal boiling point without the necessity of a

vacuum-pump equipment operating at 22.8 mmHg. If boiled at 204°C, this compound will undergo considerable decomposition. However, the heat requirements of the steam-distillation process are great since such a large amount of water must be evaporated simultaneously. Alternatives would be (1) to operate at a different total pressure in the presence of liquid water where the ratio of the vapor pressures of the substances may be more favorable and (2) to sparge superheated steam (or other insoluble gas) through the mixture in the absence of liquid water and to vaporize the ethylaniline by allowing it to saturate the steam.

Negative Deviations from Ideality

When the total pressure of a system at equilibrium is less than the ideal value, the system is said to deviate negatively from Raoult's law. Such a situation is shown in Fig. 9.9 at constant temperature. Note that in this case, as with positive deviations, where neither vapor association nor liquid dissociation occurs, the partial pressures of the constituents of the solution approach ideality as their concentrations approach 100 percent. The constant-pressure diagram for such a case has the same general appearance as the diagrams shown in Fig. 9.2.

Maximum-boiling mixtures—azeotropes When the difference in vapor pressures of the components is not too great and in addition the negative deviations are large, the curve for total pressure against composition may pass through a minimum, as in Fig. 9.10a. This condition gives rise to a maximum in the boiling temperatures, as at point *L* (Fig. 9.10b), and a condition of azeotropism. The equilibrium vapor is leaner in the more volatile substance for liquids whose *x* is less than the azeotropic composition and greater if *x* is larger. Solutions on either side of the azeotrope, if boiled in an open vessel with escape of the vapor, will ultimately leave a residual liquid of the azeotropic composition in the vessel.

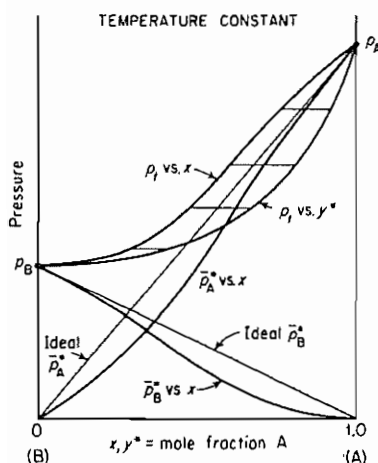


Figure 9.9 Negative deviations from ideality.

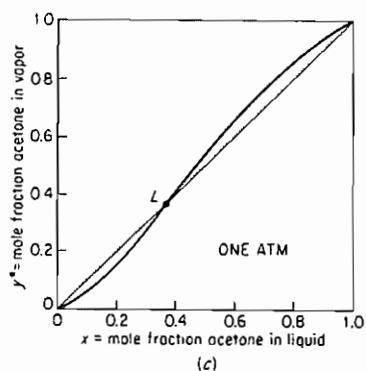
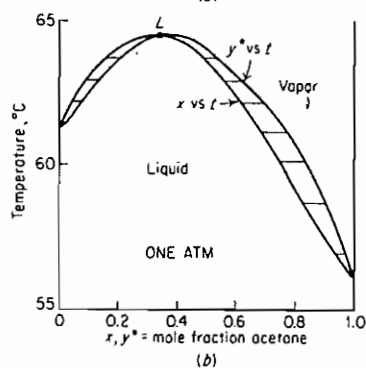
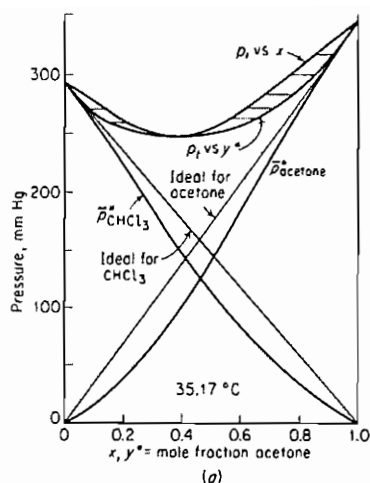


Figure 9.10 Maximum-boiling azeotropism in the system acetone-chloroform: (a) at constant temperature; (b) and (c) at constant pressure.

Maximum-boiling azeotropes are less common than the minimum type. One which is very well known is that of hydrochloric acid–water (11.1 mole percent HCl, 110°C, at 1 std atm), which can be prepared simply by boiling a solution of any strength of the acid in an open vessel. This is one method of standardizing hydrochloric acid.

Enthalpy-Concentration Diagrams

Binary vapor-liquid equilibria can also be plotted on coordinates of enthalpy vs. concentration at constant pressure. Liquid-solution enthalpies include both sensible heat and the heat of mixing the components

$$H_L = C_L(t_L - t_0)M_{av} + \Delta H_S \quad (9.10)$$

where C_L is the heat capacity of the solution, energy/(mol) (degree), and ΔH_S is the heat of solution at t_0 and the prevailing concentration referred to the pure liquid components, energy/mol solution. For saturated liquids, t_L is the bubble point corresponding to the liquid concentration at the prevailing pressure. Heat-of-solution data vary in form, and some adjustment of the units of tabulated data may be necessary. If heat is evolved on mixing, ΔH_S will be negative, and for ideal solutions it is zero. For ideal solutions, the heat capacity is the weighted average of those for the pure components.

For present purposes, saturated-vapor enthalpies can be calculated adequately by assuming that the unmixed liquids are heated separately as liquids to the gas temperature t_G (the dew point), each vaporized at this temperature, and the vapors mixed

$$H_G = y[C_{L,A}M_A(t_G - t_0) + \lambda_A M_A] + (1 - y)[C_{L,B}M_B(t_G - t_0) + \lambda_B M_B] \quad (9.11)$$

where λ_A and λ_B are latent heats of vaporization of pure substances at t_G in energy per mole and $C_{L,A}$ and $C_{L,B}$ are heat capacities of pure liquids, energy/(mole)(degree).

In the upper part of Fig. 9.11, which represents a typical binary mixture, the enthalpies of saturated vapors at their dew points have been plotted vs. y and those of the saturated liquids at their bubble points vs. x . The vertical distances between the two curves at $x = 0$ and 1 represent, respectively, the molar latent heats of B and A. The heat required for complete vaporization of solution C is $H_D - H_C$ energy/mole solution. Equilibrium liquids and vapors may be joined by tie lines, of which line EF is typical. The relation between this equilibrium phase diagram and the xy plot is shown in the lower part of Fig. 9.11. Here the point G represents the tie line EF, located on the lower plot in the manner shown. Other tie lines, when projected to the xy plot, produce the complete equilibrium-distribution curve.

Characteristics of the H_{xy} and xy diagrams [46] Let point M on Fig. 9.11 represent M mol of a mixture of enthalpy H_M and concentration z_M , and

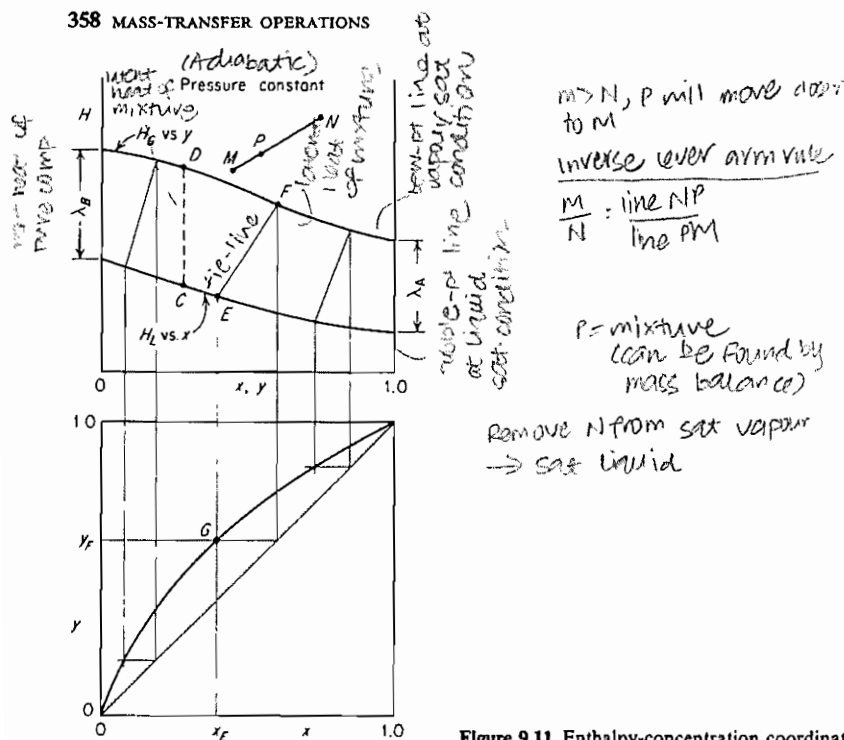


Figure 9.11 Enthalpy-concentration coordinates.

similarly N is N mol of a mixture of properties H_N , z_N . Adiabatic mixing of M and N will produce P mol of a mixture of enthalpy H_P and concentration z_P . A total material balance is

$$M + N = P \quad (9.12)$$

and a balance for component A is

$$Mz_M + Nz_N = Pz_P \quad (9.13)$$

An enthalpy balance is

$$MH_M + NH_N = PH_P \quad (9.14)$$

Elimination of P between Eqs. (9.12) and (9.13) and between (9.12) and (9.14) yields

$$\frac{M}{N} = \frac{z_N - z_P}{z_P - z_M} = \frac{H_N - H_P}{H_P - H_M} \quad (9.15)$$

This is the equation of a straight line on the enthalpy-concentration plot, passing through points (H_M, z_M) , (H_N, z_N) , and (H_P, z_P) . Point P is therefore on the straight line MN , located so that $M/N = \text{line } NP / \text{line } PM$. Similarly if mixture N were removed adiabatically from mixture P , the mixture M would result.

Consider now mixture $C(H_C, z_C)$ in Fig. 9.12. It will be useful to describe such a mixture in terms of saturated vapors and liquids, since distillation is

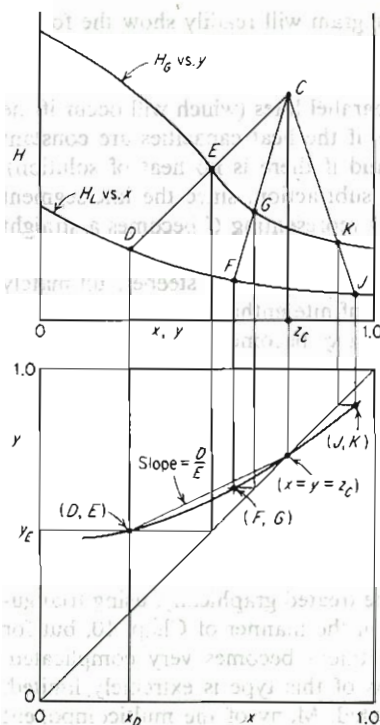


Figure 9.12 Relation between the diagrams.

mostly concerned with such mixtures. C can be considered the result of adiabatically removing saturated liquid D from saturated vapor E (DE is not a tie line), and x_D and y_E can be located on the lower diagram as shown. But C can equally well be considered as having been produced by adiabatically subtracting F from G , or J from K , or indeed by such a combination of saturated liquids and vapors given by any line from C which intersects the saturated-enthalpy curves. These, when projected to the lower diagram, form the curve shown there. Thus any point C on the Hxy diagram can be represented by the difference between saturated vapors and liquids and in turn also by a curve on the xy plot. For the combination $E-D = C$, a material balance shows

$$\frac{D}{E} = \frac{z_C - y_E}{z_C - x_D} = \frac{\text{line } CE}{\text{line } CD} \quad (9.16)$$

This is the equation on the xy diagram of the chord of slope D/E drawn between point (y_E, x_D) and $y = x = z_C$ on the 45° line. Similarly, the ratios F/G and J/K would be shown by the slopes of chords drawn from these points to $y = x = z_C$.

Consideration of the geometry of the diagram will readily show the following:

1. If the H_{Gy} and H_{Lx} curves are straight parallel lines (which will occur if the molar latent heats of A and B are equal, if the heat capacities are constant over the prevailing temperature range, and if there is no heat of solution), then $D/E = F/G = J/K$ for adiabatic subtraction, since the line-segment ratios are then equal, and the curve on xy representing C becomes a straight line.
2. If point C is moved upward, the curve on xy becomes steeper, ultimately coinciding with the 45° line when C is at infinite enthalpy.
3. If point C is on the H_{Gy} curve, the curve on xy becomes a horizontal straight line; if C is on the H_{Lx} curve, the curve on xy becomes a vertical straight line.

These concepts will be useful in understanding the applications of these diagrams.

Multicomponent Systems

Nonideal systems of three components can be treated graphically, using triangular coordinates to express the compositions in the manner of Chap. 10, but for more than three components graphical treatment becomes very complicated. Actually our knowledge of nonideal systems of this type is extremely limited, and relatively few data have been accumulated. Many of the multicomponent systems of industrial importance can be considered nearly ideal in the liquid phase for all practical purposes. This is particularly true for hydrocarbon mixtures of the same homologous series, e.g., those of the paraffin series or the lower-boiling aromatic hydrocarbons.[†] In such cases Raoult's law, or its equivalent in terms of fugacities, can be applied and the equilibria calculated from the properties of the pure components. But it is generally unsafe to predict detailed behavior of a multicomponent system from consideration of the pure components alone, or even from a knowledge of the simple binary systems that may be formed from the components. For example, three-component systems sometimes form ternary azeotropes whose equilibrium vapor and liquid have identical compositions. But the fact that one, two, or three binary azeotropes exist between the components does not make the formation of a ternary azeotrope certain, and a ternary azeotrope need not necessarily coincide with the composition of minimum or maximum bubble-point temperature for the system at constant pressure. The complexities of the systems make the digital computer helpful in dealing with the data [44].

[†] Hydrocarbons of different molecular structure, however, frequently form such nonideal solutions that azeotropism occurs, for example, in the binary systems hexane-methyl cyclopentane, hexane-benzene, and benzene-cyclohexane.

For multicomponent systems particularly, it is customary to describe the equilibrium data by means of the distribution coefficient $m^\dagger$. For component J,

$$m_J = \frac{y_J^*}{x_J} \quad (9.17)$$

where in general m_J depends upon the temperature, pressure, and composition of the mixture. The relative volatility α_{IJ} of component I with respect to J is

$$\alpha_{IJ} = \frac{y_I^*/x_I}{y_J^*/x_J} = \frac{m_I}{m_J} \quad (9.18)$$

For *ideal* solutions at moderate pressure, m_J is independent of composition and depends only upon the temperature (as this affects vapor pressure) and the total pressure:

$$m_J = \frac{p_J}{p_t} \quad (9.19)$$

and
$$\alpha_{IJ} = \frac{p_I}{p_J} \quad (9.20)$$

Bubble point For the bubble-point vapor,

$$\sum y_i^* = 1.0 \quad (9.21)$$

or
$$m_A x_A + m_B x_B + m_C x_C + \dots = 1.0 \quad (9.22)$$

With component J as a reference component,

$$\frac{m_A x_A}{m_J} + \frac{m_B x_B}{m_J} + \frac{m_C x_C}{m_J} + \dots = \frac{1.0}{m_J} \quad (9.23)$$

or
$$\alpha_{AJ} x_A + \alpha_{BJ} x_B + \alpha_{CJ} x_C + \dots = \sum \alpha_{i,J} x_i = \frac{1}{m_J} \quad (9.24)$$

The equilibrium bubble-point vapor composition is given by

$$y_i = \frac{\alpha_{i,J} x_i}{\sum \alpha_{i,J} x_i} \quad (9.25)$$

The liquid composition and total pressure having been fixed, the calculation of the temperature is made by trial to satisfy Eq. (9.24). Convergence is rapid since α 's vary only slowly with changing temperature (see Illustration 9.3).

Dew point For the dew-point liquid,

$$\sum x_i^* = 1.0 \quad (9.26)$$

$$\frac{y_A}{m_A} + \frac{y_B}{m_B} + \frac{y_C}{m_C} + \dots = 1.0 \quad (9.27)$$

[†] K rather than m is generally used to denote the distribution coefficient, but here we shall use K for the overall mass-transfer coefficient.

With component J as a reference component,

$$\frac{m_J y_A}{m_A} + \frac{m_J y_B}{m_B} + \frac{m_J y_C}{m_C} + \dots = m_J \quad (9.28)$$

$$\frac{y_A}{\alpha_{AJ}} + \frac{y_B}{\alpha_{BJ}} + \frac{y_C}{\alpha_{CJ}} + \dots = \sum \frac{y_i}{\alpha_{i,J}} = m_J \quad (9.29)$$

The dew-point liquid composition is given by

$$x_i = \frac{y_i / \alpha_{i,J}}{\sum (y_i / \alpha_{i,J})} \quad (9.30)$$

Illustration 9.3 demonstrates the method of computation.

Illustration 9.3 A solution of hydrocarbons at a total pressure 350 kN/m² (50.8 lb_f/in²) has the analysis $n\text{-C}_3\text{H}_8 = 5.0$, $n\text{-C}_4\text{H}_{10} = 30.0$, $n\text{-C}_5\text{H}_{12} = 40.0$, $n\text{-C}_6\text{H}_{14} = 25.0$ mol%. Compute the bubble point and the dew point.

SOLUTION Values of m will be taken from the DePriester nomograph [*Chem. Eng. Prog. Symp. Ser.*, 49(7), 1 (1953)]; also "Chemical Engineers' Handbook," 5th ed., fig. 13-6b, which assumes ideal liquid solutions.

Bubble point Column 1 lists the mole-fraction compositions. As a first estimate of the bubble point, 60°C is chosen, and column 2 lists the corresponding m 's at 60°C, 350 kN/m². The reference component is chosen to be the pentane, and column 3 lists the relative volatilities ($\alpha_{C_3, C_5} = 4.70/0.62 = 7.58$, etc.). $\sum \alpha_{i, C_5} x_i = 1.702$ (column 4), whence [Eq. (9.23)] $m_{C_5} = 1/1.702 = 0.588$. The corresponding temperature, from the nomograph, is 56.8°C (135°F). The calculations are repeated for this temperature in columns 5 to 7. $m_{C_5} = 1/1.707 = 0.586$, corresponding to $t = 56.7^\circ\text{C}$, which checks the 56.8°C assumed. This is the bubble point, and the corresponding bubble-point vapor composition is given in column 8 ($0.391/1.707 = 0.229$, etc.).

	x_i	m_i 60°C	α_{i, C_5}	$\alpha_{i, C_5} x_i$	m_i 56.8°C	α_{i, C_5}	$\alpha_{i, C_5} x_i$	y_i
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$n\text{-C}_3$	0.05	4.70	7.58	0.379	4.60	7.82	0.391	0.229
$n\text{-C}_4$	0.30	1.70	2.74	0.822	1.60	2.72	0.816	0.478
$n\text{-C}_5$	0.40	0.62	1.00	0.400	0.588	1.00	0.400	0.234
$n\text{-C}_6$	0.25	0.25	0.403	0.1008	0.235	0.40	0.100	0.0586
				1.702			1.707	1.000

	y_i	m_i 80°C	α_{i, C_5}	$y_i / \alpha_{i, C_5}$	m_i 83.7°C	α_{i, C_5}	$y_i / \alpha_{i, C_5}$	x_i
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$n\text{-C}_3$	0.05	6.30	6.56	0.00762	6.60	6.11	0.00818	0.0074
$n\text{-C}_4$	0.30	2.50	2.60	0.1154	2.70	2.50	0.120	0.1088
$n\text{-C}_5$	0.40	0.96	1.00	0.400	1.08	1.00	0.400	0.3626
$n\text{-C}_6$	0.25	0.43	0.448	0.558	0.47	0.435	0.575	0.5213
				1.081			1.103	1.000

Dew point The procedure is similar to that for the bubble point. The first estimate is 80°C (176°F), whence $\sum y_i / \alpha_i C_i = 1.081$ (column 4 = m_{C_3} [Eq. (9.29)]). The corresponding temperature is 83.7°C (183°F). Repetition leads to $m_{C_3} = 1.103$, corresponding to 84.0°C, which is the dew point. The corresponding dew-point-liquid compositions are listed in column 8 ($0.00818/1.103 = 0.0074$, etc.).

SINGLE-STAGE OPERATION—FLASH VAPORIZATION

Flash vaporization, or equilibrium distillation as it is sometimes called, is a single-stage operation wherein a liquid mixture is partially vaporized, the vapor allowed to come to equilibrium with the residual liquid, and the resulting vapor and liquid phases are separated and removed from the apparatus. It may be batchwise or continuous.

A typical flowsheet is shown schematically in Fig. 9.13 for continuous operation. Here the liquid feed is heated in a conventional tubular heat exchanger or by passing it through the heated tubes of a fuel-fired furnace. The pressure is then reduced, vapor forms at the expense of the liquid adiabatically, and the mixture is introduced into a vapor-liquid separating vessel. The separator shown is of the cyclone type, where the feed is introduced tangentially into a covered annular space. The liquid portion of the mixture is thrown by centrifugal force to the outer wall and leaves at the bottom, while the vapor rises through the central chimney and leaves at the top. The vapor may then pass to a condenser, not shown in the figure. Particularly for flash vaporization of a volatile substance from a relatively nonvolatile one, operation in the separator can be carried out under reduced pressure, but not so low that ordinary cooling water will not condense the vapor product.

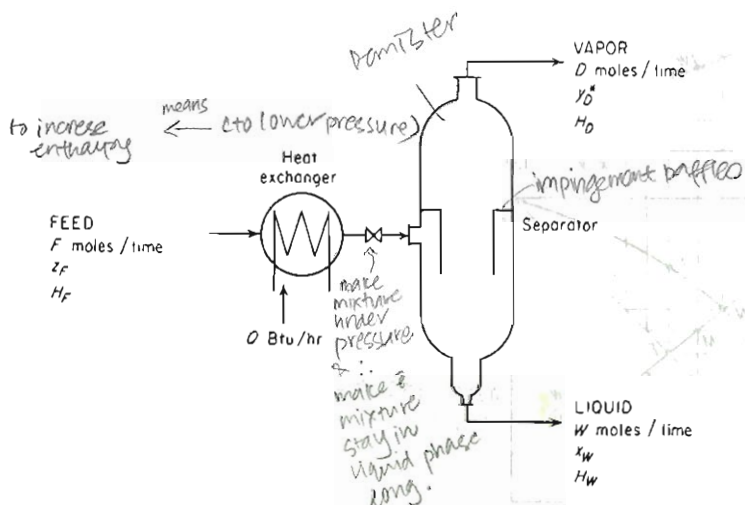


Figure 9.13 Continuous flash vaporization.

The product, D mol/time, richer in the more volatile substance, is in this case entirely a vapor. The material and enthalpy balances are

$$F = D + W \quad (9.31)$$

$$Fz_F = Dy_D + Wx_W \quad (9.32)$$

$$FH_F + Q = DH_D + WH_W \quad (9.33)$$

Solved simultaneously, these yield

$$-\frac{W}{D} = \frac{y_D - z_F}{x_W - z_F} = \frac{H_D - (H_F + Q/F)}{H_W - (H_F + Q/F)} \quad (9.34)$$

On the H_{xy} diagram, this represents a straight line through points of coordinates (H_D, y_D) representing D , (H_W, x_W) representing W , and $(H_F + Q/F, z_F)$ representing the feed mixture after it leaves the heat exchanger of Fig. 9.13. It is shown on the upper part of Fig. 9.14 as the line DW . The two left-hand members of Eq. (9.34) represent the usual single-stage operating line on distribution coordinates, of negative slope as for all single-stage (cocurrent) operations (see Chap. 5), passing through compositions representing the influent and effluent streams, points F and M on the lower figure. If the effluent streams were in equilibrium, the device would be an equilibrium stage and the products

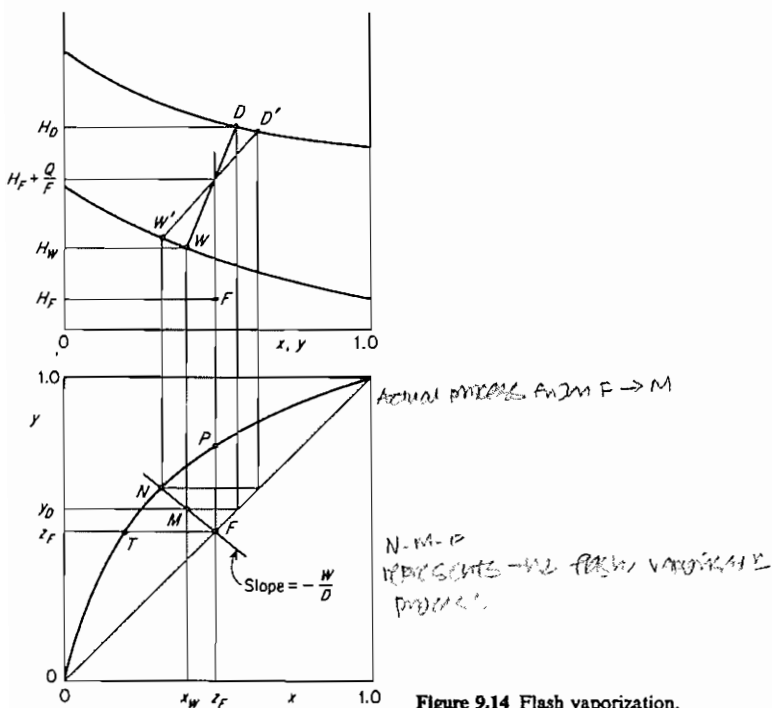


Figure 9.14 Flash vaporization.

D' and W' would be on a tie line in the upper figure and on the equilibrium curve at N on the lower figure. The richest vapor, but infinitesimal in amount, is that corresponding to P at the bubble point of the feed; and the leanest liquid, but also infinitesimal in amount, is that corresponding to T at the dew point of the feed. The compositions of the actual products will be between these limits, depending upon the extent of vaporization of the feed and the stage efficiency.

Partial Condensation

All the equations apply equally well to the case where the feed is a vapor and Q , the heat removed in the heat exchanger to produce incomplete condensation, is taken as negative. On the upper part of Fig. 9.14, point F is then either a saturated or superheated vapor.

Illustration 9.4 A liquid mixture containing 50 mol % n -heptane (A), 50 mol % n -octane (B), at 30°C , is to be continuously flash-vaporized at 1 std atm pressure to vaporize 60 mol % of the feed. What will be the composition of the vapor and liquid and the temperature in the separator for an equilibrium stage?

SOLUTION Basis: $F = 100$ mol feed, $z_F = 0.50$. $D = 60$ mol, $W = 40$ mol, $-W/D = -40/60 = -0.667$.

The equilibrium data were determined in Illustration 9.1 and are plotted in Fig. 9.15. The point representing the feed composition is plotted at P , and the operating line is drawn with a slope -0.667 to intersect the equilibrium curve at T , where $y_D^* = 0.575$ mole fraction heptane and $x_W = 0.387$ mole fraction heptane. The temperature at T is 113°C .

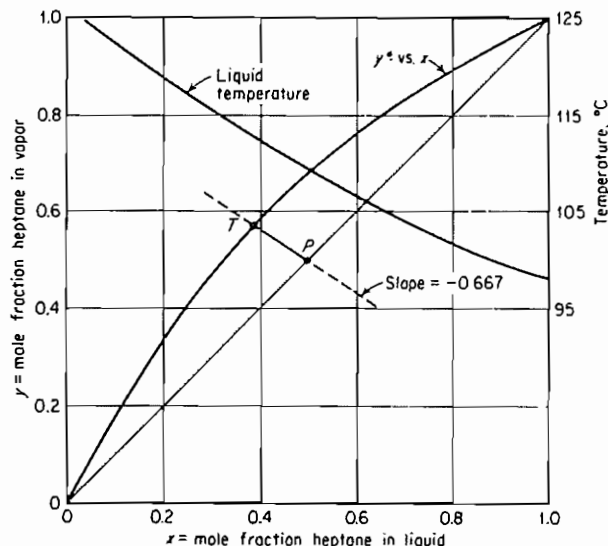


Figure 9.15 Solution to Illustration 9.4.

Multicomponent Systems—Ideal solutions

For mixtures leaving an equilibrium stage containing components A, B, C, etc., the equilibrium relation for any component J can be written

$$y_{J,D}^* = m_J x_{J,W} \quad (9.35)$$

Equation (9.34) also applies for each of the components, and when combined with Eq. (9.35) for any component J, for an equilibrium stage gives

$$\frac{W}{D} = \frac{m_J x_{J,W} - z_{J,F}}{z_{J,F} - x_{J,W}} = \frac{y_{J,D}^* - z_{J,F}}{z_{J,F} - y_{J,D}^*/m_J} \quad (9.36)$$

This provides the following expression, useful for equilibrium vaporization,

$$y_{J,D}^* = \frac{z_{J,F}(W/D + 1)}{1 + W/Dm_J} \quad (9.37)$$

$$\sum y_D^* = 1.0 \quad (9.38)$$

and for condensation,

$$x_{J,W} = \frac{z_{J,F}(W/D + 1)}{m_J + W/D} \quad (9.39)$$

$$\sum x_W = 1.0 \quad (9.40)$$

Thus Eq. (9.37) can be used for each of the components with appropriate values of m and z_F , and the sum of the y_D^* 's so calculated must equal unity if the correct conditions of W/D , temperature, and pressure have been chosen. A similar interpretation is used for Eqs. (9.39) and (9.40). These expressions reduce in the limit to the bubble point ($W/D = \infty$) and dew point ($W/D = 0$), Eqs. (9.22) and (9.30), respectively.

Illustration 9.5 A liquid containing 50 mol % benzene (A), 25 mol % toluene (B), and 25 mol % *o*-xylene (C) is flash-vaporized at 1 std atm pressure and 100°C. Compute the amounts of liquid and vapor products and their composition.

SOLUTION For Raoult's law, $y^* = px/p_i$, so that for each component $m = p/p_i$. $p_i = 760$ mmHg. In the following table, column 2 lists the vapor pressures p at 100°C for each substance and column 3 the corresponding value of m . The feed composition is listed in column 4. A value of W/D is arbitrarily chosen as 3.0, and Eq. (9.37) used to compute y_D^* 's in column 5. Since the sum of the y_D^* 's is not unity, a new value of W/D is chosen until finally (column 6) $W/D = 2.08$ is seen to be correct.

Basis: $F = 100$ mol.

$$100 = W + D \quad \frac{W}{D} = 2.08$$

Therefore

$$D = 32.5 \text{ mol} \quad W = 67.5 \text{ mol}$$

The composition of the residual liquid can be found by material balance or by equilibrium relation, as in column 7.

Substance	$p =$ vapor pressure, mmHg	$m = \frac{p}{760}$	z_F	$\frac{W}{D} = 3.0$	$\frac{W}{D} = 2.08$	$x_W = \frac{Fz_F - Dy_D^*}{W}$
				$\frac{z_F(W/D + 1)}{1 + W/D_m} = y_D^*$	y_D^*	$= \frac{y_D^*}{m}$
(1)	(2)	(3)	(4)	(5)	(6)	(7)
A	1370	1.803	0.50	$\frac{0.5(3 + 1)}{1 + 3/1.803} = 0.750$	0.715	0.397
B	550	0.724	0.25	0.1940	0.1983	0.274
C	200	0.263	0.25	0.0805	0.0865	0.329
				$\Sigma = 1.0245$	0.9998	1.000

Successive flash vaporizations can be made on the residual liquids in a series of single-stage operations, whereupon the separation will be better than that obtained if the same amount of vapor were formed in a single operation. As the amount of vapor formed in each stage becomes smaller and the total number of vaporizations larger, the operation approaches differential distillation in the limit.

DIFFERENTIAL, OR SIMPLE, DISTILLATION

If during an infinite number of successive flash vaporizations of a liquid only an infinitesimal portion of the liquid were flashed each time, the net result would be equivalent to a differential, or simple, distillation.

In practice this can only be approximated. A batch of liquid is charged to a kettle or still fitted with some sort of heating device such as a steam jacket, as in Fig. 9.16. The charge is boiled slowly, and the vapors are withdrawn as rapidly as they form to a condenser, where they are liquefied, and the condensate (distillate) is collected in the receiver. The apparatus is essentially a large-scale replica of the ordinary laboratory distillation flask and condenser. The first portion of the distillate will be the richest in the more volatile substance, and as distillation proceeds, the vaporized product becomes leaner. The distillate can therefore be collected in several separate batches, called *cuts*, to give a series of distilled products of various purities. Thus, for example, if a ternary mixture contained a small amount of a very volatile substance A, a majority of substance B of intermediate volatility, and a small amount of C of low volatility, the first cut, which would be small, would contain the majority of A. A large second cut would contain the majority of B reasonably pure but nevertheless contaminated with A and C, and the residue left in the kettle would be largely C. While all three cuts would contain all three substances, nevertheless some separation would have been obtained.

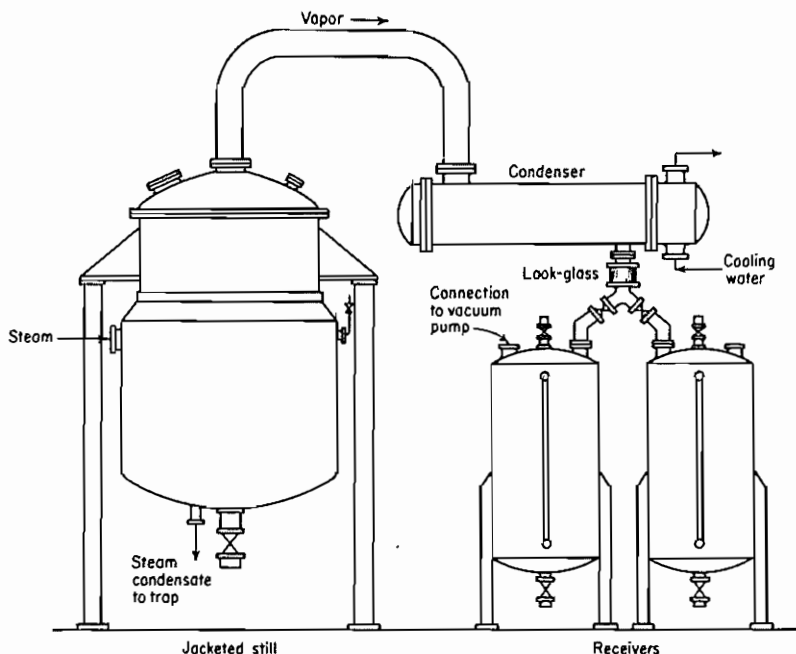


Figure 9.16 Batch still.

For such an operation to approach even approximately the theoretical characteristics of a differential distillation, it would have to proceed infinitely slowly so that the vapor issuing from the liquid would at all times be in equilibrium with the liquid. All entrainment would have to be eliminated, and there could be no cooling and condensation of the vapor before it entered the condenser. Despite the fact that these conditions are substantially impossible to attain, it is nevertheless useful to study the limiting results which a differential distillation could produce as a standard for comparison.

Binary Mixtures

The vapor issuing from a true differential distillation is at any time in equilibrium with the liquid from which it rises but changes continuously in composition. The mathematical approach must therefore be differential. Assume that at any time during the course of the distillation there are L mol of liquid in the still of composition x mole fraction A and that an amount dD mol of distillate is vaporized, of mole fraction y^* in equilibrium with the liquid. Then

we have the following material balances:

	Total material	Component A
Moles in	0	0
Moles out	dD	$y^* dD$
Moles accumulated	dL	$d(Lx) = Ldx + x dL$
In—out = accumulation	$0 - dD = dL$	$0 - y^* dD = Ldx + x dL$

The last two equations become

$$y^* dL = Ldx + x dL \quad (9.41)$$

$$\int_W^F \frac{dL}{L} = \ln \frac{F}{W} = \int_{x_W}^{x_F} \frac{dx}{y^* - x} \quad (9.42)$$

where F is the moles of charge of composition x_F and W the moles of residual liquid of composition x_W . This is known as the *Rayleigh equation*, after Lord Rayleigh, who first derived it. It can be used to determine F , W , x_F , or x_W when three of these are known. Integration of the right-hand side of Eq. (9.42), unless an algebraic equilibrium relationship between y^* and x is available, is done graphically by plotting $1/(y^* - x)$ as ordinate against x as abscissa and determining the area under the curve between the indicated limits. The data for this are taken from the vapor-liquid equilibrium relationship. The *composited* distillate composition $y_{D,av}$ can be determined by a simple material balance,

$$Fx_F = Dy_{D,av} + Wx_W \quad (9.43)$$

Differential Condensation

This is a similar operation where a vapor feed is slowly condensed under equilibrium conditions and the condensate withdrawn as rapidly as it forms. As in distillation, the results can only be approximated in practice. A derivation similar to that above leads to

$$\ln \frac{F}{D} = \int_{y_F}^{y_D} \frac{dy}{y - x^*} \quad (9.44)$$

where F is the moles of feed vapor of composition y_F and D the vaporous residue of composition y_D .

Constant Relative Volatility

If Eq. (9.2) can describe the equilibrium relation at constant pressure by use of some average relative volatility α over the concentration range involved, this can be substituted in Eq. (9.42) to yield

$$\ln \frac{F}{W} = \frac{1}{\alpha - 1} \ln \frac{x_F(1 - x_W)}{x_W(1 - x_F)} + \ln \frac{1 - x_W}{1 - x_F} \quad (9.45)$$

and graphical integration can be avoided. This can be rearranged to another useful form

$$\log \frac{Fx_F}{Wx_W} = \alpha \log \frac{F(1-x_F)}{W(1-x_W)} \quad (9.46)$$

which relates the number of moles of A remaining in the residue, Wx_W , to that of B remaining, $W(1-x_W)$. These expressions are most likely to be valid for ideal mixtures, for which α is most nearly constant.

Illustration 9.6 Suppose the liquid of Illustration 9.4 [50 mol % *n*-heptane (A), 50 mol % *n*-octane (B)] were subjected to a differential distillation at atmospheric pressure, with 60 mol % of the liquid distilled. Compute the composition of the composited distillate and the residue.

SOLUTION Basis: $F = 100$ mol. $x_F = 0.50$, $D = 60$ mol, $W = 40$ mol. Eq. (9.42):

$$\ln \frac{100}{40} = 0.916 = \int_{x_W}^{0.50} \frac{dx}{y^* - x}$$

The equilibrium data are given in Illustrations 9.1 and 9.4. From these, the following are calculated:

x	0.50	0.46	0.42	0.38	0.34	0.32
y^*	0.689	0.648	0.608	0.567	0.523	0.497
$1/(y^* - x)$	5.29	5.32	5.32	5.35	5.50	5.65

x as abscissa is plotted against $1/(y^* - x)$ as ordinate, and the area under the curve obtained beginning at $x_F = 0.50$. When the area equals 0.916, integration is stopped; this occurs at $x_W = 0.33$ mole fraction heptane in the residue. The composited distillate composition is obtained through Eq. (9.43),

$$100(0.50) = 60y_{D,av} + 40(0.33) \\ y_{D,av} = 0.614 \text{ mole fraction heptane}$$

Note that, for the same percentage vaporization, the separation in this case is better than that obtained by flash vaporization; i.e., each product is purer in its majority component.

Alternatively, since for this system the average $\alpha = 2.16$ at 1 atm (Illustration 9.1), Eq. (9.46):

$$\log \frac{100(0.5)}{40x_W} = 2.16 \log \frac{100(1-0.5)}{40(1-x_W)}$$

from which by trial and error $x_W = 0.33$.

Multicomponent Systems—Ideal Solutions

For multicomponent systems forming ideal liquid solutions, Eq. (9.46) can be written for any two components. Ordinarily one component is chosen on which to base the relative volatilities, whereupon Eq. (9.46) is written once for each of the others. For example, for substance J, with relative volatility based on substance B,

$$\log \frac{Fx_{J,F}}{Wx_{J,W}} = \alpha_{JB} \log \frac{Fx_{B,F}}{Wx_{B,W}} \quad (9.47)$$

and $\sum x_{j,w} = 1.0$ (9.48)

where $x_{j,F}$ is the mole fraction of j in the feed and $x_{j,w}$ that in the residue.

Illustration 9.7 A liquid containing 50 mol % benzene (A), 25 mol % toluene (B), and 25 mol % *o*-xylene (C) is differentially distilled at 1 atm, with vaporization of 32.5 mol % of the charge. Raoult's law applies. Compute the distillate and residue compositions. Note that this is the same degree of vaporization as in Illustration 9.5.

SOLUTION The average temperature will be somewhat higher than the bubble point of the feed (95°C, see Illustration 9.3) but is unknown. It will be taken as 100°C. Corrections can later be made by computing the bubble point of the residue and repeating the work at the average temperature, but α 's vary little with moderate changes in temperature. The vapor pressures at 100°C are tabulated and α 's calculated relative to toluene, as follows:

Substance	$p = \text{vapor pressure, } 100^\circ\text{C, mmHg}$	α	x_F
A	1370	$1370/550 = 2.49$	0.50
B	550	1.0	0.25
C	200	0.364	0.25

Basis: $F = 100$ mol, $D = 32.5$ mol, $W = 67.5$ mol. Eq. (9.47):

For A: $\log \frac{100(0.50)}{67.5x_{A,w}} = 2.49 \log \frac{100(0.25)}{67.5x_{B,w}}$

For C: $\log \frac{100(0.25)}{67.5x_{C,w}} = 0.364 \log \frac{100(0.25)}{67.5x_{B,w}}$

Eq. (9.48):

$$x_{A,w} + x_{B,w} + x_{C,w} = 1.0$$

Solving simultaneously by assuming values of $x_{B,w}$, computing $x_{A,w}$ and $x_{C,w}$, and checking their sum until it equals unity gives $x_{A,w} = 0.385$, $x_{B,w} = 0.285$, $x_{C,w} = 0.335$. The sum is 1.005, which is taken as satisfactory.

The composited distillate composition is computed by material balances. For A,

$$100(0.50) = 32.5y_{A,D} + 67.5(0.385) \quad y_{A,D} = 0.742$$

Similarly, $y_{B,D} = 0.178$ and $y_{C,D} = 0.075$

Note the improved separation over that obtained by flash vaporization (Illustration 9.5).

CONTINUOUS RECTIFICATION—BINARY SYSTEMS

Continuous rectification, or fractionation, is a multistage countercurrent distillation operation. For a binary solution, with certain exceptions it is ordinarily possible by this method to separate the solution into its components, recovering each in any state of purity desired.

Rectification is probably the most frequently used separation method we have, although it is relatively new. While simple distillation was known in the first century, and perhaps earlier, it was not until about 1830 that Aeneas Coffey

of Dublin invented the multistage, countercurrent rectifier for distilling ethanol from fermented grain mash [56]. His still was fitted with trays and down-spouts, and produced a distillate containing up to 95 percent ethanol, the azeotrope composition. We cannot do better today except by special techniques.

The Fractionation Operation

In order to understand how such an operation is carried out, recall the discussion of reboiled absorbers in Chap. 8 and Fig. 8.28. There, because the liquid leaving the bottom of an absorber is at best in equilibrium with the feed and may therefore contain substantial concentrations of volatile component, trays installed below the feed point were provided with vapor generated by a reboiler to strip out the volatile component from the liquid. This component then entered the vapor and left the tower at the top. The upper section of the tower served to wash the gas free of less volatile component, which entered the liquid to leave at the bottom.

So, too, with distillation. Refer to Fig. 9.17. Here the feed is introduced more or less centrally into a vertical cascade of stages. Vapor rising in the section above the feed (called the *absorption, enriching, or rectifying* section) is washed with liquid to remove or absorb the less volatile component. Since no extraneous material is added, as in the case of absorption, the washing liquid in this case is provided by condensing the vapor issuing from the top, which is rich in more volatile component. The liquid returned to the top of the tower is called *reflux*, and the material permanently removed is the *distillate*, which may be a vapor or a liquid, rich in more volatile component. In the section below the feed (*stripping or exhausting* section), the liquid is stripped of volatile component by vapor produced at the bottom by partial vaporization of the bottom liquid in the reboiler. The liquid removed, rich in less volatile component, is the *residue*, or *bottoms*. Inside the tower, the liquids and vapors are always at their bubble points and dew points, respectively, so that the highest temperatures are at the bottom, the lowest at the top. The entire device is called a *fractionator*.

The purities obtained for the two withdrawn products will depend upon the liquid/gas ratios used and the number of ideal stages provided in the two sections of the tower, and the interrelation of these must now be established. The cross-sectional area of the tower, however, is governed entirely by the quantities of materials handled, in accordance with the principles of Chap. 6.

Overall Enthalpy Balances

In Fig. 9.17, the theoretical trays are numbered from the top down, and subscripts generally indicate the tray from which a stream originates: for example, L_n is mol liquid/time falling from the n th tray. A bar over the quantity indicates that it applies to the section of the column below the point of introduction of the feed. The distillate product may be liquid, vapor, or a mixture. The reflux, however, must be liquid. The molar ratio of reflux to

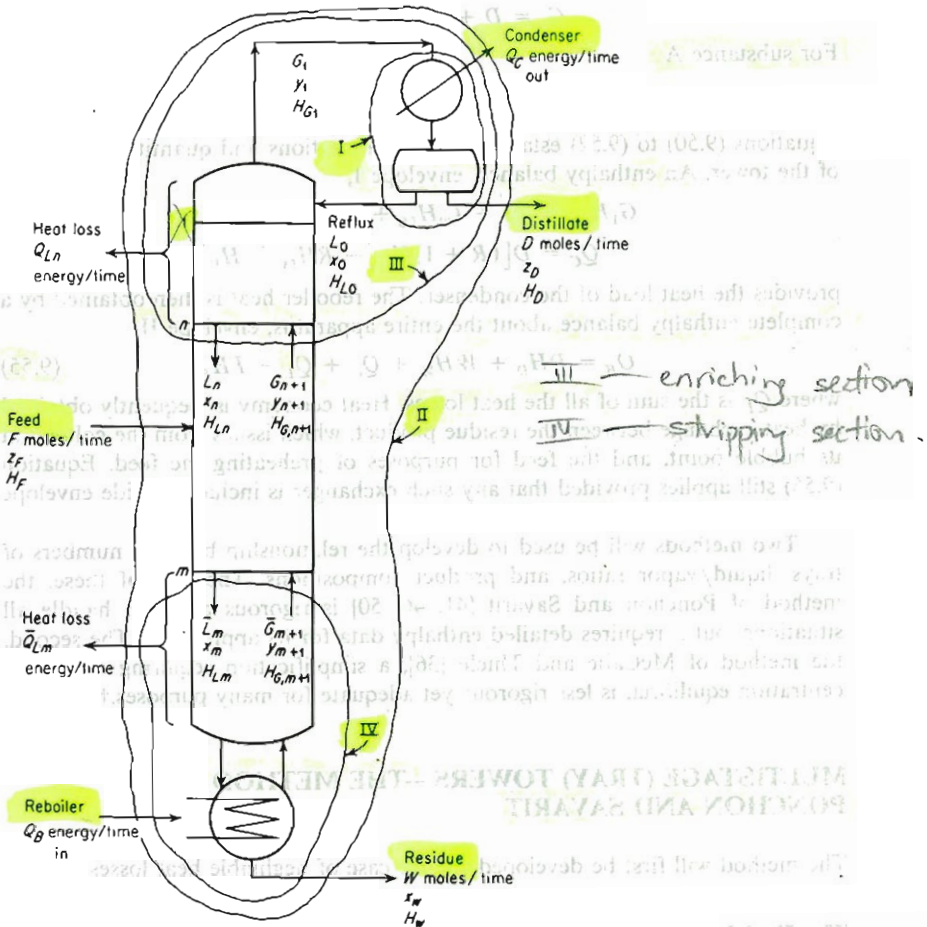


Figure 9.17 Material and enthalpy balances of a fractionator.

withdrawn distillate is the **reflux ratio**, sometimes called the **external reflux ratio**,

$$R = \frac{L_0}{D} \quad (9.49)$$

which is specified in accordance with principles to be established later.†

Consider the condenser, envelope I (Fig. 9.17). A total material balance is

$$G_1 = D + L_0 \quad (9.50)$$

† The ratio L/G is sometimes called the **internal reflux ratio**; L/F is also used for certain reflux correlations.

$$\text{or} \quad G_1 = D + RD = D(R + 1) \quad (9.51)$$

For substance A

$$G_1 y_1 = Dz_D + L_0 x_0 \quad (9.52)$$

Equations (9.50) to (9.52) establish the concentrations and quantities at the top of the tower. An enthalpy balance, envelope I,

$$G_1 H_{G1} = Q_C + L_0 H_{L0} + DH_D \quad (9.53)$$

$$Q_C = D[(R + 1)H_{G1} - RH_{L0} - H_D] \quad (9.54)$$

provides the heat load of the condenser. The reboiler heat is then obtained by a complete enthalpy balance about the entire apparatus, envelope II,

$$Q_B = DH_D + WH_W + Q_C + Q_L - FH_F \quad (9.55)$$

where Q_L is the sum of all the heat losses. Heat economy is frequently obtained by heat exchange between the residue product, which issues from the column at its bubble point, and the feed for purposes of preheating the feed. Equation (9.55) still applies provided that any such exchanger is included inside envelope II.

Two methods will be used to develop the relationship between numbers of trays, liquid/vapor ratios, and product compositions. The first of these, the method of Ponchon and Savarit [41, 46, 50] is rigorous and can handle all situations, but it requires detailed enthalpy data for its application. The second, the method of McCabe and Thiele [36], a simplification requiring only concentration equilibria, is less rigorous yet adequate for many purposes.†

MULTISTAGE (TRAY) TOWERS—THE METHOD OF PONCHON AND SAVARIT

The method will first be developed for the case of negligible heat losses.

The Enriching Section III

Consider the enriching section through tray n , envelope III, Fig. 9.17. Tray n is any tray in this section. Material balances for the section are, for total material,

$$G_{n+1} = L_n + D \quad (9.56)$$

and for component A,

$$G_{n+1} y_{n+1} = L_n x_n + Dz_D \quad (9.57)$$

$$G_{n+1} y_{n+1} - L_n x_n = (Dz_D) \text{ net gain } A \quad (9.58)$$

The left-hand side of Eq. (9.58) represents the difference in rate of flow of

† The treatment of each method is complete in itself, independent of the other. For instructional purposes, they may be considered in either order, or one may be omitted entirely.

component A, up - down, or the net flow upward. Since for a given distillation the right-hand side is constant, it follows that the difference, or net rate of flow of A upward, is constant, independent of tray number in this section of the tower, and equal to that permanently withdrawn at the top.

An enthalpy balance, envelope III, with heat loss negligible, is

$$(III) \quad G_{n+1}H_{G_{n+1}} = L_nH_{L_n} + \underbrace{Q_c + DH_D}_{\text{heat removed}} \quad (9.59)$$

Let Q' be the heat removed in the condenser and the permanently removed distillate, per mole of distillate. Then

$$Q' = \frac{Q_c + DH_D}{D} = \frac{Q_c}{D} + H_D \quad (9.60)$$

and

$$G_{n+1}H_{G_{n+1}} - L_nH_{L_n} = \underbrace{(DQ')}_{\text{constant}} \quad (9.61)$$

The left-hand side of Eq. (9.61) represents the difference in rate of flow of heat, up - down, or the net flow upward. Since for a given set of circumstances the right-hand side is constant, the difference, or net rate of flow upward, is constant, independent of tray number in this section of the tower, and equal to that permanently taken out at the top with the distillate and at the condenser.

Elimination of D between Eqs. (9.56) and (9.57) and between Eqs. (9.56) and (9.61) yields

$$\frac{L_n}{G_{n+1}} = \frac{z_D - y_{n+1}}{z_D - x_n} = \frac{Q' - H_{G_{n+1}}}{Q' - H_{L_n}} \quad (9.62)$$

L_n/G_{n+1} is called the *internal reflux ratio*.

On the Hxy diagram, Eq. (9.62) is the equation of a straight line through $(H_{G_{n+1}}, y_{n+1})$ at G_{n+1} , (H_{L_n}, x_n) at L_n , and (Q', z_D) at Δ_D . The last is called a *difference point*, since its coordinates represent differences in rates of flow:

$$\Delta_D \left\{ \begin{aligned} Q' &= \frac{\text{difference in heat flow, up - down}}{\text{net moles total substance out}} = \frac{\text{net heat out}}{\text{net moles out}} \\ z_D &= \frac{\text{difference in flow of component A, up - down}}{\text{net moles total substance out}} = \frac{\text{net moles A out}}{\text{net moles out}} \end{aligned} \right.$$

Δ_D then represents a fictitious stream, in amount equal to the net flow outward (in this case D) and of properties (Q', z_D) such that

$$G_{n+1} - L_n = \Delta_D \quad (9.63)$$

On the xy diagram, Eq. (9.62) is the equation of a straight line of slope L_n/G_{n+1} , through (y_{n+1}, x_n) and $y = x = z_D$. These are plotted on Fig. 9.18, where both diagrams are shown.

Figure 9.18 is drawn for a total condenser. The distillate D and reflux L_0 then have identical coordinates and are plotted at point D . The location shown indicates that they are below the bubble point. If they were at the bubble point, D would be on the saturated-liquid curve. The saturated vapor G_1 from the top tray, when totally condensed, has the same composition as D and L_0 . Liquid L_1

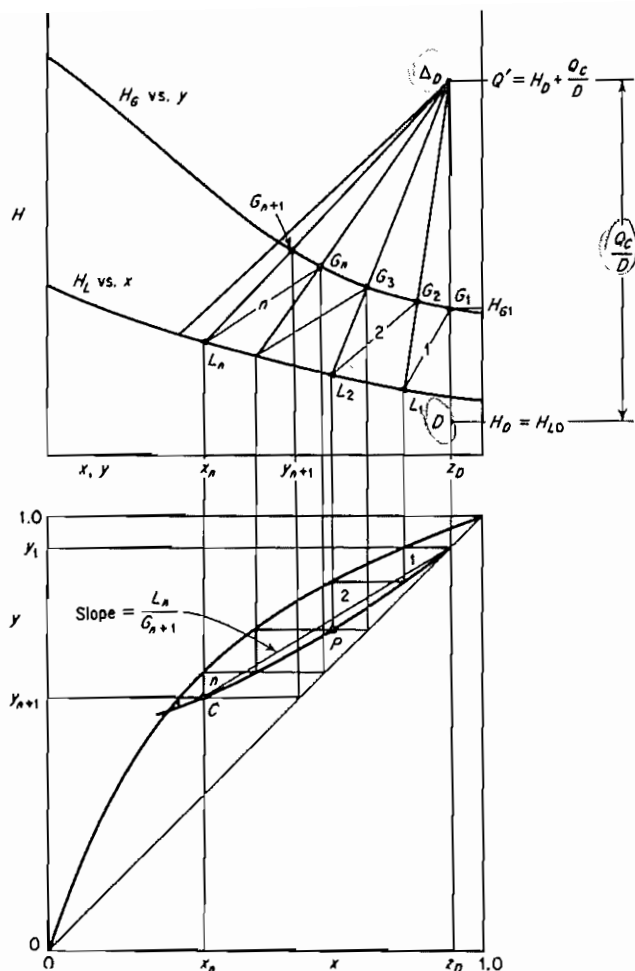


Figure 9.18 Enriching section, total condenser, reflux below the bubble point.

leaving ideal tray 1 is in equilibrium with G_1 and is located at the end of tie line 1. Since Eq. (9.62) applies to all trays in this section, G_2 can be located on the saturated-vapor curve by a line drawn from L_1 to Δ_D ; tie line 2 through G_2 locates L_2 , etc. Thus, alternate tie lines (each representing the effluents from an ideal tray) and construction lines through Δ_D provide the stepwise changes in concentration occurring in the enriching section. Intersections of the lines radiating from Δ_D with the saturated-enthalpy curves, such as points G_3 and L_2 ,

when projected to the lower diagram, produce points such as P . These in turn produce the operating curve CP , which passes through $y = x = z_D$. The tie lines, when projected downward, produce the equilibrium-distribution curve, and the stepwise nature of the concentration changes with tray number then becomes obvious. The difference point Δ_D is used in this manner for all trays in the enriching section, working downward until the feed tray is reached.

Enriching trays can thus be located on the Hxy diagram alone by alternating construction lines to Δ_D and tie lines, each tie line representing an ideal tray. As an alternative, random lines radiating from Δ_D can be drawn, their intersections with the H_Gy and H_Lx curves plotted on the xy diagram to produce the operating curve, and the trays determined by the step construction typical of such diagrams.

At any tray n (compare Fig. 9.12) the L_n/G_{n+1} ratio is given by the ratio of line lengths $\Delta_D G_{n+1}/\Delta_D L_n$ on the upper diagram of Fig. 9.18 or by the slope of the chord as shown on the lower diagram. Elimination of G_{n+1} between Eqs. (9.56) and (9.62) provides

$$\frac{L_n}{D} = \frac{Q' - H_{G_{n+1}}}{H_{G_{n+1}} - H_{L_n}} = \frac{z_D - y_{n+1}}{y_{n+1} - x_n} \quad (9.64)$$

Applying this to the top tray provides the external reflux ratio, which is usually the one specified:

$$R = \frac{L_0}{D} = \frac{Q' - H_{G_1}}{H_{G_1} - H_{L_0}} = \frac{\text{line } \Delta_D G_1}{\text{line } G_1 L_0} = \frac{\text{line } \Delta_D G_1}{\text{line } G_1 D} \quad (9.65)$$

For a given reflux ratio, the line-length ratio of Eq. (9.65) can be used to locate Δ_D vertically on Fig. 9.18, and the ordinate Q' can then be used to compute the condenser heat load.

In some cases a *partial condenser* is used, as in Fig. 9.19. Here a saturated vapor distillate D is withdrawn, and the condensate provides the reflux. This is frequently done when the pressure required for complete condensation of the vapor G_1 , at reasonable condenser temperatures, would be too large. The Δ_D is plotted at an abscissa y_D corresponding to the composition of the withdrawn distillate. Assuming that an equilibrium condensation is realized, reflux L_0 is at the end of the tie line C . G_1 is located by the construction line $L_0 \Delta_D$, etc. In the lower diagram, the line MN solves the equilibrium-condensation problem (compare Fig. 9.14). The reflux ratio $R = L_0/D = \text{line } \Delta_D G_1 / \text{line } G_1 L_0$, by application of Eq. (9.65). It is seen that the equilibrium partial condenser provides one equilibrium tray's worth of rectification. However, it is safest not to rely on such complete enrichment by the condenser but instead to provide trays in the tower equivalent to all the stages required.

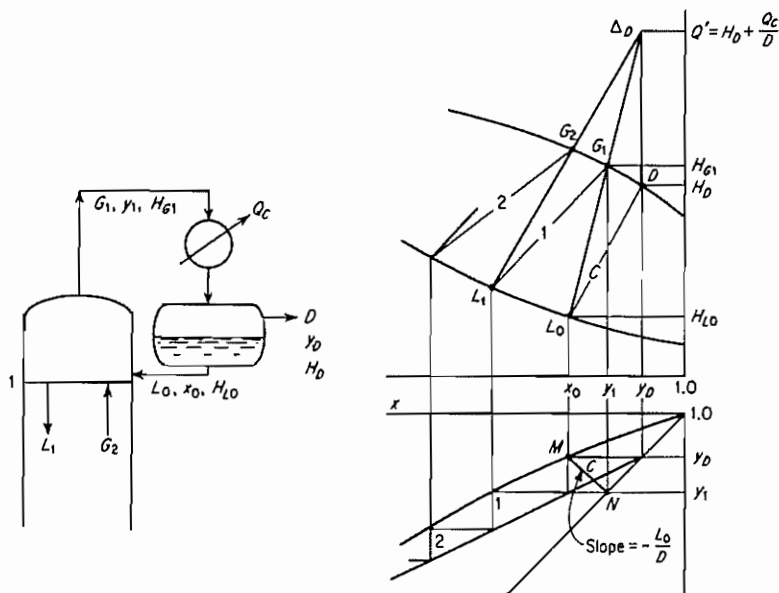


Figure 9.19 Partial condenser.

The Stripping Section

Consider the envelope IV, Fig. 9.17, where tray m is any tray in the stripping section. A balance for total material is

$$\bar{L}_m = \bar{G}_{m+1} + W \quad (9.66)$$

and, for component A,

$$\bar{L}_m x_m = \bar{G}_{m+1} y_{m+1} + W x_w \quad (9.67)$$

$$\bar{L}_m x_m - \bar{G}_{m+1} y_{m+1} = W x_w \quad (9.68)$$

The left-hand side of Eq. (9.68) represents the difference in rate of flow of component A, down – up, or the net flow downward. Since the right-hand side is a constant for a given distillation, the difference is independent of tray number in this section of the tower and equal to the rate of permanent removal of A out the bottom. An enthalpy balance is

$$\bar{L}_m H_{L_m} + Q_B = \bar{G}_{m+1} H_{G_{m+1}} + W H_w \quad (9.69)$$

Define Q'' as the net flow of heat outward at the bottom, per mole of residue

$$Q'' = \frac{W H_w - \bar{G}_{m+1} H_{G_{m+1}}}{W} = H_w - \frac{Q_B}{W} \quad (9.70)$$

whence
$$\bar{L}_m H_{L_m} - \bar{G}_{m+1} H_{G_{m+1}} = W Q'' \quad (9.71)$$

The left-hand side of Eq. (9.71) is the difference in rate of flow of heat, down - up, which then equals the constant net rate of heat flow out the bottom for all trays in this section.

Elimination of W between Eqs. (9.66) and (9.67) and between Eqs. (9.66) and (9.71) provides

$$\frac{\bar{L}_m}{\bar{G}_{m+1}} = \frac{y_{m+1} - x_w}{x_m - x_w} = \frac{H_{G_{m+1}} - Q''}{H_{L_m} - Q''} \quad (9.72)$$

On the Hxy diagram, Eq. (9.72) is a straight line through $(H_{G_{m+1}}, y_{m+1})$ at $\bar{G}_{m+1}$, (H_{L_m}, x_m) at $\bar{L}_m$, and (Q'', x_w) at Δ_w . Δ_w is a difference point, whose coordinates mean

$$\Delta_w \left\{ \begin{aligned} Q'' &= \frac{\text{difference in heat flow, down - up}}{\text{net moles of total substance out}} = \frac{\text{net heat out}}{\text{net moles out}} \\ x_w &= \frac{\text{difference in flow of component A, down - up}}{\text{net moles of total substance out}} = \frac{\text{moles A out}}{\text{net moles out}} \end{aligned} \right.$$

Thus, Δ_w is a fictitious stream, in amount equal to the net flow outward (in this case W), of properties (Q'', x_w) ,

$$\bar{L}_m - \bar{G}_{m+1} = \Delta_w \quad (9.73)$$

On the xy diagram, Eq. (9.72) is a straight line of slope $\bar{L}_m/\bar{G}_{m+1}$, through (y_{m+1}, x_m) and $y = x = x_w$. These straight lines are plotted in Fig. 9.20 for both diagrams.

Since Eq. (9.72) applies to all trays of the stripping section, the line on the Hxy plot of Fig. 9.20 from $\bar{G}_{N_p+1}$ (vapor leaving the reboiler and entering the bottom tray N_p of the tower) to Δ_w intersects the saturated-liquid-enthalpy curve at $\bar{L}_{N_p}$, the liquid leaving the bottom tray. Vapor $\bar{G}_{N_p}$ leaving the bottom tray is in equilibrium with liquid $\bar{L}_{N_p}$ and is located on the tie line N_p . Tie lines projected to the xy diagram produce points on the equilibrium curve, and lines through Δ_w provide points such as T on the operating curve. Substitution of Eq. (9.66) into Eq. (9.72) provides

$$\frac{\bar{L}_m}{W} = \frac{H_{G_{m+1}} - Q''}{H_{G_{m+1}} - H_{L_m}} = \frac{y_{m+1} - x_w}{y_{m+1} - x_m} \quad (9.74)$$

The diagrams have been drawn for the type of reboiler shown in Fig. 9.17, where the vapor leaving the reboiler is in equilibrium with the residue, the reboiler thus providing an equilibrium stage of enrichment (tie line B , Fig. 9.20). Other methods of applying heat at the bottom of the still are considered later.

Stripping-section trays can thus be determined entirely on the Hxy diagram by alternating construction lines to Δ_w and tie lines, each tie line accounting for an equilibrium stage. Alternatively, random lines radiating from Δ_w can be drawn, their intersections with curves $H_G y$ and $H_L x$ plotted on the xy diagram

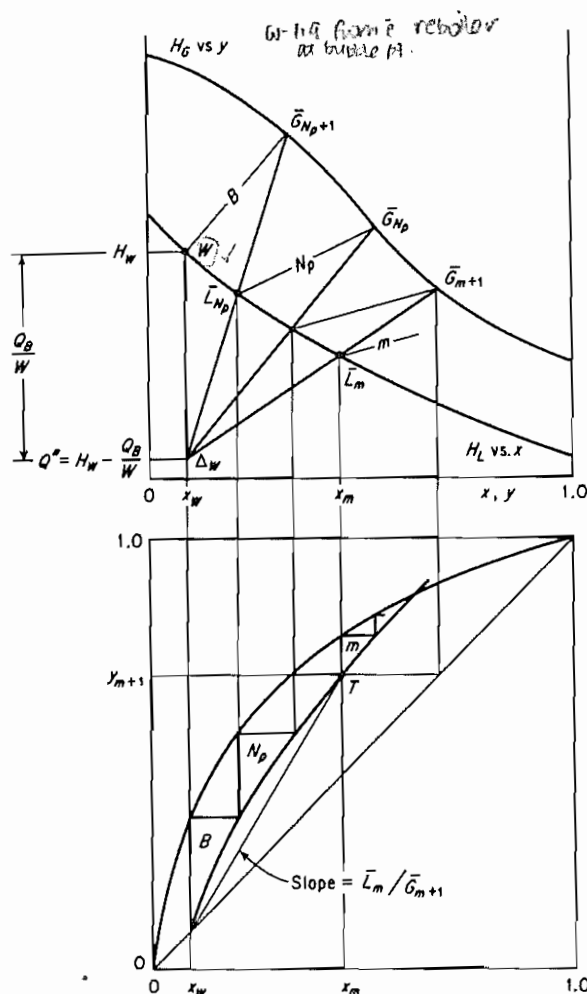


Figure 9.20 Stripping section.

to produce the operating curve, and the stages determined by the usual step construction.

The Complete Fractionator

Envelope II of Fig. 9.17 can be used for material balances over the entire device

$$F = D + W \quad (9.75)$$

$$Fz_F = Dz_D + Wx_W \quad (9.76)$$

Equation (9.55) is a complete enthalpy balance. If, in the absence of heat losses

($Q_L = 0$), the definitions of Q' and Q'' are substituted into Eq. (9.55), it becomes

$$FH_F = DQ' + WQ'' \quad (9.77)$$

If F is eliminated from Eqs. (9.75) to (9.77), there results

$$\frac{D}{W} = \frac{z_F - x_W}{z_D - z_F} = \frac{H_F - Q''}{Q' - H_F} \quad (9.78)$$

This is the equation of a straight line on the Hxy diagram, through (Q' , z_D) at Δ_D , (H_F , z_F) at F , and (Q'' , x_W) at Δ_W , as plotted in Fig. 9.21. In other words,

$$F = \Delta_D + \Delta_W \quad (9.79)$$

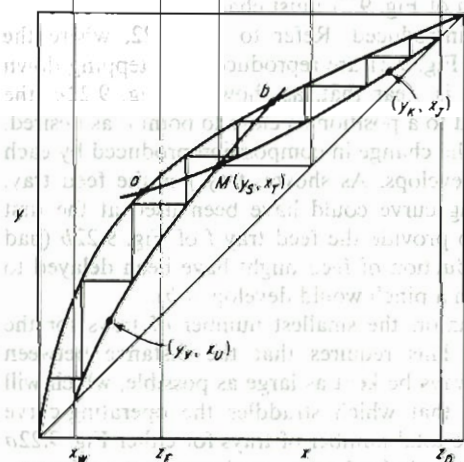
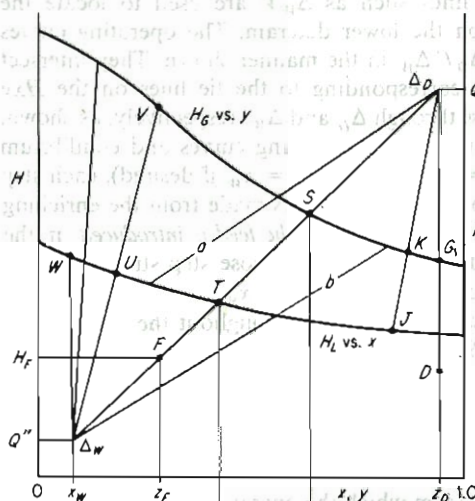


Figure 9.21 The entire fractionator. Feed below the bubble point and a total condenser.

The location of F , representing the feed, on Fig. 9.21 shows the feed in this case to be a liquid below the bubble point. In other situations, F may be on the saturated-liquid or vapor curve, between them, or above the saturated-vapor curve. In any event, the two Δ points and F must lie on a single straight line.

The construction for trays is now clear. After locating F and the concentration abscissas z_D and x_W corresponding to the products on the Hxy diagram, Δ_D is located vertically on line $x = z_D$ by computation of Q' or by the line-length ratio of Eq. (9.65) using the specified reflux ratio R . The line $\Delta_D F$ extended to $x = x_W$ locates Δ_W , whose ordinate can be used to compute Q_B . Random lines such as $\Delta_D J$ are drawn from Δ_D to locate the enriching-section operating curve on the xy diagram, and random lines such as $\Delta_W V$ are used to locate the stripping-section operating curve on the lower diagram. The operating curves intersect at M , related to the line $\Delta_D F \Delta_W$ in the manner shown. They intersect the equilibrium curve at a and b , corresponding to the tie lines on the Hxy diagram which, when extended, pass through Δ_D and Δ_W , respectively, as shown. Steps are drawn on the xy diagram between operating curves and equilibrium curve, beginning usually at $x = y = z_D$ (or at $x = y = x_W$ if desired), each step representing an equilibrium stage or tray. A change is made from the enriching to the stripping operating curve at the tray on which the feed is introduced; in the case shown the feed is to be introduced on the tray whose step straddles point M . The step construction is then continued to $x = y = x_W$.

Liquid and vapor flow rates can be computed throughout the fractionator from the line-length ratios [Eqs. (9.62), (9.64), (9.72), and (9.74)] on the Hxy diagram.

Feed-Tray Location

The material and enthalpy balances from which the operating curves are derived dictate that the stepwise construction of Fig. 9.21 must change operating lines at the tray where the feed is to be introduced. Refer to Fig. 9.22, where the equilibrium and operating curves of Fig. 9.21 are reproduced. In stepping down from the top of the fractionator, it is clear that, as shown in Fig. 9.22a, the enriching curve could have been used to a position as close to point a as desired. As point a is approached, however, the change in composition produced by each tray diminishes, and at a a pinch develops. As shown, tray f is the feed tray. Alternatively, the stripping operating curve could have been used at the first opportunity after passing point b , to provide the feed tray f of Fig. 9.22b (had the construction begun at x_W , introduction of feed might have been delayed to as near point b as desired, whereupon a pinch would develop at b).

In the design of a new fractionator, the smallest number of trays for the circumstances at hand is desired. This requires that the distance between operating and equilibrium curves always be kept as large as possible, which will occur if the feed tray is taken as that which straddles the operating-curve intersection at M , as in Fig. 9.21. The total number of trays for either Fig. 9.22a or b is of necessity larger. Delayed or early feed entry, as shown in these figures,

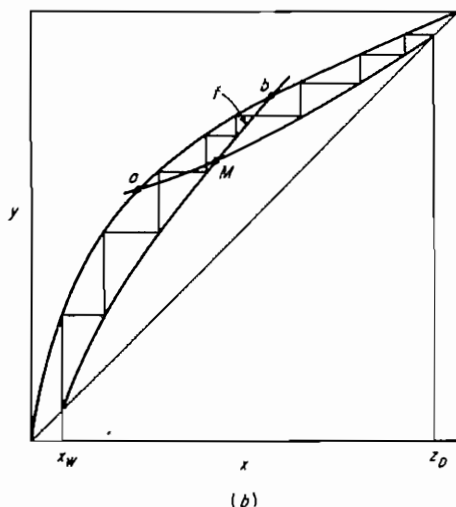
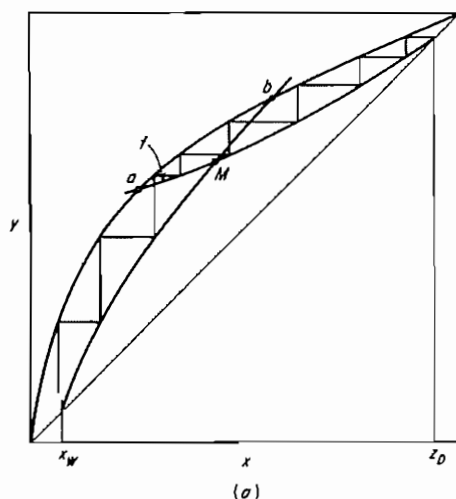


Figure 9.22 Delayed and early feed entries.

is used only where a separation is being adapted to an existing tower equipped with a feed-tray entry nozzle on a particular tray, which must then be used.

Consider again the feed tray of Fig. 9.21. It is understood that if the feed is all liquid, it is introduced above the tray in such a manner that it enters the tray along with the liquid from the tray above. Conversely, if the feed is all vapor, it is introduced underneath the feed tray. Should the feed be mixed liquid and vapor, in principle it should be separated outside the column and the liquid portion introduced above, the vapor portion below, the feed tray. This is rarely

done, and the mixed feed is usually introduced into the column without prior separation for reasons of economy. This will have only a small influence on the number of trays required [6].

Increased Reflux Ratio

As the reflux ratio $R = L_0/D$ is increased, the Δ_D difference point on Fig. 9.21 must be located at higher values of Q' . Since Δ_D , F , and Δ_W are always on the same line, increasing the reflux ratio lowers the location of Δ_W . These changes result in larger values of L_n/G_{n+1} and smaller values of $\bar{L}_m/\bar{G}_{m+1}$, and the operating curves on the xy diagram move closer to the 45° diagonal. Fewer trays are then required, but Q_C , Q_W , L , $\bar{L}$, G , and $\bar{G}$ all increase; condenser and reboiler surfaces and tower cross section must be increased to accommodate the larger loads.

Total Reflux

Ultimately, when $R = \infty$, $L_n/G_{n+1} = \bar{L}_m/\bar{G}_{m+1} = 1$, the operating curves both coincide with the 45° line on the xy plot, the Δ points are at infinity on the Hxy plot, and the number of trays required is the minimum value, N_m . This is shown in Fig. 9.23. The condition can be realized practically by returning all the distillate to the top tray as reflux and reboiling all the residue, whereupon the feed to the tower must be stopped.

Constant Relative Volatility

A useful analytical expression for the minimum number of theoretical stages can be obtained for cases where the relative volatility is reasonably constant [13, 63]. Applying Eq. (9.2) to the residue product gives

$$\frac{y_W}{1-y_W} = \alpha_W \frac{x_W}{1-x_W} \quad (9.80)$$

where α_W is the relative volatility at the reboiler. At total reflux the operating line coincides with the 45° diagonal so that $y_W = x_{N_m}$. Therefore

$$\frac{x_{N_m}}{1-x_{N_m}} = \alpha_W \frac{x_W}{1-x_W} \quad (9.81)$$

Similarly for the last tray of the column, where α_{N_m} pertains,

$$\frac{y_{N_m}}{1-y_{N_m}} = \alpha_{N_m} \frac{x_{N_m}}{1-x_{N_m}} = \alpha_{N_m} \alpha_W \frac{x_W}{1-x_W} \quad (9.82)$$

This procedure can be continued up the column until ultimately

$$\frac{y_1}{1-y_1} = \frac{x_D}{1-x_D} = \alpha_1 \alpha_2 \cdots \alpha_{N_m} \alpha_W \frac{x_W}{1-x_W} \quad (9.83)$$

If some average relative volatility α_{av} can be used,

$$\frac{x_D}{1-x_D} = \alpha_{av}^{N_m+1} \frac{x_W}{1-x_W} \quad (9.84)$$

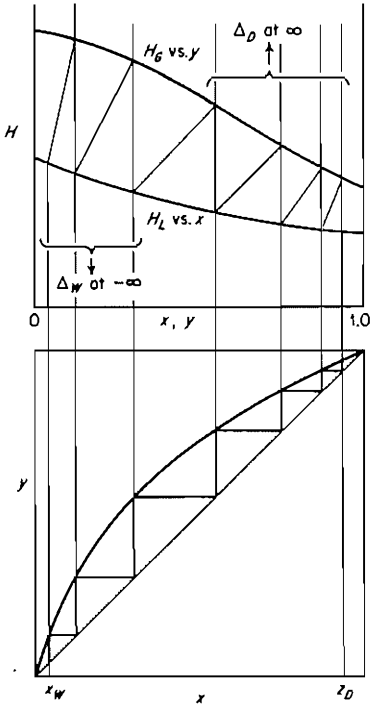


Figure 9.23 Total reflux and minimum stages.

or

$$N_m + 1 = \frac{\log \frac{x_D}{1 - x_D} \frac{1 - x_W}{x_W}}{\log \alpha_{av}} \quad (9.85)$$

which is known as *Fenske's equation*. The total minimum number of theoretical stages to produce products x_D and x_W is $N_m + 1$, which then includes the reboiler. For small variations in α , α_{av} can be taken as the geometric average of the values for the overhead and bottom products, $\sqrt{\alpha_1 \alpha_W}$. The expression can be used only with nearly ideal mixtures, for which α is nearly constant.

Minimum Reflux Ratio

The minimum reflux ratio R_m is the maximum ratio which will require an infinite number of trays for the separation desired, and it corresponds to the minimum reboiler heat load and condenser cooling load for the separation.

Refer to Fig. 9.24a, where the lightly drawn lines are tie lines which have been extended to intersect lines $x = z_D$ and $x = x_W$. It is clear that if Δ_D were located at point K, alternate tie lines and construction lines to Δ_D at the tie line k would coincide, and an infinite number of stages would be required to reach tie line k from the top of the tower. The same is true if Δ_W is located at point J.

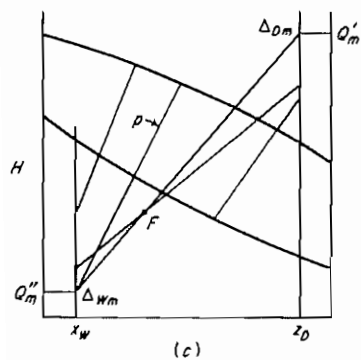
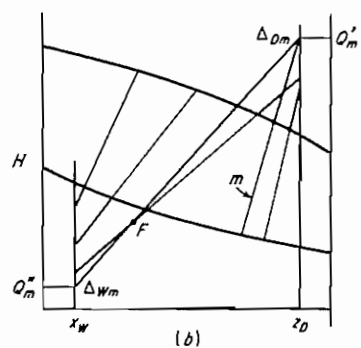
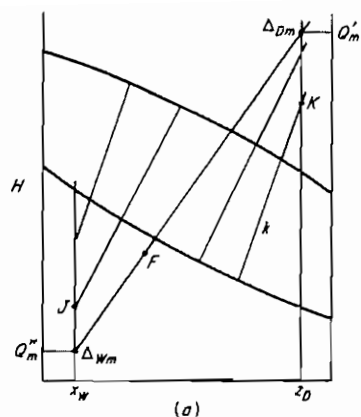


Figure 9.24 Minimum reflux ratio.